OS-299 (7-08)	TRANSMITTAL LETTER	**PUBLICATION:** Pub. 46 - October 2010 Edition
pennsylvania DEPARTMENT OF TRANSPORTATION		**DATE:** 10/07/10

SUBJECT:

Publication 46, Traffic Engineering Manual (October 2010 Edition)
Revisions to Chapter 6 (Temporary Traffic Control) and Chapter 11 (Traffic Studies)

INFORMATION AND SPECIAL INSTRUCTIONS:

This updated Pub 46 (October 2010 Edition) replaces the Pub. 46,(March 2008 Edition). The only revisions from the March 2008 Edition are in Chapter 6 (Temporary Traffic Control) & Chapter 11 (Traffic Studies), as shown below. The other chapters are the same. You may access & print the updated Pub 46 (Oct 2010) on the Department's website. This updated Pub 46 (Oct 2010) goes into effect immediately.

The following is a list of some important changes.

Additions

SECTION	Sub- Sec	CHANGE
List of Exhibits	Table of Content	Removed Exhibits 6.9-A and 6.9-B. Changed Exhibit 6.9-C to 6.9-A.
Appendix	Table of Content	Added 6F. Example, Local Road Detour MOU with Local Municipalities.
All		Removed the Month/Year Edition reference in each page's header.
6.1	6.1.3	Definitions – Added definition of "Shadow Vehicle".
6.3		Section has been completely updated to better clarify and simplify the WZSM policy, particularly as it relates to the "Significant Project" determination process. Please review carefully. Regarding Section 6.3 (WZSM policy), any existing project design that is too far along in the WZ alternatives analysis and impacts stage may continue to finish the design utilizing the pre-revision Significant Project process.
6.9	6.9.1	Temporary Concrete Barrier Types –deleted Exhibit 6.9-A, bullets (b) and (c), and Exhibit 6.9-B (Temporary Retrofit of Existing Barrier).
	6.9.5	Changed Exhibit 6.9-C to Exhibit 6.9-A.
6.10	6.10.2.3	Flashing Warning Lights on Type III Barricades - deleted bullet (b).
6.12	6.12.1	Workers' Attire – Revised entire section of 6.12.1 to reflect the new final rule for fire fighters in work zones or at an incident within the right-of-way of a federal aid roadway.
6.14	6.14.3.3	Late Merge Concept - Modified/New language was added.

11.1	11.1.2	Definitions - Three definitions were added on pge 11.1-4.
11.17		Traffic Studies (11.17 & Appendix 11-G) - This new section has been added as per SOL 470-08-7.
Appendix	6E 6F	Police Arrest Form - Page 6E-1 revised. Added Example, Local Road Detour MOU with Local Municipalities.

CANCEL AND DESTROY THE FOLLOWING:

Publication PUB 46 (March 2008 Edition).

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pennsylvania

DEPARTMENT OF TRANSPORTATION

Traffic Engineering Manual

Bureau of Highway Safety and Traffic Engineering

Publication 46

October 2010

Commonwealth of Pennsylvania
Department of Transportation

Traffic Engineering Manual

Publication 46
March 2010

List of Chapters

There are hyperlinks throughout this document that should provide network connections to other publications, regulations, Vehicle Code, etc. There are also hyperlinks that reference other sections, exhibits, or appendices within the same chapter, and these should assist you in navigating within the manual.

Although not obvious by their color, the Table of Contents and the List of Exhibits within the individual chapters also work as hyperlinks. Simply, left click on the section or exhibit number, title, or page number and your computer should take you to the proper page.

You may also find the bookmarks on the left side of the screen helpful.

This October 2010 Edition only made revisions to Chapters 6 and 11. The other chapters are the same as what appeared in the previous March 2008 Edition.

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pennsylvania
DEPARTMENT OF TRANSPORTATION

Publication 46
Chapter 1

General

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1.1 Overview

1.1.1 Purpose

The purpose of traffic control devices on streets and highways is to promote highway safety and efficiency by providing for the orderly movement of all road users. Traffic control devices notify users of regulations, and provide warning and guidance needed for the reasonably safe and efficient operation of all elements of the traffic stream.

Traffic control devices and their supports should not bear any advertising messages that are not related to traffic control. It is important to recognize, however, that Tourist-Oriented Directional Signs (TODS) and Specific Services Signs are not classified as advertising, but rather motorist service signing.

1.1.2 State and Federal Mandates

The Department adopted the Federal Highway Administration's (FHWA's) *Manual on Uniform Traffic Control Devices (MUTCD)* by reference when it adopted §212.2 (67 Pa. Code §212.2) on February 3, 2006 at 36 Pa.B. 537.

The *MUTCD* is the national standard for all traffic control traffic devices installed on any street or highway in the United States. FHWA and the National Committee on Uniform Traffic Control Devices (NCUTCD) jointly develop the *MUTCD*, and the U.S. Secretary of Transportation ultimately approves it.

Throughout the *MUTCD*, it uses the words "standard," "guidance," option," and "support" to provide the information required to make appropriate decisions regarding the use of traffic control devices on streets and highways.

The *MUTCD* defines these terms as follows:

1. "Standard" – a statement of required mandatory or specifically prohibitive practice regarding a traffic control device. The verb "shall" is typically used. Options may modify standards.

2. "Guidance" – a statement of recommended, but not a mandatory, practice in typical situations. Deviations are allowable if, engineering judgment or engineering study indicates the deviation to be appropriate. The verb "should" is typically used.

3. "Option" – a statement of practice that carries no requirement or recommendation. The verb "may" is typically used.

4. "Support" – an informational statement that carries no degree of mandate, recommendation, authorization, prohibition or an enforceable condition. Support statements use the verbs "shall," "should," and "may."

The *MUTCD* and this manual both reference many other publications. Section 1.1.10 identifies both these national publications, and many PennDOT manuals, standards and specifications.

This manual supplements the *MUTCD* and the other referenced manuals, and is designed for the use of PennDOT employees and those working for PennDOT, who are responsible to design, place and maintain signs on all State highways.

1.1.3 Principles of Traffic Control Devices

As noted in Section 1A.02 of the *MUTCD*, to be effective, a traffic control device should meet five basic requirements:

A. Fulfill a need;

B. Command attention;

C. Convey a clear, simple meaning;

D. Command respect from road users; and

E. Give adequate time for proper response.

Therefore, it is important to consider the design, placement, operation, maintenance, and uniformity of all traffic control devices in order to maximize their ability to meet these five requirements. Also, carefully consider vehicle speed as an element that governs the design, operation, placement, and location of virtually all traffic control devices.

The proper use of traffic control devices should provide the reasonable and prudent road user with the information necessary to use the streets, highways, pedestrian facilities, and bikeways, both safely and lawfully. Uniformity of the meaning and application of traffic control devices is vital to their effectiveness.

1.1.4 Engineering Study and Engineering Judgment

Always base the decision to use a particular device at a specific location on an engineering study and the application of engineering judgment. Thus, while this manual provides guidance and some engineering studies, it is not a substitute for engineering judgment.

Exercise engineering judgment in the selection and application of traffic control devices, as well as in the location and design of the roads and streets that the devices complement. Therefore, a qualified traffic engineer should make the final decision concerning the application of traffic restrictions.

1.1.5 Requests to Experiment with Unique Traffic Control Devices

If personnel from an Engineering District or the Central Office wish to experiment with a traffic control device, or request an official change to or an interpretation of the requirements of the *MUTCD* or Publication 212, they should summit a request in writing to the Traffic Engineering and Operations Division, Bureau of Highway Safety and Traffic Engineering, defining the proposal.

The request must include information in accordance with Section 1A.10 of the *MUTCD* (relating to interpretations, experimentation, changes and interim approvals), and identify the information that will be compiled during any experiment identified in the request. If deemed appropriate, the Traffic Engineering and Operations Division will prepare a request to the FHWA.

If a request to experiment is approved, the Department will typically be obligated to collect data and develop a follow-up report for FHWA.

1.1.6 Older Drivers

Another increasingly important consideration is our aging population. As people age, vision declines, physical fitness and flexibility diminish, the ability to focus attention decreases, and the time to react to unexpected circumstances increases. Each of these changes affects one’s ability to safely drive or use a crosswalk. For our older population to maintain their mobility without compromising safety, as transportation professionals we must consider their needs when applying traffic control devices.

Some specific difficulties that older drivers experience include:

1. Declining eyesight affects their ability to see signs, pavement markings, pedestrians, and other vehicles.

2. Difficulty reading signs and pavement markings at night due to glare from on-coming headlights, especially when the roads are wet.

3. Decreased physical fitness and their inability to turn their heads, which may make seeing some signs more difficult.

4. Slower response to unexpected conditions.

5. Incorrectly interpreting some signs and pavement markings.

6. Difficulty making left turns due to improper positioning in the turn lane and the inability to judge the distance of oncoming vehicles.

7. Slower gait, shorter steps, and slower reaction time requiring longer time for pedestrian crossing.

The Department desires to assist the older population whenever possible, which in turn generally assists everyone. FHWA's *Highway Design Handbook for Older Drivers and Pedestrians* is a good resource for helping the older driver.

Therefore, consider incorporating the following enhancements into normal Department practices:

- Overhead Street Name sign at all new and revised traffic signals.
- Advance Street Name plaques for major intersections.
- Larger and brighter signs.
- Brighter pavement markings and the use of raised pavement markers (RPMs).
- Positive offset for opposing left-turn lanes.
- Protected/prohibited left-turn signal phasing at locations where there are a large number of elderly drivers.
- Longer flashing "Don't Walk" pedestrian phase based on a walking speed of 3 feet per second in areas with a large number of elderly pedestrians.

1.1.7 Applicability of the *MUTCD*

In the *Federal Register* on December 14, 2006 (at 71 FR 75111), FHWA revised the language in 23 CFR 655.603(a) to clarify the applicability of the *MUTCD* by defining the phrase *"open to public travel,"* as used to define the term *"public road."*

- The *MUTCD* applies to all toll roads and roads within shopping centers, parking lots, airports, sports arenas, and other similar business and/or recreation facilities that are privately owned but where the public is allowed to travel without access restrictions.
- The *MUTCD* does not apply at military bases and other gated properties where access is restricted, and at private railroad grade crossings.

1.1.8 Acceptable Abbreviations for Use on Traffic Control Devices

Section 1A.14 of the *MUTCD* provides several tables with acceptable and unacceptable abbreviations for use on traffic control devices (typically signs and pavement markings). Exhibit 1.1-A combines Table 1A-1 and Table 1A-2 from the *MUTCD*, and adds several additional abbreviations that are relevant.

1.1.9 Proprietary Items

When preparing specifications for a project, specifying the use of a brand name is proprietary. Similarly, using one company's specification that other companies cannot match is also proprietary.

Therefore, when an adequate generic specification cannot be prepared for a traffic control device, specify at least three brand names followed by "or approved equal." If three brand names are unavailable or if the District or municipality desires a certain brand name, specific approval is necessary. Acquire approval by FHWA for Federal Oversight projects and the Bureau Director for Non-Federal (100 percent State), PennDOT Oversight NHS, and PennDOT Oversight Non-NHS projects.

For projects that involve proprietary items dealing with traffic control devices, submit an approval request to the Director of the Bureau of Highway Safety and Traffic Engineering, similar to the example in Appendix 1A.

When submitting a request, provide justification for deviation regarding the use of brand names, include the following information:

1. Letter from local government with justification, if applicable.

2. Reasons why a "generic" material description cannot be used, or at least three companies cannot be specified. Examples of acceptable justifications are:

    - The item is essential for synchronization with existing highway facilities and no suitable alternative exists.

    - The item is being used for research or for relatively short sections of road for experimental purposes.

    - No other items exist that are of acceptable quality.

3. If the item is not integral to construction, indicate the disposition of the item after construction.

4. See Section 4.6.2 for additional justification for traffic signal equipment.

To obtain approval for Federal Oversight projects, Districts are requested to submit the approval request with justification to the Bureau Director for coordination with the FHWA when necessary. The Bureau Director will coordinate with the District concerning the approval/disapproval of the item. The District is responsible to scan and link the approval/disapproval in ECMS. The approval letter should be submitted so that approvals can be obtained before PS&E submittal or project advertisement.

Adherence to the above will ensure that approvals are obtained in a timely manner.

This process is in accordance with 23 CFR 635.411. For further clarification, please reference *"Questions and Answers Regarding Title 23 CFR 635.411"* located at www.fhwa.dot.gov/programadmin/contracts/011106qa.cfm .

1.1.10 Laws, Regulations, Forms and Other Publications

Other useful sources of information include the following:

1. *Access to and Occupancy of Highways by Driveways and Local Roads* (67 Pa. Code Chapter 441), which regulates the location, design, construction, maintenance and drainage of access driveways and local roads within State highway right-of-way – available at http://www.pacode.com/secure/data/067/chapter441/chap441toc.html, and as PennDOT Publication 441.
2. *Americans with Disabilities Act and Architectural Barriers Act Accessibility Guidelines*, United States Access Board, July 23, 2004 – available at http://www.access-board.gov/ada-aba/final.pdf.
3. *Approved Construction Materials – Bulletin 15*, (PennDOT Pub. 35). A listing of approved materials and manufacturers – available at ftp://ftp.dot.state.pa.us/public/pdf/BOCM_MTD_LAB/PUBLICATIONS/PUB_35/BULLETIN_15.pdf.
4. *Clearview Typeface Supplement*, dated September 2, 2004, FHWA – available at http://mutcd.fhwa.dot.gov/pdfs/clearfont/CTSEng.pdf.
5. Engineering and Traffic Study Forms are available at (ftp://ftp.dot.state.pa.us/public/PubsForms/Forms/)
6. *Guidelines for the Maintenance of Traffic Signal Systems*, PennDOT Pub. 191, 1989 Edition.
7. *Guidelines for the Selection of Supplemental Guide Signs for Traffic Generators Adjacent to Freeways*, 2001 Edition (AASHTO).
8. *Highway Design Handbook for Older Drivers and Pedestrians*, Federal Highway Administration, Publication No. FHWA-RD-01-1-3, May 2001 – available at http://www.tfhrc.gov/humanfac/01103/coverfront.htm#toc.

9. *Maintenance Manual*, PennDOT Pub. 23, July 2004. See Chapter 15, entitled "Weight Restriction in Highways (Posted Highways) – available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%2023/PUB%2023.pdf.

10. *Manual of Transportation Engineering Studies*, 1994 Edition (ITE).

11. *Manual on Uniform Traffic Control Devices (MUTCD)*, 2003 Edition or latest edition (FHWA) – available at http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/pdf_index.htm.

12. *Official Traffic-Control Devices*, (67 Pa. Code Chapter 212), a regulation that adopted the *MUTCD* and establishes additional study requirements, warrants, principles and guidelines for the Commonwealth, and is available at http://www.pacode.com/secure/data/067/chapter212/chap212toc.html. The regulation with an appendix, which contains some additional guidance on engineering and traffic studies, is also available as PennDOT Pub. 212, at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20212.pdf.

13. *Pavement Marking Handbook*, an unnumbered PennDOT manual that provides detailed guidance for Department work force in the day-to-day operation of the Department's pavement marking program, including the operation of the truck-mounted and small paint machines.

14. *Pennsylvania Drivers Manual*, which includes guidance regarding the purpose and meaning of various types of traffic control devices – available at http://www.dot10.state.pa.us/pdotforms/pa_forms_manuals/padriversman.pdf.

15. *A Policy on Geometric Design of Highways and Streets, Fifth Edition, 2004 (AASHTO)*. Also known as the "Green Book."

16. *Project Office Manual or POM (PennDOT Pub. 2)* – available at ftp://ftp.dot.state.pa.us/public/pdf/BOCM_MTD_LAB/PUBLICATIONS/POM/POM_CURRENT/POMcover.pdf.

17. *Recommended Practice for Roadway Sign Lighting, RP-19 01, Sign Lighting Committee of the IESNA Roadway Lighting Committee*, Illuminating Engineering Society (IES).

18. *Retroreflective Sheeting Identification Guide*, September 2005 (FHWA) – available at http://safety.fhwa.dot.gov/roadway_dept/retro/sign/retrore_sheet_id.htm.

19. *Roadside Design Guide*, latest Edition (AASHTO).

20. *Roundabouts: An Informational Guide*, Pub. No. FHWA RD 00 067, June 2000 (FHWA) – available at http://www.tfhrc.gov/safety/00-0671.pdf.

21. *School Trip Safety Program Guidelines*, 1984 Edition (ITE).

22. *Specifications (PennDOT Pub. 408)*, detailed specifications for the construction and installation of traffic delineation devices – available at http://www.dot.state.pa.us/Internet/Bureaus/pdDesign.nsf/ConstructionSpecs408and7?OpenForm.

23. *Standard Highway Signs and Markings Book*, (FHWA) – available at http://mutcd.fhwa.dot.gov/ser-shs_millennium.htm.

24. *Traffic Control – Pavement Markings and Signing Standards* (PennDOT Pub. 111M), PennDOT's 8600 and 8700 series, available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20111M.pdf.

25. *Traffic Control Devices Handbook*, 2001 Edition (ITE).

26. *Traffic Engineering Handbook*, 1999 Edition (ITE).

27. *Travel Better, Travel Longer: A Pocket Guide to Improve Traffic Control and Mobility for Our Older Population*, FHWA OP 03 098, 2003 (FHWA) – available online at http://mutcd.fhwa.dot.gov/pdfs/PocketGuide0404.pdf.

28. *Trip Generation Manual*, Latest Edition (ITE).

29. *Uniform Vehicle Code (UVC) and Model Traffic Ordinance*, 2000 Edition, National Committee on Uniform Traffic Laws and Ordinances.

30. *Vehicle Code (75 Pa.C.S.)*. Title 75 of the Pennsylvania Consolidated Statutes, which sets forth the laws relative to travel on streets and highways, and specific actions required by the Department. The Vehicle Code is available online at http://www.dot4.state.pa.us/vehicle_code/index.shtml.

31. *Work Zone Traffic Control (PennDOT Pub. 213)*, a series of drawings for temporary traffic control – available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20213.pdf.

The Bureau of Highway Safety and Traffic Engineering maintains a comprehensive list of hyperlinks for traffic engineers at P:\bhste_shared\hyperlinks\common-hyperlinks.doc. Submit any suggestions for additions or changes to the list to the Chief, Traffic Engineering and Operations Division.

8. *Maintenance Manual*, PennDOT Pub. 23, July 2004. See Chapter 15, entitled "Weight Restriction in Highways (Posted Highways) – available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%2023/PUB%2023.pdf.

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19. *Roundabouts: An Informational Guide*, Pub. No. FHWA RD 00 067, June 2000 (FHWA) – available at http://www.tfhrc.gov/safety/00-0671.pdf.

20. *School Trip Safety Program Guidelines*, 1984 Edition (ITE).

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22. *Standard Highway Signs and Markings Book*, (FHWA) – available at http://mutcd.fhwa.dot.gov/ser-shs_millennium.htm.

23. *Traffic Control – Pavement Markings and Signing Standards* (PennDOT Pub. 111M), PennDOT's 8600 and 8700 series, available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20111M.pdf.

24. *Traffic Control Devices Handbook*, 2001 Edition (ITE).

25. *Traffic Engineering Handbook*, 1999 Edition (ITE).

26. *Travel Better, Travel Longer: A Pocket Guide to Improve Traffic Control and Mobility for Our Older Population*, FHWA OP 03 098, 2003 (FHWA) – available online at http://mutcd.fhwa.dot.gov/pdfs/PocketGuide0404.pdf.

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Exhibit 1.1-A Acceptable Abbreviations for Traffic Control Devices

Word Message	Standard Abbreviation	Required Prompt Word
Access	ACCS	Road, RD
Afternoon / Evening	PM	
Ahead	AHD	Fog*
Air Force Base	AFB	[name]*
Alternate	ALT, Alt	
Avenue	AVE, Ave	
Bicycle	BIKE	
Blocked	BLKD	Lane*, LN*
Boulevard	BLVD, Blvd	
Bridge	BRDG	[Name]*
Cannot	CANT	
CB Radio	CB	
Center	CNTR	
Chemical	CHEM	Spill
Circle	CIR, Cir	
Civil Defense	CD	
Compressed Natural	CNG	
Condition	COND	Traffic*, TRAF*
Congested	CONG	Traffic*, TRAF*
Construction	CONST	Ahead
Court	CT, Ct	
Crossing (other than highway-rail)	XING	
DE Traffic Route	DE	[Number]
Diesel Fuel	D	
Do Not	DONT	
Downtown	DWNTN	Traffic, TRAF
Drive	DR, Dr	
East	E	
Eastbound	E-BND	
Electric Vehicle	EV	
Emergency	EMER	
Entrance, Enter	ENT	

Word Message	Standard Abbreviation	Required Prompt Word
Exit	EX, EXT	Next*
Express	EXP	Lane, LN
Expressway	Expw	
Feet	FT, Ft	
FM Radio	FM	
Freeway	FRWY, FWY	
Friday	FRI	
Frontage	FRNTG	Road, RD
Hazardous	HAZ	Driving
Hazardous Material	HAZMAT	
High Occupancy Vehicle	HOV	
Highway	HWY, Hwy	
Highway-Rail Grade Crossing Pavement Marking	RXR	
Hospital	H	
Hour(s)	HR	
Information	INFO, Info	
Inherently Low Emission Vehicle	ILEV	
Interstate Traffic Route	I-	[Number]
It Is	ITS	
Junction / Intersection	JCT	
Lane	LN	
Left	LFT	
Liquid Propane Gas	LP-GAS	
Local	LOC	Traffic, TRAF
Lower	LWR	Level
Maintenance	MAINT	
Major	MAJ	Accident

Word Message	Standard Abbreviation	Required Prompt Word
MD Traffic Route	CO	[Number]
Mile(s)	MI	
Miles Per Hour	MPH	[Number]*
Minor	MNR	Accident
Minute(s)	MIN	
Monday	MON	
Morning / Late Night	AM	
NJ Traffic Route	NJ	[Number]
Normal	NORM	
North	N	
Northbound	N-BND	
NY Traffic Route	NY	[Number]
OH Traffic Route	OH	[Number]
Oversized	OVRSZ	Load
PA Traffic Route	PA	[Number]
Parking	PKING	
Parkway	PKWY	
Pavement	PVMT	Wet*
Pedestrian	PED	
Place	PL	
Pounds	LBS	
Prepare	PREP	To Stop
Quality	QLTY	Air*
Right	RHT	
Road	RD, Rd	
Roadwork	RDWK	Ahead, [Distance]
Route	RT, RTE, Rte	Best*
Saturday	SAT	
Service	SERV, Serv	
Shoulder	SHLDR	
Slippery	SLIP	
South	S	

Word Message	Standard Abbreviation	Required Prompt Word
Southbound	S-BND	
Speed	SPD	
Street	ST, St	
Sunday	SUN	
Telephone	PHONE, Phone	
Temporary	TEMP	
Terrace	TER	
Thursday	THURS	
Tires With Lugs	LUGS	
Tons of Weight	T	
Township	TWNSHP	Limits
Traffic	TRAF	
Trail	TR	
Travelers	TRAVLRS	
Tuesday	TUES	
Turnpike	TRNPK	[Name]*
Two-Way Intersection	2-WAY	
Two-Wheeled Vehicles	CYCLES	
University	UNIV, Univ	
Upper	UPR	Level
US Traffic Route	US	[Number]
Vehicle(s)	VEH	
Warning	WARN	
Wednesday	WED	
West	W	
Westbound	W-BND	
Will Not	WONT	
WV Traffic Route	WV	[Number]

* The prompt word should precede the abbreviation

1.1.11 Definitions

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

ADT – Average daily traffic – The total volume of traffic during a number of whole days, more than 1 day and less than 1 year, divided by the number of days in that period.

Active work zone – The portion of a work zone where construction, maintenance or utility workers are on the roadway or on the shoulder of the highway, and workers are adjacent to an active travel lane. Workers are not considered adjacent to an open travel lane if they are protected by a traffic barrier and no ingress or egress to the work zone exists through an opening in the traffic barrier.

Advisory speed – The recommended speed for vehicles operating on a section of highway based on the highway design, operating characteristics and conditions. When posted, the speed is displayed as a warning sign; that is, either a black-on-yellow or a black-on-orange sign.

Angle parking – Parking, other than parallel parking, which is designed and designated so that the longitudinal axis of the vehicle is not parallel with the edge of the roadway.

Assemblage –

(i) An organized gathering of people without vehicles, or with vehicles that are stationary, which encroaches onto a street or highway and interferes with the movement of pedestrian or vehicular traffic.

(ii) The term includes street fairs, block parties and other recreational events.

Bureau – The Bureau of Highway Safety and Traffic Engineering, which is the office of the Department responsible for traffic regulations and statewide policies regarding traffic control devices.

Conventional road, *conventional highway* – A highway other than an expressway or a freeway.

Corner sight distance –

(i) Available corner sight distance—The maximum measured distance along a crossing highway which a driver stopped at a side road or driveway along that highway can continuously see another vehicle approaching. For the purpose of measuring the available sight distance, the height of both the driver’s eye and the approaching vehicle should be assumed to be 3.5

feet above the road surface. In addition, the driver's eye should be assumed to be 10 feet back from the near edge of the highway or the near edge of the closest travel lane if parking is permitted along the highway.

(ii) Minimum corner sight distance—The minimum required corner sight distance based on engineering and traffic studies, to ensure the safe operation of an intersection. The minimum value is a function of the speed of the approaching vehicles and the prevailing geometrics.

Crash –

(i) A collision involving one or more vehicles.

(ii) Unless the context clearly indicates otherwise, the term only includes those collisions that require a police report; that is, the collision involves one of the following:

- Injury to or death of any person.
- Damage to any vehicle involved to the extent that it cannot be driven under its own power in its customary manner without further damage or hazard to the vehicle, to other traffic elements, or to the roadway, and therefore requires towing.

Delineator – A retroreflective device mounted on the road surface or at the side of the roadway in a series to indicate the alignment of the roadway, especially at night or in adverse weather.

Department – The Department of Transportation of the Commonwealth.

Divided highway – A highway divided into two or more roadways and so constructed as to impede vehicular traffic between the roadways by providing an intervening space, physical barrier or clearly indicated dividing section.

85th percentile speed – The speed on a roadway at or below which 85 percent of the motor vehicles travel.

Engineering and traffic study – An orderly examination or analysis of physical features and traffic conditions on or along a highway, conducted in accordance with this chapter for the purpose of ascertaining the need or lack of need of specific traffic restrictions, and the application of traffic control devices.

Expressway – A divided arterial highway for through traffic with partial control of access and generally with grade separations at major intersections.

Freeway – A limited access highway to which the only means of ingress and egress is by interchange ramps.

Grade – The up or down slope in the longitudinal direction of the highway, expressed in percent, which is the number of units of change in elevation per 100 units of horizontal distance. An upward slope is a positive grade; a downward slope is a negative grade.

Highway –

(i) The entire width between the boundary lines of every way publicly maintained when any part thereof is open to the use of the public for purposes of vehicular travel.

(ii) The term includes a roadway open to the use of the public for vehicular travel on grounds of a college or university, or public or private school, or public or historical park.

Local authorities –

(i) County, municipal and other local boards or bodies having authority to enact regulations relating to traffic.

(ii) The term includes airport authorities except when those authorities are within counties of the first class or counties of the second class.

(iii) The term also includes State agencies, boards and commissions other than the Department, and governing bodies of colleges, universities, public and private schools, public and historical parks.

MUTCD – The current edition of the Manual on Uniform Traffic Control Devices, as adopted by the Federal Highway Administration (FHWA), and available on their website.

Narrow bridge or underpass – A bridge, culvert or underpass with a two-way roadway clearance width of 16 to 18 feet, or any bridge, culvert or underpass having a roadway clearance less than the width of the approach travel lanes.

Night, *nighttime* – The time from 1/2 hour after sunset to 1/2 hour before sunrise.

Normal speed limit – The regulatory speed limit or the 85th percentile speed that existed before temporary traffic control was established, for example, prior to the beginning of a work zone.

Numbered traffic route – A highway that has been assigned an Interstate, United States or Pennsylvania route number, consisting of one, two, or three digits, sometimes with an additional designation such as business route, truck route or other similar designation.

Private parking lot – A privately owned parking lot open to the public for parking with or without restriction or charge.

Procession –

(i) An organized group of individuals, or individuals with vehicles, animals or objects, moving along a highway on the roadway, berm or shoulder in a manner that interferes with the normal movement of traffic.

(ii) The term includes walks, runs, parades and marches.

Retroreflective sheeting –

(i) Material which allows a large portion of the light coming from a point source to be returned directly back to a location near its origin, and is used to enhance the nighttime reflectivity of traffic control signs, delineators, barricades and other devices.

(ii) The term includes materials with enclosed glass bead lens and microprismatic retroreflective sheeting.

Roadway – That portion of a highway improved, designed or ordinarily used for vehicular travel, exclusive of the sidewalk, berm or shoulder. If a highway includes two or more separate roadways, the term refers to each roadway separately but not to all roadways collectively.

Safe-running speed – The average speed for a portion of highway determined by making a minimum of five test runs while periodically recording the speed at different locations while driving at a speed which is reasonable and prudent, giving consideration to the available corner and stopping sight distance, spacing of intersections, roadside development and other conditions.

School – A public, private or parochial facility for the education of students in grades kindergarten through 12.

School zone – A portion of a highway that at least partially abuts a school property or extends beyond the school property line that is used by students to walk to or from school or to or from a school bus pick-up or drop-off location at a school.

Secretary – The Secretary of the Department.

Special activity –

(i) An organized vehicle race, speed competition or contest, drag race or acceleration contest, test of physical endurance, exhibition of speed or acceleration, or any other type of event conducted for the purpose of making a speed record.

(ii) The term includes those races defined in 75 Pa.C.S. §3367 (relating to racing on highways).

State-designated highway, *State highway* – A highway or bridge on the system of highways and bridges over which the Department has assumed or has been legislatively given jurisdiction.

Stopping sight distance – The length of highway over which a 2-foot high object on the roadway is continuously visible to the driver, with the driver's eye height assumed to be 3.5 feet above the road surface.

TTC, *temporary traffic control* – An area of a highway where road user conditions are changed because of a work zone or incident by use of temporary traffic control devices, flaggers, police officers or other authorized personnel.

TTC plan – A plan for maintaining traffic through or around a work zone.

Through highway –

(i) A highway or portion of a highway on which vehicular traffic is given preferential right-of-way, and at the entrances to which vehicular traffic from intersecting highways is required by law to yield the right-of-way in obedience to a STOP (R1-1) sign, YIELD (R1-2) sign or other traffic control device when the signs or devices are erected as provided in this chapter.

(ii) The term includes all expressways and freeways.

Traffic calming – The combination of primarily physical measures taken to reduce the negative effects of motor vehicle use, alter driver behavior and improve conditions for non-motorized street users. The primary objectives of traffic calming measures are to reduce speeding and to reduce the volume of cut-through traffic on neighborhood streets.

Traffic control devices – Signs, signals, markings and devices consistent with this chapter placed or erected by authority of a public body or official having jurisdiction, for the purpose of regulating, warning or guiding traffic.

Traffic restriction – A restriction designated by a traffic control device to regulate the speed, direction, movement, placement or kind of traffic using any highway.

Traffic signal –

(i) A power-operated traffic control device other than a sign, warning light, flashing arrow panel or steady-burn electric lamp.

(ii) The term includes traffic control signals, pedestrian signals, beacons, in-roadway warning lights, lane-use-control signals, movable bridge signals, emergency traffic signals, firehouse warning devices, ramp and highway metering signals and weigh station signals.

Travel lane –

(i) A lane of a highway which is used for travel by vehicles.

(ii) A lane in which parking is permitted during off-peak hours but is restricted for use as a travel lane during peak hours to obtain greater traffic movement.

Warrant – A description of the threshold conditions to be used in evaluating the potential safety and operational benefits of traffic control devices based upon average or normal conditions.

Work zone – The area of a highway where construction, maintenance or utility work activities are being conducted, and in which traffic control devices are required in accordance with this chapter.

Chapter 1 Appendix

Appendix | **PAGE**

OS-2 (11-08)

pennsylvania
DEPARTMENT OF TRANSPORTATION
www.dot.state.pa.us

DATE: "[Enter Date Here]"

SUBJECT: SR[SR] Section [Sec] ECMS # [ECMS]
Proprietary Item Approval Request

To: "[Enter Bureau Director Name here]" ,P.E.
Director
Bureau of "[Enter Bureau here]"

FROM: "[Enter Name here]"
District Executive
Engineering District XX-0

We are providing the following information for Proprietary Item Approval.

[INSERT A SHORT PROJECT DESCRIPTION]

[INSERT THE PROPRIETARY ITEMS AND A JUSTIFICATION (in accordance with 23CFR and Publication 51, Chapter 3 Section B)]

(This is a sample justification. Please incorporate the appropriate CFR references (1, 2, or 3) and justifications pertaining to your proprietary item request.)
The items that we seek approval for are:
1.[Insert Item]

This request is in compliance with the Code of Federal Regulations Title 23 – Highways, Part 635-Construction and Maintenance, Subsection 411 – Material or product selection. These items were selected to maintain (system continuity and/or unavailability of alternate same product) with the current products used by the (municipality/township/city).

[INSERT THESE PARAGRAPHS]

This project (is / is not) a Federal oversight project, therefore it (should / should not) be forwarded to FHWA for approval.

The District concurs with the justification provided herewith (by the municipality/township/city) and requests your approval for the subject proprietary item(s).

If you have any questions on this matter or require any additional information, please contact, (Name) at (###) ###-####.

Attachment(s)

Approval ________________________________ Date__________
Director, Bureau of "[Enter Bureau Here]"

Engineering District XX-0 | Address | City, PA _____ | (___) ___-____

Exhibit 3A Example Template letter for Proprietary Item Approval.

pennsylvania
DEPARTMENT OF TRANSPORTATION

Publication 46
Chapter 2

Signing

Pennsylvania Department of Transportation
Chapter 2 of Publication 46
Table of Contents

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List of Exhibits

Appendix

2.1 Signing Overview

2.1.1 Purpose

One cannot over emphasized the importance of good signing since national studies indicate that deficient signing is the number one complaint of 60 percent of drivers and is the third leading cause of crashes. In addition, sign improvements have one of the highest benefit-to-cost ratio of all safety improvements.

The purpose of this chapter is to consolidate policies for the application, installation and maintenance of traffic signs either by reference or by inclusion herein.

2.1.2 Classification of Signs

As noted in Section 2A.05 of the *MUTCD*, there are only three classifications of signs:

A. Regulatory signs give notice of traffic laws or restrictions.

B. Warning signs give notice of a situation that might not be readily apparent.

C. Guide signs show route designations, destinations, directions, distances, services, points of interest and other geographical, recreational, or cultural information.

2.1.3 Design of Signs

The primary purpose of the *MUTCD* is to improve safety and reduce driver frustration by promoting uniformity in the design and application of traffic control devices. FHWA is also working internationally to share and borrow ideas so that uniformity is much broader than just in the United States.

Uniform designs and applications of traffic signs help everyone, because as drivers we can see and understand the sign messages, and the systematic advance placement of warning signs provide sufficient notice for us to take appropriate actions.

To that end, the *MUTCD* establishes the basic framework for the design and application of signs, and the *Standard Highway Signs and Markings* book provides detailed drawings of the standard signs and alphabets.

Sign Nomenclature

The *MUTCD* assigns a unique nomenclature to all common types of traffic signs. PennDOT uses the nomenclature in the *MUTCD*, but like other states, PennDOT also has some additional traffic signs that they have approved for unique applications, and for which they have assigned their own nomenclature. The first letter in sign nomenclature conforms to the following:

- Regulatory signs – R.
- Warning signs – W, except school signs start with the letter S.
- Guide signs – a variety of letters, but most commonly D, G, I, or M.

Sign names used in this manual may look awkward because some are in all capital letters while others are in title case. This mix of styles is common because the *MUTCD* and most other sign manuals generally use the following practice:

1. Uppercase legends (capitals) for sign names when the sign name and the sign legend message are the same (e.g., STOP, YIELD, and DO NOT ENTER signs).

2. Title case for symbol signs and whenever the sign name and sign message are not the same (e.g., Speed Limit, Turn, and Intersection signs).

What is Retroreflectivity?

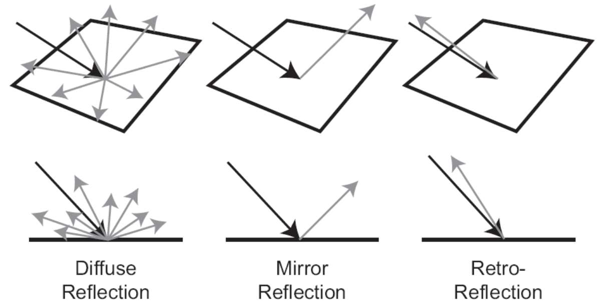
The *MUTCD* requires traffic signs to be either retroreflective or illuminated to show the same shape and color both day and night. Since it is more cost effective to make signs retroreflective than it is to illuminate them, PennDOT requires retroreflective sheeting material on all signs.

Most objects reflect light. The most common type of reflection is "diffuse reflection" where light scatters after striking rough surfaces such as trees, clothing and carpet. Only a very small amount of the diffused light reflects back toward the light source.

Another type of reflection is "mirror reflection" that occurs when light strikes smooth or glossy surfaces, and the light reflects off the surface at an equal but opposite angle. Mirror reflection frequently occurs at night on wet roads when the headlights of approaching vehicles create extensive glare. Sign faces also produce some mirror reflection due to their glossy surfaces, and for this reason; it is a good practice to rotate signs away from the driver.

In contrast, "retroreflection" (see Exhibit 2.1-A) is the unique ability of a surface to reflect light back toward the light source, and "retroreflectivity" is the measurable property of a material to redirect light back to its source.

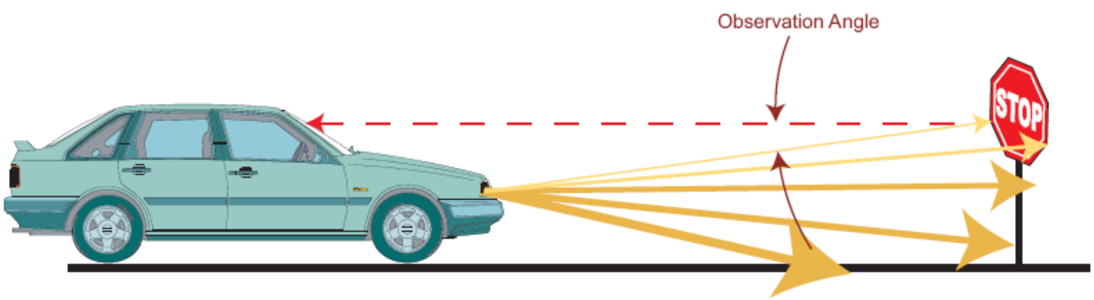
Exhibit 2.1-A Types of Retroreflection



Retroreflective Sheeting Materials

To make signs retroreflective, sign manufacturers apply retroreflective sheeting, which contains either microscopic glass beads or cube corner reflectors, to the face of each sign. If the sheeting manufacturers could make all glass beads and cube corner reflectors perfectly shaped, all reflected light would return directly to the light source (headlights). Although retroreflective sheeting does not have perfectly shaped lenses, drivers do see more reflected light the closer their eyes are to the headlights. As illustrated in Exhibit 2.1-B, the angle formed between the headlights, the sign and the driver's eyes is the observation angle, and the smaller the angle the higher the retroreflectivity.

Exhibit 2.1-B Graphic Illustration of the Observation Angle



Retroreflective materials are also more efficient when the light source is approximately perpendicular to the sign face; therefore, it is important to have signs oriented to face approaching traffic.

The ability to see traffic signs at night is a function of the following:

- Driver's night vision.
- Intensity and light distribution of the headlights.

- Distance, mounting height, and orientation of the sign in relation to the vehicle's headlights.
- Location of driver's eyes with respect to the headlights.
- Type, color and age of the retroreflective material.

Why is Retroreflectivity Important?

The nighttime visibility of signs and pavement markings is essential for highway safety. National studies show that 50 percent or more of all fatal crashes occur at night despite lower travel volumes. In fact, the average fatality rate (fatalities per 100 million vehicle-miles of travel) is about three times higher during the night than during the day.

Some of the factors that contribute to higher nighttime crash rates include:

- After age 20, the human eye needs about twice as much light approximately every 13 years in order to read. For example, compared to a 20-year old driver, a 33-year old driver needs twice as much light, a 46-year old driver needs four times as much light, a 59-year old driver needs eight times as much light, and a 72-year old driver needs 16 times as much light.
- The number of visual clues that delineate the roadway alignment are reduced at night.
- Glare from opposing traffic further reduces the number of visual clues.
- Rain, snow, fog, dew and frost reduce visibility distances.
- There are more intoxicated and sleepy drivers.

Some traffic signs may look almost new during the day but are completely ineffective at night. This nighttime visibility problem is usually a function of the type and age of the retroreflective material.

Initially, only one type of retroreflective sheeting material was available, but as technology developed, brighter and more durable materials became available. Exhibit 2.1-C shows eight types of retroreflective materials currently manufactured for permanent-type signs, and new more-efficient types are rapidly evolving. Please note that Types V and VI sheeting are not included because they are not for permanent signs (Type V sheeting is for delineation and Type VI sheeting is for temporary roll-up signs).

Exhibit 2.1-C Retroreflective Materials for Permanent Signs

Type Retroreflective Material*	Common Name	Life Expectancy (years)	General Comments
I	Engineering Grade	7	These two types of materials are no longer approved for use
II	Super-Engineering Grade	7-10	
III	High-Intensity Grade	10+	Encapsulated lens or microprismatic materials
IV	High-Performance Grade	10+	Microprismatic materials
VII, VIII, IX & X	Super-High Intensity or Very High Intensity Grades	12+	Microprismatic materials

When is Sign Lighting Required?

In 1993, PennDOT started using Type III or higher type retroreflective sheeting for all new traffic signs. Because the Department has elected to use higher types of retroreflective sheeting materials, the need for sign lighting should be minimal. In general, consider sign lighting only for overhead freeway signs as discussed in Section 2.12.8.

2.1.4 Minimum Retroreflectivity

In 1993, Congress directed the U.S. Secretary of Transportation to include minimum retroreflectivity values for traffic signs in the *MUTCD*. Following extensive research and public input, FHWA adopted minimum retroreflectivity values for most traffic signs on December 21, 2007, and incorporated them into the *MUTCD* (Revision 2 of the 2003 Edition). Specifically, Section 2A-09 and Table 2A-1 of the *MUTCD* contain the new criteria. Table 2A-1 is included herein as Exhibit 2.1-D.

The Department discontinued using Type I (Engineering Grade) and Type II (Super Engineering Grade) materials in 1993, but most local authorities continued using these materials. However, in 2004, the Department canceled the approvals of all Type I and Type II materials because: (1) the Department was aware of the on-going research; and (2) the fact that the higher grade materials were more cost effective then the cheaper materials. Therefore, no one should be using Type I or Type II materials on any public highway within the Commonwealth.

Exhibit 2.1-D Minimum Maintained Retroreflectivity Levels

Sign Color	Sheeting Type (ASTM D4956-04)				Additional Criteria
	Beaded Sheeting			Prismatic Sheeting	
	I	II	III	III, IV, VI, VII, VIII, IX, X	
White on Green	W*; G ≥ 7	W*; G ≥ 15	W*; G ≥ 25	W ≥ 250; G ≥ 25	Overhead
	W*; G ≥ 7	W ≥ 120; G ≥ 15			Ground-mounted
Black on Yellow or Black on Orange	Y*; O*	Y ≥ 50; O ≥ 50			②
	Y*; O*	Y ≥ 75; O ≥ 75			③
White on Red	W ≥ 35; R ≥ 7				④
Black on White	W ≥ 50				–

① The minimum maintained retroreflectivity levels shown in this table are in units of $cd/lx/m^2$ measured at an observation angle of 0.2° and an entrance angle of -4.0°.
② For text and fine symbol signs measuring at least 1200 mm (48 in) and for all sizes of bold symbol signs
③ For text and fine symbol signs measuring less than 1200 mm (48 in)
④ Minimum Sign Contrast Ratio ≥ 3:1 (white retroreflectivity ÷ red retroreflectivity)
* This sheeting type should not be used for this color for this application.

Bold Symbol Signs

- W1-1, -2 – Turn and Curve
- W1-3, -4 – Reverse Turn and Curve
- W1-5 – Winding Road
- W1-6, -7 – Large Arrow
- W1-8 – Chevron
- W1-10 – Intersection in Curve
- W1-11 – Hairpin Curve
- W1-15 – 270 Degree Loop
- W2-1 – Cross Road
- W2-2, -3 – Side Road
- W2-4, -5 – T and Y Intersection
- W2-6 – Circular Intersection
- W3-1 – Stop Ahead
- W3-2 – Yield Ahead
- W3-3 – Signal Ahead
- W4-1 – Merge
- W4-2 – Lane Ends
- W4-3 – Added Lane
- W4-5 – Entering Roadway Merge
- W4-6 – Entering Roadway Added Lane
- W6-1, -2 – Divided Highway Begins and Ends
- W6-3 – Two-Way Traffic
- W10-1, -2, -3, -4, -11, -12 – Highway-Railroad Advance Warning
- W11-2 – Pedestrian Crossing
- W11-3 – Deer Crossing
- W11-4 – Cattle Crossing
- W11-5 – Farm Equipment
- W11-6 – Snowmobile Crossing
- W11-7 – Equestrian Crossing
- W11-8 – Fire Station
- W11-10 – Truck Crossing
- W12-1 – Double Arrow
- W16-5p, -6p, -7p – Pointing Arrow Plaques
- W20-7a – Flagger
- W21-1a – Worker

Fine Symbol Signs – Symbol signs not listed as Bold Symbol Signs.

Special Cases

- W3-1 – Stop Ahead: Red retroreflectivity ≥ 7
- W3-2 – Yield Ahead: Red retroreflectivity ≥ 7; White retroreflectivity ≥ 35
- W3-3 – Signal Ahead: Red retroreflectivity ≥ 7; Green retroreflectivity ≥ 7
- W3-5 – Speed Reduction: White retroreflectivity ≥ 50
- For non-diamond shaped signs such W14-3 (No Passing Zone), W4-4p (Cross Traffic Does Not Stop), or W13-1, -2, -3, -5 (Speed Advisory Plaques), use largest sign dimension to determine proper minimum retroreflectivity level.

NOTE: Type I and II materials both have a uniform appearance similar to metallic paint, whereas all Type III, IV, VII, VIII, IX and X materials have a pattern of hexagons, diamonds or circular shapes measuring about one-eighth inch across. Therefore, it is easy to recognize the inferior Type I and Type II materials. FHWA's Retroreflective Sheeting Identification Guide – September 2005 (available at http://safety.fhwa.dot.gov/roadway_dept/retro/sign/retrore_sheet_id.htm), is a handy tool to help determine the grade and manufacturer of most sheeting materials.

2.1.5 Recouping Costs Incurred from Crashes

Sign maintenance is frequently required because of highway crashes. Therefore, if the responsibility party is identified, record the vehicle information and provide it to the Reimbursable Activities Unit in the Bureau of Maintenance and Operations so that the Department can receive reimbursement from the individual or their insurance carrier.

If a logo sign is damaged, the Department should contact the Logo Signing Trust at 717-232-8880 (fax 717-232-8948).

2.1.6 Laws, Regulations, and Other Publications

Adopt-A-Highway Operational Manual. A PennDOT manual defining the procedures and types of signs available for the Adopt-A-Highway Program.

Approved Construction Materials – Bulletin 15 (Publication 35). A listing of approved materials and manufacturers, including sign materials and sign manufacturers.

Maintenance Manual (Publication 23). A manual prescribing the planning, scheduling, equipment, materials, and labor required to accomplish the Department's highway maintenance program.

Manual on Uniform Traffic Control Devices (*MUTCD*). A manual adopted by the Federal Highway Administration, and which establishes national guidelines for traffic-control devices, including signs. Specifically, Part 2 addresses traffic signs, and is available at http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/pdf_index.htm.

Official Traffic-Control Devices (Publication 212). Regulations published as Chapter 212 of Title 67 of the PA Code. This regulation adopts the Federal Highway Administration's Manual on Uniform Traffic Control Devices (*MUTCD*) and establishes additional study requirements, warrants, principles and guidelines not included in the *MUTCD*. The purpose of Publication 212 is to establish greater uniformity for the design, location and operation of all official traffic signs, signals, markings, and other traffic-control devices within this Commonwealth.

Official Traffic Signs (Publication 236M). A compilation of official traffic signs approved either by regulation (Publication 212) or by signature of the State Traffic Engineer. The standards include the justification for the signs and the design details necessary to fabricate the signs.

A Policy on Geometric Design of Highways and Streets. The universal standard for geometric design of highways and streets, as published by the American Association of State Transportation Officials (AASHTO). This manual is also commonly called the "Green Book."

Sign Foreman's Manual (Publication 108). The Sign Foreman's Manual depicts the various responsibilities of the sign foreman.

Specifications (Publication 408). Specifications referenced for all Department construction projects, which includes requirements for the installation of signs and sign accessories.

Standard Highway Signs and Markings (SHS) book. FHWA's compilation of sign designs approved for national use, which is also the basis for designing the signs in Publication 236. In addition to sign designs, the book includes recreational or cultural interest symbols for potential use on TODS.

Traffic Control – Pavement Marking and Signing Standards (Publication 111M). Standard drawings specifying the types, dimensions, locations and lighting of signs on expressways and freeways, and the legend spacing and sign supports for signs on all highways. Also includes details for pavement markings, snowplowable raised pavement markers, delineation, and Type III barricades.

2.2 Official Signs and Approved Sign Manufacturers

2.2.1 Official Signs

The following signs are “official signs” and may be installed in highway rights-of-way within the Commonwealth:

- Signs included in the Handbook of Approved Signs (Publication 236M).
- Signs included in the Traffic Control Signing Standards (Publication 111M).

Signs that were approved at the time of their manufacture but are no longer listed in the above referenced publications may continue to be used until they fulfill their useful life unless directed otherwise.

2.2.2 Approved Sign Manufacturers

(a) Types of Manufacturers. The following sign manufacturers are approved to make official traffic signs:

- PennDOT’s Sign Shop, located at 21st and Herr Streets, Harrisburg.
- Department of Conservation of Natural Resources (DCNR) Sign Shop.
- Department-approved commercial sign manufacturers listed in Publication 35 or other lists made available from the Traffic Engineering and Operations Division (TEOD).
- Department-approved municipal sign manufacturers. A list is available from TEOD.

(b) Sign Identification. The manufacturer may apply their name or logo on the face of the sign. Manufacturers other than the Department's Sign Shop shall identify their name or their Department approval number on the front or back of each sign manufactured for use along a public highway within the Commonwealth in accordance with the following:

> Sign identifications on the front of the sign shall be confined to either a 1-inch diameter circle on the lower edge of the sign on or near the sign border, or within a 0.5-inch by 2-inch rectangle on or near the border on the lower half of the sign. The identification may be screened on the sign or durably affixed by a transparent sticker.

Sign identification on the back of the sign shall be on the lower half of the sign, positioned at a location that will not be covered by a sign post.

2.3 Approval and Erection Responsibilities

2.3.1 Signing Responsibilities

Effective sign maintenance is important from a customer satisfaction perspective, and from a safety aspect in reducing crashes. Therefore, careful management of sign maintenance at all levels throughout the Department is essential.

Altogether many organizations within the Department are involved in signing and each has their necessary function and area of responsibility. The flow line of information and direction starts with Department regulations and policies and ends with the actual installation and maintenance of the signs.

(a) Traffic Engineering and Operations Division (TEOD).

- Central Office. TEOD's staff in the Central Office is responsible for the following: establish signing policies and guidelines; develop standards, specifications, and regulations related to signing; develop annual contracts for the purchase of miscellaneous signs, sign accessories and work zone traffic-control devices for use by the county, maintenance districts, and raw materials for use by the Sign Shop; assist in development and implementation of sign-related safety programs; monitor the statewide use and inventory of signs and sign accessories; perform quality assurance reviews of sign installations; and oversee the sign inventory component of our SAP Plant Maintenance system.

- Sign Shop. The Sign Shop manufactures signs, and stocks select signs and sign accessories for distribution to the county maintenance districts. The Sign Shop's general goal is to have all signs which are manufactured or stockpiled at the Sign Shop and all sign accessories available for pickup by the counties within 2 weeks after receipt of the orders. On request, the Sign Shop will fill any priority order as soon as their schedule permits.

(b) Engineering District.

- District Bridge Unit. Capture signs associated with bridges, and coordinate with the District Traffic Unit when changes are necessary.

- District Traffic Unit. The District Traffic Unit determines what signs and delineation devices should be in place along all State highways within the Engineering District. They also prepare a notification of changes, e.g., install of new signs or the relocation or removal of existing signs, via SAP Plant Maintenance. The Traffic Unit may design custom-made signs for inclusion with SAP Plant Maintenance sign orders and take retroreflective readings of select signs for recording in SAP Plant Maintenance. The District Traffic Unit prepares maintenance customer service records related to signing and when required, assigns priorities or due dates for signing work. They provide technical assistance and guidance to the Maintenance Districts in matters related to signing. Ensure that 20 percent of all signs in the Engineering District are reviewed annually.

- District Plant Maintenance Coordinator. The District Plant Maintenance Coordinator trains and directs county personnel to use the computer for inventory control and sign orders; provides guidance for establishing reorder points and reorder quantities; validates and upgrades county input data to insure accurate records; and supervises the scheduling and reporting of periodic physical inventory checks.

- District Maintenance Unit. The District Maintenance Unit prepares contracts and lease arrangements for equipment not owned by the Department. Some of this equipment, such as cranes, bucket trucks and auger trucks, may be useful for signing projects.

- District Permit Unit. Ensure that the District Traffic Unit is aware of all signs required as part of any permit.

(c) County Maintenance District Responsibilities.

- Administrative Staff.

    - Identify traffic-control devices that are damaged or in need of repair.

    - Conduct routine reviews of signs.

    - Prepare maintenance customer service records for the routine maintenance of existing signs.

    - Establish a work schedule for the sign foreman and create SAP Plant Maintenance work orders.

    - Ensure that the issued work orders are completed in a timely and satisfactory fashion.

    - Transport signing materials from the Sign Shop to the County.

   - Provide direction to the sign foreman and other maintenance foremen for handling minor items on signing such as replacement of missing bolts on signs.

   - Inspect work performed by the sign crew.

- Storekeeper**.**

   - Maintain storage location "01" with signs and sign accessories for the routine maintenance and replacement of signs.

   - Place material and sign orders via SAP Plant Maintenance.

   - Receive into inventory and issue signs and accessories to foremen on an as-needed basis.

   - Recommend inventory adjustments.

   - Perform MRP runs to ensure adequate supply of on-hand sign inventories.

- Sign Foreman**.**

   - Complete sign work orders issued by the Assistance County Maintenance Manager.

   - Maintain existing signs and delineation devices in a systematic manner, including trimming trees and bushes to provide visibility to the traffic-control devices.

   - Inform the storekeeper or other responsible persons of existing and anticipated material needs.

   - Complete M-206 Customer Service Record Forms.

In the spring of 1997, the Department used the Value Adding Management (VAM) process on the sign replacement process. To improve efficiency and reduce cost, each county is now empowered to oversee routine sign maintenance.

Therefore, sign maintenance responsibilities are as follows:

1. County personnel (ACMM and Sign Foreman) are now responsible for identifying worn-out and vandalized signs. The District Traffic Unit will assist to ensure 20 percent of signs are reviewed annually. The County should conduct a systematic review with other work functions and with improved recognition of deteriorated signs. County personnel have been trained in:

- Identification of worn out/vandalized signs which require replacement
- How to conduct systematic reviews of signs during routine work activities

The District Traffic Unit is still responsible for performing all engineering studies required to identify new or revised signing needs.

2. Ordering of signs is performed by the County, so that the County inventory can be controlled at a low level.

3. If a District/County is in need of a sign(s) on a priority basis, the county can call the Sign Shop manager (717-346-9910) and arrange to have the signs made quickly. Districts and counties should use this service on rare occasions when unexpected and unanticipated events occur.

4. A sign inventory record system of field placements is a useful tool in evaluating and managing signs. The inventory record is necessary so that the Sign Foremen and ACMM can readily detect missing signs and serves as an important management tool for the overall sign program. Therefore, it is essential to maintain the inventory in SAP Plant Maintenance.

5. Districts and counties are strongly encouraged to replace old sign trucks with bucket sign trucks having vertical sign storage bins and automated post-driving and pulling equipment. Productivity gains of around 50 percent have been realized where crews have transitioned from manual to automated equipment. Vertical storage assists in preventing damage to signs fabricated from encapsulated and micro-prismatic sheeting materials.

6. Customer surveys indicate that visible and retroreflective highway signage is an important service. County managers are strongly encouraged to dedicate sign crews to sign upgrades and minimize the time the sign crew is used for other work functions, including placement of work zone control.

7. The on-hand sign inventories in the counties are to be kept to a minimum. Use signs in inventory before ordering equivalent new signs from the Sign Shop. Each county should generally have no more than a 4- to 6-week inventory of signs for replacement due to knockdowns (e.g., STOP, Curve, Speed Limit, etc. signs). A listing of recommended on-hand sign inventories is in the Sign Foreman's Manual (Publication 108).

2.3.2 Signs Erected by Others

§212.6 of Publication 212 identifies the signs that local authorities or others may install on State highways and on local highway approaches to State highways with and without Department approval. However, the Department may enter into an agreement with municipalities to alter their responsibilities.

Signs installed by others that require Department approval need inventoried in SAP Plant Maintenance.

2.3.3 Issuance of Sign Work Orders

Sign work orders are the responsibility of the County Maintenance District.

2.4 Regulatory Signs

2.4.1 Requirement for Engineering and Traffic Study

The decision to use an approved traffic control device that imposes restrictions that are not normal rules of the road should be based on an engineering and traffic study. Applicable studies are included in 67 Pa. Code Chapter 212 (Publication 212).

2.4.2 Establishment of Priorities

Signs should be used only where warranted by facts and field studies. Signs are essential where special regulations apply at specific places or at specific times only, or where hazards are not self-evident.

Normally, regulatory signs are not necessary to confirm rules of the road.

2.4.3 STOP Sign Installations

Give the installation of STOP (R1-1) signs the highest priority of all signs. When downed R1-1 signs are reported, sign crews should respond as soon as possible since the downed or missing stop sign represents a high risk for crashes. Due to this liability, when replacing a downed or missing R1-1 sign, sign crews should ensure that motorists are aware of the situation at the uncontrolled intersection by providing flagging, a temporary STOP sign on a stand, or some other appropriate traffic control measure.

2.4.4 Signing Responsibilities at Intersections with Local Roads

On February 4, 2006, the Department's new 67 Pa. Code Chapter 212 (relating to official traffic control devices) was adopted in the Pennsylvania Bulletin. Specifically, §212.5(d) (relating to traffic-control devices on local highway approaches to intersections with State highways) establishes sign installation and maintenance responsibilities at intersections between State highways and local roadways.

The Department Responsibilities

The Department is responsible for the installation, revision, maintenance, and removal of STOP (R1-1) and YIELD (R1-2) signs on State highway approaches to local roadways.

However, when a Department employee discovers a damaged or missing STOP (R1-1) or YIELD (R1-2) sign on a local roadway approach to a State highway, the employee should notify the responsible municipality. If the municipality cannot respond in a timely fashion, the county should replace the sign with a temporary or permanent installation to ensure that safety is not compromised. The county is to notify the local authorities of the action taken.

County Maintenance offices should document all communication and work associated with the replacement of the damaged or missing sign on the local road approach.

<u>Local Authorities Responsibilities</u>

In accordance with §212.5(d)(3) (regarding traffic control devices on local highway approaches to intersections with State highways), at existing intersections, local authorities (or permittees) are responsible for installing, removing and maintaining STOP (R1-1) and YIELD (R1-2) signs on local highway approaches to State highways.

Furthermore, the maintenance of STOP (R1-1) and YIELD (R1-2) signs previously installed by the Department on local roadway approaches to State highways is now the responsibility of local authorities, even if the signs are within Department right-of-way.

Local authorities (or permittees) are responsible for determining the need for any Stop Ahead (W3-1) and Yield Ahead (W3-2) signs on local highway approaches to State highways. They are then responsible for installing and maintaining any warranted signs.

In addition to the STOP and YIELD signs, local authorities (or permittees) are also responsible to install and maintain the following traffic control devices:

- Stop lines and yield lines.
- No Right Turn (R3-1), No Left Turn (R3-2), NO TURNS (R3-3), Left Turn (R3-5), LEFT LANE MUST TURN LEFT (R3-7L), One-Way (R6-Series) and other similar type traffic restriction, prohibitions, or lane control signs.

Finally, local authorities (or permittees) are required to obtain written Department approval before implementing any revised traffic-control scheme on a State highway or at the intersection of a local highway and a State highway except where Department regulations require otherwise.

2.4.5 In-Street Pedestrian Crossing Signs (R1-6)

Publication 35 identifies the types of In-Street Pedestrian Crossing (R1-6) signs approved for use in Pennsylvania in the section entitled “Traffic Accommodations and Control.”

R1-6

As noted in the sign standard in Publication 236M, the R1-6 sign may be positioned on the centerline of low-speed roadways near an unsignalized marked crosswalk. The standard forbids the use of R1-6 signs on roadways with a speed limit greater than 35 mph and on narrow roadways with a clear roadway width less than 20 feet. If there may be conflicts from long wheelbase vehicles, like buses or trucks, you may place the sign up to 50 feet from the crosswalk.

In the past, BHSTE purchased some R1-6 signs for use by local authorities in efforts to reduce pedestrian crashes. Distribution of the “free devices” was a function of pedestrian crash rates, community safety programs, and the municipality’s willingness to conduct after studies to measure motorist compliance. In these cases, the Sign Shop forwarded the necessary signs to the appropriate County Maintenance Districts for pickup by the municipalities.

The District Bicycle/Pedestrian Coordinator should be aware of any existing programs for local authorities.

2.4.6 Posting Regulatory Speed Limits

2.4.6.1 Posting of Speed Limits at Intersections

Use the following guidelines if a problem exists in the enforcement of a speed limit that ends at an intersection:

1. Ending a speed limit on the stem of a T-intersection. Since §3362(b) of the Vehicle Code (relating to posting of speed limit) requires that official traffic-control devices be posted at the beginning and end of each speed zone, it is necessary to install signs indicating the end of a speed limit on the stem of a T-intersection in one of two ways:

    - The preferred method is to install a Speed Limit (R2-1) sign indicating the appropriate speed limit on the other two legs within a reasonable distance beyond the intersection, e.g., within 200 feet of the intersection.

- A less desirable method is to install an END (R2-10) sign over a Speed Limit (R2-1) sign in advance of the T-intersection. However, this method may be necessary if there is no speed limit on the other roadway and it has not been determined through test runs that it is safe for travel at 55 mph.

2. Ending the speed limit at a four-way intersection. If a posted speed limit does not exist along the highway beyond the intersection and it has not been determined through test runs that the following section of highway is safe for travel at 55 mph, Districts may end the speed limit by installing an END (R2-10) sign above a Speed Limit (R2-1) sign in advance of the intersection.

   If the highway beyond the intersection and the intersecting highway have posted speed limits or are safe for travel at 55 mph, Speed Limit (R2-1) signs indicating the speed limit along the three other legs of the intersection may be installed within a reasonable distance from the intersection, e.g., within 200 feet of the intersection.

3. Speed limit for turning vehicles at intersections. Neither the Vehicle Code nor regulations require that drivers turning from one highway onto another highway be advised of the speed limit along the highway they are entering. Therefore, it is possible that a turning driver could travel up to one-half mile before knowing what the speed limit is along the highway they have entered.

   Normally, this is not a problem if the speed limit along the entered highway is higher that the speed limit along the highway the driver is leaving. However, this can create a possible "speed entrapment" condition if a lower speed limit exists along the second highway. In view of this, Districts should consider the installation of Speed Limit (R2-1) signs within a reasonable distance from intersections if the following conditions exist:

   - The speed limit along the second highway is lower then the speed limit along the first highway.

   - The normal one-half mile spacing of speed limit signs has not provided a sign within a reasonable distance from the intersection, e.g., within 500 to 1,000 feet of the intersection.

- There is a high number of turning, non-local vehicles.

2.4.6.2 Posting of Regulatory Speed Limits and Advisory Speeds

When the speed on an Advisory Speed (W13-1) plaque is lower than a posted regulatory speed limit, give priority to the installation of the warning signs with advisory speeds. For example:

1. Do not install a R2-1 sign within the area covered by the W13-1 plaque, or within a distance in feet in advance of the warning sign equal to 10 times the regulatory speed limit in miles per hour (e.g., 550 feet for a 55-mph speed limit).

2. An R2-1 sign may be installed immediately following the section of roadway covered by the warning signs (e.g., within 200 feet after the end of a curve).

Since this posting of signs may result in distances between R2-1 signs that are greater than one-half mile apart, it may be possible to install a supplemental R2-1 sign at an intermediate location. In any event, it is more in the interest of motorists' safety to tell drivers the safe speed than to possibly confuse them by installing conflicting regulatory and advisory signs.

2.4.6.3 Posting of Regulatory Speed Limits and School Speed Limits

Whenever possible, do not install conflicting Speed Limit (R2-1) signs within a 15-mph school zone speed limit, or within a distance of 10 times the speed limit, in feet, of the beginning of the school speed limit.

2.4.7 PASS WITH CARE Signs

PASS WITH CARE (R4-2) signs may be installed only at the end of no-passing zones.

To reduce sign clutter, generally reserve these signs for the locations where the police have identified a serious problem that the signs could correct. These locations may include areas with frequent passing maneuvers across a solid yellow line, or sections of highway where police wish to enforce the restriction.

2.4.8 ONE-WAY Sign Installations

Property owners are initially responsible for installing and maintaining ONE-WAY (R6-1, R6-2) signs to notify motorists that they are exiting a non-residential driveway onto a one-way highway. This responsibility (in the case of driveways) is uniform for new and long existing one-way highways and driveways as well as for permitted and non-permitted driveways. If a problem is identified, the owner should be contacted and the responsibility documented through a Highway Occupancy Permit or Supplemental Permit, recorded in the County Office of the Recorder of Deeds.

The Office of Chief Counsel advises that despite the permittee's responsibility in the first instance to install the sign, the Department may still have the ultimate responsibility, in the face of possible tort liability, to assure the safety of the traveling public if the permittee fails to erect a sign following a Department directive.

For example, if the absence of a ONE-WAY sign at a business driveway is a "dangerous condition," then the Department would be obligated to take corrective action. If the property owner does not take any action following a proper notification on our part, the District should install the sign and bill the permittee to recover its costs.

2.4.9 Weight Limits

Advance Signs

§4902(e) of the Vehicle Code (relating to erection of signs) requires the erection of restriction signs within 25 feet of each end of a bridge or portion of highway with a weight restriction. Moreover, when the restriction does not begin or end at an intersection with an unrestricted highway, signs are also required at the intersection nearest each end of the restricted bridge or portion of highway to avoid entrapment.

If multi-restrictions exist along a highway between intersections with unrestricted highways (e.g., various bridge weight limits), it is necessary to install advance signing for each restriction unless the first restriction is the most limiting. This signing is required regardless of whether the unrestricted intersecting highway is a State highway or a local roadway or whether the roadway would afford the driver the best alternate route around the restriction. If the closest unrestricted intersecting highway will not provide a suitable alternate route around the restriction, consider an installation of additional advance informational signs at the last intersecting highway that will provide a suitable alternate route.

Prompt Posting of Bridge Weight Limits

When approved, it is extremely important to promptly post all bridge weight limits. To expedite the posting, each county should maintain an inventory of the commonly used R12-1, R12-1-1, and R12-4 signs (see Section 2.18.1).

In lieu of stocking only completely finished signs, the county may elect to stock some partially finished signs (i.e., without the numerals), and to stock some cutout pressure-sensitive numerals. The numerals should consist of 5" E-Series digits for the standard size R12-1 and R12-1-1 signs and 3" E-Series digits for the standard size R12-4 signs.

On request, the Sign Shop will supply packs of five numerals of the same type as a single commodity. Since cutout pressure-sensitive legend has a tendency to dry out; therefore, counties should stock no more than a normal 12-month supply. Store legend upside down to help prevent curling, and use the oldest legend first.

Counties should apply pressure-sensitive legend at the location illustrated in the official sign standard, using the procedure outlined in Exhibit 2.4-A.

2.4.10 Signing on Lower-Volume Roads

Part 5 of the *MUTCD* is entitled, "Traffic Control Devices for Low-Volume Roads." However, in accordance with Paragraph B in Section 5A.01 of the *MUTCD*, the definition of "low-volume road" excludes all state highways.

Although Part 5 of the *MUTCD* is not applicable to this manual, it is still permissible to modify signing practices on lower-volume State highways. For example, since these roads typically have mostly local traffic, directional and guide signs are generally not required.

Moreover, because of the lower traffic volumes, there are typically fewer crashes on these roads even though the vertical and horizontal alignment may be worse than the higher-volume roads. Therefore, use engineering judgment to determine the recommended number and types of regulatory and warning signs. Alternate methods to provide adequate signing are as follows:

1. No posted regulatory speed limit, but all significant curves and turns identified by warning signs with advisory speeds plaques when appropriate.

2. A reduced regulatory speed limit with fewer curve and turn signs.

3. No posted regulatory speed limit, but WINDING ROAD NEXT (___) MILES (W1-5-1) sign or NARROW ROAD NEXT (___) MILES (W5-1-1) sign with advisory speed plaques as appropriate after all intersections.

Exhibit 2.4-A “Field” Application of Pressure-Sensitive Legend

1. Store legend at room temperature.
2. Apply legend at the storeroom since it is difficult to apply during hot, cold or humid conditions.
3. Wash hands and the sign face with clean soapy water and thoroughly dry to insure good adhesion of the legend.
4. Use a china pencil, soft lead pencil or non-permanent felt-tip pen to draw horizontal lines on the sign face to delineate the top and the bottom of the legend in accordance with the dimensions on the approved sign standard. These lines will also be valuable to help align the legend. (Vertical lines may also be valuable in orienting some legend.)
5. Without removing the backing paper, position the characters at their relative locations. Space between characters should generally be about 125 percent of the stroke width (the width of the lines in the legend) – a slightly smaller space should be used before and after a “4” or a “7”. Characters which have a curve at the top should extend slightly above the top transverse line and characters which have a curve at the bottom should extend slightly below the lower transverse line.
6. Pull the backing paper off part of the character and align that part before removing the balance of the paper. It is best to start with the bottom of characters with straight bottoms (e.g., 2’s, A’s, L’s, etc.), and the top of characters with straight tops (e.g., 5’s, 7’s, F’s, etc.). After the first part of the character is in place, remove the balance of the paper and allow the character to flow into place without stretching the characters.
7. Use your fingernails or a special plastic blade as a squeegee to apply pressure to the characters to affix them to the sign. The legend will normally be removable for a short time by using a blade or knife to lift a corner, but if removed, apply new legend instead of trying to reapply the same legend.
8. Use a damp cloth to remove felt-tip or china pencil markings. Do not erase lead pencil markings.
9. Take pride in your work – the Department’s image is judged by it!

2.5 Warning Signs

2.5.1 General

Guidelines for installing warning signs are included in Sections 2C.10 of the *MUTCD.*

Consider the installation of oversize warning signs if one or more of the following exists:

(a) The condition is properly signed and/or delineated, but crashes or incidents related to the condition addressed by the warning sign continue to occur.

(b) Inadequate contrast exists between the sign and the environment when a standard size sign is used.

(c) The location is on a high-speed (45 mph or higher) highway with four or more lanes.

2.5.2 Advance Placement of Warning Signs

Since the primary purpose of warning signs is to gain attention of the unfamiliar motorist, the placement of warning signs is important. The placement must allow these drivers sufficient time to see the warning sign, understand the intent, identify the potential hazard, decide what action must be taken, and then to perform any necessary maneuver.

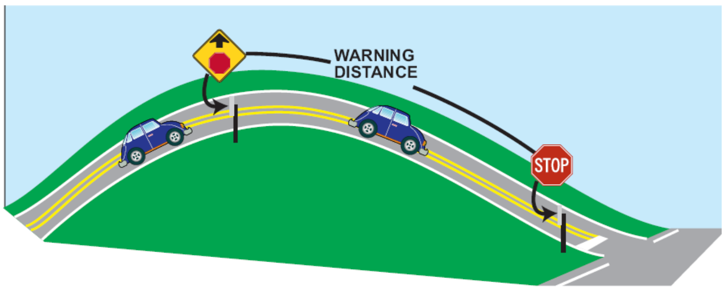


Table 2C-4 in the *MUTCD* (see Exhibit 2.5-A) provides the recommended advance sign placement distances. However, it is important to note that Condition A is only for those situations where motorists may have to change lanes in heavy traffic. Examples of applicable signs include:

- Merge (W4-1).
- Lane Reduction Transition (W4-2L, W4-2R).
- Entering Roadway Merge (W4-5).
- RIGHT LANE ENDS (W9-1).

Condition B is for all other advance placement distances and these values are typically much smaller than the values historically used by traffic engineers. The reason for the change is that FHWA has reconciled their advance distances to match the stopping sight distances in Table 3-1 of AASHTO's *A Policy on Geometric Design of Highways and Streets*, using a 2.5-second reaction time and a deceleration rate of 10 feet/second^2. Moreover, Engineering Districts should keep in mind that these are minimum distances and they may want to use larger values for the following reasons:

- The advance distances assume that drivers will always use their brakes to decelerate to a posted advisory speed, thereby wasting energy.

- The lower advance posting distances may violate drivers' expectations, especially if at the same time more realistic advisory speeds are used as suggested in Section 2.5.3.2.

Also, Districts should base the minimum advance distance on the "0 mph" advisory speed for the Stop Ahead, Yield Ahead, Signal Ahead, Advance Railroad Crossing, and Intersection Warning signs because a driver may wish to turn at an intersection or may need to stop due to other turning traffic.

A few warning signs are not placed in advance of the situation, but instead rely on the visibility of the sign from a distance. Examples include:

- Chevron Alignment (W1-8) sign.

- NO PASSING ZONE (W14-3) pennant.

- Pedestrian Crossing (W11-2) and School Advance Warning (S1-1) signs, when physically placed at the crosswalks with a Diagonal Arrow (W16-7p) sign.

- Double Arrow (W12-1) sign, i.e., as used on the approach end of an island where traffic can pass on both sides.

Exhibit 2.5-A Guidelines for Advance Placement of Warning Signs

Posted or 85th-Percentile Speed	Advance Placement Distance[1]								
	Condition A: Speed reduction and lane changing in heavy traffic[2]	Condition B: Deceleration to the listed advisory speed (mph) for the condition[4]							
		0[3]	10	20	30	40	50	60	70
20 mph	225 ft	N/A[5]	N/A[5]						
25 mph	325 ft	N/A[5]	N/A[5]	N/A[5]					
30 mph	450 ft	N/A[5]	N/A[5]	N/A[5]					
35 mph	550 ft	N/A[5]	N/A[5]	N/A[5]	N/A[5]				
40 mph	650 ft	125 ft	N/A[5]	N/A[5]	N/A[5]				
45 mph	750 ft	175 ft	125 ft	N/A[5]	N/A[5]	N/A[5]			
50 mph	850 ft	250 ft	200 ft	150 ft	100 ft	N/A[5]			
55 mph	950 ft	325 ft	275 ft	225 ft	175 ft	100 ft	N/A[5]		
60 mph	1100 ft	400 ft	350 ft	300 ft	250 ft	175 ft	N/A[5]		
65 mph	1200 ft	475 ft	425 ft	400 ft	350 ft	275 ft	175 ft	N/A[5]	
70 mph	1250 ft	550 ft	525 ft	500 ft	425 ft	350 ft	250 ft	150 ft	
75 mph	1350 ft	650 ft	625 ft	600 ft	525 ft	450 ft	350 ft	250 ft	100 ft

Notes:

1. The distances are adjusted for a sign legibility distance of 175 ft for Condition A. The distances for Condition B have been adjusted for a sign legibility distance of 250 ft, which is appropriate for an alignment warning symbol sign.
2. Typical conditions are locations where the road user must use extra time to adjust speed and change lanes in heavy traffic because of a complex driving situation. Typical signs are Merge and Right Lane Ends. The distances are determined by providing the driver a PIEV time of 14.0 to 14.5 seconds for vehicle maneuvers (2001 AASHTO Policy, Exhibit 3-3, Decision Sight Distance, Avoidance Maneuver E) minus the legibility distance of 175 ft for the appropriate sign.
3. Typical condition is the warning of a potential stop situation. Typical signs are Stop Ahead, Yield Ahead, Signal Ahead, and Intersection Warning signs. The distances are based on the 2001 AASHTO Policy, Stopping Sight Distance, Exhibit 3-1, providing a PIEV time of 2.5 seconds, a deceleration rate of 11.2 ft/second^2, minus the sign legibility distance of 175 ft.
4. Typical conditions are locations where the road user must decrease speed to maneuver through the warned condition. Typical signs are Turn, Curve, Reverse Turn, or Reverse Curve. The distance is determined by providing a 2.5 second PIEV time, a vehicle deceleration rate of 10 ft/second^2, minus the sign legibility distance of 250 ft.
5. No suggested distances are provided for these speeds, as the placement location is dependent on site conditions and other signing to provide an adequate advance warning for the driver.

2.5.3 Signing Curves and Turns

2.5.3.1 Curve and Turn Signs

The legal speed limit of the highway should be used when evaluating the need for advance Turn (W1-1) or Curve (W1-2) signs. All curves and turns with a recommended safe speed less than the legal speed for the highway should normally be signed with an appropriate curve or turn sign. An exception to the installation of a curve or turn sign may be a ramp to or from a freeway or expressway where an advisory exit speed or ramp speed sign exists and flexible delineator posts or Chevron Alignment (W1-8) signs exist.

Curve or turn signs should normally be installed a minimum distance in advance of the curve or turn equal to the appropriate values in Condition B in Exhibit 2.5-A. To use this table, you need to know the legal or 85th percentile speed, plus the recommended advisory speed around the curve or turn.

2.5.3.2 Advisory Speed Signs

An Advisory Speed (W13-1) plaque may be installed below a Curve or Turn sign if the recommended safe speed for the curve or turn is less than the legal speed for the highway.

The safe speed on curves may be determined by making several trial runs through the curve in a car equipped with a ball-bank indicator in accordance with the following guidelines:

(a) Mount the ball-bank indicator transversely in the car at an orientation to give a "zero reading" when the car is level.

(b) For the first trial run, drive the car in the center of the lane at a speed that is a multiple of 5 mph that provides a maximum ball-bank indicator reading less than the appropriate value in Exhibit 2.5-B.

(c) If necessary, make succeeding observations at higher 5 mph increments until the reading on the ball-bank indicator equals or exceeds the appropriate value in Exhibit 2.5-B. The safe speed on the curve is the highest speed that does not exceed the appropriate value in Exhibit 2.5-B while consistently driving in the center of the travel lane.

(d) On two-way roadways, conduct test runs in each direction of travel since the safe speed may be different for the different directions of travel.

Exhibit 2.5-B Maximum Ball-Bank Indicator Readings

Posted Speed Limit (mph)	Ball–Bank Indicator (degrees)
20 or less	16
25 and 30	14
35 or more	12

2.5.3.3 Additional Signing and Delineation at Curves and Turns

In addition to the advance Curve or Turn sign discussed in Section 2.5.3.1 and Section 2.5.3.2, additional signing and/or delineation of curves and turns should be considered if one or more of the following exists:

(a) Crash lists indicate that there are "run-off-the road," "hit-fixed-object," or other curve-related crashes.

(b) There is physical evidence of errant vehicles leaving the road in the form of shoulder rutting, guide rail damage, scars on adjacent trees, or other markings on the shoulder that appear to be made by vehicles.

(c) The curve or turn is "hidden" from drivers and the roadway alignment is not evident such as a combination horizontal and an over-vertical curve, an overhead utility line that diverges from the highway, or other features that could mislead drivers.

(d) Day or night test drives of the highway indicate that additional signing and/or delineation is required to adequately indicate the travel path for drivers.

The additional signing and/or delineation could consist of the Large Single Arrow (W1-6) sign, Chevron Alignment (W1-8) sign, or Flexible Delineator Posts. These devices also provide day and night target value, especially the Large Single Arrow and the Chevron Alignment signs. However, these devices should not generally be installed at a curve or turn unless an advance Curve or Turn sign exists. Exceptions are:

1. On a ramp where an Advisory Exit Speed (W13-2) or Ramp Speed (W13-3) sign exists.

2. On a ramp, freeway, or expressway where delineators are required in accordance with the Sign Foreman's Manual (Publication 108) and the *MUTCD*.

3. At locations identified in Paragraph (a) above, but the recommended safe speed for the curve or turn as determined by ball-bank readings is equal to or higher than the legal speed limit for the highway.

If it is determined that the installation of one or more of these devices is desirable, consider the following guidelines:

A. <u>Large Single Arrow (W1-6) Sign</u>. This sign should be considered for use on curves and turns that are relatively short in length. Normally curves and turns up to about 300 or 350 feet in length can be satisfactorily signed with one W1-6 sign in each direction. It is also possible to sign longer curves with a single W1-6 sign, but engineering judgment based on field conditions must be used in making this decision.

   You may also consider the W1-6 sign for use on compound curves, reverse curves and turns, winding roads, and other locations where a severe change in alignment occurs.

B. <u>Chevron Alignment (W1-8) Sign</u>. Based on the results of a study in Virginia, "Evaluation of Curve Delineation Signs," published in Transportation Research Record 1010, consider this sign for curves or turns that are greater than 7 degrees. In addition, consider the W1-8 sign when:

   - Standard delineation is in place, but there is still a high incidence of daytime and/or nighttime "run-off-the-road" crashes.

   - Standard delineation does not, or would not, show the roadway alignment; e.g., combination horizontal and over vertical curve.

   Do not use the W1-8 sign if a turn has inadequate length for proper spacing of the W1-8 sign.

   When used, a minimum of two signs should always be visible. Do not install the first W1-8 sign before the P.C. and the last sign beyond the P.T. When applicable, W1-8 signs may be installed on back-to-back installations as described in the Sign Foreman's Manual (Publication 108).

When W1-8 signs are used, Exhibit 2.5-C shows recommended spacing based on three different methods in accordance with TTI Report FHWA/TX 04/0 4052 1, entitled *Simplifying Delineator and Chevron Applications for Horizontal Curves*.

Exhibit 2.5-C Suggested Spacing for W1-8 Signs

Method 1	Method 2	Method 3	Chevron Spacing (feet)
Curve Radius (feet)	**Degree-of-Curve***	**Curve Advisory Speed (mph)***	
< 200	> 28.6	≤ 15	40
200 - 400	14.3 -28.6	20 - 30	80
401 - 700	8.2 – 14.2	35 - 45	120
701 – 1250	4.6 – 8.1	50 - 60	160
> 1250	< 4.6	> 60	200

* "Degree-of-Curve" (D) is the measurement, in degrees, of the change in alignment over a 100-foot section of roadway. The degree-of-curve can be calculated by the formula D=5729.6/radius.

C. Flexible Delineator Posts. Based on the results of the study in Virginia, these devices should be considered for use on curves that are less than or equal to 7 degrees. They may be considered for use on curves or turns which are greater than 7 degrees, when it has been determined that the W1-8 sign should not be used.

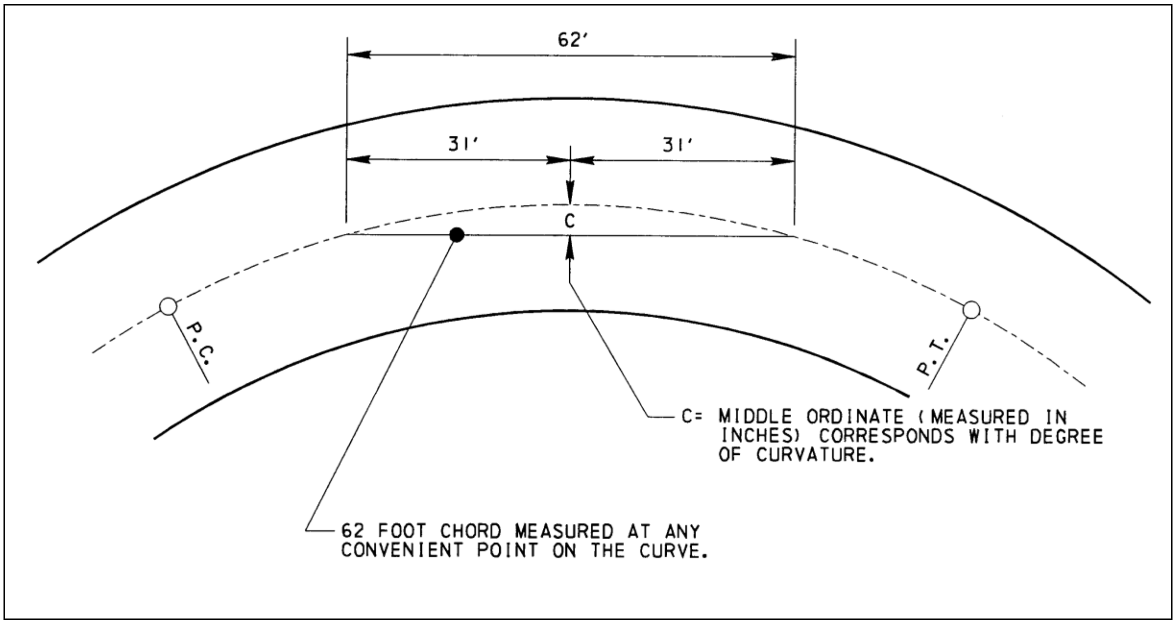
D. Combination of Signs and/or Delineation Devices. A combination of devices discussed above may be used to delineate a curve or turn (or combination of curves or turns), if a field review indicates the need for a combination of devices to adequately advise drivers of the roadway alignment.

2.5.3.4 Degree-of-Curve

If you do not know the degree-of-curve or the radius of a curve, you can estimate the degree-of-curve by two methods. First, you can take the total change in direction of the curve and divide by the length of the curve in hundreds of feet to calculate the degree-of-curve. For example, if you have a right angle (90 degree) curve that measures 1,000 feet from the beginning of the curve (P.C.) to the end of the curve (P.T.), the curve would be a 9-degree curve (i.e., 90/10 = 9). A compass that includes degrees and a distance-measuring instrument (DMI) are of value.

Although labor intensive, the second method involves stretching a 62-foot string between two points along the roadway's centerline or an edge line, and measuring the distance from the center of the string to the line. This distance is the middle ordinate, and when measured in inches, it very closely approximates the "degree-of-curve" for curves with a degree-of-curve up to approximately a 45-degree curve (e.g., a 10-inch middle ordinate equals a 10-degree curve, a 20-inch middle ordinate equals a 20-degree curve, etc.). Exhibit 2.5-D illustrates this method.

Exhibit 2.5-D. Estimating the Degree of Curve



2.5.4 Stop Ahead Signs

Stop Ahead (W3-1) signs should be installed in advance of STOP (R1-1) signs when one or more of the following exists:

(a) Because of physical conditions, the R1-1 sign is not continuously visible for the required distance specified for the W3-1 sign in Publication 236.

(b) Although the R1-1 sign is visible for the minimum distance in Publication 236, one of the following exists:

- A "running-the-stop-sign" crash experience exists.
- The view of the R1-1 sign is occasionally blocked by moving or stopped vehicles.

- The highway has a multi-lane, high-speed (45 mph or higher) approach to the R1-1 sign. (Note, if a divided highway and the median is wide enough, the STOP Sign and/or the Stop Ahead Signs should be installed on both sides of the roadway.)

- There is extensive environmental interference.

When used, install the Stop Ahead (W3-1) sign in advance of the R1-1 sign in accordance with the distance indicated in Condition B in Exhibit 2.5-A.

2.5.5 Share the Road Sign

The Share-the-Road Sign (W16-1) is available for installation on appropriate State highways throughout the Commonwealth. The purpose of the sign is to promote cooperation, understanding, and mutual safety between motorists and bicyclists on roadways where sharing roadway space is required.

Requests for the W16-1 sign may come from any legitimate source, including the following internal or external sources:

(a) Department designers or consultants may independently suggest the installation of the signs as part of the project development process. In addition, Department personnel may suggest locations for the signs as a stand-alone project.

(b) Non-department personnel may suggest locations for installation without solicitation from the Department. These suggestions may be included as part of a larger project or as a stand-alone project. Forums for this input may be District Bicycle/Pedestrian Advisory Committees, MPO/LDD Bicycle/Pedestrian Advisory Committees, or other sources. However, it is important to note that the Department will not provide signs to local municipalities for installation on local roads.

All requests for W16-1 signs on State highways should go to the District Bicycle/Pedestrian Coordinator for review. The criteria for road selection should include roads that possess any or all of the following:

- Highways promoted as a cycling route by a local or state agency, or that demonstrate a need based on the traveling patterns of local cyclists or a car-bike crash history.

- Prior to bottlenecks such as narrow bridges or underpasses, and short stretches of roads that lack paved shoulders.

- On sections of highway that have numerous commercial driveways, such as in a cluster of suburban strip malls.

- Sections of highway where lanes are greater than 14 feet wide and motorists may be tempted to travel two abreast and crowd cyclists off the road.

- On narrow highways where cyclists can only proceed safely if, they use the full lane width.

If the Bicycle/Pedestrian Coordinator determines that a request is justified, counties may order W16-1 signs from the Sign Shop. If installed by the Department, the Department is responsible for maintenance of the signs.

2.5.6 Advance Street Name Signs

On multilane roads and roads with a speed limit greater than 35 mph, Districts are encouraged to use either the Single-Line Advance Street Name (W16-8) or Double-Line Advance Street Name (W16-8A) sign, with appropriate arrows, as necessary, below any of the following advance warning signs:

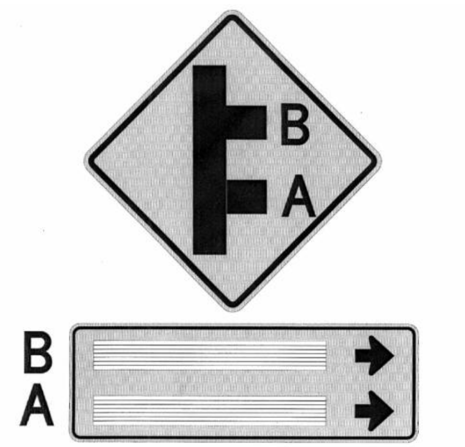
- Any W1-series sign with a side road.
- Cross Road (W2-1) sign.
- Offset Side Road (W2-1-1L, W2-1-1R) sign.
- Side Road (W2-2) sign.
- Double Side Road (W2-2D).
- 45º Side Road (W2-3L, W2-3R) sign.
- Curve – Side Road (W2-3-1L, W2-3-1R) sign.
- “T” Symbol (W2-4) sign.
- “Y” Symbol (W2-5) sign.
- “Y” Symbol Secondary (W2-5-1L, W2-5-1R) sign.
- Stop Ahead (W3-1) sign.
- Signal Ahead (W3-3) sign.

> Note: The decision to erect the above-listed warning signs should be based on their justification in Publication 236, and not solely to facilitate the installation of Advance Street Name Signs. As an alternate, you may install the Single-Line Advance Street Name (D3-2) sign or the Double-Line Advance Street Name (D3-3) sign in lieu of the W16-8 or W16-8A sign.

If a Double-Line Advance Street Name (W16-8A) sign is installed below a Double Side Road (W2-2D) sign, the consensus is that the order of destinations on the W16-8A sign should correspond to the warning sign graphic as illustrated in Exhibit 2.5-E.

Therefore, the first destination on the W16-8A sign is the second side road as approached by the driver, and the second destination on the W16-8A sign is the first side road as approached by the driver.

Exhibit 2.5-E Order of Destinations for Double Side Road Sign



Using a W16-8A sign below a W2-2D sign is a challenge. The problem is created by a conflict in how drivers read signs, i.e., we read intersection and alignment signs from the bottom to the top, whereas we read legend signs from the top to the bottom.

Therefore, if you observe problems, it may be best not to install this combination of signs.

2.6 Guide Signs and Route Markers

2.6.1 General

Guide signs are necessary to inform motorists of intersecting routes; to direct them to cities, towns, villages, or other important destinations; to identify nearby rivers, streams, parks, forests, and historical sites; and generally to give such information as will help them along their way in the most simple, direct manner possible.

Numbered traffic routes and directional signs facilitate travel by enabling motorists to reach their intended destination when using an accurate transportation map. Proper directional signing consists of Route Markers and Route Marker auxiliaries; Destination signs; Distance signs; and, where necessary, Advance Street Name signs.

Install Route Markers and Route Marker auxiliaries in sign assemblies to identify the numbered traffic route and provide additional guidance (such as general direction of the route and other information required to follow a designated numbered traffic route). Destination and Distance signs provide directions and distances to communities and points of interest that may be reached by following certain roads. Advance Street Name signs provide advance notice of the names of intersecting major streets and highways.

2.6.2 Definitions

The following words and terms, when used in this policy shall have the following meanings, unless the context clearly indicates otherwise:

Advance Route Turn Assembly – An assembly consisting of a Cardinal Direction Marker and other Route Marker auxiliaries if needed, a Route Marker, and an Advance Turn Arrow.

Advance Street Name Sign – A sign used in advance of an intersection that displays the street or road names for the next cross or side road, and may contain directional arrows as appropriate. The sign may be either:

- White legend on a green background and installed either separately or as a part of other destination signs (D1 Series) sign.
- Black legend on a yellow background (W16-8 or W16-8A sign) installed below an intersection-related warning signs as discussed in Section 2.5.6.

Advance Turn Arrow – A marker which displays a right or left arrow, the shaft of which is bent at a right angle or at a 45-degree angle. It is to be mounted below the Route Marker in Advance Route Turn Assemblies.

BUSINESS Marker – An M4-3 marker used to designate an alternate route that branches from a regular numbered route, passes through a business area, and rejoins the regularly numbered route beyond that area. If a Cardinal Direction Marker is used, it shall be mounted above the Business Marker.

BY-PASS Marker – An M4-2 marker used to designate a route that branches from the regular numbered route, bypasses a part of a city or congested area, and then rejoins the regular numbered route.

Cardinal Direction Marker – A marker with the legend EAST, WEST, NORTH, or SOUTH mounted above a route marker to indicate the general direction of the entire route.

Confirming or Reassurance Assembly – An assembly consisting of a Cardinal Direction Marker and a Route Marker which is placed along a numbered traffic route to reassure motorists that they are on their desired route.

Control city – These are major communities on or along a numbered traffic route that are familiar to a majority of drivers from either Pennsylvania or other states. These destinations must be shown on the Department's Official Transportation Map and should have a population of at least 10,000 or be a county seat. The control city may be in an adjacent State if there are no communities along the route in Pennsylvania that meet these minimum criteria. Control cities for Interstate highways are listed in **Exhibit 2.12-A**.

Destination Sign – A sign with the names of communities, points of interest, or street or road names with appropriate distances and directional arrows, if required. Destination signs are D1-Series signs, typically D1-1 through D1-3 signs. If distances are included, they should be to the central business district, or if none, to the center of the community.

DETOUR Marker – An M4-8 marker used to designate a temporary route that branches from a regular numbered route, bypasses a section of a route that is closed or blocked by construction or traffic emergency, and rejoins the regular numbered route beyond that section. When used, the Detour Marker will always be mounted at the top of the assembly.

Directional Arrow – A marker which displays a single-headed arrow pointing in the general direction (left, right, straight, 45 degrees left or right) that a route may be followed. The Directional Arrow should always be below the route marker in directional assemblies.

Directional Assembly – An assembly consisting of a Cardinal Direction Marker and other Route Marker auxiliaries if needed, a Route Marker, and a Directional Arrow.

Distance Sign – A sign carrying the names of not more than three communities or points of interest, and the distance (to the nearest mile) to those places. Distance signs are D2-Series signs, typically D2-1 through D2-3 signs.

END Marker – An M4-6 marker used to indicate that the numbered traffic route is ending. The marker should be mounted either directly above the Route Marker, or above a marker for an alternate route that is part of the designation of the route being terminated.

Junction Assembly – An assembly consisting of a Junction Marker and a Route Marker.

Junction Marker -- An M2-1 or M2-1-1 marker with the abbreviated legend JCT mounted at the top of an assembly.

Minor terminal – Communities or nationally known points of interest, except control cities, that are served by the numbered traffic route. Minor terminals can be used to guide motorists and/or to provide supplemental information relative to the position of motorists along their intended travel route. These destinations must be shown on the Department's Official Transportation Map, and may include county seats and communities where there are junctions of major traffic routes.

Numbered traffic route – A highway that has been assigned an Interstate, United States, or Pennsylvania route number to aid motorists in their travels. Any changes to Interstate or United States numbered traffic routes require approval of the American Association of State Transportation Officials (AASHTO).

Route Marker – Markers used to identify and mark a numbered traffic route.

Route Marker Assembly – An assembly which consists of a Route Marker and Route Marker Auxiliaries which further identify the route and indicate direction. Assemblies for two or more routes, or for different directions on the same route, are normally mounted in groups on a common support.

Route Marker Auxiliaries – Supplemental signs used with Route Markers to provide additional information about the numbered traffic route. Route Marker Auxiliaries shall match the color combination of the respective marker which they supplement. Route Marker Auxiliaries include, but are not limited to, Junction Markers, Cardinal Direction Markers, BY-PASS Markers, BUSINESS Markers, TRUCK Markers, TO markers, END Markers, DETOUR Markers, Advance Turn Arrows, and Directional Arrows.

Special Road Facility Symbol – A route marker designating the Pennsylvania Turnpike, or any other Department-approved symbol marker designating routes.

TO Marker – An M4-5 or M4-5-1 marker used to provide directional guidance to a particular road facility from other highways in the vicinity. When used, it is mounted at the top of an assembly.

Trailblazer Assembly – An assembly consisting of a TO Marker, a Cardinal Direction Marker if needed, a Route Marker or special road facility symbol, and a single-headed directional arrow pointed along the route leading to the road. This assembly provides directional guidance to a particular road from other highways in the vicinity.

TRUCK Marker – An M4-4 marker used to designate an alternate route for trucks that branches from a regular numbered route, bypasses an area which is congested or where height or weight limitations have been established, and rejoins the regularly numbered route beyond that area.

2.6.3 Clearview Font

The *MUTCD* currently defers to the *Standard Highway Signs and Markings* (SHS) book for the design of letters, numerals, route shields, and spacing. However, on September 2, 2004, FHWA issued interim approval for the optional use of the Clearview font for positive legends (i.e., white legend) on all guide signs. Research shows that the Clearview font generally improves the maximum nighttime sign legibility distance by approximately 12 percent for the same size sign. (See the interim approval at http://mutcd.fhwa.dot.gov/pdfs/ia_clearview_font.pdf.)

In accordance with the interim approval, FHWA will grant statewide interim approval to use the Clearview font for white legend on any guide sign to any highway agency that submits an appropriate written request. The Department obtained FHWA approval to use the Clearview font for not only the Department, but also for all local authorities within the Commonwealth.

Exhibit 2.6-A shows the transition from Highway Gothic fonts, as mentioned in the *MUTCD*, to Clearview fonts.

Exhibit 2.6-A Transitioning to the Clearview Font

SHS Standard Alphabet	Clearview "W" Series	Typical Abbreviation
Series B	Clearview 1-W	CV 1W
Series C	Clearview 2-W	CV 2W
Series D	Clearview 3-W	CV 3W
Series E	Clearview 4-W	CV 4W
Series E- Modified	Clearview 5-W & 5-W-R*	CV 5W & 5WR
Series F	Clearview 6-W	CV 6W

***** Clearview 5-W-R has slightly tighter letter space than 5-W. The Department is using Clearview 5-W-R instead of 5-W.

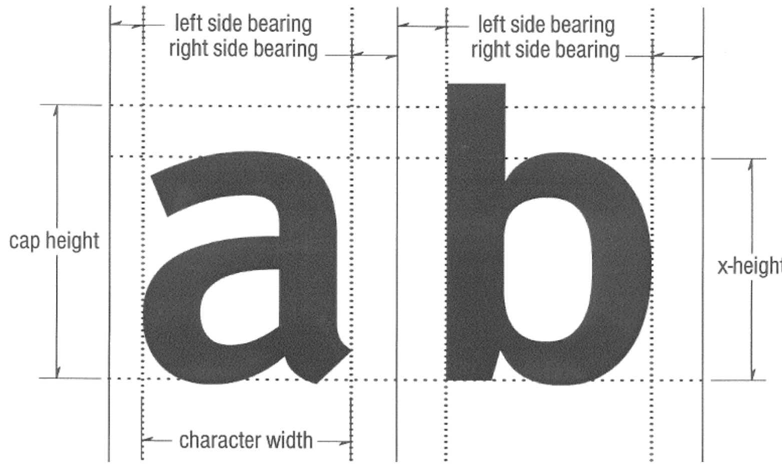
Studies indicate that upper/lower-case Clearview 3-W font has a legibility distance of about 29 percent greater than all capital letters using the Highway Gothic Series D font with the same size footprint. Therefore, since the *MUTCD* allows upper/lower-case legend for place names, there is no reason not to use it for all designations and distances on guide signs.

When designing signs with a Clearview font, it is important to note that the height of the lower-case legend is approximately 82 percent of the height of the upper-case legend. Consequently, you should ignore all references in the *MUTCD* indicating that lower-case letters are to be 75 percent of the height of the upper-case legend, or that the height of the upper-case letters are to be approximately 1.33 times the loop height of lower-case letters. Also of interest is the fact that lower-case letters with top extenders, e.g., b, d, f, h, etc., extend about 8 percent above the top of the upper-case letters.

The height of upper/lower-case legend in this manual is always specified with a single dimension such as "16CV 5WR" for 16-inch Clearview 5-W-R legend, instead of "16/12 EM" for 16/12″ E-Modified legend.

Spacing for Clearview shall conform to the criteria in FHWA's "Clearview Typeface Supplement" at http://mutcd.fhwa.dot.gov/pdfs/clearfont/CTSEng.pdf. The spacing for the Clearview font is different from Highway Gothic, but calculated in the same manner as the current spacing charts included in Section 9 of FHWA's *Standard Highway Signs and Markings* book. Specifically, calculate the spacing by using a combination of a "right side bearing" associated with the first of two letters and a "left side bearing" associated with the second letter as illustrated in Exhibit 2.6 B.

Exhibit 2.6-B Inter-letter Spacing of Clearview Font



The normal letter and numerical height for guide signs on two-way, two-lane conventional roadways is 6CV 3W, and on four-lane undivided conventional roadways should be 8CV 3W, both of which previously were "D" series.

The Clearview font is a TrueType font and is compatible with PCs, sign design software, and computerized sign makers.

2.6.4 Numbered Traffic Routes

2.6.4.1 Types of Traffic Routes

The main purpose of numbered traffic routes is to identify the best travel path between population centers and to indicate these "best paths" on the Official Transportation Map. There are three categories of numbered traffic routes: Interstate (I), United States (US), and Pennsylvania (PA).

All numbered traffic routes are established, eliminated, extended or relocated by the Secretary of Transportation through the Chief, Traffic Engineering and Operations Division (TEOD), Bureau of Highway Safety and Traffic Engineering (BHSTE). However, the American Association of State Highway and Transportation Officials (AASHTO) must also approve the establishment, elimination, or relocation of all Interstate and US numbered traffic routes.

<u>Types of Auxiliary Classification</u>

In addition to the regular numbered traffic route, the Department may assign the following additional auxiliary classifications to segments of US and PA traffic routes:

1. Business Routes. A business route is a route principally within the corporate limits of a larger municipality, usually traversing the central business district, which provides the traveling public an opportunity to travel through the business part of a municipality while the regular route does not pass through the congested part of the municipality. Business routes are often established on the old alignment of the regular route when the regular route is relocated around the municipality. The Business Route connects with the regular numbered route at the opposite sides of the municipality.

2. Truck Routes. A truck route is sometimes desirable to divert through truck traffic around the business or congested areas of larger municipalities when the regular route goes through such areas. Such truck routes not only help to facilitate the flow of passenger vehicles through the congested areas, but also help to reduce air and noise pollution, and assist truck drivers since they can frequently minimize their delays. In addition, truck routes may sometime be required to avoid locations that have insufficient vertical or horizontal clearances for trucks, or bridges or highways with weight restrictions.

3. By-Pass Routes. A by-pass traffic route classification is sometimes assigned to a new roadway that bypasses a congested urban area. However, this classification generally confuses the through drivers since they are forced to make a decision between the regular numbered route and the by-pass. Moreover, most unfamiliar drivers are reluctant to leave a regular numbered route to follow a by-pass route. For these reasons, it is generally better to relocate the regular numbered traffic route to the new alignment and, if necessary, assign a business designation or a new traffic route number to the old alignment. The by-pass route connects with the regular numbered route at the opposite sides of the community.

4. Alternate Routes. Alternate traffic routes provide a route that parallels the regular route and sometimes offers advantages as a scenic view. However, because the unfamiliar drivers do not know whether the regular numbered routes or the alternate route is the best route, avoid using this designation whenever possible.

2.6.4.2 Establishment, Elimination, or Revision of Any Numbered Traffic Route

Public Relations

Review proposed traffic route changes with county planning agencies, municipal officials and the local chambers of commerce, particularly in urban areas and other locations where businesses use the numbered traffic route in their advertising or as an address. This will not only facilitate good public relations but will provide advance information to businesses so that they can plan and implement any required changes in their advertising and/or address.

General Guidelines

To be of maximum benefit to the public, all numbered traffic routes must either be continuous across the Commonwealth, or must end at another numbered traffic routes. All numbered traffic routes should also be continuous, that is, they should not stop and restart at a subsequent location.

Engineering Districts should work with adjoining states to attempt to extend Pennsylvania's numbered traffic routes into their state and vice versa, and to keep the same number whenever possible. For example, Engineering District 6-0 has provided excellent coordination – PA 41 becomes DE 41, PA 52 becomes DE 52, PA 272 becomes MD 272, PA 896 becomes DE 896, etc.

District Requests

In order to facilitate the approval of a traffic route change (establishment, elimination, or revision), the Engineering District must submit the following to the Traffic Engineering and Operations Division:

1. A request similar to the example in Appendix 2A. This request would include a brief justification and a tentative date for the implementation of the proposed change. (Whenever possible, reserve the implementation of major traffic route changes to coincide with the publication of a new edition of the map.)

   This request should also provide a route description for those portions of highway where a proposed elimination or relocation would occur:

   - Provide the route description from south to north, or west to east for two and three-lane two-way conventional roads.

For one-way roadways and all roadways with four or more lanes, provide the description in both directions of travel. Use the format shown in Appendix 2A to request a change.

- Use the currently assigned SRs and segments. If the SR nmbers and/or segments are to be changed, coordinate all efforts through the District RMS Coordinator and the Roadway Management Division of the Bureau of Maintenance and Operations (BOMO).

2. One copy of a reproducible map or sketch indicating the proposed establishment and/or elimination. As an alternate to the reproducible map or sketch, provide a minimum of nine color-coded copies of a map. It is desirable to have all control points (transition points from one SR to another - or other official identification data if not an SR, beginning and end points, etc.) identified by SR Number, segment and offset (or other identification data if not a State Route).

3. A listing of all physical limitations of the proposed establishment or relocation, similar to the example in Appendix 2B.

4. The local authority must complete three original agreements (see Appendix 2C) between the local authority and the Department for the establishment of all proposed traffic routes over a local road. A copy of a resolution designating signature authority (see Appendix 2D) must also be attached to each agreement

2.6.4.3 Requests to Change US and Interstate Routes

Approvals for the establishment, elimination or revision of Interstate, US, and US Bicycle numbered traffic routes comes under the jurisdiction of AASHTO's "Special Committee on U.S. Route Numbering." Although all changes technically come under their control, the Department is only seeking their approval when the change involves any of the following:

1. Another state.

2. The relocation of the junction with another Interstate or US numbered route.

3. The beginning or ending location of the route.

4. The length of the route changes by more than 0.5 mile.

If any of the above applies, the Engineering District needs to provide either an electronic copy or seven paper copies of the AASHTO application for their "Special Committee on U.S. Route Numbering" to the Traffic Engineering and Operations Division (TEOD). Districts may submit an AASHTO application to TEOD at anytime, but in order to be considered at the Committee's annual fall meeting, TEOD generally needs the applications by the end of July.

The homepage for AASHTO's "Special Committee on U.S. Route Numbering" is at http://cms.transportation.org/?siteid=68&pageid=1538. This site includes the committee's *"United States Numbered Highways"* book, which in turn includes their established policies and a route-by-route description of all US numbered traffic routes. This homepage also includes a link for the application to make changes in an Interstate Route, US Route, or US Bike Route.

2.6.5 Route Marker Assemblies

2.6.5.1 General

Erect Route Marker Assemblies along numbered traffic routes on all approaches to the intersection of other numbered traffic routes, and may be erected on the approaches to numbered routes on unnumbered major State highways that carry an appreciable amount of traffic destined for the numbered route.

Where two or more numbered traffic routes follow the same section of highway, the Route Markers for Interstate, U.S., or Pennsylvania routes shall be mounted in that order from the left in horizontal arrangements and from the top in vertical arrangements. Moreover, within these three types of routes, Route Markers for lower-numbered routes shall be above or to the left of the higher-numbered routes.

Within groups of assemblies, information for routes intersecting from the left shall be to the left in horizontal arrangements and at the center of vertical arrangements. Similarly, information for routes intersecting from the right shall be at the right or bottom, and for straight-through routes at the center or top.

2.6.5.2 Junction Assembly

Erect the Junction Assembly along numbered traffic routes in advance of every intersection where another numbered traffic route intersects or joins the route being traveled. The Junction Assembly may also be installed along major unnumbered State highways in advance of their intersection or junction with a numbered traffic route. The route marker shall carry the number of the intersected or joined route. Where two or more routes are to be indicated, one Junction Marker can be used for the assembly and the route markers grouped in a single mounting.

Examples of Junction Assemblies are shown in Figure 2.D-6 in the *MUTCD*.

2.6.5.3 Advance Route Turn Assembly

An Advance Route Turn Assembly shall be installed on numbered traffic routes in advance of an intersection where a turn must be made to remain on the indicated route or to follow an intersecting route. The Advance Route Turn Assembly on numbered traffic route approaches may be omitted when either of the following conditions exists:

(a) All traffic on the approach must stop at a stop sign, and the approach is a single lane, without separate turning lanes.

(b) There is inadequate longitudinal space or other physical reasons why the assembly cannot be installed.

On multilane highway approaches to an interchange or intersection, an Advance Route Turn Assembly is essential so that drivers may position their vehicles in the correct lane to follow the desired numbered traffic route. An assembly which contains an Advance Turn Arrow should not be placed where there is an intersection (or major driveway which looks like a street or roadway) between the sign assembly and the designated turn. Sufficient distance should be allowed between the assembly and any preceding intersection (or major driveway which looks like a street or roadway) that could be mistaken for the indicated turn. Where two or more routes turn, the signing may be installed as a group on a common support.

Examples of Advance Route Turn Assemblies are shown in Figure 2D-6 in the *MUTCD*.

2.6.5.4 Directional Assembly

A Directional Assembly shall be installed on numbered traffic routes at all intersections where a turn must be made to remain on a route, to follow an intersecting route, or to indicate a straight-ahead movement to follow an intersecting route. Uses of Directional Assemblies are:

- Straight-through movements of a numbered traffic route should be indicated by a Directional Assembly with a Route Marker displaying the number of the continuing route, a Cardinal Direction Marker, and a vertical arrow. A Directional Assembly should not be used for a straight-through movement in the absence of other assemblies indicating right or left turns, as the Confirming Marker beyond the intersection normally provides adequate guidance.

- Turn movements (as indicated by an Advance Route Turn Assembly) shall be marked by a Directional Assembly with a Route Marker displaying the number of the intersected or continuing route, a Cardinal Direction Marker, and a single-headed arrow pointing in the direction of the turn.

- The beginning of a route (as indicated in advance by a Junction Assembly) shall be marked by a Directional Assembly with a Route Marker displaying the number of that route, a Cardinal Direction Marker, and a single-headed arrow pointed in the direction of the turn.

- The end of a route shall be mounted by a directional assembly with an END Marker and a Route Marker displaying the number of the route.

- An intersecting route (as indicated in advance by a Junction Assembly) shall be marked by one or two Directional Assemblies (one if one direction or two if two directions), each with a Route Marker displaying the number of the intersected route, a Cardinal Direction Marker, and a single-headed arrow pointed in the direction of movement on that route.

Examples of Directional Assemblies are shown in Figure 2D-6 of the *MUTCD*, and recommended locations for directional assemblies are as follows:

(a) Directional Assemblies should be located on the near right-hand corner of the intersection.

(b) At major intersections and at Y or offset intersections, it is often desirable to install additional assemblies on the far right-hand or left-hand corner to confirm the near-side assemblies.

(c) When the near right-hand corner position is not practical for Directional Assemblies, the far right-hand corner is the preferred alternative, with oversize signs if necessary for legibility.

(d) If cross traffic interferes with the visibility of a Directional Assembly, place the assembly where it can be read at close range as determined by engineering judgment.

2.6.6 Additional Route and Destination Guidance Signs

2.6.6.1 Confirming or Reassurance Assemblies

The Confirming Assembly shall be installed just beyond intersections of numbered traffic routes and should be installed beyond other major intersecting streets or highways. Care should be taken to erect the Confirming Assembly where it highly visible to drivers. Examples of confirming assemblies are shown in each example in Figure 2D-6 of the *MUTCD*.

2.6.6.2 Trailblazer Assemblies

Trailblazer Assemblies are located at strategic locations to indicate the direction to the nearest or most convenient point of access to a specific numbered traffic route or other special road facility. Trailblazer Assemblies indicate that the road or street where it is posted is not a part of the indicated route, but that the driver is merely being directed progressively to the route.

When a Trailblazer to a highway facility is installed, it is essential that, from that point, additional trailblazers or other appropriate directional signing, consistent with this policy, be provided at other key downstream locations en route to that highway facility.

Trailblazer Assemblies may be erected with other route marker assemblies, or alone, in the immediate vicinity of designated facilities. In addition, Trailblazer Assemblies may be used to guide drivers to a freeway on-ramp from a partial interchange where access to both directions of the freeway is not available.

2.6.6.3 Destination Signs

Destination Signs may be installed at, or in advance of, intersections to direct motorists to cities, boroughs, towns, villages, or other important destinations. These signs normally carry the names of two or three major terminal destinations – one straight ahead along the route and one in each direction along the intersecting roadway. The destination that is straight ahead is optional; however, if used, it is the first name on the sign, the destination to the left is second, and the destination to the right is on the bottom. (More than one destination may be shown to the left or right;

however, the maximum number of destinations on a sign should not exceed three.)

If there is more than one destination shown in any direction, the name of the nearest destination shall appear above that of any further away. In the case of overlapping routes, only one destination in each direction should be shown for each route.

The distance to a destination, in the nearest number of whole miles, may also be shown on the destination sign. Distances should be measured to the center of the destination.

If adequate longitudinal space does not exist for proper spacing of all sign assemblies, the destination sign may be installed on the same posts with the advance route turn or directional assemblies.

Examples of destination signs are shown in Figure 2D-6 of the *MUTCD*.

2.6.6.4 Distance Signs

Distance signs should be installed on all numbered traffic routes leaving a city, borough, or town and beyond intersections of numbered traffic routes in rural areas. These signs normally carry the names of two or three destination points. The top name should be that of the next place along the route having a post office, railroad station, route number or the name of an intersected highway, or other significant geographical identity that is shown on the Department's Official Transportation Map. The lowermost name should be the next control city along the route. If three destinations are shown, the middle line should be a minor terminal prior to the control city. If more than one minor terminal exists prior to the control city, the name on the middle line may be varied on successive distance signs.

The control city should remain the same on all successive signs throughout the length of the route until that destination is reached. An exception to this is when the route divides at some distance ahead to serve two control cities of similar importance. If the two destinations cannot appear on the same sign, the names of the two control cities may be alternated on succeeding signs. On the route continuing into another State, destination(s) in the adjacent State should be shown.

Examples of distance signs are shown in Figure 2D-6 of the *MUTCD*.

2.6.6.5 Advance Street Name Signs

Advance Street Name (D3-2, D3-3) signs are encouraged on numbered traffic routes when the speed limit is greater than 35 mph, and the roadway has two or more through travel lanes in each direction.

If an Intersection Series warning sign, or a Stop Ahead (W3-1) or Signal Ahead (W3-3) sign is used in advance of the intersection, the preferred scenario is to use a Single-Line Advance Street Name (W16-8) or Double-Line Advance Street Name (W16-8A) sign below the warning sign as discussed in Section 2.5.6 instead of installing an independently-mounted D3-series sign.

2.6.7 Longitudinal Placement

The guidelines shown in Exhibit 2.6-C should be considered when locating Route Marker Assemblies, Destination Signs, Distance Signs, and Advance Street Name Signs in the field, since the exact longitudinal placement of these signs cannot be firmly established. It is more important that these signs be readable at the right time and place than to be located with absolute uniformity.

Typical installations of numbered traffic route signing are shown in the three drawings in Figure 2D-6 of the *MUTCD.*

Route Marker Assemblies, including the Route Marker and all auxiliary signs, are treated as a single sign for vertical clearance purposes.

2.6.8 Sign Sizes

The standard sizes for Route Markers used on conventional roads shall be 24 inches by 24-inches for one- and two-digit route numbers, and 30 inches by 24-inches for three-digit route numbers. The use of 36 x 36-inch route markers for one- or two-digit route numbers and 45 x 36-inch Route Markers for three-digit route numbers is encouraged for roadways with three or more lanes where space permits.

On freeways and expressways, 36 x 36-inch Route Markers shall be used for one- or two-digit route numbers and 45 x 36-inch Route Markers shall be used for three-digit route numbers.

Route Marker auxiliaries shall be the appropriate size for the route marker as specified in Publication 236M.

Advance Street Name (D3-2, D3-3) signs, Destination Signs, and Distance Signs shall be a minimum width of 72 inches, except where space will not permit.

Advance Street Name Signs mounted with warning signs shall be 48 inches wide when used with 30″x30″ or 36″x36″ warning signs, and 72 inches wide when used with 48″x48″ warning signs.

Exhibit 2.6-C Longitudinal Placement

- You may adjust the distances shown below to accommodate actual field conditions and to obtain proper spacing between signs.
- When installing more than one sign or sign assembly, the minimum desirable distance between sign installations should be 200 feet in rural areas and 150 feet in urban areas.
- When the speed limit or prevailing speeds are above 45 mph, greater distances from the intersection and greater spacing between sign installations are desirable.

Sign or Sign Assembly	Urban Area	Rural Area
Junction Assembly	• In block before intersection.	• Not less than 400 feet in advance of intersection.
Advance Route Turn Assembly	• Approximately 300 feet in advance of intersection. • May omit when all traffic on the approach must stop at a stop sign, and the approach is a single lane, without separate turning lanes. • May omit because of inadequate longitudinal space or other physical reasons.	• Not less than 400 feet in advance of intersection. • May omit when all traffic on the approach must stop at a stop sign, and the approach is a single lane, without separate turning lanes. • May omit because of inadequate longitudinal space or other physical reasons.
Destination Sign	• Not less than 150 feet in advance of intersection. • At the intersection on a one-lane approach where all traffic must stop at stop sign. • May combine with advance route turn or directional assemblies if space is limited.	• Not less than 200 feet in advance of intersection. • At the intersection on a one-lane approach where all traffic must stop at stop sign. • May combine with advance route turn or directional assemblies if space is limited.
Directional Assembly	• Near right-hand corner of intersection. • Alternate location may be far right-hand corner or most suitable location based on engineering judgment.	• Near right-hand corner of intersection. • Alternate location may be far right-hand corner or most suitable location based on engineering judgment.
Confirming Assembly	• Normally no more than 100 feet beyond the far shoulder or curb line of all intersection numbered traffic routes and major intersecting streets or highways. • Beyond the built-up area of any city, borough, or town. • Should be installed at intervals approximately one-half mile apart.	• Normally no more than 300 feet beyond the far shoulder or curb line of all intersection numbered traffic routes and major intersection streets or highways. • Beyond the built-up area of any city, borough, or town. • Should install at intervals approximately 2 miles apart in built-up areas. • May install at intervals of approximately 5 miles in areas with little or no development and only minor intersections.
Distance Sign	• Approximately 500 feet outside the limits of a city, borough, or town, or at the edge of the built-up area if this area extends beyond the corporate limits. • Where overlapping routes separate a short distance from the corporate limits, the distance sign at the corporate limits may be omitted, and instead should be erected approximately 300 feet beyond the separation of the two routes. • May be combined with confirming assembly if space is limited.	• Approximately 300 feet beyond the far shoulder or curb line of all intersecting numbered traffic routes when these numbered routes are more than one mile apart. • May combine with confirming assembly if space is limited.
Intersection-Related Warning Signs (W2 Series, W3-1, & W3-3)	• As specified in Table 2C-4 of MUTCD. • Erection shall be based on engineering justification, and they shall not be installed solely for the purpose of facilitating advance street name signing. • Normally, do not W2 Series signs when junction or advance route turn assemblies are present.	
Advance Street Name Sign	• When mounted with warning sign, see intersection-related warning sign category above. • When mounted alone, see destination sign category above.	

2.7 Tourist Oriented Directional Signs (TODS) Policy

2.7.1 Purpose, Authority, and Authorization

2.7.1.1 Purpose

The purpose of this policy is to establish guidelines for the installation of signs within State highway right-of-way to guide travelers to businesses, services, and attractions in which the traveling public would have reasonable interest. These guidelines include the eligibility, location, design, installation, cost, and maintenance of these signs, which herein after will be referred to as "TODS."

2.7.1.2 Authority

The provisions of this chapter are promulgated under 75 Pa.C.S. §6125(d).

2.7.1.3 Authorization

Only Department approved TODS may be installed within the State highway right-of-way. However, the authorization of signs is not an endorsement of the applicant's facilities.

2.7.2 Definitions

The following words and terms, when used in this policy, have the following meanings, unless the context clearly indicates otherwise:

Attraction – Any facility which qualifies for signing under this policy.

Conventional road – Any free-access public highway other than a freeway or expressway.

Department – The Department of Transportation of the Commonwealth of Pennsylvania.

Expressway – A divided arterial highway for through traffic with partial control of access and with interchanges at junctions with high-volume highways. For purposes of this policy, sections of expressway with at-grade intersections will be considered as a "conventional road," and sections of expressway with interchanges will be considered as "freeway."

Freeway – A divided highway with full control of access to which the only means of ingress and egress is by interchange ramps.

Local authorities – County, municipal and other local boards or bodies, and State agencies other than the Department having authority to enact regulations relating to traffic. The term includes governing bodies of colleges, universities, public and private schools, public and historical parks, and airport authorities except where those authorities are located within cities of the first class or cities of the second class.

Official traffic control devices – Signs, signals, markings, and devices consistent with 75 Pa.C.S. (relating to Vehicle Code) and Department regulations, placed or erected by authority of a public body or official having jurisdiction for the purpose of regulating, warning or guiding traffic.

On-premise sign – A sign which is erected upon the same real property that the business, facility or point of interest is located. The signs shall only advertise the business, facility or point of interest located thereon.

Rural area – Any geographic area which is not included in an urban area on the Department's County Functional Classification Maps.

Rural conventional road – Any public conventional road in a rural area.

Seasonal business – Any business which is not operated on a year-round basis. The business shall be responsible for covering or removing signs as outlined in Section 2.7.10.

Secretary – The Secretary of the Department.

Signing district – A geographical area for which a governmental sponsor is willing to enter into an agreement with the Department to coordinate, provide, install and maintain all signing authorized by and in conformance with this policy after approval by the Department, without bias to any businesses and at no cost to the Department. In order to ensure that the sponsor has the support of the attractions, a public hearing is required. When the signing district concept is used, alternate sign designs may be authorized.

TODS – An official sign that is located within the right-of-way of a conventional road that indicates the name of an attraction and gives directional guidance to the same.

TODS Assembly – A single TODS installation consisting of sign posts, anchor posts, and a maximum of three individual signs.

Urban area – Any geographic area with a population of 5,000 or more inhabitants, with boundaries fixed by State and local officials in cooperation with each other, approved by the Secretary, and designated as an urban area on the Department's County Functional Classification Maps.

Urban conventional road – Any public conventional road in an urban area.

2.7.3 General Eligibility Requirements

2.7.3.1 General

The attraction shall be open to all persons regardless of race, color, religion, ancestry, national origin, sex, age or handicap; be neat, clean and pleasing in appearance; be maintained in good repair; and comply with all Federal, State and local regulations and statutes for public accommodations concerning health, sanitation and safety. Pursuant to federal regulations promulgated under the authority of The Americans With Disabilities Act, 28 C.F.R. §35.101, et seq., the participant understands and agrees that no individual with a disability shall, on the basis of the disability, be excluded from the attraction.

2.7.3.2 Distance to Services

Except as otherwise provided in this policy, on all conventional roads, the maximum distance from the intersection for which attractions can be trail blazed and qualify for TODS shall be 5.0 miles.

2.7.3.3 Local Ordinance

TODS shall not be installed when prohibited by local ordinance.

2.7.3.4 Admission Charges

If a general admission is charged, it shall be collected upon entry and any other charges shall be clearly displayed, at the place of entry, so as to be readily visible to an ordinarily observant person.

2.7.3.5 Hours of Operation

Attractions other than arenas, schools, colleges/universities, campgrounds, cultural centers, fairgrounds, farmer's markets, religious sites, roadside farm markets, and military bases shall maintain regular hours and schedules and be open to the public at least 6 days each week for at least 3 consecutive months of the year. In addition, farmers markets and roadside farm markets shall maintain regular hours and schedules and be open to the public at least 2 days each week during the normal business season.

2.7.3.6 Other Signs

TODS will not be authorized if an illegal advertising sign exists along any State highway for that specific business, or if a legal advertising sign exists on the same highway approach as the request for a TODS. In addition, if the attraction has in place any other Department-approved signing, additional signing or redundant signing will not be authorized on the same highway approach.

2.7.3.7 Sufficient Space

Sufficient space exists in accordance with Sections 2.7.6.2 and 2.7.6.3 to install signs at all locations along the route to the attraction where a turn is required.

2.7.3.8 On-premise Sign

The attraction shall have an on-premise sign identifying the name of the facility. If the facility or its on-premise signing is readily visible from the highway, a TODS shall not be placed immediately in advance of the business.

2.7.3.9 Parking Accommodations

The attraction shall have adequate on-premise or available on-street parking for patrons.

2.7.3.10 Road System

The location of the attraction shall not require motorists to perform any illegal movements or U-turns, and the roads shall be capable of handling the anticipated traffic volume and types of traffic. Motorists shall be able to readily return to the highway and proceed in the original direction of travel after visiting the attraction. This may result in the attraction being required to install signing to guide the motorist to their original direction of travel.

2.7.3.11 Route Continuity

TODSs will be installed in advance of all necessary turns subsequent to the initial TODS installation. If a sign is required on a local roadway between a State highway and the attraction, the local authorities must authorize and are responsible for the installation of the sign (by themselves or by others) on their roadway prior to the installation of signs on any State highway that would direct motorists to that local roadway.

2.7.4 Additional Eligibility Requirements

2.7.4.1 General

Additional eligibility requirements may apply depending on the type of highway and the type of area where the signs are to be installed. The requirements are less restrictive for signs installed along rural conventional roads than for signs installed along urban conventional roads.

2.7.4.2 Rural Conventional roads

TODSs may be authorized along any rural conventional road for the eligible types of attractions as defined in either Section 2.7.5.1 or Section 2.7.5.2, which meet the general eligibility requirements under Section 2.7.3, and are approved by the local municipalities within which TODSs are to be located. The approval of the local authorities is not required for a TODS installed to direct motorists to attractions operated by State or federal agencies.

2.7.4.3 All Conventional roads

TODSs may be authorized along any conventional road either urban or rural for eligible types of attractions as defined in Section 2.7.5.2, which meet the general eligibility requirements under Section 2.7.3, the attendance provisions of Section 2.7.5.3, and are approved by the local municipalities within which the TODSs are to be located. The approval of the local authorities is not required for a TODS installed to direct motorists to attractions operated by State or Federal agencies or signs installed under a signing region as defined in Section 2.7.9.

2.7.5 Eligible Types of Attractions

2.7.5.1 Rural Conventional roads

An attraction which satisfies one or more of the following definitions and the provisions of Section 2.7.3 may be considered for a TODS only along rural conventional roads:

> *Bed and breakfast* – A private residence located in a rural area that contains 10 or fewer bedrooms used for providing overnight

accommodations to the public, and in which breakfast is the only meal served and is included in the charge for the room.

Observatory – A facility designed and equipped for making observations of astronomical, meteorological, or other natural phenomena.

Recreational activity – Boating, rafting activities, spelunking, and other recreational activities recognized by the Department.

Specialty shop – A shop offering goods or services of unique interest to tourists, and which derives the major portion of its income during the normal business season from motorists that do not reside in the immediate area. The goods or services shall be readily available to tourists, without the need for scheduling appointments or return trips. The Department's appropriate Engineering District Office will screen these shops to determine eligibility.

Unique natural area – A naturally occurring area which is of outstanding interest to the general public, such as a waterfall or a cavern.

2.7.5.2 All Conventional roads

An attraction which satisfies one or more of the following definitions, the provisions of Section 2.7.3, and the any minimum attendance provisions of this section may be considered for TODSs along any conventional road either urban or rural.

Airport – A public-use facility licensed by the Department for the landing and takeoff of aircraft, and for receiving and discharging passengers and cargo.

Amusement park – A permanent area which is open to the general public for entertainment rides.

Arena – A stadium, sports complex, auditorium, civic center or racetrack, which has a seating capacity of at least 5,000.

Business district – An area within a city or borough which is officially designated as a business district by the local officials.

Campground – An area reserved for at least 20 tents or recreational vehicles, which possesses a valid permit from the Pennsylvania

Department of Environmental Protection and is open to the public with continuous operation for at least 6 months per year.

College or university – An institution which is approved by a nationally-recognized accreditation agency and which grants degrees.

Commerce park – A group of commercial manufacturing facilities, at least 25 acres in size, recognized and signed as a commerce park by the local authorities.

Cultural center – A facility for the performing arts, exhibits, or concerts which has a minimum occupancy capacity of 250 people.

Facility, tour location – A facility, such as a plant, factory, or institution which conducts daily or weekly public tours on a regularly scheduled basis year-round.

Fairground – A commercially-operated tract of land where fairs or exhibitions are held, and which has permanent buildings included but not limited to livestock exhibition pens, exhibition halls, bandstands, etc.

Farmer's market – A building, structure, or place used by two or more farmers or an association of farmers for the purpose of selling farm and food products, including but not limited to fruits, vegetables, flowers, meats and baked goods, at least 2 days per week throughout the year or harvest season.

Golf course – A facility open to the public and offering at least nine holes of play. Miniature golf courses, driving ranges, chip and putt courses, and indoor golf shall not be eligible.

Historical site – A structure or area recognized by the Pennsylvania Historical and Museum Commission as a historic attraction that has national or state significance and is cited in publications or magazines. Historic districts shall provide the public with a single, central location, such as a self-service kiosk or welcome center, where motorists can obtain information regarding the historic district.

Hospital – A hospital approved by the Department of Health which offers continuous emergency care for the general human public, and with a doctor on duty at all times. When applicable, the term "trauma center" may be used on the sign.

Institution – A health care center operated by a county, State, or federal government.

Library – A building that stores books, manuscripts, historic documents, and other information sources for use by the public, but not for sale.

Military base – A facility operated by the State or federal government for training or support of military troops, or for inventorying and warehousing military equipment.

Municipal buildings – A building housing the primary offices for city, borough, township, or county authorities.

Museum – A facility that cares for and exhibits works of artistic, historical, or scientific value for the public.

Off-track betting facility – A facility that provides off-premise wagering as authorized by Act 1988-127.

Park – An area which is open to the general public with on-premise parking for activities such as picnicking, hiking, swimming, boating, basketball, tennis, etc.

Religious site – A facility of unique religious nature, including a shrine, grotto, or similar type facility.

Resort – A facility with at least 75 rooms and those recreational amenities normally present at a facility, and which is the main focal point of a vacation. In order to qualify as a resort, the facility should be situated to take advantage of natural, historic or recreational attractions. Meals and the use of recreational facilities should normally be included in the room rate.

Roadside farm market – A stationary retail sales establishment operated by one or more farmers for the purpose of selling farm and food products directly to consumers. Operations by which the consumer harvests their own farm or food products shall be considered roadside farm markets. Roadside farm markets shall be open at least 2 days per week throughout the harvest season or year. On-premise or legal on-street parking shall be available as provided in Section 2.7.3.11.

School – Any facility for education wherein a resident of the Commonwealth can legally fulfill compulsory school requirements for any one or more grades, including kindergarten through grade 12.

Shopping center – A group of 30 or more specialty shops (antique, craft, outlet, farmers' market, etc.) or retail stores with ample parking facilities.

Ski area – A downhill skiing area with equipment rentals, or a cross country ski area with equipment rentals and a minimum of 5 miles of marked and groomed trails.

State and national park, recreation area, forest or cemetery – An area so designated and under the jurisdiction of the National Park Service, the Veterans Administration, or Pennsylvania Department of Conservation and Natural Resources. The mileage requirement for these facilities may be extended a reasonable distance to accommodate driver need.

Transportation terminal – A bus or railroad passenger terminal, complete with ticket sales.

Visitor information – A facility approved by the Department of Commerce as a visitor information center, and which is open at least 6 months each year, including 10 hours each day between Memorial Day and Labor day, and 8 hours each day during the balance of the open season. The facility shall have an attendant on duty during open hours, and provide free access to travel literature, rest rooms and drinking water. Centers other than those owned and operated by the Commonwealth of Pennsylvania must be administered or approved by the appropriate local tourist promotion agency.

Winery – A licensed site which produces a maximum of 200,000 gallons of wine per year. Sites shall maintain a minimum of 3,000 vines or 5 acres of vineyard in the Commonwealth; be open to the public for tours, tasting and sales a minimum of 1,500 hours per year and provide an educational format for informing visitors about wine and wine processing.

Zoological/botanical park – A collection of unique living animals or plants.

2.7.5.3 Minimum Attendance

In urban areas, attractions other than airports, transportation terminals, campgrounds, hospitals, commerce parks, roadside farm markets, schools, transportation terminals, visitor information centers, and publicly owned or operated facilities, shall have a minimum of 15,000 visitors per year. In urban areas, new facilities or those open to the public less than 12 months shall have invested at least $250,000 and anticipate at least the above minimum number of visitors. There is no minimum attendance required for attractions in rural areas.

2.7.6 Location, Spacing, and Design of TODSs for Conventional roads

2.7.6.1 General

TODSs may be installed to direct traffic to each entrance of an eligible attraction beginning at the nearest access point from a conventional road with an average of at least 2,000 vehicles per day. Signs shall not be authorized to direct motorists onto or off of any freeway or expressway. Signs with straight ahead arrows will not be authorized, except where the Department deems necessary to provide positive guidance.

2.7.6.2 Location.

Install TODSs in advance of the intersection where a motorist leaves the primary highway system and at all subsequent locations where the motorist is required to turn in order to travel to the attraction. When the attraction, or the attraction's on-premise sign, is readily visible from the highway, do not install a TODS immediately in advance of the attraction. All TODSs should be on the right-hand side of the highway and where sufficient space is available.

TODSs should be located to take advantage of natural terrain, to minimize the impact on the scenic environment, and to avoid visual conflict with other signs within the highway right-of-way. Department-approved breakaway sign supports shall be used.
When an at-grade intersection on a primary highway is replaced with an interchange, the location shall no longer qualify for TODSs and any TODS previously erected shall be removed by the Department.

TODSs shall be located so as not to interfere with, obstruct, or divert driver's attention from any official traffic control device. Official traffic control devices placed at intersection approaches subsequent to the placement of TODSs shall have precedence as to location and may require the relocation of TODSs. In general, TODSs shall be installed at least 200 feet from other official traffic control devices.

TODSs shall be positioned in such a manner that does not restrict drivers' vision when entering the highway from side roads or driveways.

TODS sign panels shall not be displayed for any business which is readily visible and identifiable within 200 feet along the highway.

2.7.6.3 Spacing

TODSs shall be located not less than 200 feet nor more than 1,320 feet in advance of a location where a turn is required from the highway. At intersections where more than one TODS assembly is required, the minimum spacing between such assemblies should be 200 feet. The maximum number of TODS assemblies at an intersection shall be two.

2.7.6.4 Design of TODSs on Conventional roads

Sign layout shall be in accordance with Exhibit 2.7-A. Each TODS shall have one or two lines of legend which should generally be limited to the name of the attraction or an abbreviation thereof. A maximum of 16 letters and spaces shall be permitted on each line unless specific approval for an increased number of letters and spaces is granted by the Department. Legends shall not include promotional advertising.

Generally, a directional arrow shall be required. If the distance to the business is 1/4 mile or greater, the distance in miles should be included below the arrow. The distance may be 1/4, 1/2, 3/4, or the nearest whole mile. When necessary, the sign may have a full-width message without a directional arrow, with a second line message such as “DRIVEWAY ON LEFT,” “LEFT 500 FEET,” “RIGHT AT STOP SIGN,” etc.

The standard sign size shall be 72″x24″. Where insufficient right-of-way exists, smaller signs measuring 48″x16″ may be authorized. All signs shall be of the same size where multiple TODSs are installed on a single sign assembly.

TODSs shall have white reflectorized legend and border on a blue reflectorized background. A brown reflectorized background may be authorized for State and National parks, recreational areas and historical sites. All signs shall be fabricated by a Department-approved sign manufacturer using a Department-approved Type III or IV retroreflective sheeting.

Generic symbols may be used on TODSs at the beginning of the legend area. Any generic symbol included in Exhibit 2.7-B, or included as a recreational or cultural interest area symbol in either the FHWA’s *Standard Highway Signs and Markings* book or the *MUTCD* is permitted for use.

2.7.6.5 Arrangement.

TODSs will normally be installed as independent sign assemblies. A maximum of six TODSs shall be authorized for installation on any approach to an intersection.

When the number of TODSs at an intersection approach is three or less, the signs shall be grouped together with signs displaying arrows pointing to the left above those pointing to the right. If any TODSs with straight-ahead arrows (as is the case where the road turns and the access is straight-ahead) are authorized, the sign for the straight-ahead attraction shall be installed above any signs for attractions to the left or to the right; except that seasonal attractions shall be mounted below all other signs regardless of orientation of directional arrow.

If the number of TODSs at an intersection approach is more than three, the signs shall be grouped as two separate sign assemblies with a maximum of three signs per assembly. The first sign assembly should generally be limited to attractions with straight-ahead or left arrows, and the second sign assembly will generally be limited to attractions with right arrows. Install seasonal businesses on the second assembly.

If more than one business exists in a given direction, the signs for a closer business shall be mounted above the more distant business.

The top of the sign assembly shall be a minimum of 9 feet above the ground. The bottom sign shall be a minimum of 5 feet above the near edge of roadway and 7 feet above the ground where pedestrian traffic may exist.

2.7.7 Sign Installation, Cost and Maintenance

2.7.7.1 Installation

The Department will normally supply and install all TODSs. However, the Department reserves the right to require that applicants arrange for signs to be supplied, installed, and maintained by a private contractor after the Department approval for the installation. At locations where sidewalks exist, local authorities will be requested to install the signs or authorize the attraction's private contractor to install the signs. The following are situations when the Department does not normally install the TODSs:

- When the signing district concept is used, the sponsor shall be responsible to have signs manufactured and installed after receiving Department approval of the proposed sign design and locations.

- At locations where sidewalks exist, local authorities will be requested to install the signs.

- When TODSs are required on local roads, signs shall be manufactured by a Department-approved sign manufacturer, and installed in accordance with Department standards.

- At the discretion of the Engineering District Office, for example, when the county has a significant backlog of signing work.

Upon completion of the TODS sign installation(s), each applicant shall notify the appropriate Engineering District. Upon an inspection by Engineering District personnel, if the installation is unacceptable, the applicant will be advised in writing to have the situation corrected. Upon re-inspection, if the installation is again unacceptable, or the applicant does not notify the Engineering District of the installation within 10 working days upon completion, the Department may remove the sign and the business will not be reimbursed for the sign costs or the administrative fee noted in the application (Exhibit 2.7-D).

2.7.7.2 Costs

Each applicant shall be responsible for all costs incurred for the TODSs and their installation. In addition, the applicant shall be responsible for all costs incurred due to the adjustment, relocation, covering or removal of TODSs to comply with the requirements set forth in this policy.

With all new applications, the applicant is required to pay a $100 administrative fee will be billed to the applicant following review and approval of the TODS request. Attractions requesting replacement signs for an approved facility are not required to pay additional administrative fees.

The attraction is directly responsible for any payments made to municipalities and private contractors for the installation of signs. The attraction is responsible for the assuring the work done by private contractors is in accordance with Department standards.

When the Department installs TODSs, the cost of each sign (D7-4) will be as follows:

72"x24" Standard TODS	$650
48"x16" Smaller TODS	$520

2.7.7.3 Maintenance

The Department will perform minor maintenance for Department-installed signs such as straightening posts or signs, and reserves the right to maintain, and adjust all signs within its right-of-way. If a replacement sign is necessary due to deterioration, crash or vandalism, the attraction shall be responsible for the sign replacement. If the replacement of a Department-installed sign is necessary, the attraction shall be responsible for the same payment as a new sign. (The Department will provide a cost estimate to the attraction for its concurrence prior to replacing a sign.)

2.7.7.4 Removal

The Department will notify attractions if they observe a TODS in a state of disrepair, and if the attraction has not taken corrective action within 60 days after notification, the Department will remove any sign with no reimbursement for sign costs or administrative fees.

2.7.7.5 Missing Signs

It is the responsibility of the owner of the attraction to review their signs and to advise the Department of any missing signs as soon as the problem exists. Failure to ensure that all signs are in place shall be cause for removal of all remaining signs.

2.7.7.6 Cooperation

Attractions requiring signs to be installed at the same location are encouraged to work with other attractions to share costs when the work is being done by private contractor(s).

2.7.8 Signing Districts

2.7.8.1 General

As opposed to signing individual facilities from the nearest access point from a conventional road with an average of at least 2,000 vehicles per day, the purpose of a signing district is to provide an overall, uniform signing concept for various facilities located in a central geographical area. A cohesive signing concept may encourage traffic flow to general destinations including, but not limited to cultural attraction areas, recreational attraction areas, shopping areas, and universities, and may then direct motorists to specific attraction locations.

2.7.8.2 Agreement

In order to establish a signing district, a governmental sponsor shall submit an application to the Department (see Exhibit 2.7-E) and agree to enter into an agreement with the Department to coordinate, obtain, erect and maintain all signs associated with the signing district. The governmental sponsor must ensure that all facilities eligible for signing under this chapter are provided an opportunity to participate in the signing district. A public meeting shall be held to provide attractions with an opportunity to become involved. The removal of existing “illegal” or permitted advertising signs shall be evaluated to avoid and reduce sign clutter on the highways.

Design Sign designs and color schemes will comply with those indicated in Section 2.7.6.4. As an alternate, signing schemes referenced in the PennDOT Toolbox for Development of a Wayfinding Signing Region may be used. See Section 2.7.9.2.

2.7.8.3 Installation

Department approval shall be obtained for the proposed sign locations. An agreement shall be executed between the parties before the manufacture or installation of any signs. The governmental sponsor shall be responsible for installation and maintenance of signs as outlined in the agreement.

2.7.9 Signing Regions

2.7.9.1 General

In contrast to signing individual facilities from the closest numbered traffic route, the purpose of a signing region is to sign a large number of varying facilities in a systematic manner. It is similar to a signing district except it covers multiple municipalities and uses participation criteria unique to the area. It is a recognition that there is a limit to the number of signs that can be installed in a single location and that regional signing areas may be a tool for enhancing tourism and economic development. A regional signing area, properly implemented, will provide good directional signing for motorists. It will not alone increase tourism, but may positively impact the tourist’s visit. The key to its success is signing all worthwhile motorist destinations in a uniform and comprehensive manner.

2.7.9.2 Procedures

(Reserved)

2.7.10 Seasonal Attractions

2.7.10.1 Location of Seasonal TODSs

The order of installation of TODS sign panels, whether seasonal or non-seasonal, shall be as prescribed by Section 2.7.6.5.

2.7.10.2 Covering or Removing Signs

When TODSs approved for businesses that are not operated on a year-round basis, the Department may require that they be removed or covered for any period of time greater than 15 days in which the business is not operating. The cover shall be of an opaque material of the same shape and size as the sign panel. The applicant shall be responsible for removal and storage of the sign panel, and posts if required, and for all costs associated with covering or removal of TODSs. Failure to comply with the need to cover or remove seasonal signs shall be justification for removal.

2.7.11 Removal of a TODS

2.7.11.1 General

The Department reserves the right to remove signs if space is needed for necessary official traffic-control devices or if the Department determines that the signing is not in the best interest of the Commonwealth or the traveling public. The attraction will not be reimbursed for the sign costs.

2.7.11.2 Removal of Signs

Except where otherwise provided in this chapter, TODSs may be removed by the Department for one or more of the following reasons:

- Failure to comply with eligibility requirements set forth in Section 2.7.3 and Section 2.7.4.
- Because of fire, crash, facility renovation, or similar causes, result in a qualified attraction becoming inoperable for a period of time exceeding 15 days, the TODSs shall be temporarily removed or covered, but the attraction shall not lose its priority, nor be required to reapply for placement of signs.
- If the facility closes for an extended period without a scheduled reopening date, or if in the opinion of the Department, the owner or responsible operator does not proceed with necessary repairs

within a reasonable time, the attraction shall lose its right to continued placement of its TODSs.

- If the facility ceases to operate in accordance with the TODS policy or changes ownership, its agreement with the Department is terminated, and the attraction shall lose its right to continued placement of TODSs until a new application is submitted and approved.

- If a signing district or a signing region is established and existing TODS or signs that are part of a previously established district or region do not provide for acceptable guidance continuity.

2.7.12 Application Procedure

2.7.12.1 Application

Attractions desiring signs shall request an application from the appropriate Engineering District Office, or for airport signing, contact the Bureau of Aviation as identified in Exhibit 2.7-C. Each applicant shall provide the following information to the Engineering District:

- A completed application form. A separate application shall be submitted for each attraction site where TODSs are proposed.

- A map or neatly drawn sketch of the area to indicate the locations of the requested TODSs and the location of the attraction.

- A notarized seal on the application attesting to the authenticity of the signatures. If TODSs are installed and it is subsequently determined that the applicant was not truthful, the signs shall be removed and retained by the Department and the attraction shall be billed for the actual removal costs.

2.7.12.2 Department Review

A representative of the Engineering District will meet with the owner or representative of the attraction to discuss the eligible signs and to identify their locations, or will document this information in a memo to the applicant and request the applicant's concurrence. Signs not installed by the Department are subject to a final inspection by the Engineering District prior to final approval.

2.7.12.3 Approval Documentation

When TODSs are approved for installation by the local authorities or by private contractors along a State highway, an authorization letter will be issued by the Engineering District, which will specify the location(s) of the sign(s) and the approved message(s). <u>All signs installed within the right-of-way of any State highway becomes the property of the Department, and are subject to removal by the Department if the applicant fails to abide by the terms of the TODS policy</u>.

2.7.12.4 Excess Number of Eligible Attractions

If applications are received for any one intersection for more than the allowable number of signs, the order of priority shall be based on the date of receipt of a properly completed application and the required fee. Once approved for TODSs, the attraction shall remain eligible for these signs unless it is declared in violation of this policy.

2.7.12.5 Applicant Appeals

A business may appeal a denial for TODS under Title 2, Pa. C.S., Sections 501-508 (relating to the Administrative Agency Law), by submitting a written request for a hearing within 30 days of the date of the denial notification. Businesses should submit appeals to:

Administrative Docket Clerk
Pennsylvania Department of Transportation
400 North Street-9th Floor
Harrisburg, PA 17120-0096

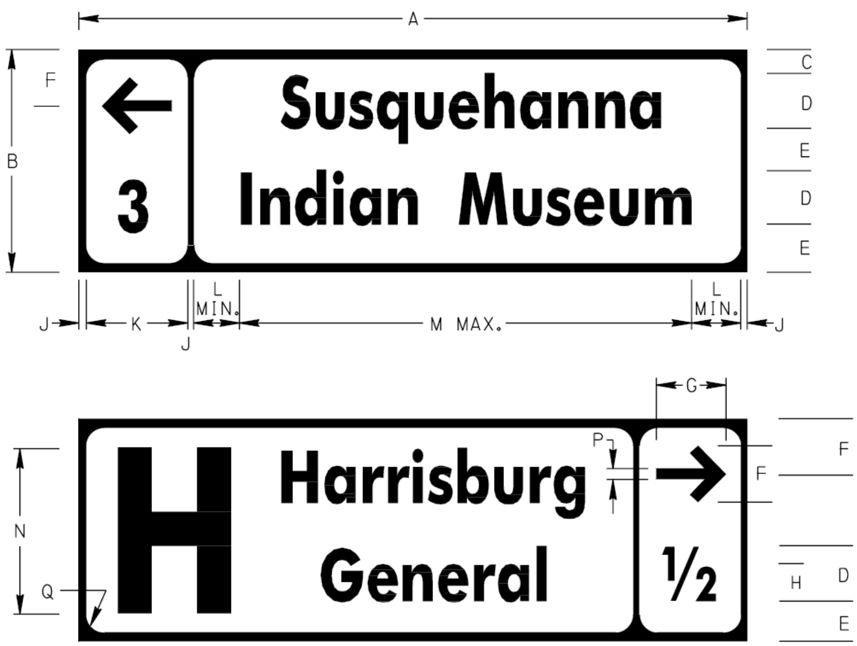
The written request shall include a filing fee made payable to the "Commonwealth of Pennsylvania" and a copy of the denial notification.

> At the time of publication, filing fees are listed at 34 Pa.B. 4081 (see http://www.pabulletin.com/secure/data/vol34/34-31/1410.html). Filing fees for appealing a TODS decision is a Level II fee, and comes under the category of "motorist information sign matters." Businesses may verify the current fee by contacting the Administrative Docket Clerk at 717-772-2741.

Exhibit 2.7-A Attraction Sign (D7-4)

(a) Justification. The Attraction Sign (D7–4) may be used on conventional highways to direct motorists to large tourist attractions in accordance with the Department's Attraction Signing Guidelines. One or two lines of legend may be used to identify the name or abbreviation of the attraction.

(b) Design. A rectangular directional box should generally be located on the left side of the sign for attractions that are straight ahead or to the left, or on the right side of the sign for attractions to the right. The box should generally include a directional arrow and a distance of 1⁄4, 1⁄2, 3⁄4 or the nearest whole mile, but the box may be eliminated if it is more appropriate to use directional information such as "DRIVEWAY ON LEFT", "LEFT 1000 FEET", etc., on the second line of legend. All legend should be "Clearview 1W, 2W or 3W" font, of the highest series possible. If necessary, the legend may be further condensed up to 35 percent. A generic symbol for hospital, campground or airport may be used in advance of the legend message.



DIMENSIONS – mm (IN)													
SIGN SIZE A x B	C	D	E	F	G	H	J	K	L	M	N	P	Q
1200 x 400 (48" x 16")	50 (2)	100 (4)	75 (3)	100 (4)	125 (5)	65 (2.6)	15 (0.6)	185 (7.4)	50 (2)	870 (34.8)	275 (11)	20 (0.8)	25 (1)
1800 x 600 (72" x 24")	90 (3.6)	150 (6)	105 (4.2)	165 (6.6)	188 (7.5)	100 (4)	20 (0.8)	280 (11.2)	75 (3)	1310 (52.4)	400 (16)	30 (1.2)	45 (1.8)

COLOR:

LEGEND AND BORDER:
WHITE (REFLECTORIZED)

BACKGROUND:
BLUE (REFLECTORIZED)

APPROVED FOR THE SECRETARY OF TRANSPORTATION

By : Glen C Rowe Date : 01–03–06
Chief, Traffic Engineering and Operations Division
Bureau of Highway Safety and Traffic Engineering

Exhibit 2.7-B Acceptable Symbols

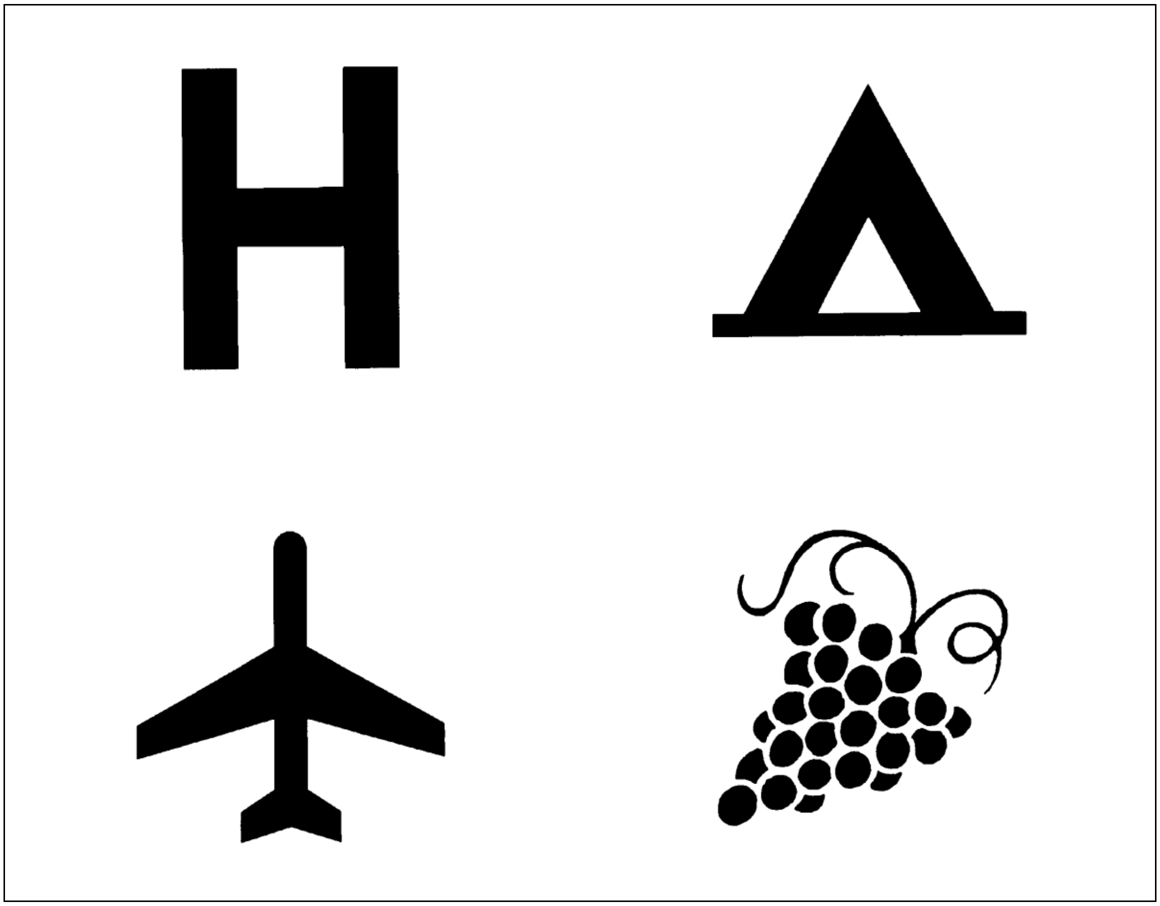


Exhibit 2.7-C District Contacts

DISTRICT	CONTACT	ADDRESS	COUNTIES
1-0	District Traffic Engineer	255 Elm Street Oil City, PA 16301 (814) 678-7181	Crawford, Erie, Forest, Mercer, Venango, Warren
2-0	District Traffic Engineer	P.O. Box 342 Clearfield, PA 16830 (814) 765-0498	Centre, Clearfield, Clinton, Cameron, McKean, Potter, Mifflin, Elk, Juniata
3-0	District Traffic Engineer	715 Jordan Ave. Montoursville, PA 17754 (570) 368-4260	Columbia, Lycoming, Montour, Northumberland, Snyder, Sullivan, Tioga, Union, Bradford
4-0	District Traffic Engineer	O'Neill Highway Dunmore, PA 18512 (570) 963-4819	Lackawanna, Luzerne, Pike, Susquehanna, Wayne, Wyoming
5-0	District Traffic Engineer	1002 Hamilton Blvd. Allentown, PA 18101-1013 (610) 817-4475	Berks, Carbon, Lehigh, Monroe, Northampton, Schuylkill
6-0	District Traffic Engineer	7000 Geerdes Boulevard King of Prussia, PA 19406 (610) 205-6550	Bucks, Chester, Delaware, Montgomery, Philadelphia
8-0	District Traffic Engineer	2140 Herr St. Harrisburg, PA 17103-1699 (717) 783-3981	Adams, Cumberland, Franklin, York, Dauphin, Lancaster, Lebanon, Perry
9-0	District Traffic Engineer	1620 N. Juniata St. Hollidaysburg, PA 16648 (814) 696-7231	Bedford, Blair, Cambria, Fulton, Huntingdon, Somerset
10-0	District Traffic Engineer	Route 286 South P.O. Box 429 Indiana, PA 15701 (724) 357-2845	Armstrong, Butler, Clarion, Indiana, Jefferson
11-0	District Traffic Engineer	45 Thoms Run Road Bridgeville, PA 15017 (412) 429-4975	Allegheny, Beaver, Lawrence
12-0	District Traffic Engineer	825 North Gallatin Ave. Ext. Uniontown, PA 15401-2105 (724) 439-7345	Greene, Washington, Westmoreland, Fayette
Bureau of Aviation	Bureau Director	Commonwealth Keystone Building, 6th Floor 400 North Street Harrisburg, PA 17120 (717) 705-1260	

Exhibit 2.7-D Application for TODS

Please print or type answers to the following questions

1.	Name of Business or Facility:
2.	Mailing Address:
3.	Name of contact person:
4.	Phone number of contact person:
5.	Directions to facility (include sketch):
6.	Number of Years in Operation:
7.	Dates and Hours of Operation:
8.	Member of the following associations:
9.	Description of Business or Facility (or enclose brochure):
10.	Does the business or facility satisfy the General Eligibility Requirements set forth in Section 2.7.3 of this policy? ____________ If no, explain discrepancies.
11.	How many visitors during the last year?
12.	Do you have billboards along any state highway? __________ If yes, please submit a sketch or map to identify the location(s) with your application.
13.	If food is prepared, what is your Department of Agriculture License Number?
14.	Do you understand your cost obligations as identified in Sections 2.7.7.2 and Section 2.7.11.1?
15.	Name of attraction as you would like it to appear on signs (maximum of 2 lines, 16 characters/spaces per line).

16. Indemnification. The Business shall indemnify, save harmless, and defend the Department from any and all claims, actions, damages, injuries, and/or expenses arising out of the Tourist Oriented Directional Signing (TODS) Program or Project or on account of any act, omission, neglect, or misconduct by the Business or a third party.

ACCORDINGLY, AS PART OF THE CONSIDERATION FOR THIS AGREEMENT, THE BUSINESS SPECIFICALLY AGREES THAT IT SHALL NOT ASSERT ANY CLAIM OF ANY NATURE WHATSOEVER AGAINST THE DEPARTMENT RESULTING FROM THIS CONTRACT OR THE ADMINISTRATION OF THE TOURIST ORIENTED DIRECTIONAL SIGNING PROGRAM.

The business further acknowledges that in the event a signing district or signing region, as discussed under Sections 2.7.8 and 2.7.9 is established, their signs may be removed in accordance with Section 2.7.11.

I hereby certify that the information provided on this application is true and correct and to the best of my knowledge (*name of the business or facility*) conforms to all Federal, State, and local regulations, including all health, sanitary and water requirements. The business or facility conforms to all municipal ordinances. It is also my understanding that if signs are installed, they may be removed by the Department as noted in Section 2.7.7.4 and Section 2.7.10. Further, the business agrees to notify the Department if the hours of operation change or if they terminate operations.

Sworn before me this day of , 20_____.

Notary:

Signature of applicant: ____

Privately operated businesses and facilities are to obtain verification from each local municipality within which TODSs are being requested to the effect that the local municipality does not have any local ordinance prohibiting their installation. The following signature blocks should be used.

17. I hereby confirm that the installation of TODS signing for the above applicant does not conflict with any local ordinance.

Name of Municipality:

Signature of municipal representative: ____

Telephone Number:

18. I hereby confirm that the installation of TODS signing for the above applicant does not conflict with any local ordinance.

Name of Municipality:

Signature of municipal representative: ____

Telephone Number:

Exhibit 2.7-E Application for Signing District

Please print or type the following information.

1. Name of Governmental Sponsor:
2. Mailing Address:
3. Name of contact person:
4. Phone number of contact person:
5. Has a map of the proposed signing district been included? ______
6. Name of Consultant:
7. Address of Consultant:
8. If consultant has not been hired, will the service of one be used? ______ If not, who will design the system?:
9. Have all illegal signs been removed? ______
10. Has an inventory of existing permitted signs been completed? ______
11. Does the governmental sponsor understand that all businesses or facilities that participate in the signing district must meet one of the definitions in Section 2.7.2 and satisfy the General Eligibility Requirements set forth in Section 2.7.3 of this policy? Yes _______ No __________
12. Do you have a copy of the PennDOT Toolbox for the Development of Wayfinding Signing Regions? ________ This Toolbox provides the fabrication details for an alternate sign design.

13. Are there any plans contemplated to expand the signing district beyond its boundaries and become a signing region? If yes, is there a timetable (attach or explain)?
14. Note: Execution of an agreement between the governmental sponsor and the Commonwealth designating the signing district must be completed before construction of the signs begins.
15. Do you understand all costs will be borne by the governmental sponsor? Further, the governmental sponsor may recoup some or all of the administrative costs of the program by the establishment of a fee structure for applicants.
16. Indemnification. The Governmental Sponsor shall indemnify, save harmless, and defend the Department from any and all claims, actions, damages, injuries, and/or expenses arising out of the subject Signing District or on account of any act, omission, neglect, or misconduct by an applicant or a third party. The governmental sponsor further acknowledges that in the event a signing region, as discussed under Section 2.7.9 is established, the signs may be removed in accordance with Section 2.7.11. I hereby certify that the information provided on this application is true and correct and to the best of my knowledge, and ____ (name of Governmental Sponsor) is fully prepared to move forward to completion of the signing district. It is also my understanding that if signs are installed, they may be removed by the Department as detailed in Section 2.7.7.4 or Section 2.7.11. Further, the business agrees to notify the Department if the hours of operation change or if they terminate operations. Sworn before me this ____________ day of ________________, 20_____. Notary: Signature of Representative: _____________________________________

2.8 Acknowledgement Signs

2.8.1 General Requirements

The Federal Highway Administration (FHWA) allows the use of signs to acknowledge the provision of highway-related services. State and local programs for acknowledgment signs are growing in popularity because they can provide additional revenue for highway maintenance programs.

Existing acknowledgment signs already installed do not have to be changed, but Engineering Districts need to consider the guidance when replacing or upgrading existing signs. Always attempt to follow good, basic engineering practices such as simplifying sign message content, reasonable sign sizes, and minimizing driver distraction.

Acknowledgment signs are a way of recognizing a company or business, or a volunteer group that provides a highway-related service. Acknowledgment signs include sponsorship signs for adopt-a-highway litter removal programs, maintenance of a parkway or interchange, and other highway maintenance or beautification sponsorship programs. Acknowledgment signs should clearly indicate the type of highway services provided by the sponsor. The FHWA recognizes a distinction between signing intended as advertising and signing intended as an acknowledgment for services provided. Advertising generally has little if any relationship to a highway service provided. An advertiser wants to get its recognizable message, company emblem, or logo before the public, and if possible, information on how or where to obtain the company's products or services. In most cases, if the sign goes beyond recognizing the company's contribution to a particular highway service at a specific highway site or includes telephone numbers or internet addresses, the sign is an advertising sign and not an acknowledgment sign.

This policy position is consistent with the principles and intent of several laws including 23 U.S.C. §1.23(b), 23 U.S.C. §109(d), and 23 U.S.C. §131. Section 1A.01 of the MUTCD states that "Traffic control devices or their supports shall not bear any advertising message or any other message that is not related to traffic control." This position is founded on safety and operational concerns, particularly as related to driver distraction. Highway signs and other traffic control devices convey crucial information. In order for road users to perceive and respond appropriately to critical information, we must ensure the conspicuity of traffic control devices to avoid compromising the safe and orderly movement of traffic.

Sign Placement

With respect to placement of traffic control signs, regulatory, warning, and guide signs have a higher priority than acknowledgement signs. In fact, acknowledgment signs are the lowest priority of information-type signs and may only be placed where adequate spacing between higher priority signs is available. In no case shall an acknowledgment sign be placed such that it obscures road users' view of other traffic control devices. Therefore, follow the following minimum spacing requirements:

1. On roads with speed limits of less than 30 mph, do not place acknowledgment signs within 150 feet of any other traffic signs, except parking regulation signs.

2. On roads with speed limits of 30 to 45 mph, do not place acknowledgment signs within 200 feet of any traffic signs, except parking regulation signs.

3. On roads with speed limits greater than 45 mph, do not place acknowledgment signs within 500 feet of any traffic control signs, except parking regulation signs.

Due to public safety concerns, do not allow acknowledgment signs at the following locations:

1. On the front, back, adjacent to or around any traffic control device, including traffic signs, signals, changeable message signs, traffic control device posts or structures, or bridge piers.

2. At key decision points where a driver's attention is more appropriately focused on traffic control devices, roadway geometry, or traffic conditions. These locations include, but are not limited to exit and entrance ramps, intersections controlled by traffic signals or by stop or yield signs, highway-rail grade crossings, work zones, and areas of limited sight distance.

If the placement of an acknowledgment sign conflicts with newly installed higher priority signs, or traffic signals, or temporary traffic control devices, or other priority devices, remove, cover or relocate the acknowledgment sign.

2.8.2 Adopt-A-Highway Signs

In the Adopt-A-Highway Program, businesses, groups of people, or individuals agree to pick up litter along a section of highway in exchange for the erection of a sign bearing their name at the beginning of the section.

The program not only helps reduce the maintenance workload, but also fosters a public awareness of the magnitude of the littering problem.

The procedures and types of signs for the program are addressed in the Adopt-A-Highway Operational Manual. In each Engineering District, the program is administered under the direction of the Adopt-A-Highway Coordinator, and the oversight of the Bureau of Maintenance and Operations (BOMO).

2.8.3 Beautification Area Signs

The Beautification Area Sign (I47-1) and the Beautification Area Sponsor Sign (I47-2) are designed to give recognition to the individual or group of individuals that provides and maintains the plantings or other beautification efforts.

2.8.4 Sponsor-a-Highway

The Sponsor-A-Highway program is a professional maintenance and marketing program intended to provide corporate sponsors that desire to assist in highway beautification the opportunity to fund litter removal services. The program is targeted towards freeways.

The Adopt-A-Highway Maintenance Corporation (AAHMC) is under contract with the Department to solicit corporations that want to pay for litter removal. They market the program, perform maintenance and install signs recognizing the contribution of the companies.

AAHMC is responsible for all aspects of the signage including installation, maintenance and removal. The Bureau of Maintenance and Operations administers the Sponsor-A-Highway program; therefore, direct any questions about the program to that office.

2.9 Operational Changes Requiring Transitional Signing

2.9.1 General Discussion of Operational Changes

As traffic patterns change at specific locations on the highway, changes are also necessary in the operational procedures and devices that control those patterns. Unfortunately, these changes may cause an increase in crash rates unless accompanied by good transitions to the new traffic controls. A transitional procedure informs drivers that traffic control will or has already changed before they arrive at the site of the new control, which should modify the driver's expectancy thereby reducing conflicts and crashes.

Sections 2.9.2 through 2.9.8 provide recommended procedures to insure a safe transition from one type of traffic control to another.

> In addition to the static signs, it is acceptable to use portable changeable message signs (PCMS) to notify drivers of operational changes.

2.9.2 Reversing STOP Signs

Step 1 Create a four-way stop situation for a period of 30 days. During this 30-day period, Stop Ahead (W3-1) signs should be erected on the previously uncontrolled approaches, and a 4-WAY (R1-3) or ALL WAY (R1-4) supplemental plaque should be added below each R1-1 signs. A Type B flashing light with a red lens may also be placed above the new R1-1 signs.

Step 2 After the 30-day period, on the approaches which originally had the STOP signs, remove the STOP (R1-1) signs, 4-WAY (R1-3) or ALL WAY (R1-4) supplemental plaques, any Stop Ahead (W3-1) signs, and any stop lines. On the other two approaches, replace the 4-WAY (R1-3) or ALL WAY (R1-4) supplemental plaques with CROSS TRAFFIC DOES NOT STOP (R1-1C) signs.

Step 3 At least 60 days after the removal of the original STOP Signs, remove the CROSS TRAFFIC DOES NOT STOP (R1-1C) signs, the flashing lights, and if not warranted, the Stop Ahead (W3-1) signs.

2.9.3 Four-Way to Two-Way Stop Control

Step 1 Remove the 4-WAY (R1-3) or ALL WAY (R1-4) supplemental plaques, any Stop Ahead (W3-1) signs, and any stop lines on the approach which originally had the STOP (R1-1) signs. On the other two approaches, replace the 4-WAY (R1-3) or ALL WAY (R1-4) supplemental plaques with STOP SIGN REMOVED FROM SIDE STREET (R1-4-1) signs. A Type B flashing light with a red lens may be placed above the two remaining R1-1 signs.

Step 2 At least 60 days after the removal of the two R1-1 signs, remove the STOP SIGN REMOVED FROM SIDE STREET (R1-4-1) signs, the flashing lights, and if not warranted, remove the Stop Ahead (W3-1) signs.

2.9.4 One-Way to Two-Way Street Operation

Step 1 Notify the general public by news media and abutting residences and/or businesses by direct mail about 30 days prior to the effective date.

Step 2 Provide adequate sight distance at all intersections and driveways by removing parking, clearing vegetation, etc. Prohibit parking on the left side of the one-way street.

Step 3 Install and cover Two-Way Traffic (W6-3) signs on both sides of the existing one-way street at its beginning and immediately beyond all intersections where major left turning movements occur. Also, install and cover Two-Way Traffic (W6-3) signs on all major approaches to the existing one-way street, but rotated 90 degrees to graphically depict traffic going both left to right and right to left.

Step 4 Eradicate any pavement markings on the one-way street or side street approaches which will be in conflict with the new two-way operation. Install centerlines on the "one-way" street.

Step 5 On the agreed upon date, uncover the Two-Way Traffic (W6-3) signs, and remove any non-applicable One-Way signs, BEGIN ONE-WAY (R6-12) signs or turn restriction signs. Install appropriate pavement marking arrows as appropriate.

Step 6 At least 3 months after the conversion, remove the Two-Way Traffic (W6-3) signs.

2.9.5 Two-Way to One-Way Street Operation

Step 1 Notify the general public by news media and abutting residences and/or businesses by direct mail about 30 days prior to the effective date.

Step 2 Install DO NOT ENTER (R5-1), WRONG WAY (R5-1A), and No Right/Left Turn signs, as appropriate.

Step 3 Block off the unused lane(s) with drums or barricades.

Step 4 Revise existing pavement markings to show new one-way direction, both on the affected street and on all side road approaches.

Step 5 Remove drums or barricades.

Step 6 At least 60 days after the conversion, remove any unwarranted DO NOT ENTER (R5-1), WRONG WAY (R5-1A), or No Right/Left Turn signs.

2.9.6 One-Way Street Reversal

Step 1 Notify the general public by news media and abutting residences and/or businesses by direct mail about 30 days prior to the effective date.

Step 2 Provide adequate sight distance at all intersections and driveways by removing parking, clearing vegetation, etc. Prohibit parking on the left side of one-way streets.

Step 3 Replace One-Way Signs with new signs showing appropriate direction. Install flags or flashing lights on the signs, if appropriate.

Step 4 Consider placing drums or barricades at cross streets and pavement markings to emphasize new turning directions.

Step 5 At least 60 days after the conversion, remove drums, barricades, and flashing lights.

2.9.7 Revised Exit Numbers

In 2001, the Department changed from the consecutive exit-numbering system to the milepost exit-numbering system. If any of the "OLD EXIT ##" signs still exist, the District Traffic Unit should provide notification to the County Maintenance District to remove the signs when other maintenance work is being performed in the area.

If it is necessary to change exit numbers, Districts are encouraged to use the following steps, similar to the approach used on the statewide program:

Step 1 Working through the District CRC and Office of Communications, notify the general public by the news media and abutting businesses by direct mail about 1 year and again about 30 days prior to the effective date, so that the residents and businesses can make necessary changes in directions, business cards, yellow pages, etc. If possible, schedule the effective date to coincide with the issuance of the new telephone directory.

Step 2 Revise the exit numbers, and add a panel to each major advance guide sign with the message "OLD EXIT XX."

Step 3 After a predetermined amount of time (2 years minimum) following the conversion, the panels with the old exit number may be removed as part of routine maintenance.

2.10 Miscellaneous Signs

2.10.1 Anti-Littering Signs

The following five signs help to remind motorists not to litter:

Sign	Size
Keep Pennsylvania Beautiful (I14-1)	48”x30”
Keep PA Beautiful Symbol (I14-2)	30”x30”, 36”x36”
Keep Pennsylvania Beautiful State Outline (I14-3)	48”x30”
Litter Fine (I14-4)	30”x24”
No Dumping Allowed (I14-5)	24”x18”

Only use Anti-Littering signs if their location will not interfere with the proper functioning of other necessary traffic control devices. Do not install these signs along sections of highway that have already been adopted under the Department's Adopt-A-Highway Program, but existing anti-littering signs may remain in place for their useful life along sections of highway that are later adopted.

The I14-1, I14-2, I14-3, and I14-4 signs can be used interchangeably for most situations. When a relatively lengthy, continuous section of highway has experienced recurrent littering, it may be desirable to post an I14-1, I14-2, or I14-3 sign followed by I14-4 signs. Following are recommendations regarding the use of anti-littering signs:

Sign	Comments
I14-1	Normally, this type of anti-littering sign is used on select on-ramps for public education purposes.
I14-2	The circular logo on this sign is consistent with the logo used for all anti-litter programs. Normally, the I14-2 sign is used along exit ramps into (entrances to) welcome centers, parking areas, rest areas, and scenic views.
I14-3	Normally, this is the first type of sign used on conventional road entry points to the Commonwealth.
I14-4	This sign is suggested for use at specific sites where littering has been a recurring problem. The sign is also suggested for entrance ramps from welcome centers, rest areas, parking areas, and scenic views to the mainline roadway.

I14-5 This sign may be used at locations that have recurring illegal dumping of heavy volumes of litter, garbage, or other waste material. Typical placement sites may include areas of excess right-of-way, graded ground adjoining a highway, and pull-off locations in remote areas.

2.10.2 Coffee Break Signs

The Department allows safety-conscious local civic organizations to offer free coffee to motorists using rest areas during specific periods such as holiday weekends. Because these activities encourage the weary motorist to stop and relax, this activity is in the best interest of highway safety.

The organization must agree in writing to offer free coffee and to indemnify and save harmless the Commonwealth and PennDOT employees, from all suits, actions and claims. Because commercial activities are prohibited in rest areas, the organization must also agree to post a sign with minimum 6-inch letters stating NO DONATIONS, and to leave the area in similar or better condition than it was prior to the coffee break.

When a “coffee break” is approved, the county should erect a SAFETY BREAK FREE COFFEE (I98-1) sign. The last line on the sign should be NEXT RIGHT or other appropriate message. The sign should be placed a minimum of 300 to 500 feet before the Rest Area Directional Sign, and should be removed, folded or covered when free coffee is not being offered.

2.10.3 Historical Markers

Concrete monuments and cast aluminum historical markers are the responsibility of the Pennsylvania Historical and Museum Commission. If these markers are in need of repair, the following office should be contacted:

Historical Marker Program
Division of History
Pennsylvania Historical and Museum Commission
Commonwealth Keystone Building, Plaza Level
400 North Street
Harrisburg, PA 17120-0053
717-787-3034

On the other hand, the cast iron historical markers for towns or rivers were installed by the Department. The Engineering Districts are encouraged to enter into agreements with local civic groups to maintain these markers. If the markers are not protected by guide rail or bridge abutments, or as a minimum non-mountable curbs in urban areas, the markers should be relocated or stockpiled for potential future use. Erect any new or refurbished installations on breakaway supports.

2.10.4 Library Signs

On State highways, libraries are responsible for providing and erecting Library (I-8) signs after receiving authorization from the Department. When requested, the District Office shall approve appropriate locations on conventional State highways, directing motorists to public libraries from the nearest numbered traffic route or other major highway.

2.10.5 Memorial Markers

The Department will only authorize or erect memorial markers on highways designated by the legislature, and on Interstate highways, authorization is contingent upon final approval and concurrence of FHWA. In order to manage this program, the Traffic Engineering Operations Division maintains a log of official named highways and bridges.

The following types of markers are available:

1. Bronze plaques may be locally procured in accordance with specifications available from the Traffic Engineering and Operations Division. Common sizes are 15"x12" and 30"x24".

2. D8-series markers have white reflectorized legend and border on a green reflectorized background. Common sizes are 36"x12" and 48"x16", depending upon the speed limit and available space. The legend is typically 3" or 4" capitals.

3. D7-series signs have 8-inch upper/lower-case white reflectorized legend on a green reflectorized background.

Signs are normally mounted at the following locations:

(a) Memorial bridges should have either a bronze plaque or a D8-series marker. Anti-theft nuts are recommended for all D8-series markers.

(b) Non-Interstate memorial highways may have D8-series markers facing traffic at the beginning of the designation and may also be installed at select locations along the highway.

(c) Interstate memorial highways may have memorial plaques placed in rest areas, scenic overlooks or at other appropriate locations where parking is provided with the signing inconspicuously located relative to vehicle operations along the highway. If installation of the memorial plaques off the mainline is not practical, the signing shall be limited to one D7-series sign at an appropriate location in each route direction. If installed on the mainline highway, memorial signs must be independent of other guide and directional signing and not adversely compromise the safety or efficiency of traffic flow.

(d) Memorial interchanges may have a D7-series sign installed on all non-Interstate approaches. Memorial signs must be independent of other guide and directional signing, and shall not adversely compromise the safety or efficiency of traffic flow, particularly with regard to entering or exiting traffic in the interchange area. Signs may be post-mounted or attached to the side of the overhead structure provided other guide signs are not already in place.

2.10.6 Motorist Alert Signs

Motorist Alert (W23-series) signs may be erected prior to the start of a construction project as a courtesy to the motorist. A Type B warning light may be added to increase the sign's effectiveness.

General messages are included on the standard sign panels so that the signs may be reused on future projects. However, the messages on the W23-1 and W23-2 signs may be tailored to the particular project. With reference to the THIS BRIDGE TO BE CLOSED FOR MAINTENANCE (W23-1) sign, suggested overlay panels using 6-inch legend are as follows:

- 36"x8" HIGHWAY or STREET panel to cover "BRIDGE"
- 60"x8" NEXT 3 MILES panel to cover "THIS BRIDGE"
- 90"x8" FOR CONSTRUCTION or FOR REPAIRS panel to cover "FOR MAINTENANCE"
- 90"x8" JUNE 28, 2007 panel to cover "NEXT WEEK"

The W23-series signs should generally be included as a special provision in applicable Department construction projects. It is desirable to stipulate that the signs become Department property at the end of the project, thus providing for their reuse.

2.10.7 Municipal Name Signs

The Department will generally install 36"x12" municipal name signs (I10-series signs) on all State highways.

When requested by local authorities, the Department will fabricate and install oversized I10-series signs if the local authorities agree to pay for the signs. The cost of new or replacement oversized signs will be as noted below, plus a one-time $100 administrative fee (one $100 fee per request of one or more signs). These signs will generally have dimensions proportionally larger than the standard I10-series sign:

48"x16" For use on two-lane highways – $400
72"x24" For use on multilane highways – $500

On conventional roads and expressways, local authorities may install custom-made name signs as an alternative to the oversized I10-series signs after approval is granted by the appropriate Engineering District. Approval is contingent upon satisfying the following requirements:

1. Sign dimensions are as follows:

Type Highway	Max. Width	Max. Area (Sq. Feet)	Max./Min. Legend Size
Two-lane	48"	12	6"/2"
Multilane	72"	24	8"/3"

2. The primary message should be something like "Welcome to ____________________", with the municipality's name composed of the largest-size legend. A smaller-size supplemental slogan with up to six words may be added, e.g., "The White Rose City", "The Christmas City", etc., and/or a symbol may be used.

3. No lights, animation, directions, distances, names of officials, or advertising are permitted.

4. All signs shall be manufactured by a Department-approved sign manufacturer using approved Type III or IV retroreflective sheeting material, and installed in accordance with Publication 111M.

5. The installation of special name signs may not be possible if insufficient space exists for the signs, or if the municipality changes too often within a short distance. (Existing traffic signs may be relocated at the municipality's expense in order to provide room for the oversize municipal signs.)

The District Executive has the authority to install or authorize the municipality to install the signs in accordance with these guidelines.

2.10.8 Recreational and Cultural Interest Symbol Signs

Recreational and cultural interest area symbol signs for State or federal parks may be used on conventional roads that pass through or are immediately adjacent to a State or federal park. They may also be used inside these areas to direct tourists to non-vehicular services such as trails and rest rooms.

Symbols may be used as legend components of a directional sign assembly, but no more than four symbols shall be used on a single assembly. The following additional guidelines shall apply:

- Recreational and cultural interest area symbols shall have white legend on a brown rectangular background. Acceptable symbols are included in the "Guide Signs Chapter" of FHWA's *Standard Highway Signs and Markings* book.

- Educational plaques may be placed underneath symbol signs whose meaning is not readily obvious from the symbol graphic itself.

- When an activity is prohibited within a recreational or cultural interest area, a red diagonal slash may be placed on the symbol sign.

2.10.9 Recycling Program Signs

The Recycling Services (I45-1) sign is designed for use within roadside rest areas to encourage the use of recycling services and to identify the group providing the services.

The Recycling Center Sign (I11) may be used to direct motorists to permanent recycling collection centers from the nearest numbered traffic route or other major highway. The center must be open to the public and consistently take a minimum of three (Act 1988-101) materials. This sign shall not be used in urban areas or on freeways and expressways.

2.10.10 Roadside Memorial Markers

Citizens frequently install roadside memorial markers adjacent to locations of fatal crashes. The Department does not grant formal approval for these markers, and the Department typically does not remove them unless they pose a safety concern. After a period of time in which the signs and ancillary material remain in a state of disrepair, they should be removed.

2.10.11 Scenic River Signs

When a creek or river has been legislatively designated as a scenic river, scenic river signs may be installed along State highways when the river is clearly visible from the highway and the Pennsylvania Department of Conservation and Natural Resources (DCNR) requests the signs. Scenic river (bearing DCNR-supplied logos) can be ordered from the Sign Shop (48"x24" D5-series) and installed on a reimbursable basis.

The following scenic rivers have been legislatively approved:

Bear Run (Act No. 88-161)
French Creek (Act 82-97)
Lehigh River (Act 82-71)
LeTort Spring Run (Act 88-42)
Lick Run (Act 82-324)
Lower Brandywine (Act 89-7)
Octoraro Creek (Act 83-43)
Pine Creek (Act 92-124)
Schuylkill River (Act 78-33 & 88-17)
Stony Creek (Act 80-18)
Tucquan Creek (Act 88-161)
Tulpehocken Creek (Act 92-118)
Yellow Breeches Creek (Act 92-118)

In addition to Scenic River Signs along State highways, the Department has agreed to cooperate with DCNR in providing overpass road name signs on canoeable rivers. These signs would be similar to the overpass signs identified in Section 2.12.10, except a smaller size may be used on rivers less than 100 feet in width.

2.10.12 Signs and Banners across State Highways

No person, municipality or corporate entity may place a sign or banner across a State highway or within the highway right of way, unless the local municipality has:

1. passed a resolution designating their intention to erect such a sign or banner, and

2. received confirmation from the Department that it has on file a copy of the resolution and all required issues have been adequately addressed.

Resolutions may be for a single event, an event that recurs on a regular basis, or multiple events throughout the year. Permanent cables across the right-of-way for erection of banners are permissible provided they are noted in the resolution. Any municipal sponsored sign or banner placed across a State highway without a resolution on file with the Department can be removed; however, the municipality should first be given the opportunity to pass a timely resolution. No sign or banner may be placed across or within the right-of-way of any limited access highway.

The Department will only consider resolutions that address the following:

- Installation location including SR, Segment/offset and vertical clearance above the roadway (minimum 17'-6").
- Size of the sign or banner, a description of the message, and the event(s) and/or organization(s) for which the banner is being erected. Events must relate to a national, state, regional or local function or charitable affair.
- Approximate date(s) of installation and removal. If the sign or banner is to be installed on a recurring basis, the occasions when it will be displayed and the approximate number of days before and after the occasion when the device will be installed and removed, respectively.
- That the municipality assumes full responsibility for erecting, maintaining and removing the device and all liability for damages occurring to any persons or property arising from any act of omission associated with the sign or banner.
- Acknowledgement that no more than 20-percent of the message will relate to naming or advertising a commercial product, enterprise, business or company regardless of whether they are sponsoring the event or banner installation.
- That traffic control will be performed in accordance with the current Publication 213.

2.10.13 SR and Segment Markers

SR and Segment markers are valuable in determining field locations for highway maintenance and information-gathering activities. Therefore, the maintenance of these markers is very important.

Any Department employee who observes that a SR or Segment marker is missing or is installed at a questionable location or orientation, should contact the District RMS Coordinator, who in turn will advise the District Traffic Engineer or the Assistant District Traffic Engineer in charge of operations so that the changes can be made. However, under no circumstances should a county install or move a marker without an order from the District RMS Coordinator.

Information on missing and incorrect SR/Segment markers is also collected every year through the STAMPP and LRS QA/QC programs. A computer-generated listing, resulting from the LRS QA/QC, will go to the District RMS Coordinator, who will then work with the appropriate District people to make the changes in the field.

Fiberglass SR and Segment markers are available from contract Legacy No. 9905-13. When ordering the markers, use Exhibit 2.10-A so that all Districts are using the same forms.

As an alternative to purchasing completely finished markers, partially-finished SR and Segment markers may be ordered from the contract. Department personnel can affix the appropriate legend as required (see Exhibit 2.4-A). The Sign Shop can provide legend using the following SAP Material Nos.:

SAP Material No.	Description
144489	1.5" numerals for Segment markers
144490	3" numerals for SR markers
144491	4" numerals for Segment markers

2.10.14 Street Name Signs

The 36"x8" Street Name Sign (D3-1) is the standard for this application. In addition, larger signs are authorized and encouraged. Overhead signs mounted on signal mast arms should generally have 8-inch upper/lower-case legend, except 10.6-inch upper/lower-case legend is recommended on roadways with speed limits greater than 35 mph.

In addition, it is very desirable to place black-on-yellow advance street name signs (D3-2 or D3-3) beneath all intersection warning signs on numbered traffic routes in rural areas. (See Section 2.5.6)

2.10.15 Welcome to Pennsylvania Signs

Welcome to Pennsylvania Signs should be erected on all roadway approaches to the Commonwealth. The standard sizes are as follows:

- Interstate highways – 264"x144" – (I13-2)
- Non-Interstate freeways – 192"x108" – (I13-2)
- Major traffic routes on conventional roads – 144"x78" – (I13-2)
- Other conventional roads – 48"x24" and 72"x36" – (I13-2A)

2.10.16 Wildflower Area Signs

The Wildflower Area Sign and plaque are designed to advise individuals that wildflowers have been planted and that picking or destroying the wildflowers is illegal. The signs are made from fiberglass reinforced plastic and may be procured from contract Legacy No. 9905-13, where they are specified as follows:

18"x24"	Wildflower Area Sign, non-reflectorized
18"x12"	Wildflower Regulation Plaque, non-reflectorized

Exhibit 2.10-A SR and Segment Markers Tabulation Sheet

District _________ **County _________**

Segment Marker (10"x10")		
Qty.	**SR No.**	**Seg. No.**

SR Markers (18"x12")					
Qty.	**1st Line**	**2nd Line**		**3rd Line**	
	SR No.	**Arrow***	**Seg. No.**	**Arrow***	**Seg. No.**

Arrow Codes:

LH = ←, RH = →, LD = ↖, RD = ↗, UA = ↑
* Arrows to the right should be after the segment number.

2.11 Excessive Use of Signs

2.11.1 Sign Clutter

The Districts are requested to regularly review existing signs to ensure that all of the signs are official and are necessary for regulatory, warning or guidance purposes. In addition to removing obsolete and unofficial signs, consider the following suggestions:

(a) Use 250-foot typical spacing between no parking signs in rural areas.

(b) Review the use of BUMP (W8-1), DIP (W8-2), Slippery When Wet (W8-5) and LOW SHOULDER (W8-9) signs to be sure that they are not left in place after the problem has been corrected.

(c) Eliminate the use of 55-mph Speed Limit Signs on non-freeways except when needed to indicate the end of a lower speed limit.

(d) Eliminate Stop Ahead (W3-1), Yield Ahead (W3-2) and Signal Ahead (W3-3) signs when the Stop (R1-1) or Yield (R1-2) sign, or the traffic signals are visible by approaching traffic for distances greater than those values noted in the justification for the advance warning sign unless there is a known crash problems. In addition, the "T" Symbol Sign (W2-4) should generally not be used on an approach which also has a Stop Ahead (W3-1) sign.

(e) Review the need for existing curve and turn signs.

2.11.2 Double Stop Signs

The installation of one STOP (R1-1) sign on top of another is in conflict with Section 2B.04 of the *MUTCD*. Therefore, if this practice exists, the Engineering District should issue the necessary work order to have one of the R1-1 signs removed. Use a larger R1-1 sign if greater emphasis is needed.

If a municipality has any double stop sign installations, advise them in writing that this practice is in conflict with the *MUTCD* and that they should revise the installation. A copy of a suggested letter is included as Exhibit 2.11-A.

2.11.3 Sign Grouping

In an effort to improve the roadside environment and to eliminate unnecessary posts, sign grouping and back-to-back installations are encouraged wherever possible. Small relocations of most signs can generally be made without compromising the design standards. Refer to Section 5.1 of the Sign Foreman's Manual, Publication 108.

Exhibit 2.11-A Sample Letter about Double Stop Signs

Dear ______________________:

It has been brought to my attention that two STOP signs installed one on top of the other exist at ___________________. I am also aware that the Federal Highway Administration has advised our Department that the use of "double stop signs" is in conflict with their *Manual on Uniform Traffic Control Devices (MUTCD)*.

We were advised by the Federal Highway Administration that double stop signs are illegal for the following reasons:

1. The STOP sign is the only octagonal-shaped sign and therefore drivers should normally recognize the sign before they are able to read the message. However, by installing one STOP sign above another, the driver is faced with an unfamiliar shape, which is not easy to recognize.

2. Drivers confronting double stop signs could also be confused, perhaps wondering if they should stop twice, or if the double stop signs really means stop and that, they do not have to stop at other STOP signs.

In view of the comments from the Federal Highway Administration, the Department fully supports their decision to forbid the use of double stop signs. Therefore, I would request that you remove one of the STOP signs or replace both signs with a larger size so that it is recognizable at a greater distance.

If you have any questions, please feel free to contact my office.

Sincerely,

2.12 Standard Freeway and Expressway Signs

2.12.1 Standards and Approvals

The types of signs unique to expressways and freeways are generally described in Publication 111M.

FHWA approval is required for signing on all Interstate highways that does not comply with existing standards:

1. For Federal Oversight projects on the Interstate system, proposed signing is normally included in the PS&E.

2. For PennDOT Oversight projects on the Interstate system, proposed highway signing that does not comply to existing standards must be sent to the Traffic Engineering and Operations Division (TEOD) for submission to the FHWA Division Office. For this submission, include two copies of plan sheets and any sign fabrication details, along with information about the types of posts and breakaway hardware.

2.12.2 Selection of Major Destinations

2.12.2.1 General

These guidelines are for selecting the most appropriate destinations for major guide signs on expressways and freeways, in order to provide the driver the best orientation possible without overloading the driver with too much information.

2.12.2.2 Definitions

Destination – The name of an incorporated community, street name, municipal center or large traffic generator, which provides the best orientation for drivers traveling on the expressway or freeway.

Map – The latest edition of the Department's Official Transportation Map.

National control city – A city designated as such by the American Association of State Highway and Transportation Officials (AASHTO) for an Interstate highway. A list of appropriate cities is included in Exhibit 2.12-A.

Exhibit 2.12-A National Control Cities in or Adjacent to Pennsylvania

Interstate	National Control Cities
70 EB	Washington, PA; New Stanton; Breezewood; Hancock
70 WB	Breezewood; Wheeling
76	Youngstown; Pittsburgh; Harrisburg; Valley Forge; Philadelphia
78	Harrisburg; Allentown; Bethlehem; Easton; New York City
79	Morgantown; Washington, PA; Pittsburgh; Erie
80	Youngstown; Sharon; Clarion; DuBois; Clearfield;Bellefonte; Williamsport; Bloomsburg; Hazleton; Stroudsburg; Delaware Water Gap, New York City
81	Hagerstown; Chambersburg; Carlisle; Harrisburg; Hazleton; Wilkes-Barre; Scranton; Binghamton
83	Baltimore; York; Harrisburg
84 EB	Milford; Port Jervis
84 WB	Scranton
90	Cleveland; Erie; Buffalo
95 NB	Wilmington; Chester; Philadelphia; Trenton
95 SB	Philadelphia, Chester, Wilmington

Population-to-distance ratio – The ratio of the population of a community identified on the map to the distance in miles from a given interchange.

Urban area – A built-up area which includes any of the following features for a given expressway or freeway:

- Four or more adjacent interchanges serving the same community.
- More than two travel lanes in each direction.
- Interchanges typically spaced at intervals of 3 miles or less.

2.12.2.3 Criteria for Selecting Major Destinations

Junction with an Interstate Highway. If the intersecting route is an Interstate highway, the closest national control city to the left and to the right should be used as the destinations on the major guide signs. A list of national control cities in or adjacent to Pennsylvania is included in Exhibit 2.12-A.

Interchange with an Expressway or Non-Interstate Freeway. If the intersecting route is an expressway or a non-Interstate freeway, the first major city, borough or municipal center identified on the map in each direction along the intersecting route should be used as the destinations on the major guide signs. Each destination should generally have a population of at least 10,000 people and be within 20 miles in urban areas, and within 50 miles in rural areas. Municipal centers should have a concentrated center, but need not be an incorporated municipality.

Non-Urban Areas – Interchange with a conventional road. In non-urban areas at interchanges with conventional roads, an effort should be made to have a destination to the left and to the right on the major guide signs.

1. If the intersecting highway is a U.S. or PA traffic route, the destinations should generally be the closest city, borough, village or home-rule municipality to the left and to the right as identified on the map. If a more distant community exists along the traffic route having a population-to-distance ratio at least 50 percent greater than any closer community identified on the map, the more distant community name may be used in lieu of the closer community name providing the more distant community:

    - is not directly linked to another interchange along the traveled route within the next 25 miles; and

    - is within 15 miles if it has a population less than 10,000 or within 20 miles if it has a population greater than 10,000.

2. If the intersecting route is not a numbered traffic route, the destinations should generally be the closest city, borough, village or home-rule municipality to the left and to the right, on or immediately adjacent to the crossing highway as identified on the map. If a more distant community exists on or immediately adjacent to the crossing route that has a population-to-distance ratio at least 50-percent greater than any closer community, the more distant community name may be used in lieu of the closer community name providing the more distant community is shown on the map and is within 5 miles, or within 10 miles if it has a population over 5,000.

3. If at least one destination cannot be selected which satisfies the criteria in paragraphs (1) or (2), a “point location” which is not shown on the map (for example, a village) may be used.

Urban Areas – Interchange with Conventional roads. In urban areas at interchanges with conventional roads, street names or municipal center names should generally be used instead of community names, after receiving written concurrence of the local authorities. In addition, a Community Next (___) Exits sign and Interchange Sequence signs should generally be used in urban areas.

2.12.2.4 Special Signing

Trailblazing. Consider trailblazing to traffic routes near the interchange by providing the traffic route shield and the “TO” supplemental message.

Large Non-Municipal Traffic Generators. Bridge and freeway names are effective destinations to provide orientation for drivers, especially in large metropolitan areas. In addition, it is permissible to use the name of a large traffic generator as the destination if the crossing roadway essentially only serves the traffic generator; examples of large traffic generators include airports, national and State parks, industrial parks, etc.

Multiple Signs. Where two or more guide signs are on the same structure, it is desirable to limit destination names to one per sign, or to a total of three in any direction of travel.

Design Considerations. The field location and layout details of the guide signs should be in conformance with Traffic Standards 8701A and 8701D (Publication 111M) respectively.

Municipal Name Signs. Signs to identify the municipal boundaries may be installed in accordance with Section 2.10.7.

2.12.3 Selection of Supplemental Destinations

Information regarding destinations accessible from an interchange other than the destinations on the major guide signs may be listed as:

1. An attraction on Logo Signs (refer to Section 2.14), or

2. A destination on a Supplemental Guide sign if determined to be appropriate by the Department.

Because Supplemental Guide Signs can only accommodate two destinations, the preferred method is also to use Logo Signs, whenever possible. However, if an destination is not eligible for logo signs under Section 2.14, it may be eligible for inclusion on a Supplemental Guide Sign. The *MUTCD* and the AASHTO *Guidelines for the Selection of Supplemental Guide Signs for Traffic Generators Adjacent to Freeways* provide details for supplemental guide signs.

2.12.4 Motorist Service Signs

Motorist Service Signs are generally installed only on freeways. Two specific types are available: (1) General Motorist Service Signs (Section 2.13), and (2) Logo Signs (Section 2.14).

Logo signs are now installed on most Interstate highways and freeways by the Pennsylvania Logo Signing Trust. Therefore, the counties and Districts should not expend unnecessary monies to replace general motorist service signs on Interstate highways unless it is known that logo signs will not be installed at the interchange.

2.12.5 Reference Location and Enhance Reference Location Signs

Signs shall be erected on all Interstate highways showing the mileage along the route with increasing order in the same direction as increasing interchange numbering. Historically, these signs were called “mile markers” or “distance markers,” but they are now referred to as “reference location signs.” These signs should be installed in accordance with the procedures identified in Publication 111M.

There are two types of “reference location signs” that may be used, but the second type is required on sections of highway where Intermediate Reference Location (D10-1a, D10-2a or D10-3a) signs are used at 1/10th or 2/10th-mile intervals as discussed in Section 2.12.6:

1. Reference Location (D10-1, D10-2, D10-3) signs as identified in Section 2D.46 of the *MUTCD*.

2. Enhanced Reference Location (D10-4) signs as identified in Section 2E.54 of the *MUTCD*.

D10-1 D10-2 D10-3

Mileage on circumferential freeways shall increase in a clockwise direction, beginning with Mileage 0 (i.e., MP 0) with the first interchange west of a radial freeway or other Interstate route or some other conspicuous landmark in the circumferential route near a south polar location.

D10-4

Mileage on spur routes should begin with Mileage 0 at the junction with the main line of the principal route. If the spur route connects two different main line routes, the principal route is the one with the same last two digits in the route number and the spur. For example, I-283 connects both I-76 and I-83, but I-83 is the principal route.

Where numbered freeways overlap, continuity of the mileage shall be established for the RMS priority route only.

When a District elects to install D10-1, D10-2, D10-3 or D10-4 signs at 1-mile intervals on non-interstate freeways, the procedures outlined below will be followed for their installation.

(a) Only full limited-access highways will be considered.

(b) Mileage on the signs will increase south-to-north and west-to-east, consistent with the existing RMS system. Mileage 0 (MP 0) will begin at the southern or western state line respectively, or at the beginning of the numbered traffic route. Determine subsequent distances by traveling the route with a distance-measuring instrument (DMI) and marking locations on the shoulder. At the same time, verify the segment/offset for subsequent entry into the RMS database.

(c) Compute distances consecutively, consistent with RMS, regardless of missing sections of limited-access highway. For example, US 15 begins at the Maryland line (MP 0) and increases for a distance of 22 miles before the limited-access portion of the highway terminates. About 15 miles farther, another limited-access portion of US 15 begins with the posting of Distance Marker MP 37.

(d) The presence of county lines will have no consequence on the mileage determination just as in the case of the Interstate System.

(e) In situations where traffic routes overlap, the standard RMS convention for assigning priority will apply, and the mileage for the priority route will be continuous on the D10-1, D10-2, D10-3 or D10-4 signs. When routes subsequently separate, base distances on each route having carried the distances continuously. For example, where US 22 and US 322 overlap, the signs will carry the mileage based on US 22. Once the routes separate, US 322 will pick-up the mileage as if it had been continuously carried through the overlapped section.

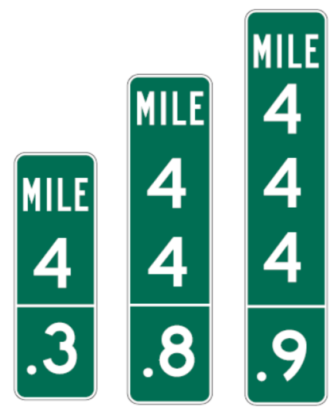
(f) Prior to the installation of any D10-1, D10-2, D10-3 or D10-4 signs, each sign must be identified on the RMS Screen 430. The District must identify each proposed sign location by county/SR/segment/offset and forward the location information directly to the Bureau of Planning and Research (BPR), Transportation Planning Division for inclusion in RMS. BPR will then notify the District when the locations of the D10-1, D10-2, D10-3 or D10-4 signs have been input to RMS. Note that the segments and offsets will not be modified to correspond with the mileage, as is the case with the Interstate System.

(g) When changes in alignment (e.g., construction of new limited-access highway, reconstruction of existing conventional road, etc.) result in recalibration of RMS distances contiguous to the section of highway where

D10-1, D10-2, D10-3 or D10-4 signs are installed, the signs will be relocated to correspond with the new RMS stationing. These revised locations must be reported to the Transportation Planning Division as noted above. Note that when realignments occur on non-freeway sections that result in actual mileage changes of the route, but do not affect the section of highway where D10-1, D10-2, D10-3 or D10-4 signs are installed, it is not required that the sign be relocated.

2.12.6 Intermediate Reference Location Signs

To assist in locating incidents, District 6-0 started using "1/10th-mile markers" on the Interstate System in the Philadelphia area in 1991. Alternate designs were subsequently used in two Districts that showed cardinal direction of travel and a small route marker, but none of these designs are currently approved. Instead, the Department has adopted the Intermediate Reference Location (D10-1a, D10-2a, and D10-3a) signs as identified in Section 2D.46 of the *MUTCD*.

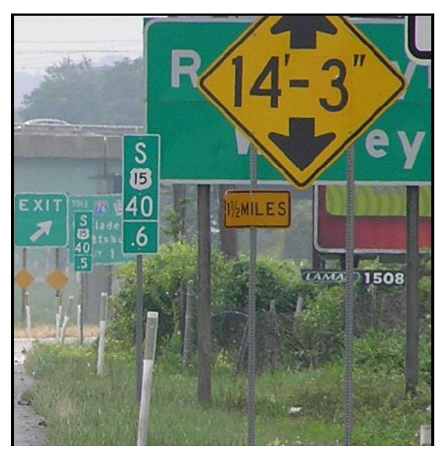


D10-1a D10-2a D10-3a

Department policy for Intermediate Reference Location (D10-1a, D10-2a, and D10-3a) signs is as follows:

1. It will be up to each Engineering District to determine when and where to install the D10-1a, D10-2a, and D10-3a signs; however, all initial installations shall be by contract.

2. The D10-1a, D10-2a and D10-3a signs should only be installed on freeways and full limited-access portions of expressways where Enhanced Reference Location (D10-4) signs currently exist or will be installed concurrently with the D10-1a, D10-2a and D10-3a signs. (This may require replacing the existing D10-1, D10-2 and D10-3 signs.)



3. Avoid installing D10-1a, D10-2a and D10-3a signs for short distances (i.e., less than 5 miles) or intermittently. Always use logical termini.

4. Intermediate Reference Location (D10-1a, D10-2a and D10-3a) signs will be installed at 2/10th-mile intervals (i.e., even-numbered tenths) except as follows:

   (a) Use 1/10th-mile intervals in urban areas where AADT exceeds 75,000 or the interchange spacing averages less than 1.5 miles

(b) Use $1/10^{th}$-mile intervals for crest vertical curves in excess of 1,000 feet where the rate of vertical curvature (K), as defined in the AASHTO Policy of Geometric Design of Streets and Highways, exceeds 380.

(c) Use $1/10^{th}$-mile intervals on horizontal curves where the sight distance is less than 1,000 feet and the D10-1a, D10-2a and D10-3a signs are mounted on the same side of the highway as the direction of the curve.

5. Install all D10-1a, D10-2a and D10-3a signs in accordance with TC-8710. Consider mounting the D10-1a, D10-2a and D10-3a signs back-to-back in the median when the median width is 30 feet or less. Otherwise, install them along the right edge of the highway.

6. When traffic routes overlap, use the standard RMS convention to assign priority.

7. For non-interstate freeways, the District must identify each proposed D10-1a, D10-2a and D10-3a sign by county/SR/segment/offset, and ensure that they are in RMS before the physical installation begins. To enter the data, access the "RMS Mile Marker Screen" through the 406-Menu Screen (#21) or through the Jump Screen (#20). Note that the segments and offsets will not be modified to correspond with the mileage as is the case with Interstate highways.

8. Deployment of D10-1a, D10-2a and D10-3a signs will include the distribution of mapping to local first responders and emergency management officials.

2.12.7 Interchange Numbering

Interchange numbering provides valuable orientation for motorists and is required on all Interstate highways. When used, interchange numbering shall be mileage-based, and conform to the mileage of the nearest Reference Location or Enhanced Reference Location signs as discussed in Section 2.12.5. Junctions with other freeways, including other Interstate highways shall be numbered.

On multi-exit interchanges, such as a cloverleaf interchange, use the suffix letters A or B. In the direction of increasing interchange numbering, the first exit shall use the suffix A and the second exit shall use the suffix B. In the opposite direction, the exits shall typically be given the suffixes in the opposite order so that both Exit A's go toward the same destination, and both Exit B's go toward the same destination.

If it is necessary to revise exit numbers, use the transitional signing procedure identified in Section 2.9.7.

2.12.8 Overhead Signs

Luminaires are not required on new overhead sign installations when a minimum 800-foot tangent sight distance exists in advance of the sign and the vertical alignment allows headlights to illuminate the sign. Districts may also disconnect existing luminaires on signs with minimum Type VIII or IX reflective backgrounds when the necessary sight distance exists.

Overhead sign structures shall not be left without any signs on the superstructure, since signs are necessary for vibration-dampening purposes.

2.12.9 Upgrading Temporary Signs

Publication 111M specifies the appropriate legend size for all common expressway and freeway signing. All signs with legend smaller than the specified size are considered temporary. Sections of Interstate highways with temporary signs are eligible for Interstate funding for sign replacement.

2.12.10 Overpass Roadway Identification Signs

The installation of Overhead Roadway Identification (I18-1) signs are encouraged on expressways and freeways to identify the numbered traffic route or local name of the roadway on the overpass bridges.

Center all I18-1 signs over the roadway approach, mounted flush to and at the same grade as the parapet on the overpass. Avoid mounting signs over expansion joints.

2.12.11 Public Relations

When the Department proposes to change signing that may cause street address or directional differences, such as changing exit numbers, exit suffixes, destination names, etc., it is imperative that adequate advance public relations be conducted with local authorities and other community organizations. The notification should be as far in advance as possible, with follow-up information as the event nears.

2.13 General Motorist Service Signs

2.13.1 Purpose and Authorization

This policy establishes uniform guidelines for the approval, design, installation and replacement of general motorist service signs for gas, food, diesel, lodging, information, camping, hospital and State police, on freeways and locations between the freeway and the motorist service. Since logo signs (Section 2.14) are anticipated for most Interstate highways, extensive changes in current general motorist service signs are not desirable except on those Interstate highways where logo signs cannot be installed.

The final authorization of signs and the selection of sign locations will rest with the Department. When the signs are to be installed on an Interstate highway, the Department's authorization is contingent upon approval of the Federal Highway Administration if the signing is not in conformance with Publication 111M. The authorization of signs should not be construed to be an endorsement of the facility or the services offered, but only means that the minimum standards and criteria are satisfied.

2.13.2 Definitions

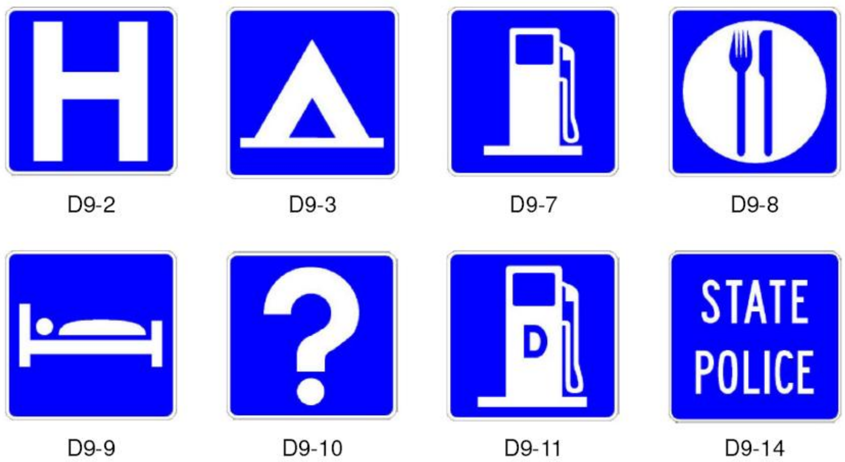
Motorist service – A gas, food, diesel, lodging, camping, information, hospital or State Police service as identified in Section 2.13.3.

Motorist service directional sign – A sign on off-ramps and on conventional roads, which displays one or more motorist service symbols or legends and a one-way or two-way directional arrow.

Motorist service panel – A sign along a freeway displaying one or more motorist service symbols or legends and the appropriate legend “NEXT RIGHT” or “SECOND RIGHT.”

Motorist service symbol – Any of the eight symbols identified in **Exhibit 2.13-A**.

Exhibit 2.13-A Motorist Service Symbols



2.13.3 Acceptable Types of Motorist Services

1. Camping. A campground licensed by the Department of Environmental Protection (DEP) with continuous operation for at least 6 months each year; accommodations for a minimum of 20 campers; an attendant available during the normal working hours; rest rooms with showers, running water and flush toilets; laundry facilities; and a public telephone. Accommodations sold on a time sharing basis or otherwise not available for general public use will not be considered toward the minimum requirements (Section 2.7 allows personalized attraction signs on non-freeway facilities, and Section 2.14 provides for logo signs on select Interstate highways.)

2. Food. A restaurant with a license issued by the Department of Health; seating for at least 20 people; continuous operation for at least 11 consecutive hours, 7 days a week; public rest rooms with sink, running water and flush toilet; and a public telephone.

3. Gas or Diesel**.** A station with gasoline or gasoline and diesel fuel; water for radiators; rest room with sink, running water and flush toilet; continuous operation for at least 16 hours per day, 7 days per week; and a public telephone.

4. Hospital. A facility with approval as a hospital by the Department of Health; and continuous emergency care capability to the general public with a doctor on duty 24 hours a day, 7 days a week.

5. Information. An establishment with approval by the Department of Commerce as a tourist or visitor information center; an open season of at least 6 months each year; open at least 10 hours per day, 7 days per week between

Memorial Day and Labor Day, and at least 8 hours per day, 7 days a week during the balance of the open season; an attendant on duty during all open hours; and free access to rest rooms, drinking water and telephone service.

6. Lodging. A hotel or motel with adequate sleeping accommodations and off-street parking for at least 20 private rooms with private baths; continuous year-round operation 24 hours a day, 7 days a week; and a public telephone.

7. State Police Station. A station manned by State Police 24 hours a day, 7 days a week.

2.13.4 Additional criteria

(a) Laws and Regulations. All motorist service facilities shall conform to all applicable Federal, State and local laws or regulations, and be open to all persons regardless of race, color, religion, ancestry, national origin sex, age or handicap.

(b) Distance from Freeway. The services shall be within the following distances from the ramp terminus and the crossing route:

- Gas and diesel – 1.0 mile, except the maximum distance for “gas” may be increased up to a total of 2.0 miles if the average distance to adjacent interchanges is more than 6 miles.

- Food, lodging, and information – 2.0 miles, except the maximum distance for “food” may be increased up to a total of 3.0 miles if the average distance to adjacent interchanges is more than 6 miles.

- Hospitals and State Police stations – 3.0 miles. (5.0 miles for Hospitals in rural areas.)

- Camping – 3.0 to 15.0 miles. If camping services are not available within the 3.0-mile limit, the limit may be extended up to 15.0 miles in increments of 3 miles until there are participants.

(c) Urban Areas. Only Information, Hospital and State Police services will be signed at interchanges within the corporate boundaries of a city, or an area which appears as an urban or metropolitan area.

(d) Continued Travel. Signs will not be authorized where a U-turn or illegal movement is required or at locations where motorists cannot conveniently return to the freeway for continuation of travel in the same direction. Signs will not be authorized at interchanges with freeways.

2.13.5 Sign Location

Motorist service panel. When motorist service panels are installed on freeways, they should be erected following the first major guide sign and, if possible, following any supplemental guide sign. A minimum 800-foot spacing will be maintained between this sign and any large guide sign.

Freeway off-ramp. A Motorist Service Directional Sign will be erected on freeway off-ramps if exiting traffic can turn in both directions on the crossing route and the motorist service is not readily visible to approaching motorists.

Conventional road. Except for camping and information services, Motorist Service Directional Signs may be installed along the crossing route and at subsequent locations to indicate directional changes for motorists traveling from the freeway. The distance to the service may also be indicated if the service is not within the immediate interchange area. (Legend-type signs with the campground or visitor center's name may be authorized at the service's expense along the crossing route and at subsequent locations in accordance with Section 2.7.) In addition to trailblazing from freeways, signs may also be installed along conventional roads for State police stations and hospitals, directing motorists from the nearest numbered traffic route or other major highway, providing the maximum distance does not exceed the appropriate base mileage identified in Section 2.13.4.b.

Local roadways. If signs are required on local roadways (municipal streets or roads) in order to trailblaze motorists from a State highway to the service, the installations shall be coordinated to ensure signing continuity.

Insufficient space. If insufficient space is available for motorist service signs, do not authorize any Motorist Service Signs will be authorized.

2.13.6 Design of New Installations

Motorist Service Panels and Motorist Service Directional Signs will be designed in accordance with Publication 111M. In urban areas (where motorist service panels are not installed) the Department may install “Information,” “Hospital,” or “State Police” motorist service symbols beneath the last advance major guide sign (the last sign with a distance to the off-ramp included on the sign) before the off-ramp. When only one motorist service symbol is warranted prior to an interchange and a motorist service panel does not exist, the appropriate symbol may be installed beneath the last advance major guide sign prior to the off-ramp. (All signs installed beneath major guide sign shall either be attached only to one of the posts or shall be attached to the major guide sign, so as to avoid interfering with any breakaway hinge.)

2.13.7 Procedures for Application

The Department will review the motorist services when a roadway is being "resigned," and may have signs erected for applicable motorist services.

Requests for service signing may be submitted to the Department at any time. The application (Exhibit 2.13-B) should be returned to the appropriate Engineering District Office. The Department will assume all costs associated with the erection of general motorist service signs.

Exhibit 2.13-B Application for General Motorist Signs

Instructions:

i. Read Section 2.13 in its totality.
ii. Complete the questions at the bottom of this sheet.
iii. Send the application to the appropriate Engineering District Office. (The District Office will review the application, determine if the facility qualifies for motorist service signing, and respond to you.)

---

1. Circle type(s) of Service:

Gas	Food	Lodging	Diesel
Camping	Hospital	Information	State Police

2. Business name and address ____

____

____Telephone No.___________________

3. Direction from the interchange .____

Distance from the interchange.____

4. Locations of proposed motorist service signs: Route, Intersecting Route, Interchange name and number:____

____

5. Applicant's name and address .____

____

____Telephone No. ___________________

6. I certify that the requirements of Section 2.13.3 and Section 2.13.4 are satisfied, and I agree to immediately advise the Department of any changes that the business does that relate to these requirements.

____
Signature

2.14 Logo Program

2.14.1 Purpose

The logo program is a traveler information service provided for motorists that travel Pennsylvania highways. Moreover, since "logo" signing is authorized as a public service, only those services and facilities that are reasonably accessible at interchanges will be signed.

Therefore, the purpose of this subchapter is to establish guidelines for the approval, design, erection, maintenance and funding of logo signing along Interstate highways and other freeways for gas, food, lodging, camping services, and general attraction destinations.

2.14.2 Authority

These guidelines are in accordance with standards issued by the Federal Highway Administration under authority of Title 23, U.S. Code Sections 109(d), 131(f) and 315, the Manual on Uniform Traffic Control Devices, 49 CFR 1.48(b), and Title 75 Pa.C.S. Section 6122. Where differences occur between these guidelines and the national standards, the more restrictive shall govern.

2.14.3 Administration

The program is currently administered by the Pennsylvania Tourism Signing Trust whose duties and responsibilities are defined in Section 2.14.12.2. All signs become the property of the Department after they are erected. Additional information is available at www.palogo.org.

2.14.4 Definitions

Attraction – A facility that is of interest to and destination for motorists and is eligible for participation in the logo program, herein referred to as a business or service.

Freeway – A highway to which the only means of ingress and egress is by interchange ramps.

General public – A service shall be deemed available to the general public if it is available to anyone, at any time, without membership or other requirements limiting use by the public at large.

Logo – A sign provided by a business or attraction to identify the business's trademark or the attraction's name. The logo is attached to a specific service sign, ramp sign or trailblazer.

Ramp sign – A small sign panel erected along an off-ramp to direct motorists to a particular service. (See Exhibit 2.14-F.)

Sign panel – The main part of a sign or trailblazer to which the individual logos are attached.

Specific Service Sign – A large sign panel installed along an Interstate highway or other freeway to indicate the specific services available at the next interchange. (See Exhibits 2.14-B, 2.14-C, and 2.14-D.)

Tourist Oriented Directional Sign (TODS) – A small directional sign (D7-4) with white legend on a blue or brown background that indicates the name of, and gives directional guidance to the attraction. These signs serve as attraction logo trailblazers for the purposes of this policy.

Trailblazer – A small sign panel similar to a ramp sign (or D7-4 TODS-type sign for attractions) that is erected on the road network accessed by way of a logo-signed interchange to direct motorists to a particular service. (See **Exhibits 2.15-F** and **2.15-H**)

2.14.5 Types of Services

Services are limited to gas, food, lodging, camping, and attractions. To qualify, services shall be open to the public regardless of their race, religion, color, sex or national origin. They shall have paved driveway entrances which are properly permitted by the Department or municipality, as applicable, except campground and attraction entrances for fairgrounds; recreational areas; state and national parks, forest or cemetery, or state game lands; and unique natural areas may be unpaved. Each facility shall have adequate on-premise signing which is clearly visible to approaching motorists and identifies the service location, and shall satisfy the following:

(a) Gas. A station for cars, or cars and trucks, which provides gasoline, oil and free public rest rooms with sinks and running water. The station shall be in continuous operation for at least 16 hours per day, 7 days a week. A telephone on or within 500 feet of the property shall be available during hours of operation.

(b) Food. A restaurant which is licensed by the Department of Agriculture or local health jurisdiction, accessible without an admission fee, and provides seating for at least 20 people within the same building, in continuous operation for at least 10 hours per day, 6 days a week, and contains public rest rooms with sinks and running water. Restaurants within shopping centers will not qualify unless they have an outside entrance directly accessing the restaurant's leased space, which is clearly labeled and readily visible and accessible to approaching motorists. Restaurants only open 6 days a week must include a supplemental message on their logo panel stating the day they are closed (see Section 2.14.9(e)).

(c) Lodging. A hotel or motel with private rooms and baths, public telephones or telephones provided in each room, adequate off-street parking, and available 24 hours a day, 7 days a week. Condominiums and time-share forms of hotel occupancy may participate, provided they are marketed to the General Public for overnight accommodations.

(d) Camping. A campground with continuous operation for at least 6 months per year and a minimum of 20 overnight sites. An attendant shall be available during the hours of operations and rest rooms with showers, running water and flush toilets shall be available. A public telephone also shall be available on the site or within 500 feet of the property. Accommodations sold on an annual or a time-sharing basis or otherwise not available for general public use will not be counted toward the minimum requirements.

(e) Attraction. An attraction must fall under one of the categories listed below. An attraction, except as otherwise provided, must have adequate legal parking accommodations, provide public restrooms with sinks and running water (Commerce Park excluded), be open a minimum of 30 days per year, and have a minimum annual per capita usage pursuant to Exhibit 2.14-A. If there is an admission charge, it must be readily visible to prospective visitors at the point of entry.

- Amusement Park. A permanent area which is open to the general public for activities such as picnicking, hiking, swimming, boating, entertainment rides, etc.

- Arena. A stadium, sports complex, auditorium, civic or convention center or racetrack, which has a capacity of at least 5,000 as determined by the Pennsylvania Department of Labor and Industry.

- Business District. An area within a municipality which is officially designated and signed as a business district by the local officials of the municipality.

- College or University. An institution which is approved by a nationally recognized accreditation agency and which grants degrees.

- Commerce Park. A group of commercial manufacturing facilities recognized and signed as a commerce park by the local authorities. Any Commerce Park that has been granted Keystone Opportunity Zone status is exempt from the requirements for minimum acreage and number of required businesses in Exhibit 2.14-A.

- Cultural Center. A facility for the performing arts, exhibits, or concerts.

Exhibit 2.14-A Eligible Attractions for Logos

Type of Attraction	Specific Criteria	Urban Area	Rural Area
College or University	Enrollment (Full & Part Time)	2,500	1,200
Business District	Number of Businesses	75	50
	Municipal Population	15,000	10,000
State or National Park, Recreational Area, Forest, State Game Land, Cemetery	Certified Attendance Figures	20,000	20,000
Amusement Park, Arena, Cultural Center, Facility- Tour Location, Fairground, Golf Course, Historic Site/Area, Museum, Observatory, Ski Area, Unique Natural Area, Zoo/Botanical Park	Certified Attendance Figures	20,000	12,000
Type of Attraction	**County Population**	**Acres**	**Businesses**
Commerce Park	< 100,000	5	3
	≤ 500,000	10	5
	≤ 1,000,000	15	10
	> 1,000,000	25	15

- Facility, Tour Location. A business that conducts daily or weekly tours on a regularly scheduled basis. Eligible attractions may be on-site tours that operate year-round at facilities such as plants, factories or institutions. Eligible attractions may also be off-site tour-providers services that operate tours daily or weekly at least six months per year for local attractions of historical, architectural, cultural or scientific interest to tourists, such as battlefields or historic districts. Tours of the off-site type are typically conducted by boat, carriage, motor coach, railway, etc. For tour-provider services, only the point of purchase for the service shall be signed.

- Fairground. A tract of land where fairs or exhibitions are held, and which has permanent buildings including, but not limited to livestock exhibition pens, exhibition halls, bandstands, etc.

- Gaming. A facility issued a Category 1, 2 or 3 slot machine license by the Pennsylvania Gaming Board under provisions of Act 71 of 2004.

- Golf Course. A facility open to the public and offering at least nine holes of play. Miniature golf courses, driving ranges, chip-and-putt courses, and indoor golf shall not be eligible.

- Historical Site or District. A structure or area recognized by the Pennsylvania Historical and Museum Commission as a historic attraction in the National Register, individual properties, or historic districts in Pennsylvania. Historic districts shall provide the public with a single, central location, such as a self-service kiosk or welcome center, where motorists can obtain information regarding the historic district.

- Museum. A facility, open to the public at least 100 days per year, in which works of artistic, historical, or scientific value are cared for and exhibited to the public.

- Observatory. A facility designed and equipped for making observations of astronomical, meteorological, or other natural phenomena.

- Recreational Area. Recreational attractions including, but not limited to, bicycling, boating, fishing, hiking, rafting, swimming, picnicking, snowmobiling, or cross country skiing.

- Shopping Center/Antique & Flea Market. A shopping center is a group of stores separated by floor to ceiling partitions, which has a minimum of 10 stores and a minimum of 400,000 square feet or has a minimum of 30 stores and a minimum of 100,000 square feet. An antique & flea market is a group of 75 or more vendors or having a total area of 30,000 square feet , that specializes in the sale of antique and/or flea market items; such applicants shall certify that they comply with Pennsylvania's sales tax laws and regulations.

- Ski Area. A downhill skiing area with equipment rentals, or a cross country ski area with equipment rentals and a minimum of 5 miles of marked and groomed trails.

- State and National Park, Forest or Cemetery, or State Game Land. An area designated by and under the jurisdiction of the National Park Service, the Veterans Administration, Pennsylvania Department of Conservation and Natural Resources, or Pennsylvania State Game Commission.

- Unique Natural Area. A naturally occurring area which is of outstanding interest to the general public, such as a waterfall or a cavern.

- Visitor Information Center. A visitors information center open at least 6 months each year, including 9 hours each day between Memorial Day and Labor Day, and 8 hours each day during the balance of the open season. The facility shall have an attendant on duty during the open hours, and provide free access to travel literature, rest rooms, and drinking water. Centers other than those owned and operated by the Commonwealth of Pennsylvania must be administered by the appropriate local tourist promotion agency.

- Winery. A licensed site which produces a maximum of 200,000 gallons of wine per year. Sites shall maintain a minimum of 3,000 vines or 5 acres of vineyard in the Commonwealth; be open to the public for tours, tasting and sales a minimum of 1,500 hours per year and provide an educational format for informing visitors about wine and wine processing.

- Zoological/Botanical Park. A facility in which living animals or plants are kept and exhibited to the public.

2.14.6 Distance to Services

The normal maximum distance that services may be located from the end of the off-ramp to qualify for a logo is as follows:

Service		Distance
Gas		1.0 mile
Food		2.0 miles
Lodging		3.0 miles
Camping		5.0 miles
Attraction	-	5.0/15.0 miles*

***** 5.0 miles from an interchange identified by the Department's Urban Boundary Classification Maps, or 15.0 miles from a rural interchange

The measurement to each business shall be along both public and private roadways and driveways delineated by pavement markings, signs and other traffic control devices. The distance is measured by computing the travel length from the terminus of the exit ramp of the most convenient interchange, to the following location:

- Gas, food, lodging and camping. The termination point is the primary building entrance for the service.
- Attractions. The termination point is the primary building entrance for the service, except the termination point for attractions with satellite parking is the gate, ticket booth or other type of primary entrance to the parking area.
- Business district or historic site. The termination point is the District boundary.
- Commerce park, recreation area, shopping center, State and National Park, Forest or Cemetery, or State Game Lands. The termination point for measuring is the entrance to the primary parking area for the facility.

The maximum distances for gas, food and lodging may be increased an additional 1.0 mile if the average distance to the two adjacent interchanges is more than 5 miles, as indicated on the Department's Official Transportation Map.

2.14.7 Unacceptable Locations for Logo Signs

Logo signs shall not be authorized at the following locations or under the following circumstances:

(a) At interchanges with other freeways.

(b) At interchanges where space for only one sign installation exists except under the following conditions:

- In urban areas, one logo sign shall be authorized, which shall be solely for the Attraction category
- In rural areas, one logo sign shall be authorized, using the three service panel, comprised of two logos for each of three-service categories. At the initial installation, the Attraction service shall be given priority, and the two other services shall be determined in accordance with Section 2.14.10.3. After the initial installation, any open service to be filled on the three-service panel shall be filled in accordance with Section 2.14.10.3 (and Attractions shall not be given specific priority, even if there were no Attractions signed after initial installation).

(c) In areas of high congestion, such as within a central business district or where long traffic delays frequently occur.

(d) When the number of turns required from the crossing route prior to the driveway of an establishment is greater than the following:

- Gas – two turns.
- Food – three turns.
- Lodging – three turns, except four turns if the facility is within one mile of the exit ramp terminal.

(e) Where a U-turn or illegal movement is required to access a business, or where it is not convenient to return to the interchange and continue in the original direction of travel.

(f) Where long sections of structure, retaining wall and/or installations of noise wall limit the ready placement of ground-mounted logo signing.

(g) At interchanges where it is necessary to direct motorists back in direction to service establishments located at a previous interchange.

(h) Where the Department determines that for safety, operational, or other explained reasons the installation of logo signs is not in the best interest of the traveling public.

(i) At any interchange approach other than that which most directly and conveniently accesses the service establishment.

(j) Where a maximum of four logo sign installations exist on any approach to an interchange.

(k) Where a trailblazer for a business would be required off the right-of-way of a State highway, unless the business obtains all required approvals and permits from the local officials for the trailblazer within 130 days from its application for logo signing.

(l) In no event shall a participant be signed at more than one interchange for each direction, for each service on a specified traffic route.

2.14.8 Continuation of General Motorist Service Signs

General motorist service signs display symbols or words for services such as “GAS,” “FOOD,” “LODGING,” “CAMPING,” “VISITOR INFO,” “DIESEL,” “HOSPITAL,” and “STATE POLICE.”. Whenever possible, these general motorist service signs should be removed as soon as logo signing is installed at a particular location.

However, if only certain types of services at an interchange participate in the logo signing program (e.g., only gas and lodging), the remaining services (e.g., food and camping) can continue to be signed via a general motorist service sign provided sufficient spacing is available along the mainline to erect the signs. In no cases, other than mentioned above, should general motorist service signs duplicate logo signing for a particular service. General motorist service symbols may be attached to the supports of mainline logo signs in the absence of separate mainline general motorist service signs. When general motorist service directional signs are placed along ramps, if possible, the signs should not be placed together with logo ramp signs on the same post.

Businesses with billboards in violation of State or federal laws or regulations, will not be authorized to participate in the Logo Sign Program.

2.14.9 Logo Requirements

(a) Design. Business logos may consist of a symbol, trademark, or a legend message identifying the name or abbreviation of the specific business. Attraction logos shall include a legend message identifying the name of the attraction. All logo designs shall be reviewed and approved in accordance with Department standards prior to fabrication. Logos which resemble any official traffic-control device or which are determined to be in poor taste by the Department will be prohibited.

(b) Size and Shape. All logos shall be rectangular in shape and conform to the following sizes:

- Mainline logos (logos directly along the Interstate highway or other freeway) shall be 60″x36″ for food services, lodging services, camping services, and attractions, and 48″x36″ for gas services.

- All ramp and trailblazer logos shall be 30″x18″, except where authorized otherwise by the Department.

(c) Legends. A legend which is not part of a regionally or nationally recognized trademark should be as large as possible, preferably with only one or two lines of messages. Only one registered trademark may be included on a logo panel. Any portion of the trademark that appears to be a supplemental message may be excluded. The maximum amount of legend shall be three lines, each having up to 12 characters (i.e., letters, numerals, or spaces). The minimum size legend shall be 8-inch for mainline logos and 4-inch for ramp and trailblazer logos.

(d) Color. Logos may use any contrasting combination of standard highway colors, i.e., white, yellow, red, blue, green, orange, brown and black. Transparent inks or electronic cuttable films may be used to correlate with standard trademarks providing the colors provide good readability during both daylight and nighttime hours. Colors that are critical to nighttime readability shall be at least as reflective as the standard silk-screened blue color, as determined by Department instrument testing. Fluorescent colors are not permitted.

(e) Supplemental Messages. When used, the minimum legend height for supplemental messages shall be 6 inches on mainline logos and 3 inches on ramp and trailblazer logos. Supplemental messages shall not extend beyond the edge of the logo and onto the sign panel. The following supplement messages may be used as applicable on logos:

- "24-HR" on gas or food logos provided the facility satisfies all eligibility criteria for all 24-hour periods subject to that criteria.
- "DIESEL" on gas logos.
- "NO TRUCKS" on gas logos (if a station has no facilities or parking for trucks).
- "SEASONAL" on attraction and camping logos for those facilities open less than 12 months a year.
- "CLOSED ____DAY" shall be used on food logos if a food establishment is only open 6 days a week.

(f) Materials. Logos shall be fabricated on an aluminum substrate with a minimum thickness of 0.080 inch. All colors in the logo shall be made from either Department approved Type III or IV retroreflective sheeting or transparent inks on Type III or IV white retroreflective sheeting.

(g) Approvals. All logo designs and supplemental messages, and any revisions thereto, shall be submitted to the Department (through the Logo Signing Trust) for review and approval. Submissions shall include sufficient layout information to determine compliance with size, shape, color, legend and material requirements. Letter sizes for all legend proposed as part of the logo design must be clearly marked. The Department may request a small retroreflective sign sample of any custom-mixed colors to determine nighttime reflectivity. No logos or supplemental messages shall be manufactured until approval is received.

2.14.10 Sign Panels

2.14.10.1 Location

(a) Separate Sign Panel. Except as provided in Section 2.14.10.2(a), (b) and (c), a separate sign panel shall be provided for each type of service for which logos are displayed. In the direction of traffic, the sequential order of sign panels shall be in the order of attraction, camping, lodging, food, and gas, except for existing installations that do not conform to this order, which installations will remain until new signing work requires relocation of such panels. Additionally, a new sign may be installed out of sequence if in the future it is installed in a combined service format, with the future additional service placed in the correct sequence. Signs shall be positioned to take advantage of natural terrain or guide rail, to have the least impact on the scenic environment and to avoid visual conflicts with other signs.

(b) Specific Service Signs. Specific service signs may be installed between the previous interchange and a point 800 feet in advance of the exit direction sign or "NEXT RIGHT" sign at the interchange from which the services are available. A minimum 800-foot spacing shall be provided between specific service signs, and between specific service signs and existing major guide signs. Excessive spacing should be avoided. Space which is closer to an exit should be fully used before specific service signs can be placed in advance of the first major guide sign.

(c) Ramp Signs. At single-exit interchanges where service facilities are not readily visible from the ramp, ramp signs (see Exhibit 2.14-F and Exhibit 2.14-G) shall be installed along the ramp or at the end of the ramp. Signs along the ramp should generally be installed on the right side of the ramp, but are permitted on the left side. A minimum 200-foot spacing shall be provided between all ramp signs, and between all ramp signs and other traffic signs on the same side of the ramp. Ramp signs are only authorized for businesses which are participating on the specific service signs.

(d) Trailblazers. Trailblazers may be installed for specific service sign participants when it is necessary to provide additional guidance to motorists after they exit from the ramp. All trailblazers shall be installed up to 300 feet before any required turn. Once the turn (or turns) is accomplished, no other confirmation trailblazers will be placed. Trailblazers for camping, lodging, food and gas are similar to ramp signs but do not include the generic type of service (see

Exhibit 2.14-F). For attraction facilities, trailblazers will be the D7-4 TODS-type signs (see Exhibit 2.14-H). At double-exit interchanges, trailblazers may be installed along the crossroad near the end of the off-ramp for all services over 1 mile from the end of the ramp; distances and arrows shall be included.

Trailblazers will be grouped at the intersection by direction (straight, left and right) and stacked totem pole style beginning at the top with camping, then food, then lodging, and with gas on the bottom. Attraction trailblazers will be separate sign assemblies and shall not be mixed on the same sign assembly with trailblazers for camping, lodging, food or gas. Trailblazers for camping, lodging, food and gas are to be stacked a maximum of six signs in height on posts in accordance with Department criteria. Trailblazers for attractions are to be stacked a maximum of three signs in height on posts in accordance with Department criteria.

For attraction trailblazers a maximum total of six signs, three on each sign assembly, shall be installed at a given location. Attraction logo trailblazers will be on separate installations from Department TODS unless an agreement between the Trust and the Department provides otherwise. Existing Department TODS immediately become the responsibility of the Logo Signing Trust when functioning as attraction logo trailblazers, and will be subject to the annual fees discussed in Section 2.14.11.2(a).

(e) Local Signing Ordinances. Logo Signs are not advertising signs, but are guide signs designed to facilitate the safe flow of vehicular traffic by providing directions to essential highway motorist services and general attractions. Section 2002(10) of the Administrative Code of 1929, 71 P.S. Section 512(10), bestows on the Department "exclusive authority and jurisdiction over all State designated highways." The Department is accordingly not subject to the mandates of local ordinances with regard to matters such as the type, size and location of signs within the right-of-way of a State highway. Nevertheless, the location of all signs will be established to avoid blocking motorists' lines of sight when entering the highway from side roads and driveways.

(f) Outdoor Advertising Sign Structures. Because Logo Signs are for the purpose of facilitating the safe flow of vehicular traffic, installation is subject to this policy and referenced statutes, regulations, policies, and handbooks. Where, however, a Logo Sign may be properly sited at various points in a given area, the location of any legally existing Outdoor Advertising Sign

Structure, as defined in 67 Pa. Code Chapter 445, relative to the Logo Sign may be considered prior to final location approval. The physical obstruction of visibility of an Outdoor Advertising Sign Structure on the same side of the traveled way and within 500 feet of the proposed location of the Logo Sign will be considered adequate justification for selecting an alternative location within the allowable range.

2.14.10.2 Composition

(a) Single-Exit Interchanges. Specific service signs shall include the name of the type of service followed by the exit number displayed in one line above the logos. Specific service signs may have up to six "gas," "food," "lodging," "camping," or "attraction" logos. Half-size specific service signs (as illustrated in Exhibit 2.14-B) may be used if full-size specific service signs are not necessary. Four-panel size specific service signs should be used if a six-panel specific service sign is not necessary. A 5-year future time frame should be considered to determine the sign size. As noted in Section 2.14.7(b), a single sign may be installed for the Attraction category at urban exits and a single three-service panel sign may be installed at rural exits.

(b) Double-Exit Interchanges. At double-exit interchanges, such as a cloverleaf interchange, specific service signs shall generally consist of two sections, one for each exit. The top section should display the logos for the first exit and the lower section should display the logos for the second exit. The name of the type of service followed by the exit number should be displayed on a line above the logos in each section. The number of logos in each section shall generally be limited to three each for "gas," "food," "lodging," "camping," or "attraction." When a type of motorist service is only at one exit, a full-size or half-size specific service sign may be used as discussed in Section 2.14.10.2(b). As noted in Section 2.14.7(b), a single sign may be installed for the Attraction category at urban exits and a single three-service panel sign may be installed at rural exits.

(c) Remote Rural Interchanges. In areas where only one or two qualified facilities are available for each of two types of services, logos for a maximum of two types of services may be displayed on the same specific service sign. The name of each type of service shall be displayed above its respective logo(s) as indicated in the bottom drawing in Exhibit 2.14-C. Logos should not be combined on a sign when it is anticipated that additional service facilities will become available during the next 5 years. When it becomes necessary to display a third logo for a type of service displayed in combination, the logos involved shall then be displayed in compliance with Section 2.14.10.2(a) and (b).

(d) Ramp Signs. Ramp signs shall conform to the general requirements of Exhibit 2.14-F and Exhibit 2.14-G. A maximum of six logos for gas, food, lodging, camping service, and attractions shall be displayed along the ramp. A maximum of three logos for each of two different types of services may be combined on the same sign panel. The name of each type of service shall be displayed above its logo(s). For services over 0.5 mile from the ramp terminal, ramp signs shall include the distance to the service (to the nearest whole mile) below the directional arrow.

(e) Dual Signing. It will not be permissible to insert wording for a convenience store or a mini-mart on a gasoline logo sign panel. Current policy does not permit such dual signing on either the small or large size gasoline logo sign panel. All gasoline logos will conform to the general requirements of Exhibit 2.14-E. A similar dual combination of signing is not permitted for other services (i.e. attraction/food, lodging/gasoline, lodging/food, etc.).

2.14.10.3 Logo Position Orientation

Logo positions on panels are determined by nearness to interchange in accordance with Section 2.14.11.3(a) and 2.14.11.3(b), beginning at the top left position on the panel and proceeding to the right, then left to right on the second and third rows, ending at the bottom right position. Existing logo orientations which are not oriented in this manner shall be permitted to remain so.

2.14.10.4 Space Allocation by Service Type

Sign space is generally allocated according to demand on a first come, first serve basis. Where the number of service types with eligible businesses fills or exceeds the capacity of the available sign space, the following procedures shall apply. The intent of these procedures is to provide directional information for the greatest variety of motorist service types rather than limit such information to service types with the highest concentration of businesses to the exclusion of other service types.

(a) New Exits in the Program. First priority shall be given to provide sign space in each service type for which there are eligible applicants. Afterwards, sign space shall be allocated proportionately to the number of eligible applicants within each service type. Allowances shall be made for adjustments to this procedure necessitated by limitations imposed by sign design formats, geographical features in the field and sign sequence requirements.

(b) Existing Exits in the Program. Where an application from an eligible business would require that an existing sign of its same service type be expanded to accommodate the additional business, resulting in the exclusion of any other unsigned service type from the program due to a lack of remaining sign space, a re-inventory shall be made of the eligible businesses of the remaining unsigned service types. Any eligible businesses will be contacted and provided an opportunity to apply for signing. After the deadline for applications, the sign space shall be allocated with priority given to any unsigned service type in proportion to the number of eligible applicants in each of the unsigned service types. If no business in an unsigned service types responds to this application opportunity, an existing service sign may be expanded even if it would result in the future exclusion of an unsigned service type.

(c) All Exits. When applications from eligible businesses are received from more service types than can be accommodated, the closest eligible applicant to the approach, as determined in accordance with Section 2.14.11.3 shall be the determining factor for sign space allocation.

2.14.11 Application and Agreements

2.14.11.1 Application

(a) Initial Contacts. If an interchange is approved by the Department for logo signs, businesses in the vicinity of the interchange will be surveyed to determine eligibility. The program and the costs involved will be explained to the eligible businesses by the administering agency.

(b) Logo Agreement. An eligible business that wishes to participate in the program and which can be accommodated will be required to enter into a "Logo Agreement" with the administering agency and pay specified "up-front costs" which will be used to pay the business' share of the total project costs (e.g., the costs of making, providing, and erecting the sign panels, attaching the logos and administering the program). The arrangement will further bind the business to pay an annual fee as discussed in Section 2.14.11.2(a). Subject to policy limitations on bumping, the same participant-entity may enter into two or more separate logo agreements with the administering agency, requiring separate sign fees, in the same or different sign classification (i.e., Food, Gas, Lodging, Attraction), at the same exit, using separate and distinct trade names/logos. Further, participants may share qualifying requirements, such as rest rooms, seating, drinking water, availability, etc. on the same business site/tract of land wherever this can be reasonably accomplished.

2.14.11.2 Annual Fee and Additional Costs

(a) Annual Fees and Compliance Forms. Participating businesses will be assessed an annual fee designed to cover preventative maintenance, the replacement of damaged sign panels, and the continuing administration of the program, and be required to return a completed compliance form satisfactory to the Administrator. The fee will be evaluated periodically by the administering agency to ensure an adequate fund for future projected expenses. Failure to pay the fee and/or return the compliance form within the specified time shall constitute breach of the Logo Agreement and will be cause for removal of the logos, and the assignment of liquidated damages incurred by the Trust because of the breach. At the time the annual fee is assessed, the businesses also shall be required to complete a business eligibility compliance form. Businesses participating in the attraction service category will not be assessed an annual fee for their trailblazer signs, but will be required to pay the full costs of repair and replacement of any attraction trailblazer as costs are incurred.

(b) Temporary Removal. If a business is closed for more than 2 weeks, its logos shall be removed, except for attraction and camping logos with the “SEASONAL” supplemental message. It will be the responsibility of the owner to notify the administering agency to remove the logos at the beginning of a closed period and to reinstall or uncover the logos upon reopening the business. A fee will be charged for temporary removal and installation.

(c) Logos. Businesses shall supply all new and replacement logos, and shall be responsible for the cost of installing replacement logos. All field work for new or replacement logos shall be performed by a Department pre-qualified contractor and authorized by the administering agency.

(d) Refurbishment. When the majority of logo sign panels need to be replaced (assumed to be every 10 to 15 years), additional fees may be assessed to cover the cost of replacing the signs or sign panels.

(e) Attraction Signs. A business that has an existing highway sign (e.g., a supplemental guide sign) may only qualify for a logo sign if they agree to fund all necessary construction costs associated with accommodating their sign.

2.14.11.3 Excess Number of Eligible Businesses

(a) General Rule. When all eligible businesses desiring logo signs cannot be accommodated, the order of opportunity to participate will be as follows:

(1) With respect to food, gas, lodging, and camping logo signs, the closest establishments will be given the first opportunity to participate.

(2) With respect to attraction logo signs, those eligible businesses will be required to submit a traffic study prepared in accordance with guidelines established by the Institute of Transportation Engineers and certified by a professional engineer licensed to do business in Pennsylvania. The businesses with the greatest Annual Average Daily Traffic (AADT) volumes will be given the first opportunity to participate.

(b) Single-Exit Interchanges. When a surplus of eligible gas, food, lodging, camping, or attraction businesses exist at single-exit interchanges, businesses in those categories within 0.5 mile to the right or straight ahead of the exit ramp terminal will be given preference, followed by the businesses within 0.5 mile to the left. (This practice will help to share the available space on the two sides of the interchange and reduce the number of left-turn movements). After all participating businesses within 0.5 mile have been signed, the closest business to the ramp terminal in either direction will be signed.

2.14.11.4 Sale or Termination of Business

(a) Participants may not reassign a Logo Sign Agreement without the prior written consent of the Trust, which consent shall be the Trust's sole discretion. Participants agreements run with the tract of land for which the application was initially made and shall not be assigned to another tract of land, except if a Participant moves its location at the same logo-signed exit, and there is no change in business entity (i.e., no transfer of business ownership), the following shall apply:

(1) If the business still qualified for signing under the Guidelines at its new location, it can remain in the Logo Program, under its existing Participant's Agreement, provided that the Participant executes an Addendum, modifying the location of the business and its signs, and pays the full cost of any removal of existing ramp or trailblazer signs, or the installation of any new ramp or trailblazer signs.

(2) If the Participant does not qualify for participation in the Logo Program at its new location, the Participant shall be removed from the logo program with no refunds, since the relocation was the result of action by the participant alone.

(3) If the business still qualifies under the Guidelines at its new location, it will still be subject to the bumping procedures as outlined in Section 2.14.11.6(b).

(b) Businesses which withdraw from the logo program because of the sale or closing of their business, or for any other reason shall not receive any reimbursement.

(c) If a participating business is sold, and the new owner wants to continue in the logo program, the new owner shall proceed as follows:

(1) If the business is sold for a different use or if the owner withdraws from the logo program, the privilege to participate in the logo program shall be offered to the next qualified business as discussed in Section 2.14.11.3, which may or may not include the new owner, depending on the qualifications set forth. If the participating business is sold for a different use and the new owner wants to qualify for the logo program, then the new owner shall follow the qualification procedures for any new business participant and shall pay the same fees as any new participant in the logo program; or

(2) If a participating business is being sold to a new owner for the same use, and the new owner wants to continue participation in the logo signing program at the same location, the existing participant and the new owner shall

apply jointly for Assignment of the existing Logo Sign Agreement, verifying that the business will continue in the same classification (i.e., gas, food, lodging, campground, or attraction), at the same location; then in such event, the existing Agreement may be assigned for the remainder of the term of such existing Logo Sign Agreement. Such Assignment shall be in a form determined by the administering agency, and subject to the payment of a fee for Assignment as determined from time to time by the administering agency. The application for Assignment by the existing participant and the new owner shall be made not later than the date of closing on the transfer of the participating business or the effective date of transfer of ownership of the participating business, whichever shall first occur, and such request for assignment shall include a verification by the participating business and the new owner of said closing date and such date of transfer of ownership.

2.14.11.5 New Businesses.

If a new business is established or if a non-participating business is interested in participating in the logo program, the business may request to participate in the program subject to the following:

(a) All new businesses will be required to pay the same costs as outlined in Sections 2.14.11.1 and 2.14.11.

(b) If the maximum number of logos is in place, applications will be considered in accordance with the priorities established in Section 2.14.11.3 and the removal provisions of Section 2.14.11.6. Businesses will not, however, be forced to vacate a sign due to another business during their first 5 years in the logo program. If a participating business is forced to vacate a sign panel due to another business, the business will be reimbursed for a depreciated portion of the up-front cost, based on a 10-year straight-line depreciation schedule.

(c) Businesses under construction, or closed businesses planning to reopen under new management, may submit applications for logo signs up to 3 months in advance of the scheduled date of the business opening.

2.14.11.6 Removal of Logo Signs

(a) Removal Necessitated by Department Action. Since the amount of available signing space at interchange areas is limited, the Department reserves the right to remove logo signs and to provide an initial cost reimbursement to participating businesses under certain circumstances. Logo sign removal may prove to be necessary under any of the following circumstances: (1) if the space is needed for necessary traffic control signs; (2) if the access control features of either the mainline or the crossing routes are changed; or (3) for other safety or operational reasons based on an engineering study.

If logo signs are to be removed for any of these reasons, the businesses will be reimbursed by the Logo Signing Trust for a portion of the up-front costs over the first 10 years. Reimbursement will be computed based on straight-line depreciation. The costs of sign removal and sign disposal will be borne by the Department.

(b) Renovations. A business will be given 6 months from the date of closing to complete renovations and reopen for business, provided that the participant maintains its logo sign contract in an active status by paying the annual fees in a timely fashion. The temporary removal provisions of Section 2.14.11.2(b) shall apply during the closed period. If renovations are not completed within 6 months, then the logo is to be removed permanently and the contract voided. If a business is closed for any purpose other than renovations, and no assignment agreement is presented for approval at the time that the business is closed pursuant to guidelines procedures, the contract immediately becomes void, and the logo sign is to be removed.

(c) Removal Caused by an Excess of Eligible Businesses. If the maximum number of logos is in place on a sign panel, new applications by other businesses for inclusion on an existing logo sign will be considered in accordance with the priorities established in Section 2.14.11.3. These priorities are consistent with standard logo signing practice, and they reflect the concept of providing maximum service to the motorist. An excessive number of eligible businesses present at a signed interchange may necessitate the removal of one or more existing participants. This removal will be accomplished according to the following:

(1) Closer Business Bumping Criteria (Gas, Food, Lodging and Camping).

(A) Implementation of "Closer Business Bumping" will be applied in sequential order by type of service beginning with the farthest participating business and proceeding inward toward the closest participating business. At single-exit interchanges, the ranking will be in accordance with Section 2.14.11.3(b) when all businesses of a type are within 0.5 mile of the ramp terminals.

(B) No replacement of a business (bumping) will take place at any interchange for any reason until the furthermost located business on any filled (and already expanded) three-or-six-panel logo sign has been participating in the logo program for a minimum of 5 years. Applications and bumping requests will not be accepted more than 60 days prior to the 5-year anniversary date, and a physical re-inventory will not commence until on or after the 5-year anniversary date.

(C) Participating businesses will be entitled to receive a full 5-year duration of sign use. In no case will a participating business be forced to vacate a logo sign for another business during the first 5 years after installation of their logo.

(D) Bumping will not be authorized where the business wishing to replace another business is already signed for another type of service at the same interchange with the same logo or essentially the same logo.

(E) Specific interchange locations and specific logo signs subject to bumping procedures (i.e., those signs deemed to be already filled to capacity with existing, participating businesses) will be determined by the Logo Signing Trust in coordination with the Department.

(F) No new “bumping” procedures shall be initiated at a specific ramp location until at least 1 year has transpired since the date of administering agency action on the last “bumping” request for the same service at the same ramp location. A “bumping” procedure will be initiated at an exit ramp location only upon written request of an eligible business operating at that exit ramp location, within the mileage distance specified by the Guidelines. Provided, however, one exception will be allowed to this 1-year policy; specifically, a brand-new business (which is located at an exit ramp qualified for the program and which business is opening for business for the first time or was constructed within such 1-year period) may initiate a "bumping" procedure before the 1-year period has transpired.

(2) Greatest AADT Volume Bumping Criteria (Attractions).

(A) Implementation of “Greatest AADT Volume Bumping” will be applied in sequential order beginning with the participating business with the lowest AADT volume and proceeding upward toward the participating business with the largest AADT volume. At single-exit interchanges, the above criteria shall also be applicable.

(B) No replacement of a business (bumping) will take place at any interchange for any reason until the business with the lowest volumes on any filled (and already expanded) three- or-six-panel logo sign has been participating in the logo program for a minimum of 5 years.

(C) Participating businesses will be entitled to receive a full 5-year duration of sign use. In no case will a participating business be forced to vacate a logo sign for another business during the first 5 years after installation of their logo.

(D) Bumping will not be authorized where the business wishing to replace another business is already signed for another type of service at the same interchange with the same logo or essentially the same logo.

(E) Specific interchange locations and specific logo signs subject to bumping procedures (i.e., those signs deemed to be already filled to capacity with existing, participating businesses) will be determined by the Logo Signing Trust in coordination with the Department.

(F) No new “bumping” procedures shall be initiated at a specific ramp location until at least 1 year has transpired since the date of administering agency action on the last “bumping” request for the same ramp location. A “bumping” procedure will be initiated at an exit ramp location only upon written request of a business operating at that exit ramp location, within the mileage distance specified by the Guidelines. Provided, however, one exception will be allowed to this 1-year policy; specifically, a brand-new business (which is located at an exit ramp qualified for the program and which business is opening for business for the first time or was constructed within such 1-year period) may initiate a “bumping” procedure before the 1-year period has transpired.

(3) Implementation Procedure. The following steps will be used when carrying out all Bumping Criteria.

(A) Whenever a business becomes aware, or is otherwise officially notified, that a specific logo sign for food, lodging, gas, camping, or attraction is filled to capacity, the business wishing to apply to replace another participant will contact the Logo Signing Trust and request information pertaining to replacement options and bumping.

(B) The first step to effect possible replacement of an existing business logo by another business will be the submission of a completed application.

(C) The Logo Signing Trust, in cooperation with the Department, will verify all qualifying data on the application and will conduct a complete resurvey of the interchange. All businesses eligible to “bump” will in turn be required to enter into a “Logo Agreement” and pay a specified fee by a specific date.

(D) A determination will be made as to what business must vacate the sign. Schedules will be established to effect as timely a removal and replacement of logo panels as possible.

(E) The business being replaced will be paid a prorated portion of the original cost by the Logo Signing Trust together with a prorated amount of their annual fee if paid. The reimbursement for the "up-front costs" will be the original cost to the business less 10 percent of that cost for each year the business logo sign being replaced has been on the panel. After 10 years, no reimbursement will be made as the life of the sign is considered to have been fully used.

(F) The effective date of logo removal and replacement under the above procedures will be the date resolved by the Trustees. This date will serve as the end of the billing period for establishing annual fee reimbursements.

2.14.11.7 Relocation of Logo Signs, New (Added) Signs

If Department projects or operations involving maintenance, design, utilities, traffic control, drainage or construction necessitate the temporary or permanent relocation of logo signs, the Department will make every effort to relocate the logo signs to an agreed upon location at Department expense. In general, the Department will first determine: (1) if the services still meet applicable guidelines for signing; (2) if the relocation of existing logo signs is possible; and (3) whether new (added) signs or changed signs are needed as a result of changes in routing. Access control, travel distance, existing signing and the route of return to the freeway will be factors in such a determination. The cost of relocating or changing existing logo signs due to Department initiated actions will be paid for entirely by the Department. The cost of installing new (added) logo signs and/or new (added) trailblazers, if determined necessary as per Department signing policy, will continue to be the responsibility of the logo applicant. Agreements and cost arrangements for new (added) signs as per Sections 2.14.11.1 and 2.14.11.2 will apply.

2.14.11.8 Construction of New Interchange or Reconstruction of Existing Interchange

When Department construction occurs which results in an exit that is deemed more operationally efficient than the exit at which a participant is currently participating and located, then the participant shall be required to transfer the location of its logo sign to the more operationally efficient interchange and be given two options: (1) the participant's logo sign will be moved to the more operationally efficient interchange, and the participant's logo signing agreement will be amended to change the description of its location; all other terms and conditions of the agreement would remain the same including the initial date of the agreement (for program longevity purposes); the participant would not be charged for moving their logo sign to the new exit; or (2) the participant could apply as a new applicant/participant at the more operationally efficient interchange under the guidelines; in such an event, the participant agreement would reflect the current date which begins a new 10-year period for amortization purposes; the participant would pay full, current "up-front costs" as defined under Section 2.14.11.1(b); they would have their 10-year "up-front costs" rebated as defined under Section 2.14.11.6(c)(3)(E). Under both alternatives, the participant's current logo sign will be removed.

2.14.11.9 Expansion of Existing Four-Logo Sign Panels to Accommodate Six-Logo Panels

Prior to January 1990, a maximum of four lodging, food, or camping logos were permitted on a single sign installation. The current Department policy now permits a maximum of six lodging, food, camping, or attraction logos for these services (see Exhibit 2.14-B).

In those locations where four participating businesses have filled a four-panel logo sign, and when additional qualifying businesses wish to join the logo program, a site review will be conducted to determine if it is feasible to expand the sign or to relocate it to a position where an expanded six-panel logo sign can be installed.

If there is a demonstrated need and if sign expansion is feasible, the administering agency upon approval will systematically replace four-panel logo sign with six-panel signs as part of future construction projects incorporating at least eight interchanges. The two extra logo spaces on the larger signs will be made available to qualifying businesses as per established guidelines and fees. No separate costs for design or construction of the expansion will be assessed when the work is performed as part of a construction project involving eight interchanges or more.

If a four-panel logo sign is filled and if a single new business wishes to participate in the logo program immediately, the option available is that the business pay the normal program fees or bear the full design and construction costs of the expansion to six panels, whichever is greater. The above option would also apply if two new businesses wish to participate in the program immediately. In such case, normal program fees will be paid by each or the full cost of the expansion will be shared by the two new businesses, whichever is greater.

The expansion of mainline logo signs may be accomplished by expanding the sign back panel either horizontally or vertically, and by relocating or extending the posts. The decision as to how best to add sign area should be made after a thorough review of site conditions and in consideration of existing logo installations, the type, number and spacing of posts, aesthetics, and structural design requirements. Design proposals for logo sign expansion shall be reviewed and approved by the Department before sign design is finalized and before any construction work takes place. Proper mounting height and required breakaway characteristics for logo signs should be followed when mainline logo signs are expanded.

The Department or the Logo Signing Trust will not at their expense reconstruct a six-panel sign to reallocate unused space for the benefit of new business applicants. The determination that a sign is filled to capacity will be made by the Department. Any reallocation of space on an existing logo sign which involves deleting services, separating exits, or moving logos from top to bottom (or vice versa) will be made by the Department. In general, split service signs (those with two types of services displayed) will be considered filled whenever each specific service panel is filled. Reallocation of space on a sign from one service to another will not be allowed unless space is available along the mainline to properly accommodate one or more new signs. All changes to logo signs will be made consistent with applicable state and national standards and will require concurrence from the Federal Highway Administration.

2.14.12 Funding and Responsibilities

2.14.12.1 General

All costs associated with the design, erection, maintenance and administration of logo signs will be assessed of all participating businesses. The program will be administered on a non-profit basis by a trust. All signs will become Department property after erection.

The program is currently administered by the PA Tourism Signing Trust, and information is available at www.palogo.org.

2.14.12.2 Duties of a Logo Signing Trust

If the logo program is administered by a logo signing trust, the trust will be responsible to:

(a) Select an engineering firm to inventory eligible exits to identify potential businesses.

(b) Contact the businesses for promotional purposes.

(c) Establish the fee schedule and enter into an agreement with the businesses on a contractual basis.

(d) Collect fees from the businesses.

(e) Obtain signed compliance forms from applicants to verify business eligibility.

(f) Authorize an engineering firm to develop construction plans for Department and Federal Highway Administration approval.

(g) Coordinate with the Department relative to sign placement and obtain concurrence from Department District Offices upon completion of a construction contract.

(h) Bid and award the construction project.

(i) Inspect and maintain the sign panels.

(j) Report to the Department inquiries and/or complaints which may be received relative to existing logo signing.

(k) Prepare an annual report and submit it to the Department.

(l) Administer the program on a day-to-day basis.

2.14.12.3 Audit

The logo program is administered by a Trust, and a financial audit shall be performed on at least a biennial basis.

2.14.12.4 Annual Report

The Department, in conjunction with the Trust, will prepare an annual report for submission to the House and Senate Transportation Committees within approximately 120 days after each fiscal year, i.e., the 12-month fiscal period used by the program Administrator. The report shall summarize the number of businesses participating in the program, the fees charged for such participation, the methodology used to determine these fee amounts and the program's annual financial statements.

2.14.13 Department Action

2.14.13.1 Department Responsibility

Although the Logo Signing Trust administers the Logo Program, the Department will cooperate with, share file information, and provide expertise to the Trust and to engineering consultants who represent the Trust. The Department, through the Engineering District Offices, will assist the Trust and its design consultants in determining suitable locations for logo signing. The Department will field review logo signing and will accept in writing the completed logo signs at the conclusion of construction. The Department will maintain file copies of plans prepared by the Trust that show the logo sign locations. The Department will be responsible for logo program guidelines and regulations. The Department will conduct Quality Assurance field reviews to inspect logo sign installations.

Due to construction and other activities, the Department may occasionally undertake the resetting of existing logo signs. Logo signs are Department property and the Logo Signing Trust merely administers the program on our behalf to include installation and maintenance. Logo signs are to be accommodated during construction projects similar to accommodations for other guide signs. Note that the design criteria used for sizing the foundations for the breakaway systems and steel posts for existing logo sign panels is often significantly different than that used for a guide sign. Logo signs are designed for future expansion while using the same support foundation design, thus, the design criteria must take future expansion into consideration. The Department should contact the Trust before undertaking design work for resetting logo signs.

2.14.13.2 Applicant Appeals

A business may appeal a denial for logo signing or bumping actions under Title 2, Pa. C.S., Sections 501-508 (relating to the Administrative Agency Law), by submitting a written request for a hearing within 30 days of the date of the denial notification. Businesses should submit appeals to:

Administrative Docket Clerk
Pennsylvania Department of Transportation
400 North Street-9th Floor
Harrisburg, PA 17120-0096

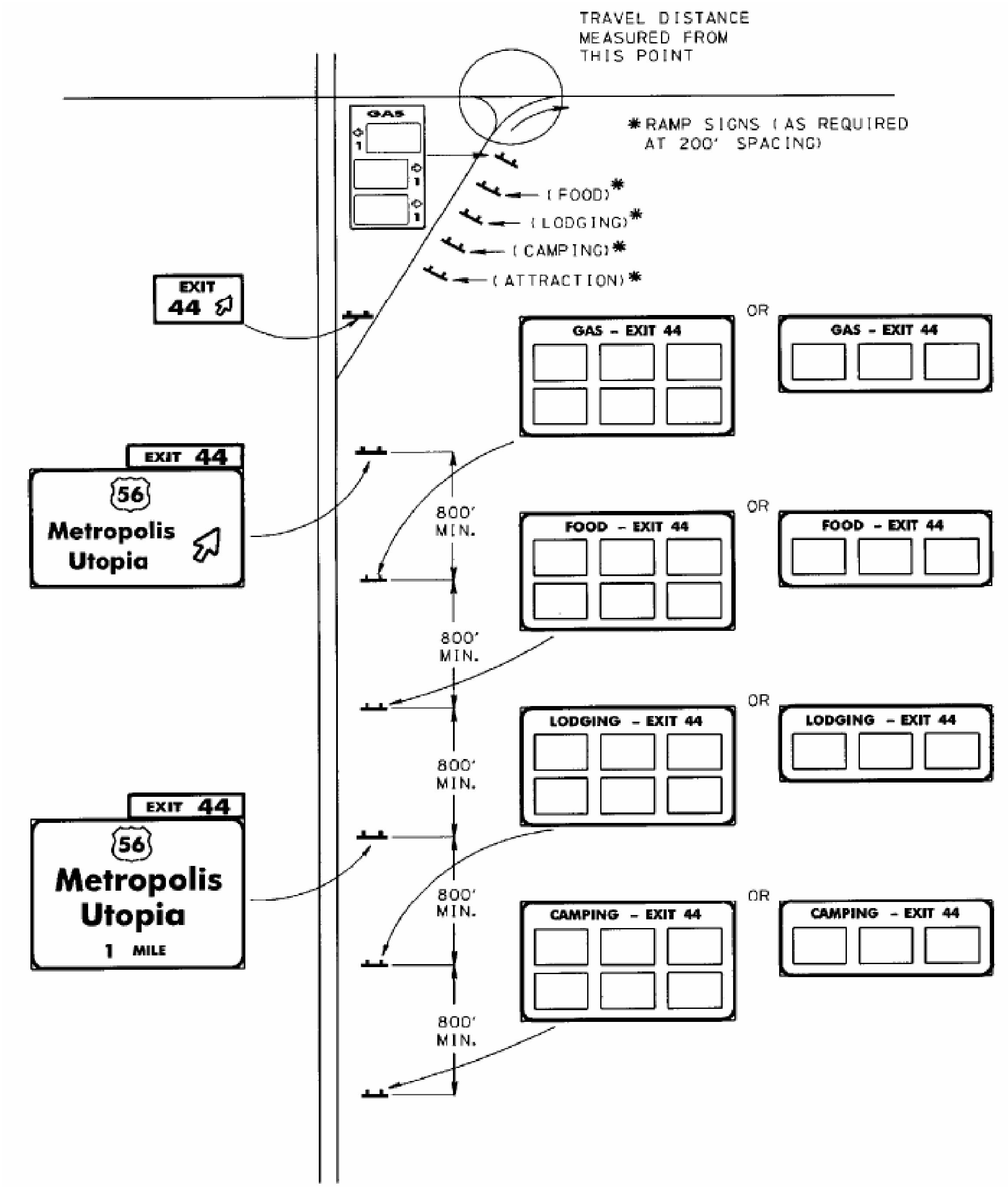
The written request shall include a filing fee made payable to the “Commonwealth of Pennsylvania” and a copy of the denial notification.

At the time of publication, filing fees are listed at 34 Pa.B. 4081 (see http://www.pabulletin.com/secure/data/vol34/34-31/1410.html). Filing fees for appealing a logo decision is a Level II fee, and comes under the category of "motorist information sign matters." Businesses may verify the current fee by contacting the Administrative Docket Clerk at 717-772-2741.

2.14.14 Changes in Program Administration

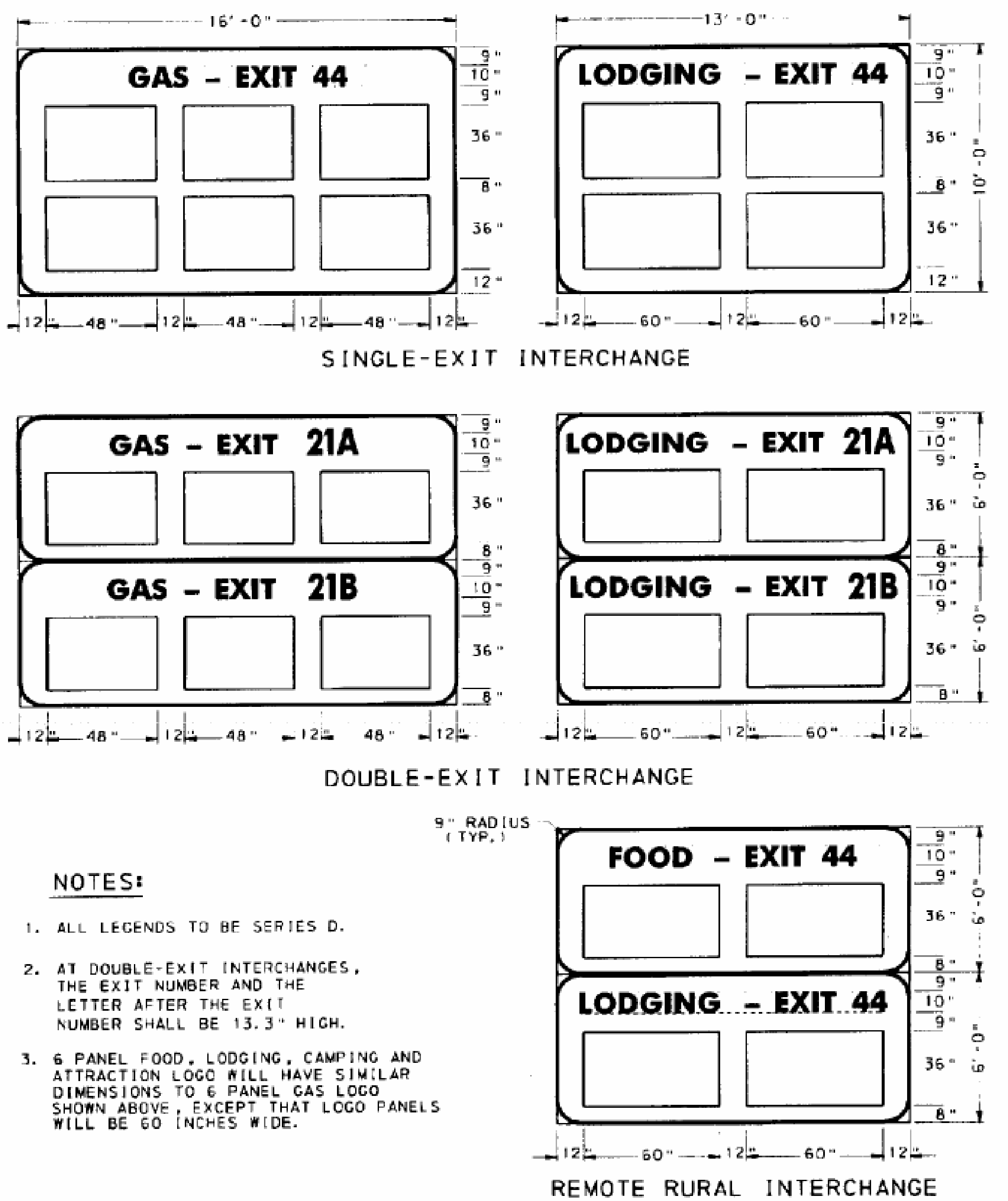
If for any reason the services of the non-profit Trust are terminated, all financial resources and records will become the Department's property for use as an on-going program.

Exhibit 2.14-B Typical Signing for Single-Exit Interchanges



Note: A maximum of four logo sign installations (with a maximum of five specific service types) are permitted at a given interchange approach. In the above figure, an attraction logo may be substituted for any of the other services, provided the appropriate sequencing of signs is maintained. <u>Each panel may be designed to accommodate two, three, four or six logos</u>.

Exhibit 2.14-C Typical Specific Service Signs



NOTES:

1. ALL LEGENDS TO BE SERIES D.
2. AT DOUBLE-EXIT INTERCHANGES, THE EXIT NUMBER AND THE LETTER AFTER THE EXIT NUMBER SHALL BE 13.3" HIGH.
3. 6 PANEL FOOD, LODGING, CAMPING AND ATTRACTION LOGO WILL HAVE SIMILAR DIMENSIONS TO 6 PANEL GAS LOGO SHOWN ABOVE, EXCEPT THAT LOGO PANELS WILL BE 60 INCHES WIDE.

Exhibit 2.14-D Typical Three-Service Sign

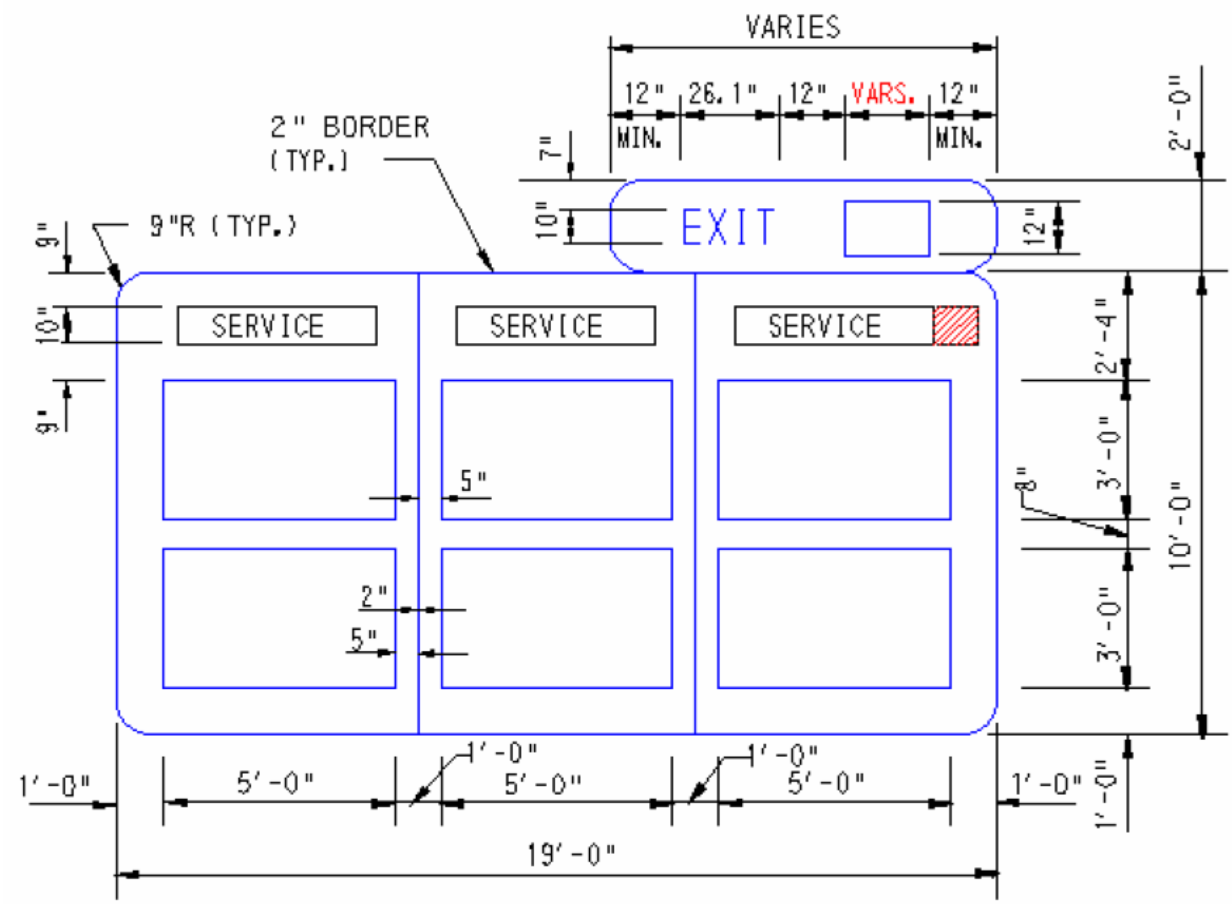
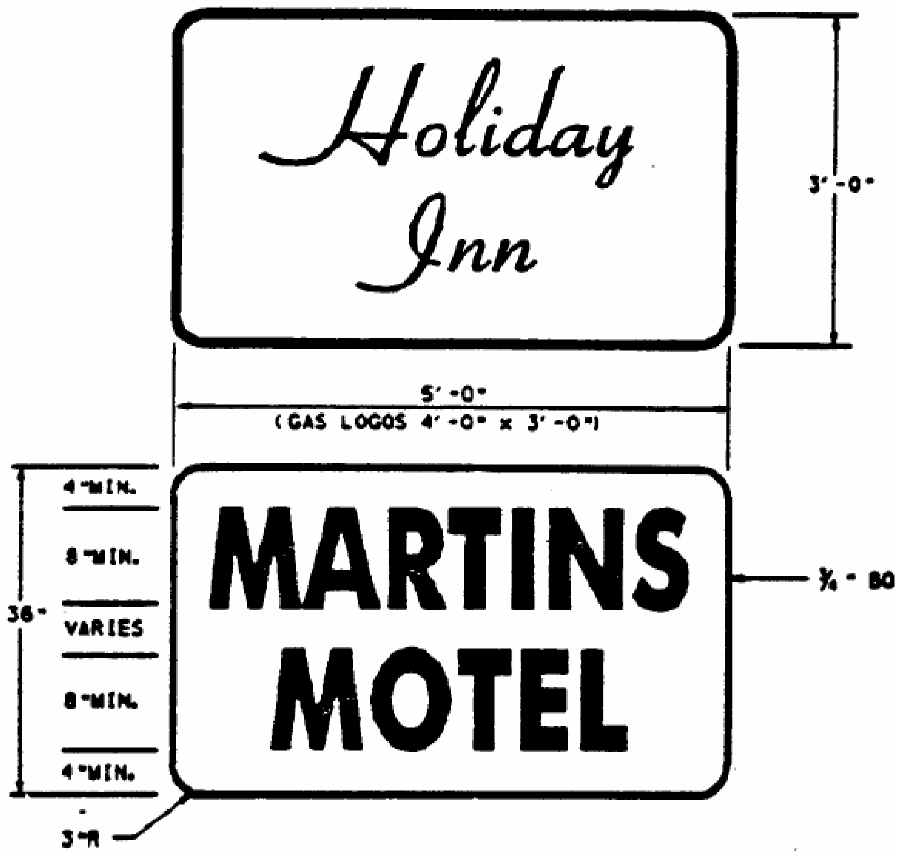
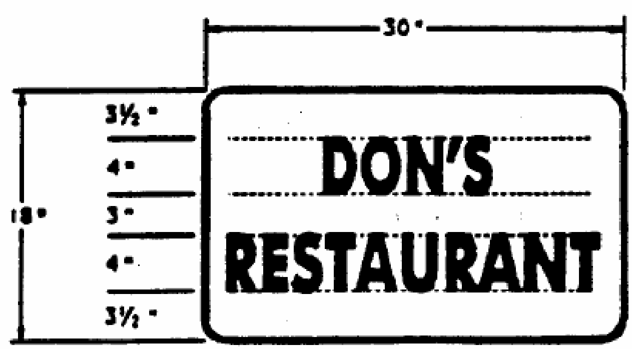


Exhibit 2.14-E Typical Logos

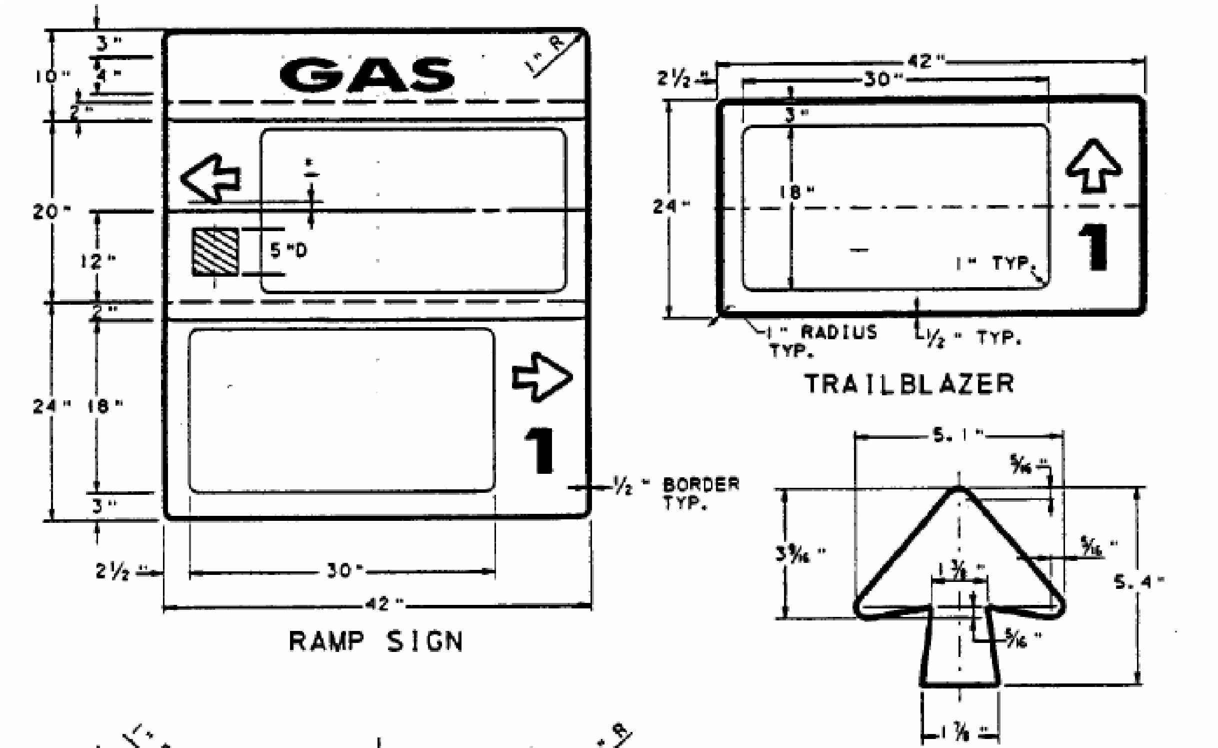


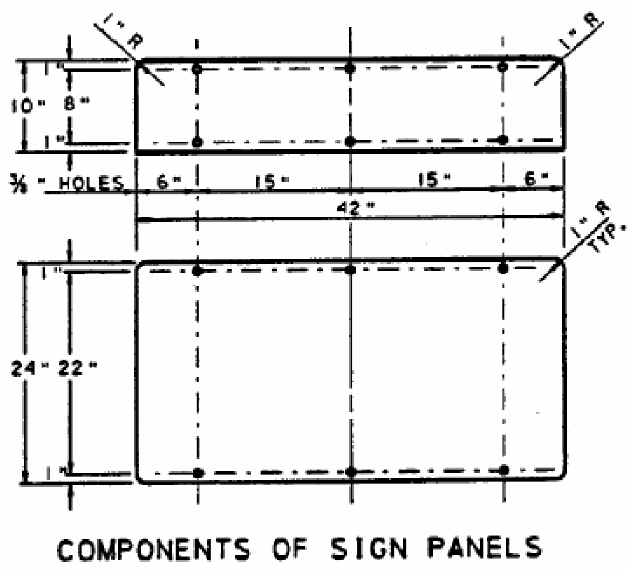
LOGOS ON SPECIFIC SERVICE SIGNS



LOGOS ON RAMP SIGNS AND TRAILBLAZERS

Exhibit 2.14-F Ramp Signs and Trailblazers





NOTES:

1. THE TYPE OF SERVICE SHALL ALWAYS BE INCLUDED ON RAMP SIGNS, BUT NOT ON TRAILBLAZERS.
2. LOGOS WITH DESTINATIONS TO THE LEFT SHOULD BE ABOVE THOSE WITH DESTINATIONS TO THE RIGHT, AND CLOSER SERVICES IN EITHER DIRECTION SHOULD BE ABOVE THE MORE DISTANT SERVICES IN THE SAME DIRECTION.
3. SIGN PANEL COMPONENTS SHALL HAVE A MINIMUM THICKNESS OF 0.100 INCH.
4. RAMP SIGNS SHALL BE MOUNTED ON THE RIGHT SIDE OF RAMP. MOUNTINGS ON LEFT SIDE OF RAMPS SHOULD ONLY OCCUR WHEN EXTENUATING CIRCUMSTANCES PRECLUDE RIGHT SIDE INSTALLATIONS.

Exhibit 2.14-G Ramp Sign Installation (4- and 6-Panel Logos)

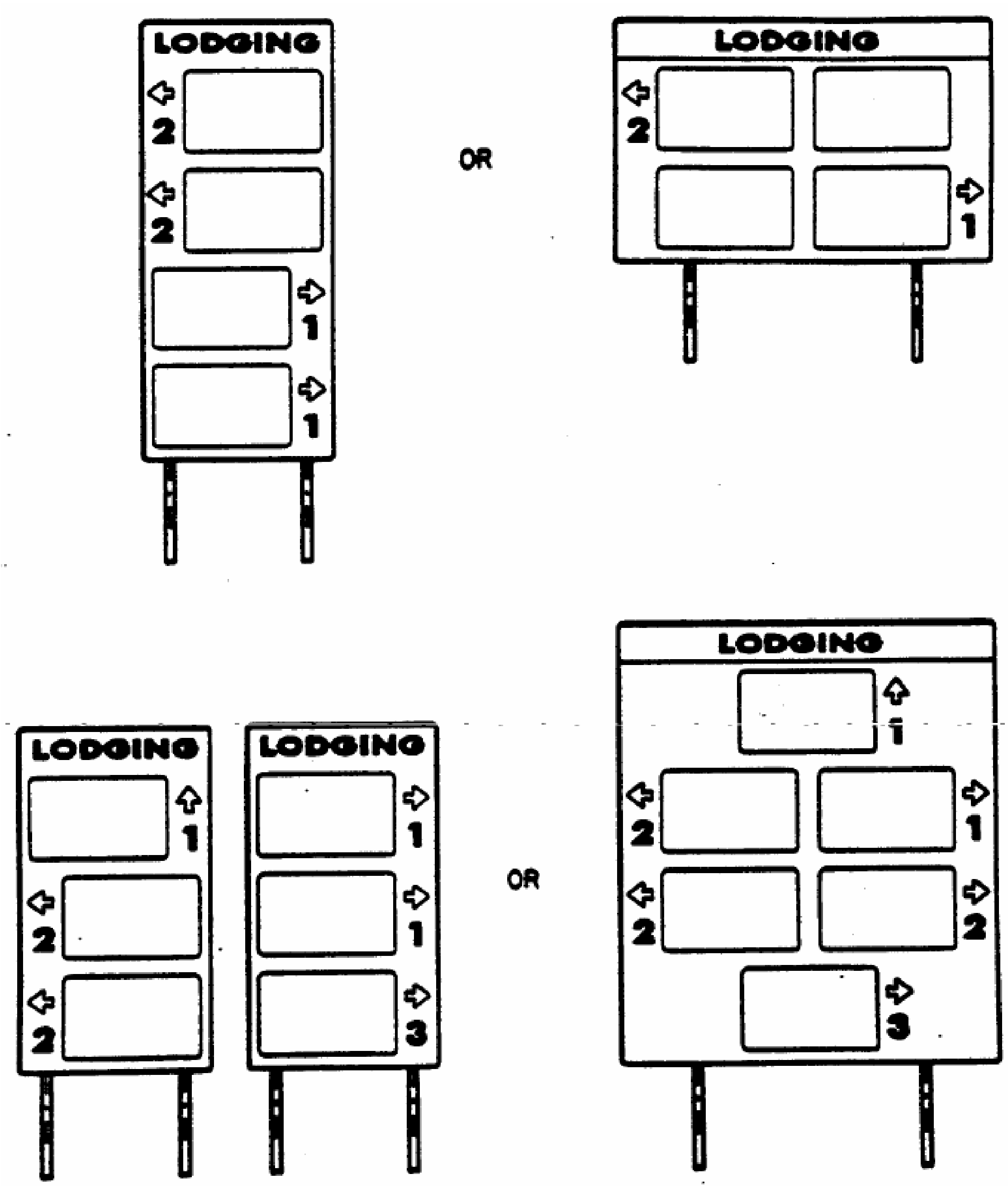
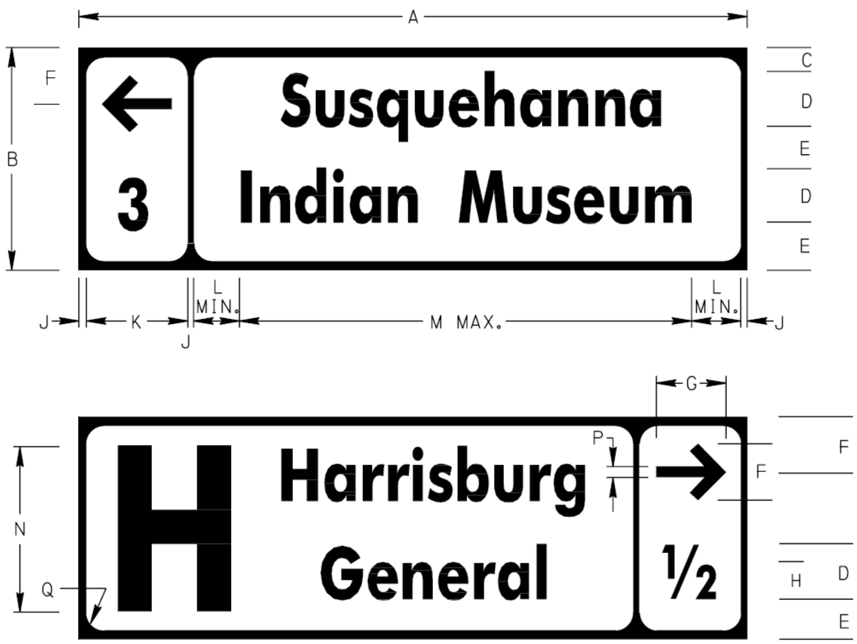


Exhibit 2.14-H Attraction Trailblazer

(a) Justification. The Attraction Sign (D7–4) may be used on conventional highways to direct motorists to large tourist attractions in accordance with the Department's Attraction Signing Guidelines. One or two lines of legend may be used to identify the name or abbreviation of the attraction.

(b) Design. A rectangular directional box should generally be located on the left side of the sign for attractions that are straight ahead or to the left, or on the right side of the sign for attractions to the right. The box should generally include a directional arrow and a distance of 1⁄4, 1⁄2, 3⁄4 or the nearest whole mile, but the box may be eliminated if it is more appropriate to use directional information such as "DRIVEWAY ON LEFT", "LEFT 1000 FEET", etc., on the second line of legend. All legend should be "Clearview 1W, 2W or 3W" font, of the highest series possible. If necessary, the legend may be further condensed up to 35 percent. A generic symbol for hospital, campground or airport may be used in advance of the legend message.



DIMENSIONS – mm (IN)													
SIGN SIZE A x B	C	D	E	F	G	H	J	K	L	M	N	P	Q
1200 x 400 (48" x 16")	50 (2)	100 (4)	75 (3)	100 (4)	125 (5)	65 (2.6)	15 (0.6)	185 (7.4)	50 (2)	870 (34.8)	275 (11)	20 (0.8)	25 (1)
1800 x 600 (72" x 24")	90 (3.6)	150 (6)	105 (4.2)	165 (6.6)	188 (7.5)	100 (4)	20 (0.8)	280 (11.2)	75 (3)	1310 (52.4)	400 (16)	30 (1.2)	45 (1.8)

COLOR:

LEGEND AND BORDER:
WHITE (REFLECTORIZED)

BACKGROUND:
BLUE (REFLECTORIZED)

APPROVED FOR THE SECRETARY OF TRANSPORTATION

By : Glen C Rowe Date : 01–03–06
Chief, Traffic Engineering and Operations Division
Bureau of Highway Safety and Traffic Engineering

2.15 Congestion Management Signs

2.15.1 General

Our highways are becoming more crowded every year and the traditional morning and afternoon "peak hours" in some metropolitan areas have expanded to encompass several hours.

Because of the congestion, highway improvements are necessary to eliminate the bottlenecks. However, since it is not physically possible to build our way out of congestion, traffic restrictions are frequently necessary to enhance highway operations.

This section identifies some techniques to help manage congestion. It is anticipated that the use of these signs and other similar types of signs will become more common in the future.

2.15.2 Park and Ride Signing

One of the major purposes of congestion management signing is to move people and goods, rather than just moving vehicles. Therefore, an important congestion management objective is to increase the number of occupants per vehicle. The following types of signs encourage carpools and vanpools, and are especially valuable in the fringe metropolitan areas:

- Park and Ride Sign (D4-2)
- Car Pool Information Sign (D12-2)

2.15.3 High Occupancy Vehicle Signs

High Occupancy Vehicle (HOV) lanes are sometimes established to give preferential treatment to high occupancy vehicles. These lanes are typically operated as inbound traffic lanes to metropolitan areas in the morning hours and as outbound traffic lanes during the afternoon hours. In addition to the following signs, blank-out or variable message signs, moveable gates and channelizing devices are normally required in order to provide a safe operating system:

- Restricted Lane Ahead Sign (R3-10)
- Preferential Lane Sign (R3-11)
- Buses and Carpools Only Sign (R3-11-1)
- Carpool (____) or More Occupants Sign (R3-11-2)

- Restricted Lane Ends Sign (R3-12)
- Restricted Lane Ahead Sign (R3-13)
- Preferential Lane Sign (R3-14)
- Restricted Lane Ends Sign (R3-15)

2.15.4 Lane Restriction Signs

In order to enhance the capacity of a highway, the following signs may prove useful:

- Trucks Use Right Lane Sign (R4-5)
- Truck Lane (_____) Feet Sign (R4-6)
- Left Lane No Buses Sign (R4-11)
- Left Lane No Trucks Sign (R4-11-1)
- No Trucks Buses Trailers in Left Lane Sign (R4-11-2)

2.15.5 Ramp Metering Signs

Ramp metering signals are in operation on some freeway ramps in the Philadelphia area and at a limited number of locations in other major metropolitan areas. The signing associated with ramp metering is determined on a case-by-case basis.

2.16 Incident Management Signs

2.16.1 General

Incidents are random events that reduce roadway capacity. A breakdown of 58,284 incidents recorded by Engineering District 6-0's Service Patrol from July 18, 2000 through June 26, 2007 is as follows:

Incidents	Percent
Disabled Vehicles	72.4
Abandoned Vehicles	11.7
Crashes	11.5
Debris	1.9
Pedestrians	0.1
Fires	0.2
Miscellaneous	2.2

On freeways, the average incident frequency is normally estimated to be 40 per million vehicle miles of travel. When incidents occur, they frequently set off a vicious chain reaction, causing congestion, driver fatigue, driver impatience, driver frustration, and secondary incidents. Over the past few years, most of the large metropolitan areas have developed an incident management program designed to detect and remove incidents and restore the roadway capacity as quickly and safely as possible.

Electronic surveillance in the form of inductance loop detectors is the primary means of detection on most freeway management projects. Other information sources include the media, cellular telephone, police patrols, maintenance vehicles, service patrols, and fleet dispatchers. Incident verification is increasingly being done by Closed Circuit TV (CCTV), using solid state cameras.

This section identifies some of the useful signs to assist in avoiding primary or secondary incidents; identifying and responding to incidents when they do occur; and keeping the motorists apprised of the situation.

2.16.2 Emergency Detour Signing

A key component of incident management is to establish a clearly defined detour route to assist motorists around a crash, natural disaster, hazardous material spill, or other unplanned incidents. Detour routes need to be instituted in a timely manner to reduce delays, queues and the potential for secondary crashes. Emergency Detour Route signs will facilitate the driver in navigating around an incident providing these signs are consistent, conspicuous, and convey clear concise messages.

Therefore, all District Traffic Units shall have pre-established emergency detour route plans for roadway segments between every interchange on all limited access highways. Emergency detour routes shall be determined with input from but not limited to: PSP, County Emergency Management, local municipalities and emergency responders, and the Pennsylvania Turnpike Commission when applicable. After development and confirmation of the detour routes, the plan should be distributed to the key stakeholders.

Moreover, on a semi-annual basis, each District is to validate the detour plans, confirm roles and responsibilities for activations and deactivations, and exchange contact information. These semi-annual meetings may also serve other purposes such as discussing seasonal maintenance operations.

In order to implement an emergency detour route in a timely manner, Emergency Detour Color trailblazer (D15-1) signs shall be used at the end of off-ramps and in advance of all required turns and important decision points along the emergency detour route. Therefore, in addition to the trailblazers, the Emergency Detour – Follow (Color) Arrow (D14-1) sign should be erected on the off-ramp either as a hinged permanent sign installation, or placed on temporary supports on an as-needed basis.

If permanent folded signs are used, it is recommended that alternate messages be incorporated on the otherwise blank front side of the sign panel to make good use of supports and to avoid the “blank sign” appearance.

To develop statewide consistency in color usage, the following rule should be applied:

Blue………..North Detour
Red………...South Detour
Green………East Detour
Black………West Detour

Recognizing that most emergency detours routes have been established prior to this policy, conformance changes or modification will not be required. The above color pattern should be used for newly established routes, except if conflicts exist, alternate colors may be used.

Emergency Color detour signs may not be practical to implement on all detour routes due to confusion or conflicts with existing signing (e.g., confusion with Pittsburgh’s colored beltway signing). Therefore, the following three types of alternatives may be used in lieu of color detour signs:

1. Additional “TO” routes – the interstate shield or route marker with a TO (M4-5 or M4-5-1) sign can be effective on directing motorists back to the interstate. This method is typically effective on simple parallel detour routes. Care must be taken to ensure that confusion with existing TO signs will not exist.

2. Changeable Message Signs (CMS) and Highway Advisory Radios (HAR) – the use of CMS and HAR may be a primary means of guiding motorist thru an emergency detour route. However, if temporary CMSs are used, a delay in positioning the devices will present a challenge. A dependable and systematic plan for activation and deactivation of temporary and/or permanent signs must exist. Refer to Sections 2.16.6 and 2.16.7 in this chapter for additional information concerning CMS and HAR.

3. Temporary Detour Signs – the timely deployment of temporary signs is a challenge. A dependable and systematic plan for activation and deactivation of signs must exist. Districts may address timeliness by:

   - Establishing a stockpile at a nearby location.
   - Erecting signs in the field in a covered or folded mode.
   - Erecting signs in the field in a meaningless or dormant mode.
   - Any combination of the above.

The above three options can only be used if colored detours cannot be practically implemented. Districts shall document their rational for not using colored detours.

Incidents affecting a regional area, such as snowstorm, may require several interchanges or long sections of highway to be closed. Typically, an emergency regional detour will be communicated to motorists with ITS devices (CMSs, HARs, and Wizards) and notices to media outlets, trucking industry, adjacent states and Transcom. It is not prudent for the Districts to pre-establish detour routes for all logistical combination of regional impacts. However, Districts shall be prepared to spontaneously layout and establish a regional detour. It is also imperative that Districts have an ongoing inventory of ITS devices so devices can be readily deployed. The District Traffic Engineer or his/her designee should provide guidance on the emergency route layout, and County Maintenance shall have a plan for systematic deployment of ITS devices.

2.16.3 Crash Investigation Site Signing

Although it is advantageous to have disabled vehicles moved from the roadway to the shoulder, additional benefits occur if vehicles can be moved to special crash investigation sites that are not visible from the mainline. These sites should accommodate vehicles involved in minor crashes in which vehicles can still be driven. The sites not only help alleviate traffic congestion, but also allow motorists to exchange insurance information in a safer environment.

Special signing to identify the location of these crash investigation sites should be included whenever these sites are available as part of the construction project.

2.16.4 Emergency Cellular Telephone Signs

Cellular telephones have proven to have substantial benefits in establishing a highway incident detection system at minimum costs to highway agencies. Because of their effectiveness, it is desirable to enter into agreements with cellular telephone companies and the Pennsylvania State Police to establish a cellular telephone reporting system with a toll-free number, especially along freeways.

Emergency cellular Telephone Signs (D12-14) may be erected after each interchange when space permits. The end plaque may be used when motorists pass beyond the coverage area and the limited coverage plaque may be used in areas where reception is intermittent.

2.16.5 Fire Hose Access Signs

Fire Hose Access Signs (D17-1) are very useful in helping the fire companies locate the location of hose connections at bridges and access doors through noise barriers. The signs should generally be oriented parallel to the direction of traffic flow.

2.16.6 Changeable Message Signs

Changeable message signs may be effectively used to provide valuable information to motorists regarding incidents, estimated delay time and alternate routing. Even if motorists can not be rerouted, they will be more content if they are made aware of the specifics.

Due to the complex nature of many highway incidents, highway advisory radio can be used to supplement changeable message signs. Alternating messages such as “ACCIDENT AHEAD” and “TUNE 530 AM” can provide almost unlimited information for the motorists.

In order for changeable message signs to be effective, the importance of monitoring traffic conditions cannot be overemphasized.

2.16.7 Highway Advisory Radio Signs

Highway Advisory Radio (HAR) is an alternative to changeable message signs. In order for HAR to be effective, HAR signs should be erected in advance of and at intervals within the range of the transmitted signal. The message broadcast on HAR must be transportation related. FHWA approval is required for the installation of HAR signs on the Interstate.

2.17 Welcome Center and Rest Area Signs

2.17.1 General

Install Welcome Center, Parking Areas and Rest Area Signs in accordance with the standards in Publication 111M (see TC-8701P, TC-8701R, and TC-8701W).

2.17.2 Supplemental Signs

Additional internal signs include the following:

- No Pets (A2-4)
- Pet Area (A2-4-2)
- Pet Area with Arrow (A-2-4-3)
- No Trespassing (A3-1)
- Roadside Rest Rules (A2-6-2)
- Recycling Services (I45-1)

2.18 Procuring Signs and Sign Accessories

2.18.1 County Sign Inventory

Minimum Inventory.

The Department cannot afford the luxury of overstocked supplies, nor can the counties afford to be out of some of the most popular or critical signs. Therefore, a suggested list of some of the most popular and critical signs are included as Exhibit 2.18-A. In most cases, the suggested reorder points for Storage Location 01 includes both a numerical value and a 2-month normal usage – the greater of these two values is the recommended “reorder point.”

Reorder Quantities.

“Reorder quantities” for Storage Location 01 should be similar to, but not greater than the reorder point.

Overstocked Items.

If a county has more than a 2-year supply of any sign or sign accessory, other counties in the district should be contacted to determine if any of the items could be transferred to them. If none of the counties in the District can use the overstocked item, Districts are encouraged to send an administrative message to all terminals, advising of the availability of the overstocked item.

If the excess stock continues, counties are encouraged to return overstocked items to the Sign Shop, except the Sign Shop will also not accept any obsolete items or signs with custom text. Although counties receive financial credit for items transferred to other counties, counties do not receive any direct credit for items returned to the Sign Shop. However, the Sign Shop recalculates the unit price of inventoried items at the Sign Shop, thereby passing the savings on to the next county that purchases one of the signs.

2.18.2 Materials from the Sign Shop

Traffic signs and most sign accessories are generally available from the Sign Shop. However, SR plaques, cutout letter and occasionally some signs will be available thru contract. BHSTE will determine where materials are to be procured and ensure that the necessary settings in Plant Maintenance are made.

Ordering Materials from the Sign Shop.

The Sign Shop provides signs for permanent sign installations, some work zone signs and other ancillary accessories. All materials are to be ordered through the SAP Plant Maintenance system. However, depending on the material, it may be ordered thru the sign inventory functional area as part of a sign work order or it may be necessary to order the material through the procurement area of SAP. Permanent signs that are inventoried as equipment records in Plant Maintenance may be ordered as part of the Sign Inventory Work Order process. Other commodities should be ordered via standard SAP procurement procedures. The Bureau of Office Services should be contacted with specific procurement questions not related to permanent signs ordered as part of a Sign Inventory Work Order.

When SAP Material Numbers exist for signs, preference should always be given to ordering signs by their specific numbers. Signs with custom text, including those with an "(x)" in the description column, have to be ordered using the SAP Plant Maintenance Sign Inventory Work Order Process with the custom information included in the long text (LT) field.

Occasionally, the District will need to include SignCAD drawings with the sign orders, but the county will order the sign in Plant Maintenance. To accomplish this, use the following procedures:

(a) The District creates a SignCAD drawing for the subject sign and places this in their District folder at P:\bhste_shared\SignCAD_Files_for_Sign_Orders\District

(b) The District creates a notification for the subject sign equipment record. In the "Subject" screen area at the bottom of the notification the path to the specific SignCAD file is placed.

(c) The ACMM will subsequently create a Sign Inventory Work Order for the subject sign that will include linking the notification to the work order (using the IW3K transaction). The sign that requires the SignCAD file will be identified as one with an N (non-stocked) in the item category (IC) field. The file path must be manually copied from the "Subject" screen area of the notification to the item's LT field on the component screen of the work order.

(d) The information contained in the LT field functions similar to the spec file in MORIS and is automatically copied into the Purchase Requisition and eventually the Stock Transport Order where it is visible for the Sign Shop.

(e) The Sign Shop will access the required SignCAD drawing in the folder specified for fabrication of the sign. The file will be deleted from the folder upon completion of the order so the District must save the original file in another folder elsewhere if they want it for future use.

(f) Orders that do not contain the necessary/sufficient information to properly fabricate the sign will be canceled. A notice of cancellation with explanation will be generated within SAP.

Some typical examples are shown in Exhibit 2.18-B.

(a) When ordering City Name, Borough Name, Township Name, etc., double check the spelling in the LT field and ensure that spaces in the names are used when, and only when, the name is more than one word.

(b) If a sign does not have its own SAP Material Number, it should be order as a R99 1 or W99 1 sign if it is a standard-size regulatory or warning sign, or otherwise as a D5 sign.

Types of Sign Fabrication.

Signs may use any of the following substrates:

1. Flat sheet aluminum.

2. Flat sheet aluminum with extruded aluminum stiffeners (all Exit Gore signs and D5-series signs larger than 96"x48").

3. Fiberglass (SR paddles only).

Sign messages may be created by any of the following methods:

1. Silk-screened message and border (e.g., DO NOT PASS, Turn, Curve, Crossroad, signs, etc.).

2. Silk-screened background (reverse-screened, e.g., STOP, YIELD, DO NOT ENTER signs).

3. Cut-out legend and border (e.g., used on custom-made signs such as destination signs, where the legend is usually cut by a computerized sign maker).

4. Electronic cuttable film with the message cut from the sheet and applied to a covered blank.

5. Combination of silk-screening and cut-out legend (e.g., used on many Weight Limit signs, most Route Marker signs, etc., where most of the sign message is silk-screened and the numerals are applied by hand).

<u>Identification of Sheeting Materials and Time of Manufacture</u>.

Each sign manufactured at the Sign Shop has a code to identify the year and quarter of the year that the sign was made and the manufacturer and class of reflective sheeting. On silk-screened signs, this information is typically screened near the border on the lower half of the signs. On signs made with cut-out legend, one sticker is generally applied – one in the lower left corner of the sign.

The code for the year is the last two digits of the year, e.g., 07, 08, 09, 10, etc., (only the last digit was used on silk-screened signs prior to 1980 and on custom-made signs prior to 1986). The letters A, B, C and D represent the first, second, third and fourth quarters of the year. The reflective sheeting manufacturer is identified by the following symbols:

Manufacturer	Symbol
Nippon	■
3M	●
Avery Denison	◆

<u>Materials Available From Statewide Contracts</u>.

Legacy Contract No.	Title
9550-10	Posts and Accessories (channel bar posts, steel square posts, wood post sleeves, W-beam posts and breakaway hardware and anti-theft bolts).
9905-09	Delineation Devices (post-mount guide rail delineators, web-mount guide rail delineators, top-mount mount barrier delineators, side-mount barrier delineators, and temporary non-plowable chip seal markers).
9905-11	Work Zone Traffic Control Devices (traffic cones, drums, and temporary pavement marking tape).
9905-13	Traffic Signs (primarily used for cutout letters and SR blanks).

Exhibit 2.18-A Suggested County Sign Safety Stock

Nomen-Clature	Size	Description	Safety Stock Quantity
R1-1	30x30	Stop	6 signs or 3 mo.
R1-2	36x36	Yield	2 signs or 3 mo.
R2-1(35)	24x30	Speed Limit (35)	2 signs or 2 mo.
R2-1(40)	24x30	Speed Limit (40)	3 signs or 2 mo.
R2-1(45)	24x30	Speed Limit (45)	2 signs or 2 mo.
R4-1	24x30	Do Not Pass	2 signs or 2 mo.
R4-7	24x30	Keep Right	3 signs or 2 mo.
R5-1	30x30	Do Not Enter	1 sign or 3 mo.
R5-1	36x36	Do Not Enter	1 sign or 3 mo.
R5-9	36x24	Wrong Way	1 sign or 3 mo.
R6-1(R)	36x12	On Way Right	2 signs or 2 mo.
R6-1(L)	36x12	On Way Left	2 signs or 2 mo.
R8-3A	24x24	No Parking Symbol	1 sign or 2 mo.
R11-2	48x30	Road Closed	1 sign or 2 mo.
R12-1	24x30	Weight Limit 10 Tons	4 mo.
R12-1	24x30	Weight Limit ( ) Tons	4 mo.
R12-1-2	24x12	Bridge	4 mo.
R12-1-1	24x18	( ) Mile Ahead	4 mo.
R12-1-1	24x18	( ) Miles Ahead	4 mo.
R12-4	24x18	Except Combinations ( ) Tons	4 mo.
W1-1R	30x30	Right Turn	2 signs or 2 mo.
W1-1L	30x30	Left Turn	2 signs or 2 mo.
W1-2R	30x30	Right Curve	2 signs or 2 mo.
W1-2L	30x30	Left Curve	2 signs or 2mo.
W1-4R	30x30	Right Reverse Curve	1 sign or 2 mo.
W1-4L	30x30	Left Reverse Curve	1 sign or 2 mo.
W1-6	48x24	Large Arrow (Single)	2 signs or 2 mo.
W1-8	18x24	Chevron Alignment Marker	3 signs or 2 mo.
W1-8	24x30	Chevron Alignment Marker	3 signs or 2 mo.
W2-1	30x30	Cross Road	2 signs or 2 mo.
W2-2	30x30	Side Road	2 signs or 2 mo.
W2-4	30x30	T Symbol	2 signs or 2 mo.
W3-1A	36x36	Stop Ahead	2 signs or 3 mo.
W3-5	36x36	Speed Reduction (40)	1 sign or 2 mo.
W3-10	48x36	No Passing Zone Pennant	2 signs or 2 mo.
W5-2A	30x30	Narrow Bridge-Underpass Sym.	2 signs or 2 mo.

W13-1(20)	18x18	Advisory Speed (20)	1 sign or 2 mo.
W13-1(25)	18x18	Advisory Speed (25)	1 sign or 2 mo.
W13-1(30)	18x18	Advisory Speed (30)	1 sign or 2 mo.
W13-1(35)	18x18	Advisory Speed (35)	1 sign or 2 mo.
W13-1(40)	18x18	Advisory Speed (40)	1 sign or 2 mo.
W16-1	18x18	Hazard Marker	3 signs or 2 mo.
W16-2R	12x36	Right Clearance Marker (B & Y)	5 signs or 2 mo.
W16-2L	12x36	Left Clearance Marker (B & Y)	5 signs or 2 mo.
W21-7	36x36	Work Area Ahead (Plastic)	1 sign or 2 mo.
W21-10	24x24	Stop and Slow Paddle	1 sign or 2 mo.
M2-1	21x15	Junction Marker (B & W)	1 sign or 2 mo.
M3-1	24x12	Card Direct Marker North	2 signs or 2 mo.
M3-2	24x12	Card Direct Marker East	2 signs or 2 mo.
M3-3	24x12	Card Direct Marker South	2 signs or 2 mo.
M3-4	24x12	Card Direct Marker West	2 signs or 2 mo.
M6-1	21x15	Directional Arrow 90 Deg.	2 signs or 2 mo.

***** In addition, stock a 4-month supply of pressure-sensitive legend for these partially finished signs.

Exhibit 2.18-B Ordering Custom Signs

Nomen-Clature	Sign Name	Example
R1-3	(X) WAY	2 WAY
R2-2-1	TRUCKS OVER (X) LBS SPEED (X)	10,000 LBS SPEED 15
R4-6	TRUCK LANE (X) FEET	1500 FEET
R12-1	WEIGHT LIMIT (X) TONS	12 TONS
R12-1-1	(X) MILES AHEAD	3 MILES
R14-16-1	VEHICLES OVER (X) PROHIBITED	10' WIDE
W1-5-1	WINDING ROAD NEXT (X) MILES	5 MILES
W5-1-1	NARROW ROAD NEXT (X) MILES	5 MILES
W12-2	LOW CLEARANCE (X)' (X)"	13' 2"
W99-1	W99-1 SPECIAL WARNING	Curve sign with Unusual Side Road Geometry
D1-1	SINGLE LINE DESTINATION	LA SCRANTON 10 *
D1-2	DOUBLE LINE DESTINATION	SAA COLUMBIA 15 * YORK 12 RA **
D2-1	DOUBLE LINE DISTANCE	YORK 9 HARRISBURG 36 **
D5	72" X 60" SPECIAL DIRECTIONAL	EXIT 13 WITH RIGHT DIAGONAL
D10-4	ENHANCED DISTANCE MARKER	WEST, US 12, MILE 221
I10-1	CITY NAME	PITTSBURGH
I10-5	COUNTY NAME	LACKAWANNA

***** SAA = straight-ahead arrow, LA = left arrow, RA = right arrow, LDA = left diagonal arrow, RDA = right diagonal arrow.

****** Put signs second-line message in second line of the spec file instructions. Similarly, put the third-line message of a three-line sign in third line of the spec file instructions

Chapter 2 Appendix

Proposed Traffic Route Change

__________ County
Proposed Relocation of PA _____

______________________________, Director
Bureau of Maintenance and Operations
ATTN: Chief, Traffic Engineering and Operations Division

District Executive
Engineering District _____

In order to eliminate traffic congestion and improve traffic flow through ________________ _________________________, we are recommend the partial elimination and the partial relocation of PA ______. This action would move PA _________ from a predominantly commercial area.

This change would take effect immediately upon securing the necessary approvals and delivery and installation of the necessary signing.

	From		To	
Co. / SR	**Segment / Offset**	**Intersection**	**Segment / Offset**	**Intersection**
		Partial Elimination of _________		
48 / 0412	0020 / 0032	SR 3002	0070 / 0016	SR 3008
		Partial Relocation of _________		
48 / 3002	0100 / 0012	SR 0412	0140 / 0492	SR 3008
		Northbound		
48 / 3008	0030 / 0082	SR 3002	0030 / 0497	SR 1001
48 / 1001	0100 / 0025	SR 3008	0230 / 0063	SR 2010
		Southbound		
48 / 1001	0231 / 0065	SR 2010	0101 / 0028	SR 3008
48 / 3008	0031 / 0499	SR 1001	0101 / 0025	SR 3002

Also enclose the following: (1) a listing of the physical limitations of the roadways for the proposed partial relocations; and (2) either one reproducible map graphically showing the location of the proposed partial elimination or partial relocation, or nine color-coded originals.

Enclosures

Physical Limitations of the Proposed Traffic Route

Date ________________

County ______________________

Traffic Route No. ______

ESTABLISHMENT ______ EXTENSION ______ RELOCATION______

SR / segment →				
ROAD WIDTH				
Roadway				
Berm				
Overall				
BASE				
Depth				
Type				
SURFACE				
Type				
Designation				
BRIDGE RESTRICTIONS				
Width (min.)				
Load (min.)				
OVERPASS				
Clearance				
CURVES				
No. over 6-degree				
Maximum				
GRADES				
No. over 6%				
Maximum length				

Does this route

(a) Save Mileage Distance? Yes ___ No ___ If yes, how much mileage? ________
(b) Re-route a present route? Yes ___ No ___

Is the complete route over State Highways? Yes ___ No ___ If no, an agreement(s) must be executed and enclosed.

Is the proposed route properly marked with signs and pavement markings? Yes ___ No ___

Submitted by ______________________________ Date ________________
(District Traffic Engineer)

Approved by _______________________________ Date _________________
(District Executive)

Municipal Agreement re Traffic Route Change

THIS AGREEMENT, made this ________ day of ____________, 20___, between the Commonwealth of Pennsylvania, acting through the Department of Transportation, hereinafter called the COMMONWEALTH,

and

____,

hereinafter called the MUNICIPALITY.

WITNESSETH:

WHEREAS, the parties desire to designate certain highways under the jurisdiction of the MUNICIPALITY as traffic routes; and,

WHEREAS, the parties desire to have such traffic routes established and marked in accordance with standards, policies and regulations of the COMMONWEALTH's Department of Transportation; and,

WHEREAS, the parties desire to enter into this Agreement for the purpose of outlining their respective obligations.

NOW, THEREFORE, the parties agree to the following:

1. Following highways under the jurisdiction of the MUNICIPALITY are hereby established as traffic routes.

Via	From Intersection	To Intersection	Direction	Traffic Route No.

These routes are more specifically shown on a map which is attached as Exhibit A and made a part of this Agreement. The establishment and maintenance of these traffic routes shall be subject to the conditions set forth below.

2. The MUNICIPALITY shall, at its cost, be responsible for policing and maintaining the roadway and provide for snow removal.

3. The COMMONWEALTH shall provide and maintain all traffic route markers. The MUNICIPALITY shall, at its cost, be responsible for providing and maintaining all other traffic-control devices including pavement markings.

4. The COMMONWEALTH's Secretary of Transportation shall have the right, in his/her sole discretion, to discontinue any or all of the above highways under the jurisdiction of the MUNICIPALITY as traffic routes. In the event that any such determination is made, the COMMONWEALTH shall remove all signs and markings indicating the highway to be a traffic route.

5. The MUNICIPALITY shall indemnify, save harmless and defend COMMONWEALTH, the Department of Transportation, their officers, agents and employees, from all suits, actions or claims of any character, name or description, brought for or on account of any injury, death or property damage as a result of the design, construction and/or maintenance of the above-noted highways, whether the same be due to the use of

defective materials, defective workmanship, neglect in the MUNICIPALITY and/or the MUNICIPALITY's contractor, their officers, agents and employees during the performance of any work on the above-noted highways under the jurisdiction of the MUNICIPALITY.

6. If the MUNICIPALITY shall fail to perform any of its obligations under this Agreement, and if the COMMONWEALTH desires to continue any or all of the above-noted highways as traffic routes, and the MUNICIPALITY shall fail to cure any defaults and performance within thirty (30) days of notice from the COMMONWEALTH, the MUNICIPALITY authorizes the COMMONWEALTH to withhold so much of the MUNICIPALITY’s Liquid Fuels Tax allocation as may be necessary to complete necessary work and/or to reimburse the COMMONWEALTH in full for all costs incurred, and does hereby and herewith authorize the COMMONWEALTH to withhold such amount and to apply such funds, or portions thereof, to remedy such default.

7. The MUNICIPALITY agrees to be bound by the Act of May 20, 1937, (P.L. 728, No. 193), as amended by Act of October 5, 1978, (No. 260), which provides, in substance, that the Board of Claims shall have jurisdiction of claims against the COMMONWEALTH arising from contracts and the power to order the interpleader or impleader to other parties when necessary for a complete determination of any claim or counterclaim in which the COMMONWEALTH is a party.

IN WITNESS WHEREOF, the parties have executed this Agreement on the date first above written.

ATTEST

COMMONWEALTH OF PENNSYLVANIA
DEPARTMENT OF TRANSPORTATION

______________________________ BY _________________________________
Deputy Secretary of Transportation

(SEAL)

ATTEST:

____________________________ ___________________________________
Title

(SEAL)

APPROVED AS TO LEGALITY AND FORM

BY ____________________________
Chief Counsel

BY ____________________________
Deputy Attorney General

* Resolution designating signature authority attached.

Resolution Designating Signature Authority

BE IT RESOLVED, by authority of the______________________________________
(Name of governing body)

_______________________________________ of the ______________________________,
(Name of Municipality)

_________________________ County, and it is hereby resolved by authority of the same,

that the ___________________________ of said Municipality Authority be authorized and
(designate official title)

directed to sign the attached Agreement on its behalf and that the ______________________
(designate official title)

be authorized and directed to attest the same.

ATTEST ____
(Name of Municipality)

_________________________________ By:__________________________________
(Signature and designation of official title) (Signature and designation of official title)

(SEAL)

I, ________________________________, ___________________________________
(Name) (Official Title)

of the ____, do hereby certify that the
(Name of governing body and municipality)

foregoing is a true and correct copy of the Resolution adopted at a regular meeting of the

____________________________________, held the_______ day of _____________, 20___.
(Name of governing body)

DATE _____________________ ____
(Signature and designation of official title

pennsylvania
DEPARTMENT OF TRANSPORTATION

Publication 46
Chapter 3

Markings

Pennsylvania Department of Transportation
Chapter 3 of Publication 46
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List of Exhibits

Appendix

3.1 General

3.1.1 Purpose

Markings are a common and expected component of the highway system. The primary purpose of markings is to provide the driver clear visual information to operate his vehicle in a variety of situations. Markings define the intended travel path during both daylight and nighttime hours, and during various weather conditions. Common types include the following:

- Pavement markings (includes lines, words, symbols, pavement markers, and curb markings).
- Object markings.
- Delineators.
- Barricades, channelizing devices, and islands.

Markings also perform a function in the overall traffic-control system. In general, markings supplement other traffic control devices and provide guidance and warning. However, in accordance with §3307(b) of the Vehicle Code, they also perform a regulatory function because no-passing zone pavement markings are required in addition to signs to enforce no-passing zones.

As with all traffic-control devices, markings should be readily recognized and understood.

3.1.2 Laws, Regulations and Other Publications

Americans with Disabilities Act and Architectural Barriers Act Accessibility Guidelines, United States Access Board, July 23, 2004. Available online at http://www.access-board.gov/ada-aba/final.pdf.

Approved Construction Materials – Bulletin 15 (Publication 35). A listing of PennDOT approved materials and manufacturers.

Guide to Roundabouts (Publication 414). This PennDOT publication is designed as a supplement to FHWA's *Roundabouts: An Informational Guide* to supplement existing design policies and procedures. The publication is available online at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20414.pdf

Manual on Uniform Traffic Control Devices (MUTCD). A manual adopted by the Federal Highway Administration and which establishes national guidelines for traffic-control devices, including signs. The *MUTCD* is available through the

Superintendent of Documents, U.S. Government Printing Office, Washington D.C. Specifically, Part 3 addresses pavement markings, and is available at http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/ch3.pdf.

Official Traffic-Control Devices (Publication 212). Regulations published as Chapter 212 of Title 67 of the PA Code. This regulation adopts the Federal Highway Administration's Manual on Uniform Traffic Control Devices (*MUTCD*) and establishes additional study requirements, warrants, principles and guidelines not included in the *MUTCD*. Specifically, §212.202 establishes extra warrants for no-passing zones that are not included in the *MUTCD*.

Pavement Marking Handbook. This PennDOT manual provides detailed guidance for Department work force in the day-to-day operation of the Department's pavement marking program, including the operation of the truck-mounted and small paint machines.

Pennsylvania Drivers Manual. This manual provides guidance for drivers, including the purpose and meaning of the various types of markings, available at http://www.dot10.state.pa.us/pdotforms/pa_forms_manuals/padriversman.pdf.

Roundabouts: An Informational Guide, Pub. No. FHWA RD 00 067, June 2000 (FHWA) – available online at http://www.tfhrc.gov/safety/00-0671.pdf.

Specifications (Publication 408). Specifications referenced for all Department construction projects, which includes requirements for the installation of signs and sign accessories. Specifically, Sections 961, 962, 963, 964 and 965 address pavement marking materials and delineation devices. Pub. 408 is also available at http://www.dot.state.pa.us/Internet/Bureaus/pdDesign.nsf/ConstructionSpecs408and7?OpenForm

Traffic Control – Pavement Markings and Signing Standards (Publication 111M). This publication includes PennDOT's 8600 and 8700-series standards. Specifically, the 8600-series deal with markings: the types, dimensions, and locations of pavement markings (TC-8600), snowplowable raised pavement markers (TC-8602), and delineators (TC-8604). The publication is available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20111M.pdf

Vehicle Code (75 Pa. C.S.). The Pennsylvania Vehicle Code is law that typically defines actions required by drivers and the Department. It discusses markings in several sections, but of specific importance is the requirement in §3307(b) that both signs and markings are required to enforce no-passing zones.

Work Zone Traffic Control (Publication 213). This publication shows drawings for temporary traffic control, including the use of temporary pavement markings.

3.1.3 Definitions

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

Delineator – A retroreflective device mounted on the road surface or at the side of the roadway in a series to indicate the alignment of the roadway, especially at night or in adverse weather.

Divided highway – A highway divided into two or more roadways and so constructed as to impede vehicular traffic between the roadways by providing an intervening space, physical barrier or clearly indicated dividing section.

85th percentile speed – The speed on a roadway at or below which 85 percent of the motor vehicles travel.

Expressway – A divided arterial highway for through traffic with partial control of access and generally with grade separations at major intersections.

Freeway – A limited access highway to which the only means of ingress and egress is by interchange ramps.

Narrow bridge or underpass – A bridge, culvert or underpass with a two-way roadway clearance width of 16 to 18 feet, or any bridge, culvert or underpass having a roadway clearance less than the width of the approach travel lanes.

School zone – A portion of a highway that at least partially abuts a school property or extends beyond the school property line that is used by students to walk to or from school or to or from a school bus pick-up or drop-off location at a school.

3.2 Pavement Markings

Pavement markings are the only traffic control device that is visible without the driver taking his or her eyes off the roadway. Since pavement markings are on the physical road surface, they are superior to all other markings in showing the intended travel path. Quality pavement markings also make drivers feel more comfortable and reduce their stress level.

Pavement markings are the only traffic control device that is visible to drivers without taking their eyes off the roadway.

The Department's Pavement Marking Handbook is the primary document for Department personnel that are directly involved with the operation of the Truck-Mounted Paint Machine Program and the Small Paint Machine Program. The handbook gives detailed guidance on how to run the Department's in-house program, including but not limited to the following areas:

- Personnel and organization.
- Annual work program.
- Pavement marking standards.
- Procurement, storage and handling of pavement marking materials.
- Truck-mounted paint machine program, including the application of waterborne paint, the quality assurance program, and health and safety issues.
- Small paint machine program.
- Traffic control and management systems.

Since detailed information for the above topics is in the Pavement Marking Handbook, see the handbook.

3.2.1 Pavement Marking Materials

This section provides a brief description of the various marking materials that are approved and available for use in Pennsylvania. The Materials and Testing Division conduct the testing of these materials on the Department's test deck and in the laboratory, with the assistance of the Traffic Engineering and Operations Division.

The only materials that can be sold for or used on any public highways in Pennsylvania are those materials listed in *Bulletin 15 (Publication 35)*. Exhibit 3.2-A lists materials used in Pennsylvania and appropriate cross-references.

Exhibit 3.2-A Pavement Marking Materials

Material	Section in Publication 408	Section in Publication 35
Hot thermoplastic	960	960
Cold plastic	961	961
Waterborne paint	962	962
Epoxy markings	964	964
Preformed thermoplastic	965	965
Snowplowable RPMs	966	966
Glass beads for traffic paint	1103.14	1103.14
Methyl methacrylate	*	Miscellaneous: Traffic Division
Polyurea	*	
Polyester	*	
Wet reflective tape	*	

* Specifications are on file with the Traffic Engineering and Operations Division

Pavement marking materials are further described as follows:

1. Waterborne Traffic Paint. Waterborne Traffic Paints (white and yellow) is the most common type of pavement marking used in Pennsylvania. This paint is pre-heated prior to application, applied by truck-mounted paint equipment for center, lane and edge lines, and normally dries in 90 seconds or less.

2. Glass Beads. Since there are various marking materials approved for use in the state, the Department has developed specifications for several types of beads for use with these materials. To view the gradation of the following bead types, go to Publication 408, Section 1103.14.

    - Type A Beads. These are the standard beads specified in AASHTO M-247, Type I, Moisture Resistant. Type A Beads can be used with all marking materials.

    - Type B Beads. This gradation is much larger than the standard beads (Type A), and was developed for use with durable materials to provide better wet-night visibility of the pavement markings. Commercially available trade names include Visibeads, Big Beads, Megalux, etc.

    - Type C Beads. These beads are another large gradation similar to the Type B Beads, but are used primarily with waterborne and polyester pavement marking materials for improved wet-night visibility.

    - Type D Beads. This gradation is only slightly different from the standard beads (Type A) with a greater percentage of beads in the bigger sieve sizes. Type D Beads may be used with waterborne paints as an alternate to Type A Beads.

3. Hot Thermoplastic. Hot thermoplastic material is generally a synthetic resin that softens when heated and hardens when cooled without changing the properties of the material. Although the material contains glass beads, an additional coating of beads is applied after installation to provide initial reflectivity. Hot thermoplastic material can be applied by screed, ribbon or spray methods. A sealer should be used on concrete and old bituminous roadways immediately prior to application of the thermoplastic. Existing marking materials should be removed for proper adherence to the roadway.

4. Cold Plastic Tape and Legends. This material is pre-formed and is white or yellow in color, has pressure-sensitive adhesive on its back surface, and is capable of being applied to bituminous and/or concrete roadway surfaces. These materials can be surface-applied or inlaid. If surface applied, a sealer will generally be required. Purchase materials via the Pavement Marking Legend Contract (No. 9905-04).

   This material is further classified into the following two categories, and each has its own specifications:

   - Cold plastic pavement marker or legend. This has an adhesive back and is 60 mils or 90 mils thick. This material is usually considered to be of the permanent type.

   - Temporary pavement marking tape. This has an adhesive back and is from 20 to 50 mils thick. Temporary tape comes as:

     1. Non-removable.

     2. Removable, Type I.

     3. Removable, Type II.

     The discerning difference between removable Type I and Type II tape is that Type II tape does not leave a discernible mark on the highway 30 days after its removal.

5. Epoxy. This material is a two-part, 100 percent solids system, which contains resins and a catalyst, and can be hot-spray applied after mixing. It is a very durable material that can be applied to both bituminous and concrete pavement after the roadway surface is thoroughly cleaned of old pavement markings or curing compound.

6. Preformed Thermoplastic. This material is a preformed polymer thermoplastic (white or yellow) with glass beads uniformly distributed throughout its cross section, and used on bituminous and concrete pavements. This material is surface applied, and usually fused in place by a propane torch. Purchase materials via the Pavement Marking Legend Contract (No. 9905-04).

7. Methyl Methacrylate. This material is a two-part compound which can be extruded, sprayed or manually applied to either concrete or bituminous pavements. It contains both glass beads and anti-skid aggregate. Existing marking materials should be removed for proper adhesion.

8. Polyurea. Polyurea-based liquid pavement marking material consists of a two-component, 100 percent solid, thermosetting material. Limited experience indicates that it is a durable pavement marking material that dries to no-track in 3 to 8 minutes at all temperatures down to about 40 degrees.

9. Polyester. Polyester is a two-component system (resin and catalyst) which is applied separately on the roadway and cures to a durable marking. It can be applied to both concrete and bituminous pavements but has not performed well on concrete. It should not be applied to new bituminous surfaces until the surface oils have disappeared (approximately 2 weeks). It can be applied over old markings.

10. Wet Reflective Striping Tape. This material is a highly reflective tape under both dry and wet roadway conditions, even when fully submersed in water. There are two types of wet reflective striping tape – “temporary” for work zones and “permanent” for normal applications.

11. Pavement Markers. Pavement markers are used as positioning guides in conjunction with other longitudinal markings without conveying information to the motorist as to passing or lane use restrictions. Their purpose is to provide delineation during darkness and/or adverse weather conditions that may render other pavement markings ineffective. These markers are available as either one-way or two-way markers depending on the number of reflective faces. In addition, there are two basic models:

    - Temporary raised pavement markers (TRPMs).

    - Snowplowable raised pavement markers (RPMs).

    There are also recessed markers, but these are not for new installations.

Waterborne paint with glass beads is the most popular pavement markings used in United States. This is the least expensive of the materials used today, but unfortunately, despite improvements in durability, it is generally the least durable of all pavement-marking materials.

Some materials are more durable than other materials, but not all materials are compatible with each other or with some road surfaces. Exhibit 3.2-B shows material compatibility between existing and new striping materials.

Exhibit 3.2-B Material Compatibility for Restriping

Original Material	New Material					
	WB Paint	**Thermo**	**Tape**	**Epoxy**	**MMA**	**Polyurea**
WB Paint	Y	Y	N	Y*	N	N
Thermo	Y	Y	N	N	N	N
Tape	N	N	N	N	N	N
Epoxy	Y	Y	N	Y**	-	-
MMA	Y	Y	N	-	Y	-
Polyurea	Y	Y	N	-	-	Y

* Epoxy may be applied over existing waterborne paint if the waterborne paint was applied at a wet-film thickness of 9 mils or less, or if the existing waterborne paint has a durability rating of "5" or less.

** Only one restripe with epoxy.

Exhibits 3.2-C and 3.2-D show recommended pavement marking materials based on national research and internal experience. An analysis of the two tables indicates that:

1. Thermoplastic is the preferred durable material on bituminous pavements.
2. Epoxy is the preferred durable material on Portland cement pavements.
3. Waterborne Paint is the most cost-effective material.

Exhibit 3.2-C Recommended Materials for Bituminous Pavements

Traffic Characteristics	Pavement Remaining Service Life*		
	0 – 2 years	**2 – 4 years**	**> 4 years**
ADT < 1,000	***Thermo, Epoxy,*** WB Paint	***Thermo, Epoxy,*** WB Paint	***Thermo***, WB Paint, Epoxy, Polyurea, MMA
1,000 < ADT < 10,000	***Thermo, Epoxy,*** WB Paint	***Thermo***, Epoxy, Polyurea, MMA	***Thermo***, Tape, Epoxy, Polyurea, MMA
ADT > 10,000	***Thermo***, Epoxy	***Thermo***, Tape, Epoxy, Polyurea, MMA	***Tape, Thermo***, Epoxy, Polyurea, MMA
Heavy Weaving or Turning	***Thermo***, Epoxy	***Thermo***, Epoxy, Polyurea, MMA	***Thermo***, Epoxy, Polyurea, MMA

***** The highest-performing material(s) are in bold text, but it is not the most cost-effective material.

Exhibit 3.2-D Recommended Materials for Portland Cement Pavements

Traffic Characteristics	Pavement Remaining Service Life*		
	0 – 2 years	**2 – 4 years**	**> 4 years**
ADT < 10,000	***Epoxy***, Thermo, WB Paint	***Epoxy***, Thermo, WB Paint, Polyurea, MMA	***Epoxy***, Thermo, Polyurea, WB Paint, MMA
10,000 < ADT < 50,000	***Epoxy***, Thermo, WB Paint, Polyurea	***Epoxy***, Thermo, Tape, Polyurea, WB Paint, MMA	***Epoxy***, Thermo, Polyurea, MMA
ADT > 50,000	***Epoxy***, Thermo	***Epoxy***, Thermo, Tape, Polyurea, MMA	***Tape***, Thermo, Polyurea, Epoxy, MMA
Heavy Weaving or Turning	***Epoxy***, Thermo, Polyurea	***Epoxy***, Thermo, Tape, Polyurea, MMA	***Epoxy***, Thermo, Tape, Polyurea, MMA

***** The highest-performing material is in bold text, but it is not the most cost-effective material.

Black Contrast Markings

White epoxy pavement markings are not as white as most other white materials, and because of this, they are frequently difficult to see on Portland cement roadways during some daylight conditions. Therefore, a recommended procedure is to apply black epoxy, flooded with black aggregate to make the white markings stand out. For skip lines, crews can apply a black line immediately after the white skip lines, using the same dimensions for both colors. Another method used in some states, is to place an 8-inch wide line as the background material and then apply the white marking over the black marking.

Black epoxy should satisfy color chip 37038 of Federal Standard 595B, and have similar quality as the white epoxy pavement markings.

3.2.2 Minimum Retroreflectivity

In 1993, Congress directed the U.S. Secretary of Transportation to revise the *MUTCD* to include minimum levels of retroreflectivity for traffic signs and pavement markings. Since then, FHWA has sponsored extensive research on the retroreflectivity needs of drivers for both signs and pavement markings, but unlike traffic signs, FHWA has not adopted or even officially proposed minimum values for pavement markings.

Recent research indicates that minimum, maintained retroreflectivity values for both white and yellow pavement marking should be between 80 and 100 mcd/m^2/lux on high-speed roadways. However, maintaining a minimum retroreflectivity for pavement markings will be difficult for the following reasons:

1. Markings are typically “manufactured” on location under varying temperature and humidity conditions, applied over existing surfaces that may be less than ideal (e.g., rough texture due to surface treatment, oil contaminants, etc.), and traffic sometimes drive on or cross over the markings before they are cured, all of which affect the initial retroreflectivity.

2. The retroreflectivity of some line segments deteriorate much faster than other segments because vehicles frequently run on the lines in heavy weaving areas, around curves, and at intersections. In addition, asphalt roadways sometime bleed and the asphalt material may track onto the pavement markings, causing permanent discoloration and loss of retroreflectivity.

3. Snowplows, sanding, chemicals, anti-skid materials, and tire studs and chains cause markings to deteriorate very quickly during the winter months.

4. If lines wear out during the winter, it may not be practical to replace them for several months.

Therefore, all of the above will be major concerns when FHWA proposes their minimum retroreflectivity recommendations. Currently, the Department does not have any minimum retroreflectivity requirements for its paint program. (The Department has, however, established minimum values for newly-applied pavement marking materials in Publication 408.)

3.2.3 Pavement Marking Retroreflectivity Database

In anticipation of a minimum retroreflectivity value requirement discussed in Section 3.2.4, and in an effort to enhance the Department’s Quality Assurance Program, BHSTE has purchased hand-held retroreflectometers (HHRs) and provided statewide training for each Engineering District.

The Bureau is also collecting retroreflectivity measurements to track the life cycle of various pavement-marking materials on different types of roadways. Specifically, the Bureau has a contract to collect retroreflectivity readings with a mobile retroreflectometer on the Interstate system and on principal arterials, i.e., maintenance functional classifications (MFCs) A, B, and C. Additionally, Engineering Districts are to collect data from predetermined collector and local roads (MFCs D, E, and F).

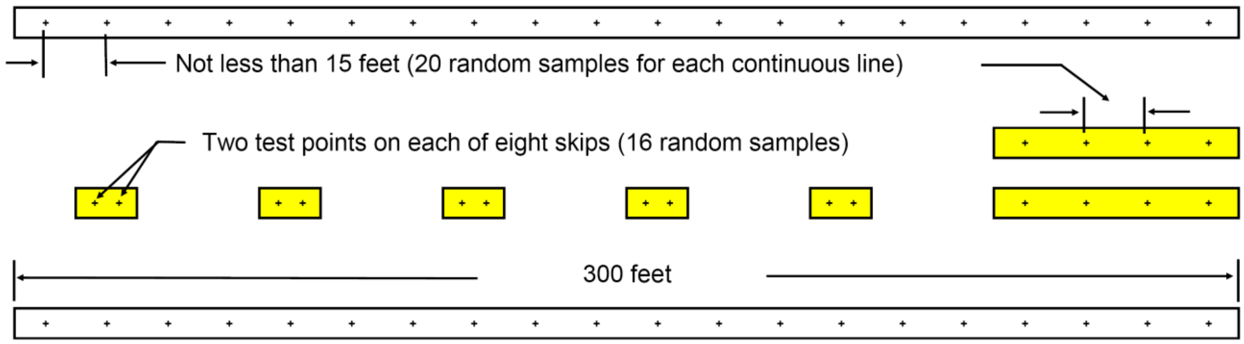
BHSTE has contacted each District to establish approximately six checkpoint areas in each District. The frequency at which measurements are collected depends on the marking material and the District's ability to collect measurements over the predetermined checkpoint areas. However, the goal is to collect readings every 2 months after application, and enter the measurements into the Department's retroreflectivity database.

These guidelines provide uniformity in the collection of this statewide data and describe how to send the data electronically to BHSTE using an Excel® spreadsheet from the HHR. (Note: This information will be included in the Materials and Testing Division's PTM #431.)

Exhibit 3.2-E shows the locations where the Districts should take the retroreflectivity readings for this database.

This cooperative effort by BHSTE and the Engineering Districts will create a retroreflectivity database that should yield answers to many pavement marking durability questions. The ultimate goal is to be able to predict the end-of-service life for pavement markings on various roadway surfaces and at locations exposed to different levels of winter maintenance, and to use these values to determine a cost-effective schedule for reapplying pavement markings.

Exhibit 3.2-E Pavement Marking Sampling



Notes:

1. Select locations that meet the criteria in PATA Nos. 7, 10c, 12 and 15 of Publication 213 for the elimination of all signs and channelizing devices when taking measurements. Use a flagger when taking readings.

2. Take all measurements in the direction of traffic flow, except on the centerline of two-lane roads where the required number of measurements will be made in each direction. Measure both single and double lines.

3.2.4 Durable Pavement Markings

Waterborne paint is the most common pavement marking material, but it also degrades faster than most other materials. Therefore, highway agencies use the expression “durable pavement marking” to describe those products that typically last much longer. In general, the term encompasses hot thermoplastic, cold-preformed thermoplastic, epoxy, tape, polyester, polyurea, and methyl methacrylate (MMA).

While it may be desirable to use durable pavement markings on all expressways, freeways and NHS highway projects, Districts can decide if durable pavement markings will be used and on which roadways. Each District will fund all durable pavement markings from their normal funding sources.

> Since waterborne paint is generally the most cost-effective pavement marking material, the real question is, “Will the paint line still be in place and the retroreflectivity consistently above the ‘minimum’ acceptable values when the Department is able to repaint the lines?” If the answer is “yes,” then waterborne paint may be the best option.

3.2.5 Pavement Markings in Construction and Resurfacing Projects

Contractors are required to mix all waterborne paints prior to using them. If other types of marking materials are used, the contractor should handle those materials as specified by the manufacturer. The line-painting equipment shall be calibrated to apply the pavement marking material at the correct application rate as required in the specifications and the quality verified by applying the marking materials as test stripes on sample plates prior to installing the actual lines.

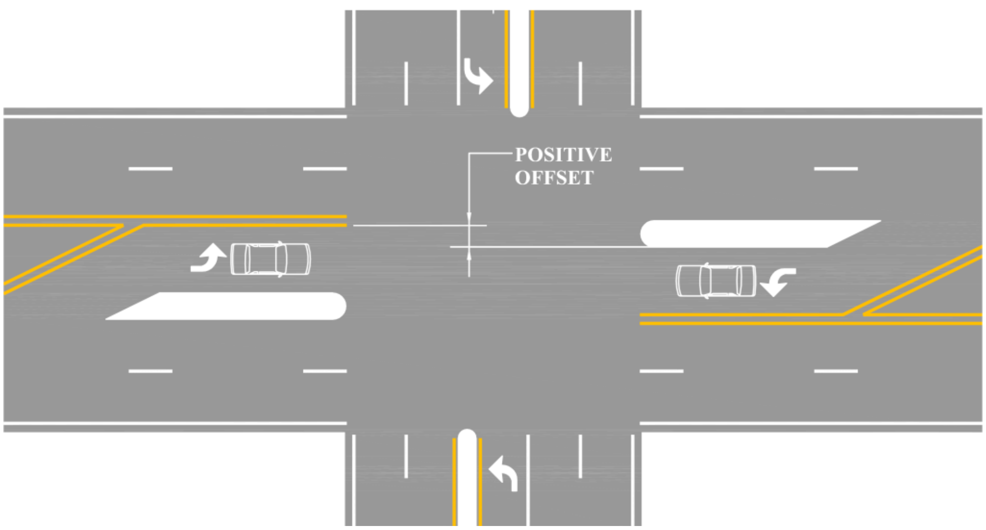
Construction inspectors are to use the "Pavement Markings, Construction Inspector Quality Assurance Review Form" as included in Appendix 3A.

3.2.6 Positive Offset at Opposing Left-Turn Lanes

At unsignalized intersections and at signalized intersections with permissive left-turn phases, whenever possible, use a positive offset at opposing left-turn lanes to reduce the sight distance obstruction caused by a vehicle in the opposing left-turn lane. A 3-foot positive offset will usually negate the sight-distance problem caused by a vehicle in the opposing left-turn lane.

On undivided highways where pavement width allows, create this positive offset by using solid lane lines to form a parallel or tapered island between the left-turn lane and the adjacent through lane, as shown in Exhibit 3.2-F. In some cases, it may be feasible to use reduced-width lanes at the intersection to accommodate the creation of an island.

Exhibit 3.2-F Positive Offset at Opposing Left-Turn Lanes



3.2.7 Raised Pavement Markers

Raised Pavement Markers (RPMs) provide important guidance to motorists at night, particularly when the roadway is wet and longitudinal pavement markings lose their effectiveness. Customer service surveys have also indicated that motorists like RPMs.

Based on research, the only acceptable permanent RPMs for new installations are the snowplowable, low-profile type that installed in a holder and listed in Publication 35. Districts with reflectors in recessed slots may continue to install replacement reflectors for the life of the slots.

The Department has established the following guidelines:

1. Install RPMs on the lane lines and within exit-ramp gore areas of all Interstate highways, freeways, and expressways. (An exception may occur if the pavement is in poor condition and will not accommodate RPMs, or if the Department plans to resurface the pavement within 4 years.)

2. As funding is available, install RPMs on high-volume NHS roadways, and on sections of other roadways where there is a higher-than-normal incidence of nighttime run-off-the-road crashes.

3. Install bridge-deck RPMs on all bridges that are 200 feet or longer in length when RPMs are on the adjacent sections of roadway.

Over the past 15 years, the Department has deployed a significant number of RPMs. Therefore, it is important to manage these devices in an effective and efficient manner such that they continue to provide effective guidance to motorists.

Design and installation details for RPMs are in Pavement Marking Standard TC-8602 and in Section 966 of Publication 408.

<u>Acceptability Criteria</u>

One difficulty in managing these devices is the typical need to rely on a subjective rating system to determine replacement. To assist in this evaluation, each District is to consider RPMs acceptable if all of the following apply:

- At least 50 percent of the reflective surface is visible at night.

- At least 90 percent of RPMs exist and are visible at night.

- On expressways and freeways, at least five RPMs are continuously visible at night using low beams.

Since most damage to RPMs occurs over the winter months, the level of replacements should be determined during the April to May timeframe each year, which generally results in a need to add or replace lenses on a cyclical basis, e.g., every 2 years.

RPM Inventory

Each District should maintain an RPM inventory and management system, and BHSTE will develop a statewide database that will be the ultimate repository for these files.

Legislative Mandate and Annual Reporting

The Department's annual budget typically has a line item that requires the Department to spend about $4 million for RPMs. Further, the Department is normally required, on an annual basis, to report the expenditures to the House and Senate Transportation Committees.

To satisfy the legislative mandates, on an on-going annual basis, Districts should annually develop their RPM program as follows:

1. By late summer, the Deputy Secretary for Highway Administration will advise each District of their funding level for the fiscal year and the appropriate contact person in the Center for Program Development and Management.

2. Districts should develop a program designed around the funding that they receive. When preparing cost estimates, use current costs for each new unit (holder with reflector) and replacement reflector. Also, include items for the engineering and construction inspection

3. By February 28, the District's Planning and Programming Engineer/Manager should create a new project in MPMS for their RPM program. The District should forward the newly created MPMS numbers to the appropriate contact person in the Center for Program Development and Management. When creating the project, the Project Class is *"Safety Improvement"* and the Improvement Type is *"Reflective Pavement Markers (HWYRPM)."* These RPM projects are 100 percent federally funded.

4. By June 15, each District should obligate the approved amount. After that date, Central Office will redistribute any money that is not obligated.

3.2.8 Stop and Yield Lines

Section 3B.16 of the *MUTCD* discusses the placement of stop and yield lines. However, some common mistakes are placing the line too far from an intersecting roadway where drivers do not have a clear view of approaching traffic, and locating the stop line directly across from the STOP sign.

In order to better clarify the use and placement of stop lines at stop-controlled intersections, consider the following:

1. Stop signs and stop lines are separate traffic control devices and should be engineered independent of each other. Stop lines are not required to supplement the installation of every stop sign. If it is determined that conditions are such that stop lines are not necessary or are unlikely to enhance the safety of the intersection, eliminate them or do not install them.

2. Use stop lines where it is necessary to provide drivers with additional guidance as to where vehicles should stop, in compliance with the stop sign.

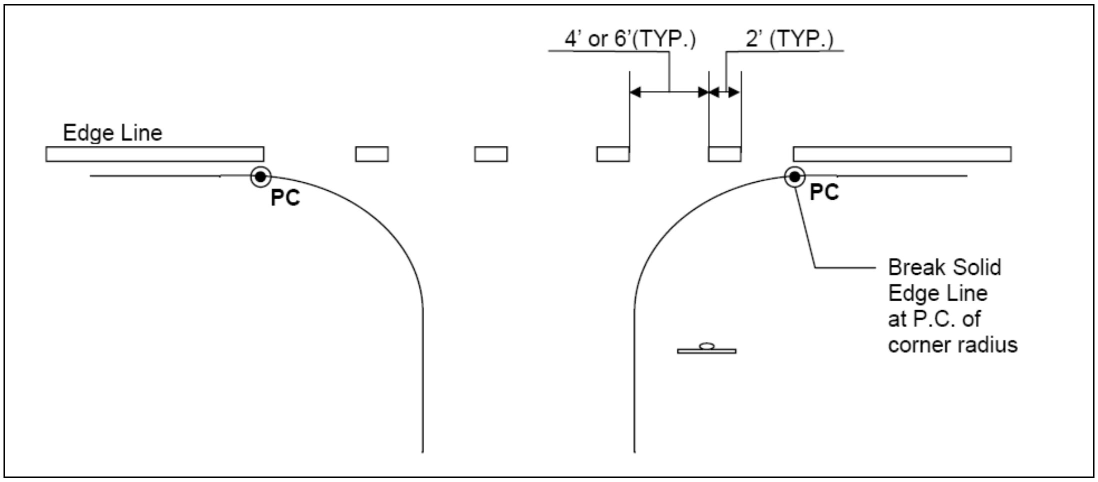
3. If using a stop line in conjunction with a stop sign, the normal position is in-line with the stop sign. However, if the stop sign cannot be located where vehicles are expected to stop (such as at wide-throat intersections), the stop line should be placed at the expected (desired) stopping point. Typically, this is as close to the edge of the intersecting roadway as possible where drivers have the best available sight lines.

4. Exact location of the desired stopping point should be determined in the field taking into account the various site-specific intersection parameters. Place stop lines no more than 30 feet or less than 4 feet from the nearest edge of the intersecting traveled way. When spotting the location of these lines, always consider the fact that a driver's eyes are about 10 feet behind the stop or yield line and at an elevation of 3.5 feet above the road surface.

5. Stop lines should not encroach onto the shoulder of the intersecting roadway. If the expected (desired) stop line location is determined to fall within the shoulder area of the intersecting road, adjust the stop line placement accordingly so as not to encroach on the shoulder area.

6. Consider use of a dotted edge line to indicate the edge of the cartway where the intersecting road meets the approach (see Exhibit 3.2-G). This will provide the driver with a visual cue of where the intersecting travel lane begins. Districts can also use stop lines in conjunction with these dotted edge lines.

Exhibit 3.2-G Dotted White Edge Lines



3.2.9 Crosswalks

Research of marked versus unmarked crosswalks at 1,000 locations in 16 states indicates that the risk of a pedestrian-vehicle crash is 3.6 times greater at uncontrolled intersections with marked crosswalk than with an unmarked crosswalk.[1] Therefore, Districts should use discretion when approving marked crosswalks at uncontrolled intersections.

Pavement Marking Standard Drawing TC-8600, Sheet 4 of 8, shows three types of crosswalk pavement markings. When crosswalks are used, coordinate with the local entity to determine the type of crosswalk marking. The minimum width of a crosswalk is 6 feet, as measured between any transverse markings.

Unsignalized mid-block crosswalks should be in accordance with Section 11.9. All crosswalks should be at approximately 90 degrees to the roadway. For added visibility, the crosswalk may be marked with wide white diagonal lines at a 45-degree angle to the crosswalk or with wide white longitudinal lines parallel to traffic in accordance with Pavement Marking Standard TC-8600.

Local authorities sometimes want to use decorative crosswalks, such as brick pavers, stampings, etc. Although not prohibited, Districts need to address funding and maintenance issues. Specifications for severable types of decorative crosswalks are included in Publication 447, New Product Evaluation for Low Volume Local Roads, available at ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20447.pdf.

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[1] Zegeer, C. V., J. R. Stewart, H. Huang and P. A. Lagerwey. Safety Effects of Marked vs. Unmarked Crosswalks at Uncontrolled Locations: Executive Summary and Recommended Guidelines. Report No. FHWA-RD-01-075, Washington DC, FHWA, March 2001.

Districts may use yield lines in advance of unsignalized mid-block crosswalk. If used, place the yield lines 20 to 50 feet in advance of the crosswalk and install a Yield Here to Pedestrians (R1-5) sign immediately adjacent to the yield line (see Figure 3B-15 in the *MUTCD*). Yield lines should conform to Pavement Marking Standard TC-8600.


UNITED STATES ACCESS BOARD

Americans with Disabilities Act
and Architectural Barriers Act
Accessibility Guidelines

July 23, 2004

UNITED STATES ACCESS BOARD
A FEDERAL AGENCY COMMITTED TO ACCESSIBLE DESIGN

3.2.10 Accessible Parking Spaces

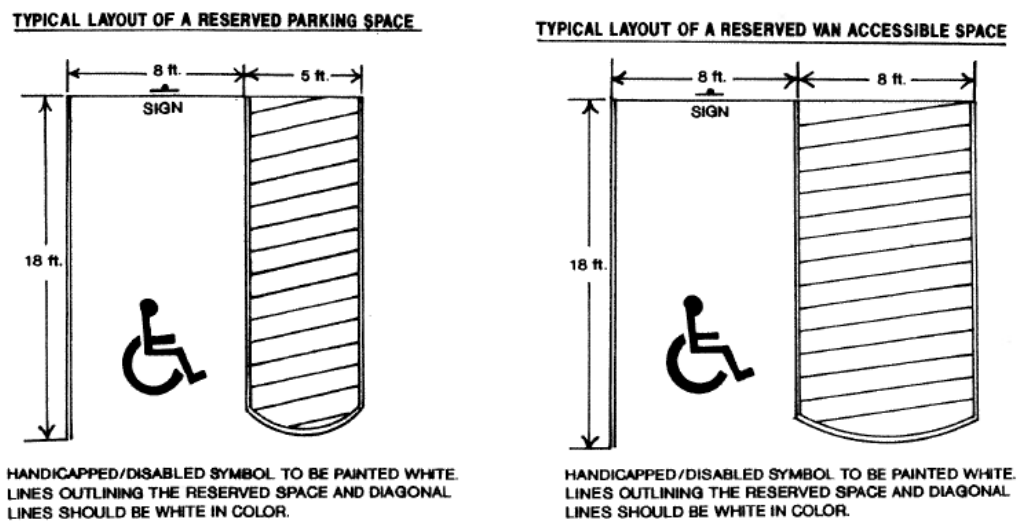
Section 208 of the Americans with Disabilities Act (ADA), 36 CFR Part 1191 (see http://www.access-board.gov/ada-aba/final.pdf) establishes the following minimum number of required ADA accessible parking spaces, based on the total number of spaces in the parking area.

Exhibit 3.2-H Minimum Accessible Parking Spaces

Total Number of Parking Spaces Provided in Parking Facility	Minimum Number of Required Accessible Parking Spaces
1 to 25	1
26 to 50	2
51 to 75	3
76 to 100	4
101 to 150	5
151 to 200	6
201 to 300	7
301 to 400	8
401 to 500	9
501 to 1,000	2 percent of total
1,001 and over	20, plus 1 for each 100, or fraction thereof, over 1,000

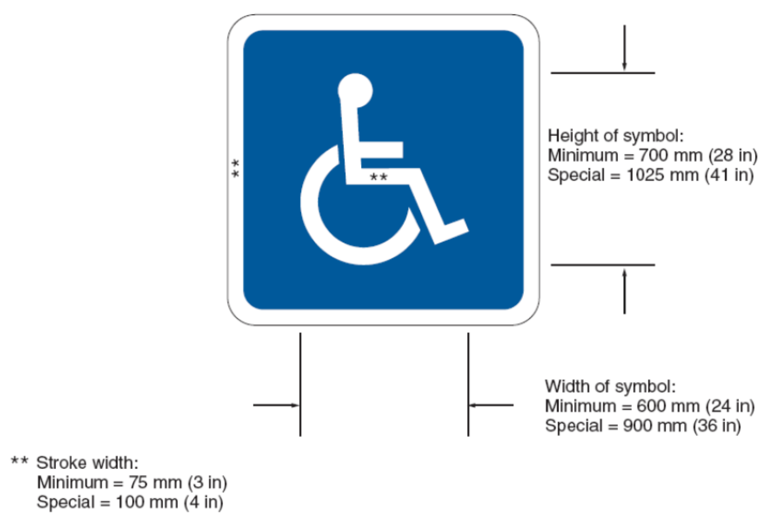
It is also important to note, that in accordance with Section 208.2.4 of the Americans with Disabilities Act, for every six ADA accessible spaces or fraction thereof, at least one shall be a van parking space. Exhibit 3.2-I shows typical layout details for parking spaces.

Exhibit 3.2-I Typical ADA Accessible Parking Space Layouts



A common practice in some commercial parking lots is to paint the entire accessible parking space blue, but painted surfaces may be very slippery when wet. Therefore, as an alternative, the symbol illustrated in Figure 3B-19 of the *MUTCD* (and included as Exhibit 3.2-J) may be used.

Exhibit 3.2-J Parking Space Marking with Blue Background



3.2.11 Pavement Word and Symbol Markings

The use of word-type pavement messages, frequently called "horizontal signing," is an effective way to communicate with drivers providing the markings are visible.

Therefore, Districts are encouraged to use horizontal signing to supplement traditional regulatory, warning, and directional signs.

Districts may supplement route marker signs with elongated route markers and directional arrows on the pavement. Preformed thermoplastic is the recommended material for words and symbol messages on bituminous surfaces, and either epoxy or tape are the recommended materials on Portland cement surfaces.

In 2006, the National Committee on Uniform Traffic Control Devices (NCUTCD) adopted a recommendation to change the *MUTCD* to allow red, white, and blue pavement markings to accommodate the Interstate shield marking.

This change will be included in the next edition of the *MUTCD.*

Words and symbols shall conform to Figures 3B-14 through 3B-21, and 3B-25 of the *MUTCD*. Of specific importance, Lane-Reduction Arrows shall comply with Figure 3B-21.

The standard height of word messages is 8 feet on all types of roads, except “SCHOOL” is 10 feet high. A maximum of three lines of message may be used, spaced at four to ten times the legend height (i.e., typical spacing of 32 to 80 feet), and should read in the direction of travel.

3.2.12 Speed Measurement Markings

To assist the Pennsylvania State Police (PSP) in speed enforcement, Engineering Districts should cooperate with PSP in the application of transverse speed measurement markings for the State Police Aerial Reconnaissance Enforcement (SPARE) program.

PSP has offered the following general criterion that is important in the selection of SPARE locations:

- The site is on a tangent or near tangent section of highway, preferably 0.9 mile in length (minimum 0.6 mile).
- Each set of SPARE markings is totally within one judicial jurisdiction.
- The SPARE site does not cross under any structures.
- Locations exist within several miles downstream where officers can park and wait, and where they can pull offenders over.
- Posted speed limit of 55 mph or higher.
- Sites are not in Philadelphia or Delaware Counties, the eastern part of Montgomery County, or at any location in close proximity of an airport.

Procedures regarding the development of new SPARE sites are as follows:

1. The local PSP barrack submits their request to their Bureau of Patrol for consideration.

2. The Bureau of Patrol forwards recommended requests to affected Engineering District.

3. After review, the District denies the request, or approves the request and prepares a layout under the direction of a registered Pennsylvania land surveyor. If the roadway is not a tangent section, survey crews should measure the distances between the markings along the outer edge of the shoulder on the inside of the curve.

4. The District prepares a site drawing that show the markings and the Speed Check Marker (I2-1) signs, and the location of the nearest SR markers and the distance to them. (See the TC-8600 Standard Drawing, Sheet 1 of 8.)

5. The District sends the original site drawing to BHSTE for processing and certification. BHSTE then forwards the site drawing and an Attestation Form to the Bureau of Patrol for their final processing and files.

When the State Police request an Engineering District to remark a site, the Engineering District should provide their best estimate of when the work will be complete, and then notify the State Police when the work is physically complete.

3.2.13 Do Not Block Intersection Markings

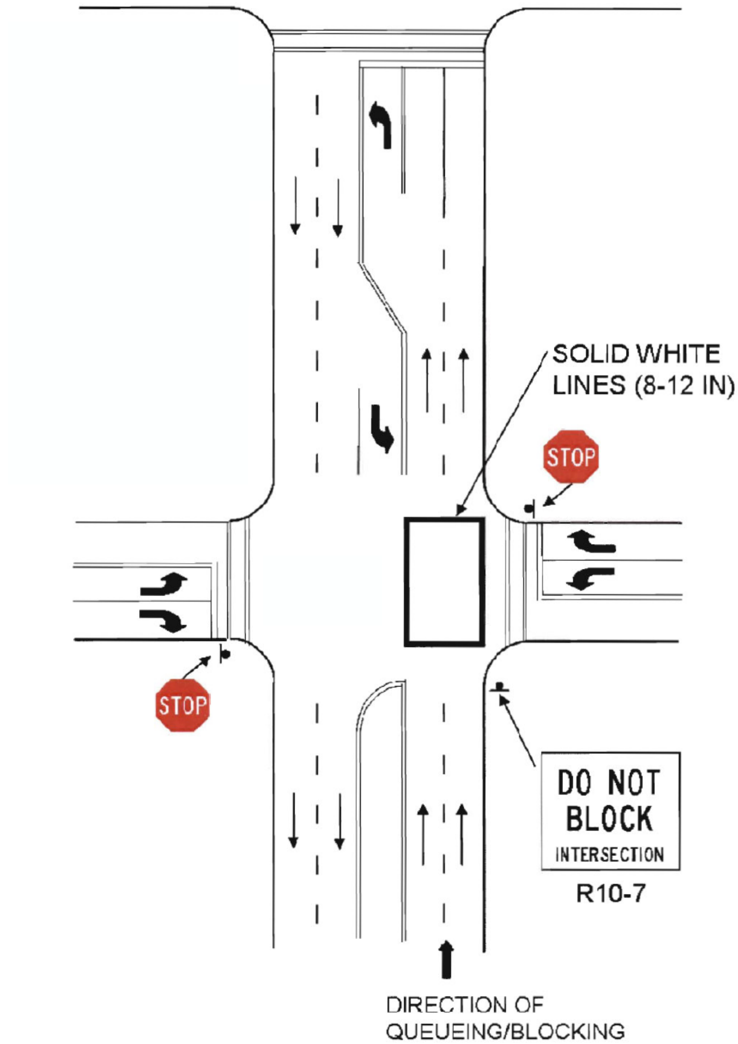
Do Not Block Intersection Markings may be used to mark the edges of an intersection area that is in close proximity to a signalized intersection, railroad crossing, or other nearby traffic control that may cause vehicles to stop within the intersection and impede other traffic entering the intersection.

If used, the Do Not Block Intersection Markings should consist of 8-inch or 12-inch solid white lines that outline the area of the intersection as illustrated in Exhibit 3.2-K.

Do not use “Do Not Block Intersection Markings” unless accompanied by a DO NOT BLOCK INTERSECTION (R10-7) or similar sign.

As the number of approach lanes increase, it is more difficult to ensure the safety of drivers either turning across the markings or entering the roadway and crossing the markings. Therefore, do not use Do Not Block Intersection Markings on roadways with more than two approach lanes.

Exhibit 3.2-K Do Not Block Intersection Markings



3.3 Object Markers

3.3.1 Object Marker Design

Section 3C.01 and Figure 3C 1 of the *MUTCD* allow several types of Object Markers, including markers with three or more yellow retroreflectors that are a minimum of 3 inches in diameter. These circular devices are typically acrylic retroreflectors that are very bright when viewed at an angle normal to the face of the retroreflector, but they have almost no retroreflectivity when viewed at an angle of 30 degrees or more from normal.

Therefore, Districts should not use Object Markers with these circular retroreflectors because the Object Markers frequently are not visible to drivers – for example, when turning at intersections, traveling around sharp turns and curves, or at any location where the markers become misaligned. Exhibit 3.3-A shows the recommended Object and End-of-Roadway markers.

Exhibit 3.3-A Object and End-of-Roadway Markers

3.4 Delineators

3.4.1 Acceptable Delineators

The only acceptable delineation sheeting material is Type V, abrasion-resistant material.

3.4.2 Delineation of Guiderail, Parapets and Barriers

Studies indicate that appropriate delineation provides substantial nighttime guidance and reduces the likelihood of nighttime crashes. Therefore, the Department offers the following guidelines to upgrade delineation to improve guidance for motorists, particularly older drivers.

As a part of this effort, Engineering Districts should:

1. Incorporate guiderail, parapet and barrier delineation into all PS&E packages.

2. Establish a program to retrofit and maintain delineation on the National Highway System (NHS) when:

    - The average daily traffic volume (ADT) is 3,000 or greater.

    - Shoulders are 6 feet or less in width.

    - There is evidence of vehicles running off the roadway and hitting fixed objects.

The spacing of these devices on new installations should be as follows:

1. At 75 feet on tangent roadway sections and on horizontal curves with less than 2 degrees of curvature.

2. At 37.5 feet on all 2-degree and sharper horizontal curves and on approaches to structures where the full width shoulder transitions to a narrower width across the structure. In these cases, parapet-mounted delineation should also be at the 37.5-foot spacing.

<u>Guiderail Delineation</u>

For W-beam guiderail, use the Type D guiderail delineator (i.e., polycarbonate, butterfly-shaped models that fit in the web of the guiderail) as shown on Sheet 2 of 4 of TC-8604. For highway maintenance crews, these delineators are currently available on statewide contract Legacy No. 9905-09 using the following SAP material numbers:

SAP Material No.	Description
304918	Guiderail web-mounted, Type D (W/B)
144433	Guiderail web-mounted, Type D (W/R)
144431	Guiderail web-mounted, Type D (W/W)
144432	Guiderail web-mounted, Type D (Y/B)
144430	Guiderail web-mounted, Type D (Y/R)

Other types of delineators such as post-mounted guiderail delineation (e.g., “RailRider”) or independently mounted flexible delineator posts are encouraged along strong-post cable guiderail systems and as a continuation of the delineation beyond the limits of the guiderail to provide information regarding continued important alignment features (sharp horizontal curve where the guiderail ends or starts midway in curve).

Delineation of Parapets and Barriers

Install barrier-mounted delineator on all types of median barrier and single-face barrier using the spacing indicated above. Install both side-mount and top-mount delineation as illustrated on Sheet 2 of 4 of TC-8604 for the traffic side of each barrier.

3.5 Islands and Median Barriers

3.5.1 Mountable Curbs and Resurfacing

Some older arterials have mountable curb medians separating traffic. Although these mountable curbs have a minimal effect on averting head-on collisions resulting from out-of-control vehicles, they do have a significant safety benefit in restricting turning movements either from the mainline or from adjacent driveways. As these arterials age and require overlays, the overlays compromise safety by allowing drivers to begin making left turns.

Therefore, the District should assess the project needs prior to resurfacing. Specifically, the District needs to answer the following question for all resurfacing projects involving mountable curbs: *"Will the removal of the mountable curb result in a substantive increase in left-turn movements?"*

1. If the answer is "no" and there is minimal or no potential for developing turning movements in the future, the project may eliminate the mountable curb. However, the District should review the potential for placing a positive standard median barrier to prevent head-on crashes. If this is not possible or appropriate, the District may replace the mountable curb with standard pavement markings.

2. If the answer is "yes," the first priority is to consider replacing the mountable curb with a positive standard median barrier to prevent head-on crashes and restrict left turn movements. This is particularly important if the operating speeds are in excess of 40 mph. Give special attention to addressing sight distance concerns at existing breaks in the median. If the District cannot adequately address these sight distance concerns, replace the mountable curb in kind, per the Roadway Construction Standards (RC-65M).

The District should document the above study assessment in the project file.

Chapter 3 Appendix

Appendix | **Page**

PAVEMENT MARKINGS

CONSTRUCTION INSPECTOR QUALITY ASSURANCE REVIEW FORM

DISTRICT__________ COUNTY____________________ PROJECT__________________________________

TRAFFIC ROUTE_______ STATE ROUTE_________ SEGMENT_____________ OFFSET___________

APPLICATION CONTRACTOR________________________ PRIME___________ SUB____________

DATE:___________________ TIME:______________ WEATHER:_______________________________

PAVEMENT TEMPERATURE_______ AMBIENT TEMPERATURE___________ % RELATIVE HUMIDITY_________

TYPE OF MARKING: Long Lines __________ Arrow _________ Legend ___________ Other_______________________

MATERIAL: Paint ________, Epoxy __________, Thermoplastic: Hot___________ or Preformed _________________

Cold Plastic: Inlaid__________ or Surface Applied__________, Other__________________________________

MANUFACTURER:____ MANUFACTURED DATE:________________

PRODUCT NAME or FORMULA:______________________________ MATERIAL CERTIFICATION PROVIDED? Y / N _______

COMPLETE THE FOLLOWING APPROPRIATE SECTIONS:

A. TRUCK MOUNTED AND SELF-PROPELLED MACHINE OPERATION (LONG LINES)

1. Is the marking being applied in accordance with the manufacturer's specifications? Yes________ No _______
2. Is the roadway surface clean and dry? Yes________ No__________
3. Is a primer being used prior to the application? Yes__________ No____________ N/A_____________
4. Are the pavement, ambient temperature and percentage of relative humidity being monitored and documented at least three times a day? Yes____________ No_________
5. Was paint mixed before using? Yes___________ No_____________
6. Are center line guns adjusted to provide a 6-inch space between lines? (+ ½ inch) Yes_____ No_____ N/A_________
7. Are lane lines 4 inches to the right of the pavement joint or seam? (+ ½ inch) Yes_______ No________ N/A________
8. Are edge lines 4 inches from the edge of the pavement? (+ ½ inch) Yes_______ No__________ N/A__________
9. What is the application speed? _____________MPH

PAINT/BINDER AND BEAD GUNS	Left Outside			Left Inside			Right		
	Yes	No	N/A	Yes	No	N/A	Yes	No	N/A
10. Are the edges of the lines clean and sharp?									
11. Are the lines being applied uniformly?									
12. Are glass beads evenly distributed over entire line?									
13. Are glass beads properly embedded (60-70%)?									
14. What is the width, in inches, of the traffic lines being applied?									
15. What is the length of the skip line pattern (Cycle should be 40')?									
16. What is the length of the skip line (10')?									
17. What is the binder temperature at the guns?									
18. What is the measured application rate of glass beads?	# / Gal.			# / Gal			# / Gal		
19. What is the quantity of binder used and length of lines applied? (Quantity of paint should be in compliance with the following charts.)	Gal. Ft.			Gal. Ft.			Gal. Ft.		

4" WIDE LINES (15 MILS)		4" WIDE LINES (12 MILS)		NOTES
100 ft.	0.312 gals.	100 ft.	0.249 gals.	1) Skip line uses 75% less paint than a solid line.
500 ft.	1.559 gals.	500 ft.	1.247 gals.	
1000 ft.	3.117 gals.	1000 ft.	2.494 gals.	2) Six inch wide lines use 50% more paint than 4 inch wide lines.
2500 ft.	7.794 gals.	2500 ft.	6.236 gals.	
5280 ft.	16.460 gals.	5250 ft.	13.170 gals.	3) Eight inch wide lines use 100% more paint than 4 inch wide lines.

To determine gallons for a 4 inch line 15 mils thick, divide feet of line by 320.7776

To determine gallons for a 4 inch line 12 mils thick, divide feet of line by 400.9112

To determine gallons of paint for legends and symbols at 15 mil thickness, divide square feet of markings by 106.93

B. MANUAL OR WALK BEHIND OPERATION (ARROWS, LEGENDS, ETC.)

1. Type of application: Hand Gun_____________ Machine ___________________
2. Is the marking properly centered in the lane? Yes____________ No________________
3. Are the edges clean and sharp? Yes_____________ No_______________
4. Does the marking have uniform paint coverage? Yes __________ No_______________
   (Paint should be sufficiently heavy to allow for 60-70% glass bead embedment.)
5. Type of glass bead applications: Hand______Gravity Fed ________Pressure Fed _______ __N/A___________
6. Are glass beads evenly distributed over the marking? Yes___________ No_______________
7. Are glass beads properly embedded (60-70%)? Yes _____________ No_______________

C. RETROREFLECTIVITY MEASUREMENTS OF APPLIED PAVEMENT MARKINGS

1. Did contractor follow PTM 431? Yes______ No______
2. Did markings meet retroreflectivity requirements? Yes______ No______

Remarks:

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____

____
Inspector

FID#______________________________

pennsylvania
DEPARTMENT OF TRANSPORTATION

Publication 46
Chapter 4

Traffic Signals

Pennsylvania Department of Transportation
Chapter 4 of Publication 46
Table of Contents

List of Exhibits

4.1 General

4.1.1 Purpose

This chapter addresses Department policies relative to the study, design, installation, operation, and maintenance of all types of traffic signals.

This chapter emphasizes the need for local authorities to understand the costs associated with owning a traffic signal. Therefore, prior to the Department undertaking any study to determine if a signal may be warranted, the Department must receive a written financial commitment from the municipal governing body as indicated in Section 4.2.1.

4.1.2 Responsibility to Approve, Install, and Maintain Traffic Signals

In accordance with §212.5 of Publication 212, it is the Department's responsibility to approve all traffic signals except:

- Within cities of the first and second class (i.e., Philadelphia and Pittsburgh) unless there is an agreement between the city and the Department.
- In other municipalities with municipal traffic engineering certification in accordance with 67 Pa. Code Chapter 205.

Moreover, in accordance with §212.5 of Publication 212 (relating to installation and maintenance responsibilities), local authorities or other agencies are responsible for installing, maintaining, and operating all traffic signals on both State and local highways, including all associated signs and markings included on Department-approved traffic signal plans.

4.1.3 Laws, Regulations, and Other Publications

(a) *Approved Construction Materials* (Publication 35, also known as Bulletin 15).

(b) *Guidelines for the Maintenance of Traffic Signal Systems* (Publication 191).

(c) *Manual on Uniform Traffic Control Devices* (*MUTCD*), latest edition, Federal Highway Administration.

(d) *Official Traffic Control Devices* (Publication 212). The text without the appendix is also available as 67 Pa. Code Chapter 212.

(e) *School Trip Safety Zone Guidelines*, latest edition, Institute of Transportation Engineers.

(f) *Specifications* (Publication 408).

(g) *Traffic Signal Design Handbook* (Publication 149).

(h) *Traffic Standards – Signals; TC-7800 Series* (Publication 148).

(i) *Travel Better, Travel Longer: A Pocket Guide to Improve Traffic Control and Mobility for Our Older Population*, FHWA, 2003, (FHWA-OP-3-098).

(j) *Trip Generation*, Institute of Transportation Engineers.

(k) *Trip Generation Handbook*, Institute of Transportation Engineers.

(l) *Vehicle Code* (75 Pa.C.S.).

(m) *Work Zone Traffic Control Guidelines* (Publication 213).

4.2 Requests for Traffic Signals

4.2.1 Application and Financial Commitment

Requests for approval to install traffic signals at a location should be submitted the District Executive who is responsible for that particular District. Each official request must include a completed application form executed by the appropriate officials and a scale drawing indicating the geometric and topographic features of the location. All requests from local authorities must also include a written financial commitment from the municipal governing body indicating that:

- They will commit the appropriate funds to construct the signal installation within 2 years of permit issuance should the Department find that the installation meets warrants for a signal; and

- They agree to operate and maintain the signal on an ongoing basis after the signal is installed.

4.2.2 Procedure

Upon receipt of a request for a new traffic signal, a change in traffic signal operation, or the installation of any other power-operated device requiring the approval of the Secretary of Transportation, the following procedure shall apply.

When a request comes from the general public, or from someone other than the local authorities or others who are authorized to operate and maintain traffic signals, the Department should acknowledge receipt of the request and advise the individual or party to refer their request to the proper local authorities since the local authorities would be responsible for all costs associated with any installation of the device, and for all future operation and maintenance costs.

When local authorities make such a request, the District will write an appropriate letter of acknowledgement. This letter may suggest a meeting between the local officials and the District Traffic Unit at the earliest possible opportunity to discuss the Department's policies and procedures as they relate to traffic signals and to determine the extent of the problem as seen by the local officials.

This meeting will:

1. Allow the local officials to explain their concerns about the location.

2. Enable the Department to review the concerns with local officials and to determine whether alternative measures should be considered.

3. Give the Department an opportunity to explain Department policies and procedures, signal warrants, and the need for uniformity. The meeting will also assist in determining the best time to perform a traffic engineering investigation; for instance, the need to conduct an investigation during the school year where school activities are a primary concern.

4. Give the Department an opportunity to provide an estimated range of costs to design, construct, operate, and maintain the traffic signal and to inform the local officials of the need for their municipal governing body to provide a written financial commitment indicating:

    - They will commit the appropriate funds to construct the signal installation within 2 years of permit issuance should the Department find that the installation meets warrants for a signal; and

    - They agree to operate and maintain the signal on an ongoing basis after the signal is installed.

After the meeting, if it appears that an official application (Form TE-952, Application for Permit to Install and Operate Traffic Signals) should be executed, the District should submit a letter to the municipality with a blank TE-952 application form and a request for a condition diagram (and guidelines for its preparation). The letter may also list a range of estimated costs for a typical installation and for annual operation and maintenance. The letter shall request a written financial commitment from the governing body. Such written financial commitment shall indicate that the municipality will commit the appropriate funds to construct the signal installation within 2 years of permit issuance should the Department determine that the location meets warrants for a signal, and that the municipality agrees to operate and maintain the signal on an ongoing basis after it is installed.

Engineering studies to determine whether a new traffic signal is warranted at a specific location will only be conducted by the Department after a written financial commitment is received from the municipal governing body.

4.2.3 Transfer of Authority to Local Officials

4.2.3.1 Purpose

The purpose of this policy is to establish criteria and guidelines governing the Department entering into a Transfer Of Authority (TOA) agreement with local authorities to authorize the local authorities to install official traffic control devices along State highways and at intersections of local highways and State highways without specific State approval as provided in §6122(c) of the Vehicle Code.

4.2.3.2 Authorization

§6122 of the Vehicle Code provides the authority to erect traffic control devices along highways and defines the limits of this authority.

§6122(a) of the Vehicle Code provides the authority for local authorities to erect traffic control devices (except for traffic signals) on local highways and requires them to obtain approval of the Department prior to erecting official traffic control devices on State highways.

§6122(c) of the Vehicle Code provides the authority for the Department to enter into agreements with local authorities transferring to them the authority to install official traffic control devices without specific State approval provided they conduct engineering and traffic studies which conform with the rules and regulations promulgated by the Department.

4.2.3.3 Criteria

The Department may consider entering into a TOA agreement with a municipality transferring to them the authority to install, revise, remove, maintain, and operate official traffic control devices on State highways and at intersections of local highways and State highways provided:

(a) The municipality has a full-time municipal traffic engineer who is an employee of the municipality, and the municipality and the municipal traffic engineer have been certified in accordance with 67 Pa. Code Chapter 205 (relating to municipal traffic engineering certification).

(b) The municipality agrees that they will not make any changes or remove any traffic control device installed as a part of any Department-administered, federally-funded program without Department approval.

(c) The municipality agrees that they will not install, revise, or remove any traffic control device on a freeway or other limited access highway without PennDOT approval.

(d) The municipality agrees that they will not install, revise, or remove any traffic control device within the limits of any Department construction project or Department-administrated, federally-funded project without Department approval.

(e) The municipality agrees to conduct the necessary engineering and traffic studies in accordance with Publication 212 and the *MUTCD*, and to base their traffic restrictions on State highways and at the intersection of local highways and State highways on the findings and results of the engineering and traffic study.

(f) The municipality agrees to meet all Department regulations, standards, and policies related to the design, location, and operation of official traffic control devices.

(g) The municipality agrees to make every effort within their financial capabilities to upgrade all traffic control devices to meet the current requirements of the *MUTCD*, Publication 212, and Publication 236M.

(h) The municipality agrees to retain in their files:

1. All data collected as a result of engineering and traffic studies which provide the justification for the installation, revision, or removal of the device.

2. A completion certificate signed by the municipal traffic engineer verifying the completion of the installation, revision, or removal of official traffic control devices or a work order approved by the municipal traffic engineer when issued and countersigned by the individual responsible for installation, revision, or removal of the traffic control devices when the required work is completed.

3. A schedule and results of periodic evaluations (at least yearly) of all official traffic control devices within the municipal boundaries including warranted update or revision recommendations.

4. A plan of all traffic signals within the municipal boundaries to include a plan view of each traffic signal installation; mode of operation and, if applicable, a movement, sequence, and timing diagram; traffic flow diagram of peak hour volumes; and a collision diagram for the 3-year period preceding the traffic signal installation or latest traffic signal update.

(i) The municipality agrees that the Department's review and approval is required for the installation, revision, removal, maintenance, and operation of an official traffic control device at or near the boundary of another political subdivision unless the municipality accepts responsibility for the design of the project and the other political subdivision has given approval.

(j) The municipality agrees to indemnify and save harmless the Commonwealth, the Department, their officers, agents, and employees from all suits, actions, or claims of any character, name, and description brought for or on account of any injuries, deaths, or damages received or sustained, during the performance of the municipality's obligations under the agreement, by any person, persons, or property, by or from the municipality, whether the same be due to the use of defective materials, defective workmanship, neglect in safeguarding the work on public interests, or by or on account of any act, omission, neglect, or misconduct of the municipality during the performance of said work.

4.2.3.4 Application Procedure

If a municipality desires to enter into a TOA agreement with the Department to transfer to them the authority to install official traffic control devices on State highways and at the intersection of local highways and State highways without specific State approval, they should submit such a request to:

Bureau of Highway Safety and Traffic Engineering
Pennsylvania Department of Transportation
P.O. Box 2047
Harrisburg, PA 17105-2047

The request from a municipality must include a copy of the municipality's official municipal traffic engineering certification, three original agreements properly executed by the municipality, and a resolution from the governing body of the municipality authorizing the execution of a resolution and a TOA agreement that will be provided to the municipality by the Department.

4.2.3.5 Approval

When a request to enter into a TOA agreement is received from a municipality, the Bureau of Highway Safety and Traffic Engineering (BHSTE) will review the request to determine if the necessary documents are included and properly executed.

If the municipality's request includes the documents required above, the BHSTE will coordinate its review with the appropriate District Traffic Unit to gain their knowledge and input, and will then process the request accordingly.

4.2.3.6 Rescission

Non-compliance by the municipality with any provision of the TOA agreement shall be a breach of the agreement and, at the Department's option, may be cause for immediate termination of the agreement.

If the Department rescinds the TOA agreement, the Department’s Bureau of Highway Safety and Traffic Engineering will provide the municipality with a written notice stating the effective date of, and the reasons for, the rescission of the agreement. From that point on, the municipality shall follow all normal Vehicle Code requirements and Department regulations, standards, and procedures governing the installation of official traffic control devices along State highways and at intersections of local and State highways. Specifically, the municipality will need Department approval for the installation, revision, and removal of all signals, including those on local highways.

4.3 Studies

4.3.1 General

Determination of the need to install, revise, or remove a traffic signal shall be based on a thorough engineering and traffic study of roadway and traffic conditions in accordance with Publication 212. Responsibilities shall be as follows:

(a) State Highways. The Engineering District will conduct studies at locations that involve State highways except for first and second class cities, municipalities having municipal traffic engineering certification, municipalities having an agreement with the Department, or for locations involving a business development. In these latter cases, the study will be conducted by that particular entity. When the Department completes a traffic signal study at the request of a municipality and determines that a traffic signal warrant is satisfied, the municipality may be responsible to design the proposed traffic signal, prepare the bid documents, and prepare the traffic signal permit drawing. The proposed signal design, bid documents, and traffic signal permits may be reviewed and approved by the Department prior to the project being advertised for bid by the municipality.

(b) Local Highways. Local authorities shall be responsible for conducting all studies for traffic signals at locations that do not involve State highways.

(c) Business Development. In accordance with Chapter 441, Section 441.6(4)(i) and Publication 282, developers shall be responsible for all studies necessary to justify traffic signals requested on any type of highway as a result of the traffic demands at a new or existing development.

(d) Department Coordination. Regardless of the party that conducts the study for traffic signals, the Department should be kept informed of all signal locations and related information.

(e) Plan. As noted in Sections 4.2.1 and 4.2.2, a scale drawing that depicts the geometric and topographic features of the location must be submitted with every completed traffic signal application form. The drawing shall be prepared in compliance with Publication 149 and other guidelines provided by the Department. The applicant shall be responsible for developing the drawing unless the signal installation is part of a Department construction project; in which case, the Department will develop the drawings as part of the project.

(f) Costs. All costs related to these studies shall be the responsibility of the entity conducting the study as indicated above.

4.3.2 Use of Warrants

There are nine warrants for traffic signals, which are as identified in §212.302(b) of Chapter 212 (relating to warrants for traffic control signals). Warrants 1 through 8 are as defined in Sections 4C.02 through 4C.09 in *Part 4 of the MUTCD*, and Warrant 9 is as defined in §212.302(b)(3) of Publication 212.

The nine warrants, and in some cases clarifying information, are as follows:

4.3.2.1 Warrant 1, Eight-Hour Vehicular Volume

See Section 4C.02 of the *MUTCD*.

4.3.2.2 Warrant 2, Four-Hour Vehicular Volume

See Section 4C.03 of the *MUTCD*.

4.3.2.3 Warrant 3, Peak Hour

See Section 4C.04 of the *MUTCD*.

4.3.2.4 Warrant 4, Pedestrian Volume

See Section 4C.05 of the *MUTCD*.

4.3.2.5 Warrant 5, School Crossing

In addition to the criteria in Section 4C.06 of the *MUTCD*, to satisfy Warrant 5 at an established school crossing, it is necessary to perform a traffic engineering study to determine the adequacy and frequency of vehicular traffic gaps. Guidelines for these "gap acceptance studies" are as follows:

(a) Conduct gap studies in accordance with the procedures developed in the Institute of Transportation Engineer’s publication, “School Trip Safety Zone Guidelines,” 1984 Edition.

(b) The study shall cover all times during the day when children use the crossing. This will involve, as a minimum, a morning and an afternoon study period.

(c) Each study period must be a minimum of 30 minutes in length; however, the study period should cover the entire time period the children use the crossing.

(d) Analyze locations where a median is involved in accordance with the following:

1. Median width of 6 feet (1.9 m) or less: This median width is not sufficient to provide adequate protection for the pedestrian; therefore, the width of the school crossing is from curb to curb.

2. Median width greater than 6 feet to a maximum of 16 feet: This width may be sufficient for adequate pedestrian protection. Determine adequacy during the study based on the number of pedestrians, the speed of traffic, the type of median, etc.

3. Median width greater than 16 feet: This size median is sufficient for adequate pedestrian protection. In this case, the width of the school crossing is from the curb to the median.

Justification for the installation of a signal under Warrant 5 requires the local officials or school officials to submit the following:

(a) Documentation clearly showing that the proposed signal installation is located at an established school crossing. A map of the local school area or municipality showing all established school crossings is desirable.

(b) A response to the following questions:

1. Is it possible to relocate the school crossing to eliminate the need for a traffic signal?

2. Can adult crossing guards or police officers more efficiently and safely control the school crossing?

(c) A completed "gap acceptance study" detailing the method of study with supporting data and analysis in the format described in the ITE publication referenced above. This study must show that the number of adequate gaps in the traffic stream during the period when the children are using the crossing is less than the number of minutes in the same period.

The design of signals installed under Warrant 5 shall include the following:

(a) Pedestrian-control signals.

(b) If at an intersection, the signal must be both traffic and pedestrian actuated.

(c) If installed within the limits of a progressive signal system, the signal must be interconnected with the system so that the timing does not disrupt the progressive movement.

(d) If installed at a non-intersection crossing, the signal shall be pedestrian-actuated. Parking and other obstructions to sight distance should be prohibited for at least 100 feet (30 m) in advance of and 20 feet (6.1 m) beyond the crosswalk.

The installation of traffic signals at school crossings may not be the absolute solution to the problem of conflicts between vehicles and school children. School and local traffic authorities must be aware of their responsibility to properly instruct children in the use of traffic signals. Moreover, the installation of a signal under the school crossing warrant does not preclude the use of school crossing guards.

4.3.2.6 Warrant 6, Coordinated Signal System

See Section 4C.07 of the MUTCD.

4.3.2.7 Warrant 7, Crash Experience

In addition to the criteria pertaining to Warrant 7 in Section 4C.08 of the MUTCD, the five or more reported crashes of types susceptible to correction that occur within a 12-month period may include both reportable crashes and non-reportable crashes that are documented in the police files, and that occurred within a 12-month period during the most recent 3 years of available crash data.

4.3.2.8 Warrant 8, Roadway Network

To qualify under Warrant 8 in the *MUTCD*), a major route must be classified as an Urban Extension, Principal Arterial, or Minor Arterial that is a reasonable connection between two Principal Arterials and/or Urban Extensions as shown on the official Functional Classification Map.

To be justified under this warrant, a copy of the location as it appears on the Functional Classification Map, with approval date along with the federal-aid route numbers, must be submitted with the request for traffic signal approval.

4.3.2.9 Warrant 9, ADT Volume Warrant

As noted in §212.302(b)(3) of Publication 212, Pennsylvania has a Warrant 9, ADT Volume Warrant, which is as follows:

(a) This warrant applies only at a proposed intersection, an intersection revised by a highway construction project, or at the driveway of a proposed commercial or residential development where vehicle counts cannot be taken. If a traffic control signal is installed under this warrant, a traffic count must be taken within 6 months of the opening of the development or within 2 years of the opening of the highway. If the traffic volumes do not satisfy this warrant, or one or more of the other eight warrants, consideration should be given to removing the traffic control signal and replacing it with appropriate alternative traffic control devices, if any are needed.

(b) This warrant is satisfied when the estimated ADT volumes on the major street and on the higher volume minor street or driveway approach to the intersection, when projected using an accepted procedure such as put forth in Trip Generation and the Trip Generation Handbook published by the Institute of Transportation Engineers, equals or exceeds the values in either Condition A or Condition B:

4.3.3 Analysis of Data

4.3.3.1 Department Participation

In order to clarify the procedure followed to determine if a traffic signal installation qualifies for state and/or federal funding, the following shall be adhered to:

(a) Prior to the start of any signal design work, justification for the traffic signals shall be prepared by the design agency and reviewed by the District Traffic Engineer. If the traffic signal is justified, the District Traffic Engineer will process the traffic signal maintenance agreement. Final design should not start until the agreement is in place.

(b) Where existing traffic signals are in operation on State Highways and are affected by the Department's construction, the following information will be required to justify continued operation:

1. A copy of the condition diagram of the intersection, and the existing traffic signal permit drawing indicating the present signal operation and installation.

2. A drawing indicating the proposed construction.

Exhibit 4.3-A ADT Volume Warrant

Condition A – ADT Volume Warrant					
Number of Lanes for Moving Traffic on Each Approach		Estimated ADT*			
Major Street	Minor Street	Major Street (Both Approaches)		Higher-Volume Minor Street (One Direction Only)	
		100%	70%**	100%	70%**
1	1	10,000	7,000	3,000	2,100
2 or more	1	12,000	8,400	3,000	2,100
2 or more	2 or more	12,000	8,400	4,000	2,800
1	2 or more	10,000	7,000	4,000	2,800

Condition B – ADT Volume Warrant					
Number of Lanes for Moving Traffic on Each Approach		Estimated ADT*			
Major Street	Minor Street	Major Street (Both Approaches)		Higher-Volume Minor Street (One Direction Only)	
		100%	70%**	100%	70%**
1	1	15,000	10,500	1,500	1,050
2 or more	1	18,000	12,600	1,500	1,050
2 or more	2 or more	18,000	12,600	2,000	1,400
1	2 or more	15,000	10,500	2,000	1,400

* Based on the volume projected to be present within 6 months of the opening of the development or within 2 years of the opening of the highway.

** May be used if the 85th percentile speed of the major street traffic exceeds 40 mph or the intersection lies within the built-up area of an isolated community having a population of less than 10,000.

3. Traffic volume data to justify continued signal operation (counts of existing and/or forecasted directional movements).

(c) Where the Department's construction creates a new intersection or involves an existing non-signalized intersection, the following information will be required to justify the installation of signals:

1. A drawing indicating the proposed construction.

2. The anticipated traffic volume that will be entering that portion of the intersection to be controlled by the proposed signals. This estimated traffic volume must be that volume that will use the intersection when it is open to traffic or within a period of 2 years from the date of opening. Therefore, in evaluating the need for a traffic signal, the District Traffic Engineering Unit must obtain the estimated AADT of the intersection for the year the facility will be open to traffic. To determine the volume 2 years after the facility is opened, use a linear interpolation between that volume expected to use the intersection when opened and the design year volumes. For existing intersections, use current volume counts.

(d) Consider any additional pertinent facts and information (such as school crossings, etc.) that the designer believes may be important in determining the need for a traffic signal.

(e) Compare the existing and projected traffic and other pertinent information with the warrants in Chapter 4C of the *MUTCD*, §212.302(b) of Publication 212, and Section 4.3.2 in this publication, to determine if minimum criteria is satisfied.

(f) A further review of traffic conditions at the intersection should be made to determine if the need for traffic signals will be created by the Department's construction. If the construction is such as to create a substantial increase in traffic volumes or a definite change in the traffic flow, the Department will include the signal in the construction plan. A definite change in traffic flow is defined as:

1. An additional approach is added which provides for an immediate increase in cross traffic. (The creation of a full intersection that eliminates two offset intersections does not constitute an additional approach.)

2. Where construction creates a marked increase in conflicting turning movements. (The creation of left-turn stand-by lanes by themselves does not constitute a marked increase in turning movements.)

(g) The Department may not participate if new development (industrial, commercial, recreational, residential, etc.) within the area is the prime traffic generator.

(h) The Department's policy is not to use State or federal funds to upgrade unwarranted signals, but the Department will allow the use of these funds to remove the signals. See Section 4.10 for a discussion on removing traffic signals.

(i) Signal systems are considered "ITS Projects." If funded with Federal Highway Trust Funds, they must comply with the DOT ITS Rule (23CFR, Parts 655 and 940).

4.3.3.2 Capacity Analysis

All studies for permits for new traffic signals, or for revised permits for existing signals that involve a change in timing, phasing, or type of controller operation, shall include a complete capacity analysis for the intersection. Districts should not approve new or revised permits if the projected operation of the intersection is reduced to an unsatisfactory level of service; i.e., Level Of Service below D in urban areas and below C in rural areas.

4.4 Permits

4.4.1 General

Written Department approval is required before a traffic signal can be installed, revised, or removed on any State or local highway except for those local authorities given this responsibility as indicated in §6122(a)(2) and §6122(c) of the Vehicle Code and §212.5 (specifically, subsections (b)(1)(ii)(B), (b)(1)(iii), and (c)(1)) of Chapter 212. Approval will be given only to local authorities, the Department, or contractors working for the Department if the location data satisfies state or federal traffic signal warrants.

4.4.2 Authority for Issuing

4.4.2.1 Purpose

This policy establishes the authority and responsibility to review, issue, and cancel permits or approvals for traffic control devices and restrictions.

4.4.2.2 Authority

The District Executive, an Assistant District Executive, or the District Traffic Engineer, may issue or cancel permits or approvals for traffic control signals providing they have a Professional Engineer's License. The Bureau of Highway Safety and Traffic Engineering can provide advice and assistance in the area of permit issuance or cancellation upon request.

The Chief or an Assistant Chief of the ITS/Congestion Management Division shall be responsible to approve traffic signal equipment and have approved materials listed in the *Approved Construction Materials* (Publication 35). In accordance with §212.8(b)(4) of Publication 212 (relating to traffic signal equipment requiring Department approval), items requiring approval include the following:

- Controller units.
- Signal heads – lane-use traffic control, pedestrian, and vehicle.
- Detectors – pedestrian and vehicle.
- Load switches.
- Flasher units.
- Time clocks.
- Relays.

- Preemption and priority control equipment.
- Electrically-powered signs – variable speed limit signs, blank-out signs, and internally illuminated signs, including School Speed Limit Signs.
- Portable traffic-control signals.
- Local intersection coordinating units.
- Dimming devices.
- In-roadway warning lights.
- Auxiliary devices and systems.

4.4.2.3 Procedures

In most cases, Engineering Districts do not need to submit traffic signal plans or special provisions to the Bureau of Highway Safety and Traffic Engineering (BHSTE) for review and approval. On the other hand, traffic signal plans or special provisions may be submitted to BHSTE on a case-by-case basis at the discretion of the Engineering District when they believe that a review by BHSTE would be appropriate or helpful.

However, in accordance with current policies, the following additional reviews are required:

(a) Bureau of Highway Safety and Traffic Engineering (BHSTE). Submit traffic signal plans or special provisions to BHSTE for approval before approval is issued by the Engineering District if the signals have alternate or experimental devices or procedures, such as red signal ahead signs, automated red light enforcement (ARLE), etc.

(b) Rail Safety Division of the Pennsylvania Public Utility Commission (PUC). Make a formal submission including the traffic signal plan to the PUC whenever there is a proposed change to an existing traffic signal with existing railroad preemption, or whenever train preemption is proposed at a new location.

If the District Grade Crossing Engineer does not have authorization to make a direct submission, then the Engineering District should submit the formal application to the Central Office Grade Crossing Section in the Bureau of Design (BOD), and request them to submit the application to the PUC.

(c) Bureau of Design. In accordance with the Federal-Aid Highways Stewardship and Oversight Agreement, FHWA approval is required for signals included in FHWA oversight projects. Therefore, the Engineering District should submit traffic signal plans and special provisions for these projects to the Bureau of Design, who will then forward them to FHWA for approval in accordance with customary project development procedures. FHWA approval is not required for other signal installations provided that they conform to the MUTCD.

4.4.2.4 Traffic Signal Permit

The appropriate municipal officials need to review and approve construction plans for traffic signals prior to the final review and approval by the Department. Local officials also need to review and sign all traffic signal permit diagrams prior to issuance by the Department. The purpose of this review is to obtain the municipality's concurrence with the proposal since they must operate and maintain the signals. The signature of the municipal engineer or other responsible official will indicate their concurrence with the proposal.

A permit data block must be located on the permit drawing to maintain a uniform record of all changes.

The District Traffic Engineering Unit should issue a signal permit for each new or revised signal under any program within a month of the bid award.

Forward a copy of the following approved traffic signal permits to the Congestion Management and Signalization Section in BHSTE:

- All new or revised permits involving railroad preemption.
- Any blanket portable traffic control signal permit.

4.4.2.5 Recordkeeping

The office that issues the permit, certificate of approval, cancellation, or the approval or denial of requests for the installation of traffic control signals is responsible for maintaining a file, including all required documentation for the justification of the devices and an up-to-date record of the permit status. (This is particularly important for projects involving federal funding since the FHWA may audit these projects.)

In accordance with Department policies, each Engineering District shall retain in their files the following traffic signal-related items for a minimum of 2 years after cancellation or removal of the signal:

- Engineering and traffic studies.
- Municipal applications and agreements.
- Traffic signal permits and wiring diagrams, including all subsequent revisions.
- As-built construction plan sheets showing the applicable intersections.

4.5 Design

4.5.1 General

Signal design encompasses a variety of items which must be considered prior to and during the development of traffic signal permit drawings, construction plans, and specifications. Drawings and specifications define the proposed work to ensure that the traffic signals will operate in a safe and efficient manner. Drawings include those items required by regulations, including, but not limited to, a plan view of the location, traffic signal layout, phasing diagram, timing plans, required signing and markings, construction details, and tabulations.

4.5.2 Signals Near Grade Crossings

The Public Utility Commission (PUC) has jurisdiction over all highway-railroad crossings, whether crossing a highway at grade, above grade, or below grade. Therefore, the following policy shall apply when traffic signals are to be located within 300 feet of a railroad or transit highway grade crossing or when railroad preemption is proposed:

(a) <u>Construction Projects when Formal Application to PUC is Required</u>.

The Department must follow the procedures of Section 7.6 of the Design Manual Part 1A (DM-1A) by making formal application to the PUC if adjusting a railroad facility, or if the highway approaches to the crossing are to be altered, relocated, or a new crossing is to be constructed. When, as a part of such alterations, a traffic signal plan is included in the Department's construction contract, Districts must submit the information required in the following three basic stages:

<u>Stage 1</u>.

- Determine if a traffic signal is required and warranted.
- Obtain from the railroad involved the information required in the Appendix of Publication 149.
- Determine if the local officials will assume responsibility for the maintenance of the signal.
- Prepare a preliminary plan for the traffic signal and obtain BHSTE's approval.

- Submit the plan to the railroad for their review and comment.

Stage 2.

- Obtain BHSTE approval of the final traffic signal plans.
- Traffic signal agreements executed by the local officials and approved by the Department or a letter of intent from the local officials shall be a part of the plans for the PUC. Districts shall submit this information to the Chief Utility Engineer in the Bureau of Design for their submission to the PUC, except when the District Grade Crossing Engineer is authorized to make submissions directly to the PUC.
- The District Traffic Unit should attend the field investigation and conference.

Stage 3.

- After the field investigation and conference, all comments pertaining to the traffic signal plan shall be resolved and a final plan prepared and approved by BHSTE, and afterwards made a part of the Department's construction plan.
- The District should obtain railroad approval in writing prior to the PUC hearing. As a result of the hearing, the traffic signal plan will then be entered as a Department exhibit.
- The PUC will grant final approval of the traffic signal as part of the PUC Order.

(b) Traffic Signal Installations Required by PUC Complaint or Investigation Docket.

In cases where the PUC under a Complaint or Investigation Docket orders the Department to prepare a study or proposal for the installation of traffic signals near grade crossings, the District will develop the necessary plans and obtain BHSTE concurrence. Upon receipt of concurrence, Districts shall forward the recommendations to the Chief Utility Engineer in the Bureau of Design for their submission to the PUC, except when the District Grade Crossing Engineer is authorized to make submissions directly to the PUC.

(c) PennDOT Construction when Formal Application to PUC is not Required.

The District should follow the above Stage 1 procedures in all other instances when existing or proposed traffic signals are included in the Department's construction contract.

However, when the final plan is accepted by BHSTE and a traffic signal agreement has been executed by the local officials or a letter of intent has been received, the District, if authorized, will submit the traffic signal plan to the PUC for approval.

Upon receipt of PUC approval, the District will include the traffic signal plan in the Department's construction plan.

(d) New Request or Revision – Construction if required is by others.

New requests or revisions to traffic signals, requested by the local officials or developed by the District, will follow the procedure set forth in "(c)" above, except that upon receipt of PUC approval, a permit will be issued to the local officials by the District Executive.

Since the traffic signals are to be a part of the Department's construction project in "(a)" and "(c)" above, the District shall issue a permit with a copy of the plan sheet to the local officials after the construction project is let and awarded. A final as-built plan with wiring diagrams for the controller, accessory equipment, detectors, signal heads, and interconnecting equipment should also be furnished to the local officials upon finalization of the installation.

4.5.3 Signal Plans

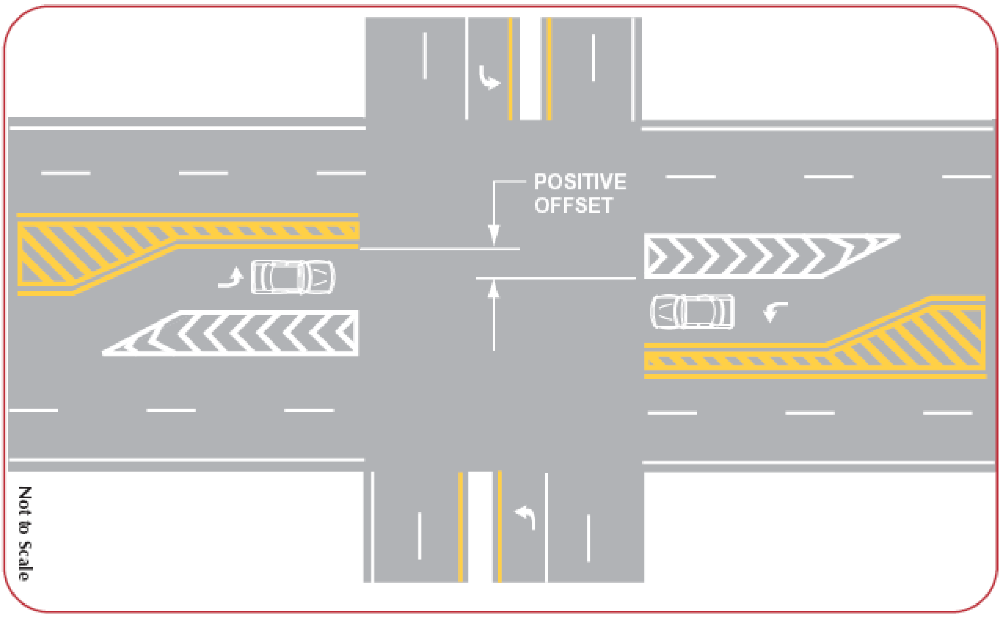
Develop preliminary and final signal plans as indicated in the *Traffic Signal Design Handbook* (Publication 149).

4.5.4 Positive Offset for Opposing Left-Turn Lanes

Whenever possible, it is desirable to design opposing left-turn lanes to provide positive offset to minimize the obstruction of sight distance. To accomplish this, align the left edge of both left-turn lanes to the left of the right edge of the opposing left-turn lane as illustrated in Exhibit 4.5-A. A minimum 3-foot positive offset will usually eliminate the sight-distance problem cause by a vehicle in the opposing left-turn lane.

On undivided highways, this positive offset can sometimes be provided by the use of solid channelizing lines in accordance with the following figure from the FHWA *Travel Better, Travel Longer: A Pocket Guide to Improve Traffic Control and Mobility for Our Older Population*.

Exhibit 4.5-A Positive Offset for Left-Turn Lanes



4.5.5 Left-Turn Signal Phasing

In addition to the left-turn signal phasing criteria in Publication 149, consider the following information and sound engineering judgment in future studies to determine the type of traffic signal phasing for left-turn movements at intersections.

Please note that in Section 1.0 (Volume) on Page C-14 of Publication 149, the conflict factor and the minimum approach volume of two left turns for each existing cycle during two or more 1-hour periods of a normal weekday are not the only criteria when determining whether a left-turn phase is required. Rather, the conflict factor and the left-turn volume are a guide and a good starting point to evaluate the use of left-turn phasing. As indicated in Publication 149, a left-turn signal phase may be considered if either the volume criteria (conflict factor/minimum left-turn volume) or other geometric and operational characteristics satisfy the various criteria.

Other criteria that should be considered when determining if a left-turn signal phase is appropriate are as follows:

- A minimum of five correctable reported crashes have occurred within a continuous 12-month period over the last 3 years. This criterion allows the consideration of both correctable, non-reportable crashes and correctable, reportable crashes over the most recent 3-year period. Non-reportable crashes must be police-verified and documented to enable consideration (i.e., a police crash report is available).

- If a left-turn phase will reduce delay and improve the overall level of service.

- When drivers are having difficulties making left turns and the municipality desires a left-turn phase, perform a capacity analysis to determine if a left-turn phase would increase delay. Whenever possible, the municipality and the Department should reach consensus on the incorporation of a left-turn phase based upon safety concerns and the level of increased delay. In general, avoid a solution that lowers an intersection into Level of Service E.

- Signalized intersections with left turn lanes (either new installations or in conjunction with a major permit change) should be analyzed to determine if a left turn phase would be beneficial to safety and traffic flow.

- A signalized intersection without a left-turn phase may have less turning traffic than adjoining or nearby signalized intersections with left-turn phases. A portion of the driving public may migrate to an intersection with a left-turn phase for safety and security reasons (i.e., they may find it easier to make the turn there). Therefore, installing a left-turn phase may increase the turning traffic at that intersection and reduce left-turn traffic at adjoining or nearby signalized intersections with existing left-turn phases.

In addition to considering the geometric and operational characteristics already mentioned in Publication 149 and some of the factors mentioned above, consider a protected/prohibited left-turn phase if any of the following conditions exist:

- Dual left-turn lanes.

- Three or more opposing through lanes.

- Multi-legged intersections with more than four approaches.

- Approaches that have experienced five or more crashes (including non-reportable) of a type that would be susceptible to correction by the protected/prohibited phasing within a continuous 12-month period over the most recent 3 years. (Non-reportable crashes must be police-verified and documented to enable consideration.)

- Approaches with significant non-correctable, sight distance deficiencies, including deficiencies created by stopped opposing left-turn vehicles.

Before implementing a protected/prohibited left-turn phase, examine the left-turn lane storage length to determine if adjustments are necessary.

This information applies to new traffic signal installations. However, if there is a known safety problem or a major permit change (such as pole replacements) at existing signals with protected/permitted left-turn phasing; review the above conditions for possible protected/prohibited phasing. Consider changes in signal phasing at such sites if there is a documented problem involving reportable and non-reportable crashes attributed to the protected/permitted left-turn phasing.

4.5.6 LED Traffic Signals

Project Use

On Department construction or highway occupancy permit projects, specify red, yellow, and green LED vehicular signal head modules, yellow and green LED arrow traffic signal modules, and LED pedestrian signal head modules for all new vehicular and pedestrian signal heads when the municipality has previously installed, or plans to install, LED signals at other intersections within their municipality. Proprietary specifications, requiring specific makes and models of LEDs, should not be included in State or federally-funded projects.

In addition, Engineering Districts should encourage the use of LED technology on projects let and funded by municipalities.

LED Minimum Useful Life

If an LED module fails to function as intended due to workmanship or material defects within the first 60 months from the date of delivery, the manufacturer shall replace or repair the module at no additional expense to the local municipality or the Department. (This is a condition on the Department's certificate of approval (COA) issued to LED manufacturers.)

The relationship of how the luminous intensity of LEDs degrades over time has not been firmly established. However, LED manufacturers are required to replace or repair, within the first 60 months from the date of delivery, any red, yellow, green, or arrow LED signal module that does not meet the luminous intensity described in the applicable specification. Therefore, it is strongly recommended that the owner and/or maintainer of the traffic signal equipment obtain the actual delivery date of each LED signal module, and then perform and document visual inspections of each LED signal head module within the first 60 months from the date of delivery. The supplier and/or manufacturer should be notified of any observed degradations during this initial 60-month period.

After 60 months from the date of delivery, the owner and/or maintainer of the traffic signal equipment should continue the visual inspections of LED signal modules. When a noticeable degradation appears, either replace the LED signal module or test it (using the 44-point or other test prescribed in the specifications) to ensure that it continues to have proper luminous intensity.

Sample Special Provisions

The following sample special provisions can be used:

1. Light Emitting Diode (LED) Vehicular Signal Head Modules. Furnish and install vehicular signal heads of the type indicated. Comply with Publication 408, Sections 955 and 1104.06, except as follows:

    - Furnish a light emitting diode (LED) [indicate size and color] module for the signal indications.

    - Provide LED modules that have a Provisional or Sale Certificate of Approval issued by the Pennsylvania Department of Transportation.

    - Install LED modules according to the manufacturer's instructions.

2. Light Emitting Diode (LED) Pedestrian Signal Head Modules. Furnish and install pedestrian signal heads of the type indicated. Comply with Publication 408, Sections 955 and 1104.06, except as follows:

    - Furnish a light emitting diode (LED) module for the [indicate size and color/symbol] signal indications.

    - Provide LED modules that have a Provisional or Sale Certificate of Approval issued by the Pennsylvania Department of Transportation.

    - Install LED modules according to the manufacturer's instructions.

4.5.7 Temporary and Portable Traffic Control Signals

Temporary and portable traffic control signals may be used to control one-lane, two-way traffic. They may also be used for other special applications such as a highway or street intersections with a temporary haul road or equipment crossing.

The design and application of portable traffic control signals must conform with the applicable requirements of the Department's certificate of approval issued to the manufacturer for portable traffic control signals, and to any special requirements defined in the Temporary Traffic Control (TTC) Plan.

The Work Zone Traffic Control Guidelines (Publication 213) provides the following traffic control figures and related information in Appendix A for temporary and portable traffic control signals:

- Figure PATA 26e L — *"Long-Term Stationary Operation Two-Lane, Two-Way Roadway – Temporary Traffic Control Signals"*
- Figure PATA 26e PL — *"Long-Term Stationary Operation Two-Lane, Two-Way Roadway – Portable Traffic Control Signals"*

- Figure PATA 26e PS — *"Short-Term Stationary Operation Two-Lane, Two-Way Roadway – Portable Traffic Control Signals"*

4.5.8 Automated Red Light Enforcement (ARLE)

§3116 of the Vehicle Code provides legal authority for a pilot program of automated red light enforcement (ARLE) in first class cities (i.e., the City of Philadelphia). All intersections to use ARLE must have advance written approval of the Secretary of Transportation based upon recommendations from local officials, the Engineering District, and the Bureau of Highway Safety and Traffic Engineering.

4.5.9 Preemption and Priority Control

(Reserved.)

4.5.10 Overhead Street Name Signs

Street name signs are invaluable to unfamiliar drivers, and overhead street name (OSN) signs that are near traffic signals are especially valuable because drivers should already be looking in that general area. Therefore, when roads have generally accepted road or street names, OSN signs (D3-4 and D3-5) are required at all new signalized intersections.

The addition of OSN signs at existing traffic signals is encouraged when revising the traffic signal permit whenever a bucket truck will be required at the intersection to make other physical changes to the signals.

The D3-4 and D3-5 sign standards depict the use of upper and lower case legend. All new OSN signs within Pennsylvania should use Clearview legend in accordance with the sign standards.

In an attempt to maximize legibility, avoid using suffixes such as "Rd" and "St" except as follows:

- For alphabetic or numeric street names (e.g., K St and 13th St).
- To distinguish between other nearby roads with the same base name (e.g., Schuylkill Dr and Schuylkill Ave), or
- When the name of the roadway is a cardinal direction (e.g., East Ave and North St), or anytime that the lack of the suffixes could be confusing.

When retrofitting existing traffic signal installations, it may be necessary to allow signs to rotate to reduce wind loading.

4.6 Equipment

4.6.1 Use of Accessory Equipment

(Reserved.)

4.6.2 Request for Proprietary Specifications

Section 1.1.9 discusses the use of proprietary items. Occasionally the Department receives requests from municipal officials to specify a particular brand of traffic signal equipment in the special provisions for a project. Although it is the policy of the Department and the Federal Highway Administration to use non-proprietary specifications on all projects, review requests of this type to determine if this equipment is essential for compatibility with existing signal equipment or if it is in the public interest to obtain major signal hardware from a particular manufacturer.

It is important to consider any proprietary specifications early in preliminary design to obtain any necessary approvals and to help the municipal officials understand all cost implications.

Therefore, if there is more than one non-proprietary product or material that will fulfill the requirements for an item of work or project, prepare the PS&E for the project to allow all such materials providing:

- All products are of satisfactory quality and are equally acceptable based on an engineering analysis.
- The anticipated prices for the related items of work are approximately the same.

To be in the public interest, the municipal officials must document that at least one of the following is true:

1. More than 75 percent of the existing traffic signal equipment is from one particular manufacturer, they have a substantial inventory of that manufacturer's spare parts, and their maintenance personnel have received extensive training and are experienced in the installation and maintenance of this brand of equipment.

> A municipality that uses a signal maintenance contractor for major controller repairs or installation may still satisfy the above criteria. However, they would not qualify if the contractor maintains equipment for various municipalities using other brands of equipment.

2. Reasons why a "generic" material description cannot be used, or at least three companies cannot be specified. Examples of acceptable justification are:

    - The item is essential for synchronization with existing highway facilities and no suitable alternatives exist.

    - The item is being used for research or for relatively short sections of road for experimental purposes.

    - No other item exists that is of acceptable quality.

After the municipal officials provide written documentation that their signal program satisfies these criteria, the District should review the information and submit it with their recommendation to BHSTE for concurrence.

4.7 Operation

4.7.1 Flashing Operation of Traffic Signals

New Traffic Signal Installations

In an effort to better acclimatize motorists to the presence of a new traffic signal installation, it is normally desirable to put the new traffic signals on flash prior to the start of the normal stop and go (red/yellow/green) operation.

Flashing new traffic signals is discussed as part of the operational testing in Section 4.8.3, and in Section 950.3 of Publication 408.

During Periods of Low Traffic Volumes

Consider flashing operation when the total vehicular volume drops below 325 (urban) or 225 (rural) vehicles per hour for a period of 4 or more consecutive hours. Although these volumes are the basic criteria used to establish the hours of flashing operation, this does not preclude the use of time-of-day to specify the periods of flashing operation for signals at shopping centers, industrial plants, or locations where periods of flashing operation are needed. For these situations, the District will indicate the hours of flashing operation on the permit diagram.

Do not use a flashing mode of operation if an engineering and traffic study indicates that a sight distance problem exists; the number or severity of crashes increased when the signals were flashed; or there was an increase in observed unsafe conflicts.

4.8 Construction

4.8.1 General

Traffic signals on state or local highways may be installed by contracts administered by the local authorities, municipal forces, contracts related to highway occupancy projects, or Department construction contracts.

Local authorities are responsible for the inspection of the traffic signals that they have installed. The Engineering District will provide inspection for signals that are included in Department contracts or are installed in conjunction with Highway Occupancy Permits issued by the Department for work on State highways.

The Engineering District will review the completed permitted signal installation to verify that it complies with the signal permit requirements for signal design, controller operation, signing, and pavement markings.

Costs associated with the installation of traffic signals shall be the responsibility of the municipality where the signal is located unless the municipality enters into an agreement with another party who assumes all or part of these costs. Signal installation costs associated with Highway Occupancy Permit projects are generally funded by the permitee. The Department will participate in the cost to install or upgrade a traffic signal only when a Department construction project creates the need for a new signal or the need to revise an existing signal.

4.8.2 Requests for Change of Signal Equipment

On some Department construction projects, the municipality has objected to the type of traffic signal equipment proposed for use by the contractor and has requested that alternate equipment be installed.

If the District agrees with the change(s) in traffic signal equipment the municipality wishes to make, and if the proposed equipment adequately provides the required functions and operation and has a certificate of approval, the request for change may proceed as follows:

- After the contract award, the municipality should request permission to make a change and to negotiate this change with the contractor. The municipality's request must include the specific item or items they wish to change.

- If the District concurs with the municipality's request, obtain all necessary approvals, after which the District may grant the municipality permission to negotiate with the contractor.

If the change requested by municipality increases the cost, the municipality must bear the entire cost increase. If the cost is less, the Department must receive the benefit of the cost reduction.

4.8.3 Operational Testing and Guarantee

Section 950.3 of *Publication 408* specifies that for presently unsignalized intersections, the initial turn-on is to be in the flashing mode for 3 to 7 days. In light of current practices, the number of days may be 3 to 5 days providing that it includes a minimum of 2 weekdays prior to beginning the normal stop-and-go mode of operation.

With reference to the 30-day operating test specified in Publication 408, it should be noted that failures which are not attributed to defective equipment or faulty workmanship should not be cause for restarting the 30-day test. As an example, damage caused by crashes not resulting from malfunctioning equipment or damage to underground electric conductors caused by pavement break-up should not be cause for restarting the 30-day test at Day 1. However, the test period should be temporarily halted for any such malfunctions and resumed only when all equipment is restored to operating condition. If the operation test is prolonged to an extent that the equipment should be serviced in accordance with the manufacturer's recommendations, the contractor shall provide that service as prescribed in Publication 408.

Moreover, Publication 408 specifies that all traffic signal installations that are part of a traffic signal system, as specified in the proposal or shown on the drawings, shall be tested as a unit. Therefore, it is imperative that each signal installation in a signal system be tested individually, and also that all signal installations within a system be tested as a whole with a separate system test.

The intent of Publication 408’s 180-day guarantee period from the date of completion of the 30-day test is for the contractor to guarantee the in-service operation of mechanical and electrical equipment, related components, and the controller assembly during this period. The contractor will be responsible for replacing faulty workmanship and repairing or replacing defective materials or equipment, and correcting malfunctions. On the other hand, damages resulting from crashes, damages not resulting from defective or malfunctioning equipment or faulty workmanship may be beyond the contractor's control and may be the responsibility of the signal owner. The aforementioned examples are not all inclusive and the existence of one or more does not automatically relieve the contractor of his responsibility. Use engineering judgment to determine the responsibility for the failures. When the failure or malfunction is the responsibility of the contractor, the Department, rather than the municipality, should notify the contractor.

4.8.4 Acceptance of Traffic Signal Materials

Accept the following material on construction projects by certification:

Traffic Signal Support, Mast Arm	951.2
Traffic Signal Support, Strain Pole	951.2
Traffic Signal Support, Pedestal	951.2
Controller Assembly	952.2
Electrical Distribution System	954.2
Signal Heads	955.2
Detectors	956.2
Interconnect Cable	957.2

In accordance with Department of Transportation Specifications, the Engineer may accept certification of materials in lieu of inspection and testing. Therefore, all traffic signal materials included in Publication 408 and the special provisions will be accepted by certification. In addition, those traffic signal materials which must have a Certificate of Approval issued by the Department shall be tabulated and submitted to the Engineer by the contractor. Acceptance of the materials tabulated for traffic signal items means only that the materials are suitable for the proposed use.

The materials tabulation shall be submitted for each project within 3 weeks of the Notice-to-Proceed. This tabulation shall include the following project information:

- State Route
- Section
- County
- Prime Contractor

Also, provide the following information for each different item of material:

- Identity of Material
- Manufacturer's Name
- Manufacturer's Model Number
- Department's Certificate of Approval Number

Accessories, such as hardware, brackets, supplies, minor devices, etc., necessary for the proper installation of traffic signals but not in Publication 408 or the special provisions are not intended to be submitted for testing and acceptance. However, all load-carrying accessory materials shall be certified as to their structural adequacy for the loads indicated.

Request catalog cuts only if they are necessary to provide clarification of special features or functions of the proposed materials.

The contractor shall submit three copies of traffic signal materials tabulation to the District Executive, attention Assistant District Executive, Construction, for review and acceptance. The review by the District staff should include contact with the appropriate municipal officials, if necessary. The District should notify the contractor by letter, of the acceptance or rejection of his submission. If rejected, indicate the reason in the letter.

4.8.5 Final Acceptance of a Traffic Signal

Final inspection of a traffic signal that has been installed on a Department construction project should be similar to that used when making a final inspection of municipally-installed signals. Therefore, the District Traffic Units should adopt the following procedure, as a minimum, to ensure municipal input is obtained prior to assigning them signal ownership:

- Notify the municipality in writing of the starting date of the 30-day test, and any extensions thereof, and advise them that they must be present. It is also recommended that the consultant/designer be present.
- Upon successful completion of the 30-day test, invite the municipality to any final inspection meeting. Verbally advise the municipality, with a written confirmation, of the Department's acceptance of the signal and that they are now responsible for the continued operation and maintenance of the signal in accordance with the provisions of the traffic signal application and permit, the traffic signal maintenance agreement, and Commonwealth regulations.

4.8.6 Guidelines for Traffic Signal Construction Inspection

4.8.6.1 Material Approval

BHSTE issues a Certificate of Approval (COA) to manufacturers authorizing the sale of specific electrically-operated traffic signal equipment. Contractors are responsible to tabulate a list of the proposed equipment within 3 weeks after the Notice to Proceed, and to submit the list to the Department. BHSTE also approves standard loadings for traffic signal supports, but shop drawings are submitted, and design calculations will be required for non-standard loadings.

Districts should accept materials when accompanied by a certification stating compliance of the material to the appropriate specification.

Three copies of warranties, guarantees, instruction manuals, wiring diagrams, field wiring diagrams, and parts lists are required and one copy of each should be retained as follows:

- Department file.

- Municipal file
- Retained in the controller cabinet.

4.8.6.2 Testing

The contractor is required to perform tests on various items during the construction of a traffic signal project. Documentation of some test results is required by the contractor, all other tests will have to be documented by the inspector. Following is a tabulation of required tests with references to the section of the specification requiring the test.

The following tests must be performed using equipment manufactured for the specified test. The equipment should contain the manufacturer's instructions on its use.

- Sec. 950.2(e) Controller Assembly Shop Test.
- Sec. 950.3 Traffic Signal Installation 30-Day Operating Test.
- Sec. 952.3(a) Malfunction Management Unit or Conflict Monitor Function Test.
- Sec. 952.3(b) Time Clock Evaluation.
- Sec. 953.3(b) 30-Day Systems and Communications Test.
- Sec. 954.3(i) Traffic Signal Circuits Test for Short Circuits, Unspecified Grounds, and Resistance to Earth Ground.
- Sec. 956.3(a) Loop Detector Leakage Resistance, Series Resistance, and Inductance Testing.
- Sec. 956.3(b) Earth’s Magnetic Flux Test and Magnetometer Detector Leakage and Series Resistance Test.
- Sec. 956 3(c) Magnetic Detector Leakage and Series Resistance Test.
- Sec. 956.3(d) Pedestrian Pushbuttons Evaluation and Pedestrian Phase Timing Test.
- Sec. 957.3(a) Traffic Signal Communications Test for Short Circuits, Unspecified Grounds, and Resistance to Ground.

4.8.6.3 Revisions.

Consult the District Traffic Engineer or designated representative prior to approving any changes from what is shown on the drawings.

4.8.6.4 Foundations

Check the following items:

- Correct location?
- Is there adequate clearance between traffic signal support and overhead utility lines at this location?
- Is excavation large enough for the required foundation?
- Prior to placing cement concrete, ensure that the following items are in-place and adequately supported:
    - Is the ground rod in place and was it resistance tested? Is an additional ground rod at least one length away from and bonded to the first?
    - Is reinforcement in place with proper clearance between it and the outside edges of foundation?
    - Are anchor bolts the correct size, and free of dirt, grease, or other foreign matter? Is the projection of bolts sufficient to adjust the rake of the support?
    - Is the conduit the proper size and quantity?
- Are the top of foundation forms the correct size and elevation, and are provisions made for chamber on edges of foundation?
- Is expansion joint filler between existing pavement and foundation?
- Are the conduits capped and threads on anchor bolts protected during placement of concrete?
- Is the cement concrete cured for 72 hours before placing support?

4.8.6.5 Traffic Signal Supports

Check the following items:

- Is the approval number on base plate of shaft and flange plate of arm?
- Is there any damage to galvanizing?
- Is support placed on the foundation with the proper rake to provide vertical set when the load is applied?
- Is the ground wire from the ground rod connected directly to the lug inside the handhole, with no other ground wires connected to

the lug? Are all other ground wire(s) in base of support connected to the aforementioned ground wire with an appropriate clamp?

- Is the cover on the handhole?
- Is the cap on top of shaft?
- Is the cap on end of arm (mast arm)?
- Are grommets protecting all holes for wire outlet in arm (mast arm)?
- Is the wire entrance near top of shaft?
- Are the span wire clamps and tether wire clamps in correct location?
- Is the proper sag in span wire with load applied?
- Are signal cables properly lashed to span wire?
- Is the ground wire connected to the span wire and strain pole?
- Are the electrical cables inside shafts supported with clamp attached to cable support in top of shaft?
- Is mortar between the foundation and base plate, with a drain hole?

4.8.6.6 Controller Assembly and Traffic Signal Systems

Since a representative of the District Traffic Engineer will be present when a traffic signal is placed in operation, the inspector should be primarily concerned with installation of the cabinet and determining that all required equipment is contained in the cabinet. To aid in identification, much of this equipment is labeled with the manufacturer's name, serial number, and model or part number.

- Was controller assembly shop tested?
- Is the cabinet located and oriented so that you can observe the equipment in the cabinet and the signals at the intersection simultaneously?
- If the cabinet is on a traffic signal support, is it at the required elevation and attached with the required hardware? Does it project into the accessible route? Does it meet ADA requirements for

detection of protruding objects by a visually impaired person? (See Section 307 in the ADA requirements, available at http://www.access-board.gov/ada-aba/final.pdf).

- If the cabinet is on a cement concrete foundation, refer to Publication 148 standard for a traffic signal support foundations. Is a drain hole provided in the foundation, and is caulking between the bottom of cabinet and the foundation?
- Is the cabinet the proper size?
- Are devices in cabinet mounted on a shelf or a rack?
- Is the cabinet equipped for maximum phase capability of the controller unit?
- Is the cabinet ventilated?
- Does the exhaust fan operate as required?
- Are there an adequate number of terminal blocks?
- Are controller modules and load switches labeled as to function?
- Are all switches, controls, and indicators clearly identified?
- Flasher?
- Time clock?
- Are switches in the police panel?
- Are there pushbuttons for simulated detector operation?
- Do you have written documentation of the malfunction management unit or conflict monitor test?
- Does the preemption work? Extensive testing is required for intersections with railroad preemption.
- Are spare and reserved conductors terminated on the ground bus?
- Electric Load Center:
    - Are the circuit breakers sized in accordance with load?

- Is a ground fault circuit interrupter provided for the duplex receptacle?
- Surge Protector?
- Radio Frequency Interference Filter?

- Grounded conductors on neutral bus?
- Grounding conductors on grounding bus?
- All stranded conductors provided with insulated spade terminals?
- Are all wire terminal screws tightened?

4.8.6.7 Electrical Distribution

Check the following items:

- Electric service.
  - Service pole location.
  - Cabinet for disconnect.
  - Fuses rating for not less than load.
  - Switch rating not less than load.
  - Surge protector.
  - Ground rod resistance tested.
  - Service wire sized in accordance with load.
- Trench. Excavated to required depth and width? Was the pavement cut prior to beginning excavation with no large or sharp rocks in first layers of backfill? Was the backfill properly compacted?
- Conduit. Is it cut normally with saw? If cut by other means, was the conduit distorted or crushed? Were bends made without kinks? Is all metallic conduit bonded?
- Grounding. Are all ground rods in a signalized intersection interconnected with ground wire? Note, in a span wire installation, the span wire is used as a ground wire between strain poles.
- Signal Cable. Was it damaged when pulled through conduit and traffic signal supports? Is the cable only spliced in the base of traffic signal support? Are all spliced conductors in base of support identified with cable tags which indicate phase and

function? Are all unused conductors terminated on grounding bus in the controller cabinet? Were all signal circuits tested for short circuits, unspecified grounds, and resistance to ground?

- Junction Box. Is a drain hole in the bottom and coarse aggregate below box for drainage? Do all junction boxes satisfy the minimum dimensions; i.e., boxes may be provided with larger dimensions?

4.8.6.8 Signal Heads

Check the following items:

- LED modules or lamps correct size?
- Where lamps are used, if filament in lamp is "U" shaped, is opening in filament in the up position? Can this be accomplished by rotating socket?
- Is the LED module or lens oriented according to manufacturer's recommendations (e.g., top of lens or module at top of signal head)?
- Are signal heads located correctly in regard to roadway; i.e., aimed correctly and plumb?
- Is the top of all overhead signals facing an approach at the same elevation?
- Are all specified visors, louvers, and backplates in place?
- Is the vertical clearance of signal heads proper?
- Are the signals wired in accordance with specified color code?
- Is a drip loop in cable provided at entrance to signal head?
- If present, were optically programmed signals programmed in accordance with the instructions provided with the signal head?
- Are arrow indications oriented in the proper direction?
- Are non-operating new signals bagged?
- Are existing signals remaining in operation until the new signals are put into operation?

4.8.6.9 Detectors

Most vehicle detectors are loop detectors; but other forms of vehicle detection such as video are seeing increased use. The following information pertains to loop detectors, but it can be as a reference for other forms of detection along with manufacturer's recommendations.

Loop Detectors

Check the following items:

- Are the sensors (loops) the correct size and located as required, avoiding all unstable pavement, cracks, and joints if possible?
- Will the loops detect all vehicles, including bicycles and motorcycles?
- Are the saw slots for sensors the correct depth and width?
- Is the inside of the slot smooth with no sharp edges?
- Does the conduit for the sensor lead from roadway to junction box or controller cabinet as required?
- Was the saw slot blown free of moisture and debris with compressed air prior to placing sealant?
- Was the sensor wire placed in slot, and held in slot with short pieces of closed-cell polyethylene foam or small pieces of cable jacket?
- Are the number of turns of sensor wire as recommended to provide required inductance?
- Were the two leads from sensor twisted together before placed in conduit?
- Was the sensor wire tested for leakage and series resistance?
- If sensor wire is not spliced to lead-in immediately, were the ends of sensor wire and tubing sealed to prevent entrance of moisture?
- Are the sensor wires securely spliced to lead-in cable?
- Was duct seal placed in the end of conduit before placing sealant?

- Was the sealant prepared in accordance with manufacturer's instructions and placed in slot without voids and without overflowing slot?

- Are there any splices in the lead-in cable (none are permitted)?

- After the sensor was spliced to lead-in cable, was the inductance measured and recorded for each sensor? Is a copy of the test in the controller cabinet?

- Were the settings on the amplifier properly adjusted?

- Are the amplifiers labeled as to function?

<u>Pedestrian Pushbuttons</u>

Located to be readily visible and accessible to a pedestrian that is in position to use the crossing.

- Trench

- Conduit: Refer to Section 954 – Electrical Distribution

- Junction Box

4.9 Maintenance

4.9.1 General

Traffic signal maintenance shall be provided by the local authorities unless they have an agreement with another entity indicating otherwise. Maintenance should be done in accordance with the Department's Publication 191, *Guidelines for the Maintenance of Traffic Signal Systems.* If state or federal funds have been used to pay for the installation of the signal, maintenance must be provided as indicated in Publication 191.

Costs associated with the operation and maintenance of traffic signals shall be the responsibility of the municipality where the signal is located unless the municipality has an agreement with another party who assumes all or part of these costs.

4.9.2 Annual Program

As described in Section 7.0 of Publication 191, the Department may periodically evaluate signal maintenance performance of the municipalities involved in State or federal-aid projects to determine how well the municipality is fulfilling its maintenance responsibilities on these projects. Municipalities may use this information as a guide in determining their capability and to aid in deciding whether to use a private contractor for traffic signal maintenance.

4.9.3 Project Reviews

Proper maintenance of traffic signal equipment is necessary to ensure the efficient and safe movement of traffic. Therefore, the Department may: (1) make itself available to aid and (2) evaluate the maintenance capability of the responsible municipality prior to the final design of any project which includes traffic signals.

FHPM 6-4-3-1 requires that all completed projects constructed with federal-aid funds be maintained at an acceptable level of physical integrity and operation, consistent with its original design standards. Failure on the part of the state or municipality to do so can result in the withholding of future federal-aid funds unless and until the appropriate corrective action is taken by the responsible agency.

4.9.4 Maintenance Evaluation Procedures

4.9.4.1 Overview

Traffic signals may not be installed or revised using state or federal funds unless the municipality is able and willing to properly operate and maintain the new equipment. A determination of their ability may be made by the Department following a review of the municipality's staff, budget, maintenance shop, spare parts inventory, and most importantly, their past performance in maintaining traffic signals.

If a municipality is incapable or unwilling to perform the required maintenance of the proposed new equipment, one or more of the following may be considered:

- The proposed new equipment may be changed to a type of equipment that provides the same functions and can be maintained by the municipality.
- The municipality may be encouraged to upgrade its staff, budget, or inventory to allow for proper maintenance.
- The municipality may retain a qualified, competent contractor to provide maintenance.

If none of the above options (or any other which will ensure proper maintenance) are possible or agreeable, signal improvement projects in the municipality may:

1. not be advanced to final design or construction using federal or state funds,
2. be deleted from the project if it is part of a larger construction project, or
3. be designed and constructed entirely with local funds.

In order to prevent wasted design costs, it is desirable to make the determination of municipal maintenance capability prior to initiation of final design. At that time, the municipality will be required to enter into an agreement that details the signal maintenance effort that must be provided throughout the useful life of the proposed improvement.

The agreement, for those municipalities who are judged to be currently providing an adequate level of maintenance, should describe the existing function (staff, budget, etc.) and require that, as a minimum, it be retained.

The agreement must list those changes that are to be initiated and provided throughout the useful life of the proposed project. Typically, the required changes will include items such as increased staffing, additional training for existing staff, budget increases, retention of a maintenance contractor, etc.

4.9.4.2 Evaluation of Maintenance Function

It is important to consider not only the adequacy of an existing maintenance function, but also the ability of the municipality to properly maintain additional signal equipment which may or may not be similar to the equipment they presently service.

"Ability to maintain" is a complex, many-faceted characteristic that cannot be simply or directly measured. Rather, each of its components must be studied in its own context and an overall rating of the function pieced together. The experience of the rater, the degree of cooperation offered by the municipality, the municipality's performance in relation to that of surrounding municipalities, and many other non-measurable factors may influence the final rating.

Recommended steps in the evaluation process are as follows:

1. Step 1 – Use of Maintenance Capability Checklist.

   As soon as possible after a project, which includes traffic signals, receives PMC approval for design, the involved municipality may be forwarded a copy of the checklist with instructions to complete and return it within a reasonable period of time.

   The completed form will be reviewed to determine if there are any obvious deficiencies in the municipality's maintenance functions.

   The use of the checklist and forms in Section 7.0 of Publication 191 will help determine if the municipality's preventative and response maintenance efforts are acceptable and at what level they are being performed. It must be remembered, however, that the level at which the response maintenance effort is being performed is not a qualitative measure of performance, but rather an indication of the type of response maintenance program being employed by the municipality.

   The checklist is not intended to be an end in itself and may be used in combination with an interview/meeting to arrive at a meaningful evaluation. Although many of the questions in the checklist may be answered with a “yes” or “no,” the complete answer to the question often can be ascertained via a discussion with the municipal officials who are directly responsible for the maintenance program.

2. Step 2 – Field Evaluation.

   The results of the checklist and interview procedure will, in many cases, paint a picture of the maintenance effort as it is perceived by the municipality. The true test, however, lies in its application, and can be more directly measured by reviewing the condition and operating characteristics of their existing signal equipment. In addition to what the checklist and interview reveal, the existing signal equipment must operate dependably and be returned to service quickly, when it malfunctions, in order for a maintenance function to be judged acceptable.

   The field review may be performed prior to, or immediately after, the interview/ meeting, but both should be done by the same individual(s). By conducting the field review prior to meeting with the municipality, however, it would be possible to identify the areas of weakness as evidenced in the field.

3. Step 3 – Ability to Maintain Additional Equipment.

   Completion of the first two steps should provide a clear picture of the type and number of traffic signals that can be properly maintained, given: (1) the number of personnel assigned to the function; (2) their level of experience and training; (3) the available inventory and equipment; and (4) the budget that supports them.

   The maintenance requirements of the proposed equipment should then be determined and compared to the municipality's capabilities.

   The two basic questions that need answers at this point are:

   1. Can the municipality provide adequate maintenance for the traffic signals proposed in this project?

   2. If not, what must the municipality do to ensure adequate signal maintenance throughout the useful life of the equipment?

      - Should the signal maintenance budget be increased? If so, by how much?

- Are additional maintenance personnel needed? How many and at what level?
- Do existing personnel require additional training?
- Should the municipality purchase specific maintenance equipment?
- Should a contractor be hired in lieu of using municipal personnel?

Based on the answers to the above questions, the signal maintenance agreement for the project should be prepared. A standard agreement is used for this purpose.

4.9.4.3 Documentation

Keep proper documentation of the procedures followed in evaluating and rating municipal maintenance functions. Documentation may include:

- Any completed maintenance capability checklist; signed and dated by the responsible municipal official.
- Brief narrative of the results of any field reviews including date, attendees, intersections reviewed, description of the visual and operating characteristics of the equipment reviewed, and recommended improvements.
- Minutes of any interviews/meetings including any amplification of answers to the checklist questions and an overall rating of the municipal maintenance function in each of the following areas that apply:
    1. Manpower – number, experience, training.
    2. Inventory – parts, equipment.
    3. Budget.
    4. Maintenance contractor – ability, dependability.
- Listing of the items that must be completed prior to construction of the project, and any other recommendations.
- Agreed upon action plan.